

# Dual ADC

The data except this specification conform to those of each product.

---

### 1. Product Type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

---

### 2. Description

The function of Dual ADC is described in this document. This function outputs 2 different gain data of same pixel. By combining these image, the image which is like high bit image or HDR image can be get. Dual ADC supports two modes. In combining-less mode, high gain and low gain lines are output alternately. In combining mode, the high gain line and the low gain line are combined inside the sensor, and then the 12-bits gradation compressed data is output.

FRAMOS Technologies Inc.  
Internal use only

Sony Semiconductor Solutions Corporation reserves the right to change products and specifications without prior notice.  
This information does not convey any license by any implication or otherwise under any patents or other right.  
Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

---

### 3. Features

- ◆ Dual ADC driving method
  - Without combination (Output high gain line and low gain line alternately)
  - With built-in combination (Output after combined high gain line and low gain line)
- ◆ Gradation compression (with built-in combination)
- ◆ AD Line identifier (without combination)
  - SAV / EAV Sync code (In case of SLVS output)
  - Data ID in Packet Header (In case of SLVS – EC output)
- ◆ Support mode
  - Normal mode (XVS / XHS sync)
  - Sequential Trigger mode
  - Fast Trigger mode
- ◆ Supported readout drive mode
  - All-pixel scan mode
  - ROI mode
  - Overlap ROI mode
- ◆ Support bit
  - Without combination: 8 bit, 10 bit and 12 bit
  - With built-in combination: 12 bit only
- ◆ Support interface
  - SLVS output (4 ch / 8 ch output)
  - SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane output)
- ◆ Range of gain setting
  - 0 dB to 24 dB: Analog Gain (0.1 dB step)
  - 24.1 dB to 48 dB: Analog Gain: 24 dB + Digital Gain: 0.1 dB to 24 dB (0.1 dB step)
  - 0 dB, 6 dB, 12 dB, 18 dB, 24 dB: Gain setting added to analog gain of high gain side

## Contents

|          |                                                |     |
|----------|------------------------------------------------|-----|
| 1.       | Product Type -----                             | 1   |
| 2.       | Description -----                              | 1   |
| 3.       | Features -----                                 | 2   |
|          | Contents -----                                 | 3   |
| 4.       | Register Map -----                             | 6   |
| 5.       | Dual ADC -----                                 | 29  |
| 5.1.     | Overview -----                                 | 29  |
| 6.       | Readout Drive Mode (without combination) ----- | 30  |
| 6.1.     | Readout Drive mode -----                       | 30  |
| 6.1.1.   | IMX530 -----                                   | 30  |
| 6.1.2.   | IMX531 -----                                   | 35  |
| 6.1.3.   | IMX532 -----                                   | 40  |
| 6.1.4.   | IMX535-AAQJ -----                              | 45  |
| 6.1.5.   | IMX535-AAMJ -----                              | 50  |
| 6.1.6.   | IMX536-AAQJ -----                              | 55  |
| 6.1.7.   | IMX536-AAMJ -----                              | 60  |
| 6.1.8.   | IMX537-AAQJ -----                              | 65  |
| 6.1.9.   | IMX537-AAMJ -----                              | 70  |
| 7.       | Drive Mode Setting (without combination) ----- | 75  |
| 7.1.     | Drive mode setting -----                       | 75  |
| 7.1.1.   | IMX530 -----                                   | 75  |
| 7.1.1.1. | All-pixel scan -----                           | 75  |
| 7.1.1.2. | ROI, Overlap ROI -----                         | 79  |
| 7.1.2.   | IMX531 -----                                   | 81  |
| 7.1.2.1. | All-pixel scan -----                           | 81  |
| 7.1.2.2. | ROI, Overlap ROI -----                         | 85  |
| 7.1.3.   | IMX532 -----                                   | 87  |
| 7.1.3.1. | All-pixel scan -----                           | 87  |
| 7.1.3.2. | ROI, Overlap ROI -----                         | 91  |
| 7.1.4.   | IMX535-AAQJ -----                              | 93  |
| 7.1.4.1. | All-pixel scan -----                           | 93  |
| 7.1.4.2. | ROI, Overlap ROI -----                         | 97  |
| 7.1.5.   | IMX535-AAMJ -----                              | 99  |
| 7.1.5.1. | All-pixel scan -----                           | 99  |
| 7.1.5.2. | ROI, Overlap ROI -----                         | 103 |
| 7.1.6.   | IMX536-AAQJ -----                              | 105 |
| 7.1.6.1. | All-pixel scan -----                           | 105 |

|           |                                                                                          |     |
|-----------|------------------------------------------------------------------------------------------|-----|
| 7.1.6.2.  | ROI, Overlap ROI-----                                                                    | 109 |
| 7.1.7.    | IMX536-AAMJ-----                                                                         | 111 |
| 7.1.7.1.  | All-pixel scan-----                                                                      | 111 |
| 7.1.7.2.  | ROI, Overlap ROI-----                                                                    | 115 |
| 7.1.8.    | IMX537-AAQJ-----                                                                         | 117 |
| 7.1.8.1.  | All-pixel scan-----                                                                      | 117 |
| 7.1.8.2.  | ROI, Overlap ROI-----                                                                    | 121 |
| 7.1.9.    | IMX537-AAMJ-----                                                                         | 123 |
| 7.1.9.1.  | All-pixel scan-----                                                                      | 123 |
| 7.1.9.2.  | ROI, Overlap ROI-----                                                                    | 127 |
| 7.2.      | The register settings for HMAX-----                                                      | 129 |
| 7.3.      | The register settings for SLVS-EC output 10 bit mode (IMX535, IMX536, IMX537 only) ----- | 129 |
| 8.        | AD Line identification method (without combination)-----                                 | 130 |
| 8.1.      | SLVS Output -----                                                                        | 130 |
| 8.2.      | SLVS – EC Output-----                                                                    | 131 |
| 9.        | Readout Drive Mode (with built-in combination)-----                                      | 132 |
| 9.1.      | Readout Drive mode -----                                                                 | 132 |
| 9.1.1.    | IMX530-----                                                                              | 132 |
| 9.1.2.    | IMX531-----                                                                              | 136 |
| 9.1.3.    | IMX532-----                                                                              | 140 |
| 9.1.4.    | IMX535-AAQJ-----                                                                         | 144 |
| 9.1.5.    | IMX535-AAMJ-----                                                                         | 148 |
| 9.1.6.    | IMX536-AAQJ-----                                                                         | 152 |
| 9.1.7.    | IMX536-AAMJ-----                                                                         | 156 |
| 9.1.8.    | IMX537-AAQJ-----                                                                         | 160 |
| 9.1.9.    | IMX537-AAMJ-----                                                                         | 164 |
| 10.       | Drive Mode Setting (with built-in combination)-----                                      | 168 |
| 10.1.     | Drive Mode Setting -----                                                                 | 168 |
| 10.1.1.   | IMX530 -----                                                                             | 168 |
| 10.1.1.1. | All-pixel scan-----                                                                      | 168 |
| 10.1.1.2. | ROI, Overlap ROI-----                                                                    | 172 |
| 10.1.2.   | IMX531 -----                                                                             | 174 |
| 10.1.2.1. | All-pixel scan-----                                                                      | 174 |
| 10.1.2.2. | ROI, Overlap ROI-----                                                                    | 178 |
| 10.1.3.   | IMX532 -----                                                                             | 180 |
| 10.1.3.1. | All-pixel scan-----                                                                      | 180 |
| 10.1.3.2. | ROI, Overlap ROI-----                                                                    | 184 |
| 10.1.4.   | IMX535-AAQJ -----                                                                        | 186 |
| 10.1.4.1. | All-pixel scan-----                                                                      | 186 |

|           |                                                                                          |     |
|-----------|------------------------------------------------------------------------------------------|-----|
| 10.1.4.2. | ROI, Overlap ROI-----                                                                    | 190 |
| 10.1.5.   | IMX535-AAMJ -----                                                                        | 192 |
| 10.1.5.1. | All-pixel scan-----                                                                      | 192 |
| 10.1.5.2. | ROI, Overlap ROI-----                                                                    | 196 |
| 10.1.6.   | IMX536-AAQJ -----                                                                        | 198 |
| 10.1.6.1. | All-pixel scan-----                                                                      | 198 |
| 10.1.6.2. | ROI, Overlap ROI-----                                                                    | 202 |
| 10.1.7.   | IMX536-AAMJ -----                                                                        | 204 |
| 10.1.7.1. | All-pixel scan-----                                                                      | 204 |
| 10.1.7.2. | ROI, Overlap ROI-----                                                                    | 208 |
| 10.1.8.   | IMX537-AAQJ -----                                                                        | 210 |
| 10.1.8.1. | All-pixel scan-----                                                                      | 210 |
| 10.1.8.2. | ROI, Overlap ROI-----                                                                    | 214 |
| 10.1.9.   | IMX537-AAMJ -----                                                                        | 216 |
| 10.1.9.1. | All-pixel scan-----                                                                      | 216 |
| 10.1.9.2. | ROI, Overlap ROI-----                                                                    | 220 |
| 10.2.     | The register settings for HMAX-----                                                      | 222 |
| 10.3.     | The register settings for SLVS-EC output 10 bit mode (IMX535, IMX536, IMX537 only) ----- | 222 |
| 11.       | Built-in combination Function (with built-in combination)-----                           | 223 |
| 11.1.     | Bit extension (D-range extension)-----                                                   | 223 |
| 11.2.     | Threshold and Data selection-----                                                        | 224 |
| 11.3.     | Gradation compression-----                                                               | 225 |
| 11.4.     | Area of Gradation compression-----                                                       | 226 |
| 11.5.     | Supports A/D 12 bits / output 8 bits-----                                                | 226 |
| 12.       | Gain Adjustment Function (without combination / with built-in combination)-----          | 227 |
| 13.       | Mode Transition-----                                                                     | 229 |
| 13.1.     | Mode Transitions-----                                                                    | 229 |
| 14.       | Note-----                                                                                | 229 |
|           | Revision History-----                                                                    | 230 |

#### 4. Register Map

The register map for Dual ADC is shown below.

Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 02h in Write mode. (I<sup>2</sup>C:30\*\*h) (IMX530, IMX531, IMX532)

| Address |                  | bit | Register Name | Description                                                                                                          | Default value after reset |                                               | Reflection timing                             |             |                                               |             |     |     |
|---------|------------------|-----|---------------|----------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------|-----------------------------------------------|-------------|-----------------------------------------------|-------------|-----|-----|
| 4-wire  | I <sup>2</sup> C |     |               |                                                                                                                      | By register               | By address                                    |                                               |             |                                               |             |     |     |
| D4h     | 30D4h            | 0   | VMAX [23:0]   | LSB                                                                                                                  | 0012EC<br>h               | ECh                                           | V                                             |             |                                               |             |     |     |
|         |                  | 1   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 2   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 3   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 4   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 5   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 6   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 7   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
| D5h     | 30D5h            | 0   |               | When sensor master mode vertical span setting.<br>(Number of operation lines count from 1)                           |                           | FRAMOS Technologies Inc.<br>Internal use only |                                               | 0012EC<br>h | 12h                                           |             |     |     |
|         |                  | 1   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 2   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 3   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 4   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 5   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 6   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 7   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
| D6h     | 30D6h            | 0   |               |                                                                                                                      |                           |                                               |                                               |             | FRAMOS Technologies Inc.<br>Internal use only | 0012EC<br>h | 00h |     |
|         |                  | 1   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 2   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 3   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 4   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 5   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 6   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 7   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
| D8h     | 30D8h            | 0   | HMAX [15:0]   | When sensor master mode and sensor slave mode, horizontal span setting.<br>(Number of operation clocks count from 1) | 00A6h                     |                                               | A6h                                           |             |                                               |             | S   |     |
|         |                  | 1   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 2   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 3   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 4   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 5   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 6   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 7   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
| D9h     | 30D9h            | 0   |               |                                                                                                                      |                           |                                               | FRAMOS Technologies Inc.<br>Internal use only | 00A6h       |                                               |             |     | 00h |
|         |                  | 1   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 2   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 3   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 4   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 5   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 6   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  | 7   |               |                                                                                                                      |                           |                                               |                                               |             |                                               |             |     |     |
|         |                  |     |               | MSB                                                                                                                  |                           |                                               |                                               |             |                                               |             |     |     |

| Address |                  | bit | Register Name | Description                                        | Default value<br>after reset |               | Reflection<br>timing |
|---------|------------------|-----|---------------|----------------------------------------------------|------------------------------|---------------|----------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                    | By<br>register               | By<br>address |                      |
| E2h     | 30E2h            | 0   | GMRWT [7:0]   | Refer to the register list in each<br>Readout mode | 0Ch                          | 0Ch           | S                    |
|         |                  | 1   |               |                                                    |                              |               |                      |
|         |                  | 2   |               |                                                    |                              |               |                      |
|         |                  | 3   |               |                                                    |                              |               |                      |
|         |                  | 4   |               |                                                    |                              |               |                      |
|         |                  | 5   |               |                                                    |                              |               |                      |
|         |                  | 6   |               |                                                    |                              |               |                      |
|         |                  | 7   |               |                                                    |                              |               |                      |
| E3h     | 30E3h            | 0   | GMTWT [7:0]   | Refer to the register list in each<br>Readout mode | 48h                          | 48h           | S                    |
|         |                  | 1   |               |                                                    |                              |               |                      |
|         |                  | 2   |               |                                                    |                              |               |                      |
|         |                  | 3   |               |                                                    |                              |               |                      |
|         |                  | 4   |               |                                                    |                              |               |                      |
|         |                  | 5   |               |                                                    |                              |               |                      |
|         |                  | 6   |               |                                                    |                              |               |                      |
|         |                  | 7   |               |                                                    |                              |               |                      |
| E5h     | 30E5h            | 0   | GAINDLY [7:0] | Refer to the register list in each<br>Readout mode | 04h                          | 04h           | S                    |
|         |                  | 1   |               |                                                    |                              |               |                      |
|         |                  | 2   |               |                                                    |                              |               |                      |
|         |                  | 3   |               |                                                    |                              |               |                      |
|         |                  | 4   |               |                                                    |                              |               |                      |
|         |                  | 5   |               |                                                    |                              |               |                      |
|         |                  | 6   |               |                                                    |                              |               |                      |
|         |                  | 7   |               |                                                    |                              |               |                      |
| E6h     | 30E6h            | 0   | GSDLY [7:0]   | Refer to the register list in each<br>Readout mode | 34h                          | 34h           | S                    |
|         |                  | 1   |               |                                                    |                              |               |                      |
|         |                  | 2   |               |                                                    |                              |               |                      |
|         |                  | 3   |               |                                                    |                              |               |                      |
|         |                  | 4   |               |                                                    |                              |               |                      |
|         |                  | 5   |               |                                                    |                              |               |                      |
|         |                  | 6   |               |                                                    |                              |               |                      |
|         |                  | 7   |               |                                                    |                              |               |                      |

Registers corresponding to Chip ID = 02h in Write mode. (I<sup>2</sup>C:30\*\*h) (IMX535, IMX536, IMX537)

| Address |                  | bit | Register Name | Description                                                                                                                     | Default value after reset |             | Reflection timing                                                                                                    |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|---------|------------------|-----|---------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------|-------------------------------------------------|---|--------------------------------------------------------------------------------------------|---------|-----|---|
| 4-wire  | I <sup>2</sup> C |     |               |                                                                                                                                 | By register               | By address  |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
| D4h     | 30D4h            | 0   | VMAX [23:0]   | LSB<br><br>When sensor master mode vertical span setting.<br>(Number of operation lines count from 1)                           | 000CD4h                   | D4h         | V                                                                                                                    |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 1   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 2   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 3   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 4   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 5   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 6   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 7   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
| D5h     | 30D5h            | 0   |               |                                                                                                                                 |                           | VMAX [23:0] |                                                                                                                      | When sensor master mode vertical span setting.<br>(Number of operation lines count from 1) | 000CD4h     | 0Ch                                             | V |                                                                                            |         |     |   |
|         |                  | 1   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 2   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 3   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 4   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 5   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 6   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 7   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
| D6h     | 30D6h            | 0   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             | VMAX [23:0]                                     |   | When sensor master mode vertical span setting.<br>(Number of operation lines count from 1) | 000CD4h | 00h | V |
|         |                  | 1   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 2   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 3   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 4   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 5   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 6   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 7   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
| D8h     | 30D8h            | 0   | HMAX [15:0]   | LSB<br><br>When sensor master mode and sensor slave mode, horizontal span setting.<br>(Number of operation clocks count from 1) | 0081h                     |             | 81h                                                                                                                  |                                                                                            |             |                                                 |   |                                                                                            |         | S   |   |
|         |                  | 1   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 2   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 3   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 4   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 5   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 6   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 7   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
| D9h     | 30D9h            | 0   |               |                                                                                                                                 |                           | HMAX [15:0] | When sensor master mode and sensor slave mode, horizontal span setting.<br>(Number of operation clocks count from 1) | 0081h                                                                                      | 00h         |                                                 | S |                                                                                            |         |     |   |
|         |                  | 1   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 2   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 3   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 4   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 5   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 6   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 7   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
| E2h     | 30E2h            | 0   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            | GMRWT [7:0] | Refer to the register list in each Readout mode |   | 10h                                                                                        | 10h     |     | S |
|         |                  | 1   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 2   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 3   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 4   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 5   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 6   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |
|         |                  | 7   |               |                                                                                                                                 |                           |             |                                                                                                                      |                                                                                            |             |                                                 |   |                                                                                            |         |     |   |

| Address |                  | bit | Register Name | Description                                     | Default value<br>after reset |               | Reflection<br>timing |
|---------|------------------|-----|---------------|-------------------------------------------------|------------------------------|---------------|----------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                 | By<br>register               | By<br>address |                      |
| E3h     | 30E3h            | 0   | GMTWT [7:0]   | Refer to the register list in each Readout mode | 70h                          | 70h           | S                    |
|         |                  | 1   |               |                                                 |                              |               |                      |
|         |                  | 2   |               |                                                 |                              |               |                      |
|         |                  | 3   |               |                                                 |                              |               |                      |
|         |                  | 4   |               |                                                 |                              |               |                      |
|         |                  | 5   |               |                                                 |                              |               |                      |
|         |                  | 6   |               |                                                 |                              |               |                      |
|         |                  | 7   |               |                                                 |                              |               |                      |
| E5h     | 30E5h            | 0   | GAINPLY [7:0] | Refer to the register list in each Readout mode | 04h                          | 04h           | S                    |
|         |                  | 1   |               |                                                 |                              |               |                      |
|         |                  | 2   |               |                                                 |                              |               |                      |
|         |                  | 3   |               |                                                 |                              |               |                      |
|         |                  | 4   |               |                                                 |                              |               |                      |
|         |                  | 5   |               |                                                 |                              |               |                      |
|         |                  | 6   |               |                                                 |                              |               |                      |
|         |                  | 7   |               |                                                 |                              |               |                      |
| E6h     | 30E6h            | 0   | GSDLY [7:0]   | Refer to the register list in each Readout mode | 30h                          | 30h           | S                    |
|         |                  | 1   |               |                                                 |                              |               |                      |
|         |                  | 2   |               |                                                 |                              |               |                      |
|         |                  | 3   |               |                                                 |                              |               |                      |
|         |                  | 4   |               |                                                 |                              |               |                      |
|         |                  | 5   |               |                                                 |                              |               |                      |
|         |                  | 6   |               |                                                 |                              |               |                      |
|         |                  | 7   |               |                                                 |                              |               |                      |

FRAMOS Technologies Inc.  
Internal use only

Registers corresponding to Chip ID = 03h in Write mode. (I<sup>2</sup>C:31\*\*h)

| Address |                  | bit | Register Name    | Description                                                                         | Default value after reset |            | Reflection timing |
|---------|------------------|-----|------------------|-------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                  |                                                                                     | By register               | By address |                   |
| 04h     | 3104h            | 0   | FID0_ROIH1ON [0] | The horizontal setting of FID0 ROI area (1, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                         | 00h        | V                 |
|         |                  | 1   | FID0_ROIV1ON [0] | The vertical setting of FID0 ROI area (x, 1) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                         |            | I                 |
|         |                  | 2   | FID0_ROIH2ON [0] | The horizontal setting of FID0 ROI area (2, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                         |            | V                 |
|         |                  | 3   | FID0_ROIV2ON [0] | The vertical setting of FID0 ROI area (x, 2) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                         |            | I                 |
|         |                  | 4   | FID0_ROIH3ON [0] | The horizontal setting of FID0 ROI area (3, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                         |            | V                 |
|         |                  | 5   | FID0_ROIV3ON [0] | The vertical setting of FID0 ROI area (x, 3) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                         |            | I                 |
|         |                  | 6   | FID0_ROIH4ON [0] | The horizontal setting of FID0 ROI area (4, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                         |            | V                 |
|         |                  | 7   | FID0_ROIV4ON [0] | The vertical setting of FID0 ROI area (x, 4) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                         |            | I                 |

| Address |                  | bit | Register Name    | Description                                                                         | Default value<br>after reset |               | Reflection<br>timing |
|---------|------------------|-----|------------------|-------------------------------------------------------------------------------------|------------------------------|---------------|----------------------|
| 4-wire  | I <sup>2</sup> C |     |                  |                                                                                     | By<br>register               | By<br>address |                      |
| 05h     | 3105h            | 0   | FID0_ROIH5ON [0] | The horizontal setting of FID0 ROI area (5, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                            | 00h           | V                    |
|         |                  | 1   | FID0_ROIV5ON [0] | The vertical setting of FID0 ROI area (x, 5) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                            |               | I                    |
|         |                  | 2   | FID0_ROIH6ON [0] | The horizontal setting of FID0 ROI area (6, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                            |               | V                    |
|         |                  | 3   | FID0_ROIV6ON [0] | The vertical setting of FID0 ROI area (x, 6) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                            |               | I                    |
|         |                  | 4   | FID0_ROIH7ON [0] | The horizontal setting of FID0 ROI area (7, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                            |               | V                    |
|         |                  | 5   | FID0_ROIV7ON [0] | The vertical setting of FID0 ROI area (x, 7) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                            |               | I                    |
|         |                  | 6   | FID0_ROIH8ON [0] | The horizontal setting of FID0 ROI area (8, y) (y = 1 to 8)<br>0: Disable 1: Enable | 0                            |               | V                    |
|         |                  | 7   | FID0_ROIV8ON [0] | The vertical setting of FID0 ROI area (x, 8) (x = 1 to 8)<br>0: Disable 1: Enable   | 0                            |               | I                    |

| Address |                  | bit | Register Name      | Description                                                                                                         | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                                                     | By register               | By address |                   |
| 20h     | 3120h            | 0   | FID0_ROIPH1 [12:0] | LSB                                                                                                                 | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                     |                           |            |                   |
| 21h     | 3121h            | 7   |                    | Designation of horizontal cropping position for FID0 on area (1, y) (y = 1 to 8)<br>*Set the value of multiple of 8 |                           |            |                   |
|         |                  | 0   |                    |                                                                                                                     | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                                                                 |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                          |                           |            |                   |
| 22h     | 3122h            | 6   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 0   | FID0_ROIPV1 [12:0] | LSB                                                                                                                 | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                     |                           |            |                   |
| 23h     | 3123h            | 7   |                    | Designation of vertical cropping position for FID0 on area (x, 1) (x = 1 to 8)<br>*Set the value of multiple of 8   |                           |            |                   |
|         |                  | 0   |                    |                                                                                                                     | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                                                                 |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                          |                           |            |                   |
| 24h     | 3124h            | 6   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 0   | FID0_ROIWH1 [12:0] | LSB                                                                                                                 | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                     |                           |            |                   |
| 25h     | 3125h            | 7   |                    | Designation of horizontal cropping size for FID0 on area (1, y) (y = 1 to 8)<br>*Set the value of multiple of 8     |                           |            |                   |
|         |                  | 0   |                    |                                                                                                                     | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                                                                 |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 6   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 0   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 1   |                    | Fixed to 0                                                                                                          |                           |            |                   |

| Address |                  | bit | Register Name      | Description                                                                                                            | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                                                        | By register               | By address |                   |
| 26h     | 3126h            | 0   | FID0_ROIWV1 [12:0] | LSB                                                                                                                    | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping size for FID0 on area (x, 1)<br>(x = 1 to 8)<br>*Set the value of multiple of 8       |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                        |                           |            |                   |
| 27h     | 3127h            | 7   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 0   |                    | MSB                                                                                                                    | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 28h     | 3128h            | 6   |                    | Fixed to 0                                                                                                             |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 29h     | 3129h            | 0   | FID0_ROIPH2 [12:0] | LSB                                                                                                                    | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping position for FID0 on area (2, y)<br>(y = 1 to 8)<br>*Set the value of multiple of 8 |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                        |                           |            |                   |
| 2Ah     | 312Ah            | 7   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 0   |                    | MSB                                                                                                                    | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 2Bh     | 312Bh            | 6   |                    | Fixed to 0                                                                                                             |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 2Ah     | 312Ah            | 0   | FID0_ROIPV2 [12:0] | LSB                                                                                                                    | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping position for FID0 on area (x, 2)<br>(x = 1 to 8)<br>*Set the value of multiple of 8   |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                        |                           |            |                   |
| 2Bh     | 312Bh            | 7   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 0   |                    | MSB                                                                                                                    | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 2Bh     | 312Bh            | 6   |                    | Fixed to 0                                                                                                             |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                             |                           |            |                   |

| Address |                  | bit | Register Name      | Description                                                                                                        | Default value after reset |            | Reflection timing |   |       |     |   |                                                                                                                        |
|---------|------------------|-----|--------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|---|-------|-----|---|------------------------------------------------------------------------------------------------------------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                                                    | By register               | By address |                   |   |       |     |   |                                                                                                                        |
| 2Ch     | 312Ch            | 0   | FID0_ROIWH2 [12:0] | LSB                                                                                                                | 0000h                     | 00h        | V                 |   |       |     |   |                                                                                                                        |
|         |                  | 1   |                    | Designation of horizontal cropping size for FID0 on area (2, y)<br>(y = 1 to 8)<br>*Set the value of multiple of 8 |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 2   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 3   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 4   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 5   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 6   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 2Dh     | 312Dh            | 7   |                    |                                                                                                                    |                           | MSB        |                   | 0 | 00h   |     |   |                                                                                                                        |
|         |                  | 0   | Fixed to 0         | —                                                                                                                  |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 1   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 2   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 3   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 4   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 5   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 2Eh     | 312Eh            | 6   |                    |                                                                                                                    | FID0_ROIWH2 [12:0]        | Fixed to 0 | 0                 | — |       |     |   |                                                                                                                        |
|         |                  | 7   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 0   | FID0_ROIWH2 [12:0] | LSB                                                                                                                |                           |            |                   |   | 0000h | 00h | I |                                                                                                                        |
|         |                  | 1   |                    |                                                                                                                    |                           |            |                   |   |       |     |   | Designation of vertical cropping size for FID0 on area (x, 2)<br>(x = 1 to 8)<br>*Set the value of multiple of 8       |
|         |                  | 2   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 3   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 4   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 5       |                  |     |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 2Fh     | 312Fh            | 6   |                    |                                                                                                                    | FID0_ROIWH2 [12:0]        | Fixed to 0 | 0                 | — |       |     |   |                                                                                                                        |
|         |                  | 7   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 0   | MSB                | 0                                                                                                                  |                           |            |                   |   | 00h   | —   |   |                                                                                                                        |
|         |                  | 1   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 2   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 3   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 4   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 30h     | 3130h            | 5   |                    |                                                                                                                    | FID0_ROIWH2 [12:0]        | Fixed to 0 | 0                 | — |       |     |   |                                                                                                                        |
|         |                  | 6   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 7   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 0   | FID0_ROIWH2 [12:0] | LSB                                                                                                                |                           |            |                   |   | 0000h | 00h | V |                                                                                                                        |
|         |                  | 1   |                    |                                                                                                                    |                           |            |                   |   |       |     |   | Designation of horizontal cropping position for FID0 on area (3, y)<br>(y = 1 to 8)<br>*Set the value of multiple of 8 |
|         |                  | 2   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 3   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 4       |                  |     |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 5       |                  |     |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 31h     | 3131h            | 6   |                    |                                                                                                                    | FID0_ROIWH2 [12:0]        | Fixed to 0 | 0                 | — |       |     |   |                                                                                                                        |
|         |                  | 7   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 0   | MSB                | 0                                                                                                                  |                           |            |                   |   | 00h   | —   |   |                                                                                                                        |
|         |                  | 1   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 2   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 3   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 4   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
| 31h     | 3131h            | 5   |                    |                                                                                                                    | FID0_ROIWH2 [12:0]        | Fixed to 0 | 0                 | — |       |     |   |                                                                                                                        |
|         |                  | 6   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |
|         |                  | 7   |                    |                                                                                                                    |                           |            |                   |   |       |     |   |                                                                                                                        |

| Address |                  | bit | Register Name      | Description                                                                                                          | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                                                      | By register               | By address |                   |
| 32h     | 3132h            | 0   | FID0_ROIPV3 [12:0] | LSB                                                                                                                  | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    | Designation of vertical cropping position for FID0 on area (x, 3)<br>(x = 1 to 8)<br>*Set the value of multiple of 8 |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                      |                           |            |                   |
| 33h     | 3133h            | 7   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 0   | MSB                | 0                                                                                                                    | 00h                       | —          |                   |
|         |                  | 1   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 4   | Fixed to 0         | 0                                                                                                                    | —                         |            |                   |
|         |                  | 5   | Fixed to 0         | 0                                                                                                                    | —                         |            |                   |
| 6       | Fixed to 0       | 0   | —                  |                                                                                                                      |                           |            |                   |
| 34h     | 3134h            | 0   | FID0_ROIWH3 [12:0] | LSB                                                                                                                  | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    | Designation of horizontal cropping size for FID0 on area (3, y)<br>(y = 1 to 8)<br>*Set the value of multiple of 8   |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                      |                           |            |                   |
| 35h     | 3135h            | 7   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 0   | MSB                | 0                                                                                                                    | 00h                       | —          |                   |
|         |                  | 1   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 4   | Fixed to 0         | 0                                                                                                                    | —                         |            |                   |
|         |                  | 5   | Fixed to 0         | 0                                                                                                                    | —                         |            |                   |
| 6       | Fixed to 0       | 0   | —                  |                                                                                                                      |                           |            |                   |
| 36h     | 3136h            | 0   | FID0_ROIWH3 [12:0] | LSB                                                                                                                  | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    | Designation of vertical cropping size for FID0 on area (x, 3)<br>(x = 1 to 8)<br>*Set the value of multiple of 8     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                      |                           |            |                   |
| 37h     | 3137h            | 7   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 0   | MSB                | 0                                                                                                                    | 00h                       | —          |                   |
|         |                  | 1   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                      |                           |            |                   |
|         |                  | 4   | Fixed to 0         | 0                                                                                                                    | —                         |            |                   |
|         |                  | 5   | Fixed to 0         | 0                                                                                                                    | —                         |            |                   |
| 6       | Fixed to 0       | 0   | —                  |                                                                                                                      |                           |            |                   |

| Address |                  | bit | Register Name      | Description                                                                      | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|----------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                  | By register               | By address |                   |
| 38h     | 3138h            | 0   | FID0_ROIPH4 [12:0] | LSB                                                                              | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping position for FID0 on area (4, y) (y = 1 to 8) |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
| 39h     | 3139h            | 7   |                    |                                                                                  |                           |            |                   |
|         |                  | 0   |                    |                                                                                  |                           | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       | 0                         |            |                   |
| 3Ah     | 313Ah            | 6   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 0   | FID0_ROIPV4 [12:0] | LSB                                                                              | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 8)   |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
| 3Bh     | 313Bh            | 7   |                    |                                                                                  |                           |            |                   |
|         |                  | 0   |                    |                                                                                  |                           | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       | 0                         |            |                   |
| 3Ch     | 313Ch            | 6   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 0   | FID0_ROIWH4 [12:0] | LSB                                                                              | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping size for FID0 on area (4, y) (y = 1 to 8)     |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
| 3Dh     | 313Dh            | 7   |                    |                                                                                  |                           |            |                   |
|         |                  | 0   |                    |                                                                                  |                           | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       | 0                         |            |                   |
| 3Dh     | 313Dh            | 6   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       | 0                         |            |                   |

| Address |                  | bit | Register Name      | Description                                                                      | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|----------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                  | By register               | By address |                   |
| 3Eh     | 313Eh            | 0   | FID0_ROIWV4 [12:0] | LSB                                                                              | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping size for FID0 on area (x, 4) (x = 1 to 8)       |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
|         |                  | 7   |                    |                                                                                  |                           |            |                   |
| 3Fh     | 313Fh            | 0   |                    |                                                                                  | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       |                           |            |                   |
|         |                  | 6   |                    | Fixed to 0                                                                       |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       |                           |            |                   |
| 40h     | 3140h            | 0   | FID0_ROIPH5 [12:0] | LSB                                                                              | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping position for FID0 on area (5, y) (y = 1 to 8) |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
|         |                  | 7   |                    |                                                                                  |                           |            |                   |
| 41h     | 3141h            | 0   |                    |                                                                                  | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       |                           |            |                   |
|         |                  | 6   |                    | Fixed to 0                                                                       |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       |                           |            |                   |
| 42h     | 3142h            | 0   | FID0_ROIPV5 [12:0] | LSB                                                                              | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping position for FID0 on area (x, 5) (x = 1 to 8)   |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
|         |                  | 7   |                    |                                                                                  |                           |            |                   |
| 43h     | 3143h            | 0   |                    |                                                                                  | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       |                           |            |                   |
|         |                  | 6   |                    | Fixed to 0                                                                       |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       |                           |            |                   |

| Address |                  | bit | Register Name      | Description                                                                                                         | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                                                     | By register               | By address |                   |
| 44h     | 3144h            | 0   | FID0_ROIWH5 [12:0] | LSB                                                                                                                 | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                     |                           |            |                   |
| 45h     | 3145h            | 7   |                    | Designation of horizontal cropping size for FID0 on area (5, y) (y = 1 to 8)<br>*Set the value of multiple of 8     |                           |            |                   |
|         |                  | 0   |                    |                                                                                                                     | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                                                                 |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                          |                           |            |                   |
| 46h     | 3146h            | 6   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 0   | FID0_ROIWH5 [12:0] | LSB                                                                                                                 | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                     |                           |            |                   |
| 47h     | 3147h            | 7   |                    | Designation of vertical cropping size for FID0 on area (x, 5) (x = 1 to 8)<br>*Set the value of multiple of 8       |                           |            |                   |
|         |                  | 0   |                    |                                                                                                                     | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                                                                 |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                          |                           |            |                   |
| 48h     | 3148h            | 6   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 0   | FID0_ROIWH5 [12:0] | LSB                                                                                                                 | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 5   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                     |                           |            |                   |
| 49h     | 3149h            | 7   |                    | Designation of horizontal cropping position for FID0 on area (6, y) (y = 1 to 8)<br>*Set the value of multiple of 8 |                           |            |                   |
|         |                  | 0   |                    |                                                                                                                     | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                     |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                                                                 |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                          |                           |            |                   |
| 49h     | 3149h            | 6   |                    | Fixed to 0                                                                                                          |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                          |                           |            |                   |

| Address |                  | bit | Register Name      | Description                                                                    | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|--------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                | By register               | By address |                   |
| 4Ah     | 314Ah            | 0   | FID0_ROIPV6 [12:0] | LSB                                                                            | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                |                           |            |                   |
|         |                  | 2   |                    |                                                                                |                           |            |                   |
|         |                  | 3   |                    |                                                                                |                           |            |                   |
|         |                  | 4   |                    |                                                                                |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping position for FID0 on area (x, 6) (x = 1 to 8) |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                |                           |            |                   |
| 4Bh     | 314Bh            | 7   |                    |                                                                                |                           |            |                   |
|         |                  | 0   |                    |                                                                                | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                |                           |            |                   |
|         |                  | 2   |                    |                                                                                |                           |            |                   |
|         |                  | 3   |                    |                                                                                |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                            |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                     |                           |            |                   |
| 4Ch     | 314Ch            | 6   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  | 0   | FID0_ROIWH6 [12:0] | LSB                                                                            | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                |                           |            |                   |
|         |                  | 2   |                    |                                                                                |                           |            |                   |
|         |                  | 3   |                    |                                                                                |                           |            |                   |
|         |                  | 4   |                    |                                                                                |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping size for FID0 on area (6, y) (y = 1 to 8)   |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                |                           |            |                   |
| 4Dh     | 314Dh            | 7   |                    |                                                                                |                           |            |                   |
|         |                  | 0   |                    |                                                                                | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                |                           |            |                   |
|         |                  | 2   |                    |                                                                                |                           |            |                   |
|         |                  | 3   |                    |                                                                                |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                            |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                     |                           |            |                   |
| 4Eh     | 314Eh            | 6   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  | 0   | FID0_ROIWW6 [12:0] | LSB                                                                            | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                |                           |            |                   |
|         |                  | 2   |                    |                                                                                |                           |            |                   |
|         |                  | 3   |                    |                                                                                |                           |            |                   |
|         |                  | 4   |                    |                                                                                |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping size for FID0 on area (x, 6) (x = 1 to 8)     |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                |                           |            |                   |
| 4Fh     | 314Fh            | 7   |                    |                                                                                |                           |            |                   |
|         |                  | 0   |                    |                                                                                | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                |                           |            |                   |
|         |                  | 2   |                    |                                                                                |                           |            |                   |
|         |                  | 3   |                    |                                                                                |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                            |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  | 6   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                     |                           |            |                   |
|         |                  |     |                    |                                                                                |                           |            |                   |
|         |                  |     |                    |                                                                                |                           |            |                   |

| Address |                  | bit | Register Name      | Description                                                                      | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|----------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                  | By register               | By address |                   |
| 50h     | 3150h            | 0   | FID0_ROIPH7 [12:0] | LSB                                                                              | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping position for FID0 on area (7, y) (y = 1 to 8) |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
| 51h     | 3151h            | 7   |                    |                                                                                  |                           |            |                   |
|         |                  | 0   |                    |                                                                                  |                           | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       | 0                         |            |                   |
| 52h     | 3152h            | 6   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 0   | FID0_ROIPV7 [12:0] | LSB                                                                              | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping position for FID0 on area (x, 7) (x = 1 to 8)   |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
| 53h     | 3153h            | 7   |                    |                                                                                  |                           |            |                   |
|         |                  | 0   |                    |                                                                                  |                           | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       | 0                         |            |                   |
| 54h     | 3154h            | 6   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 0   | FID0_ROIWH7 [12:0] | LSB                                                                              | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    |                                                                                  |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping size for FID0 on area (7, y) (y = 1 to 8)     |                           |            |                   |
|         |                  | 6   |                    | *Set the value of multiple of 8                                                  |                           |            |                   |
| 55h     | 3155h            | 7   |                    |                                                                                  |                           |            |                   |
|         |                  | 0   |                    |                                                                                  |                           | 00h        | —                 |
|         |                  | 1   |                    |                                                                                  |                           |            |                   |
|         |                  | 2   |                    |                                                                                  |                           |            |                   |
|         |                  | 3   |                    |                                                                                  |                           |            |                   |
|         |                  | 4   |                    | MSB                                                                              |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                       | 0                         |            |                   |
| 56h     | 3156h            | 6   |                    | Fixed to 0                                                                       | 0                         |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                       | 0                         |            |                   |

| Address |                  | bit | Register Name      | Description                                                                                                            | Default value after reset |            | Reflection timing |
|---------|------------------|-----|--------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                    |                                                                                                                        | By register               | By address |                   |
| 56h     | 3156h            | 0   | FID0_ROIWV7 [12:0] | LSB                                                                                                                    | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping size for FID0 on area (x, 7)<br>(x = 1 to 8)<br>*Set the value of multiple of 8       |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                        |                           |            |                   |
| 57h     | 3157h            | 7   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 0   |                    | MSB                                                                                                                    | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 58h     | 3158h            | 6   |                    | Fixed to 0                                                                                                             |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 59h     | 3159h            | 0   | FID0_ROIPH8 [12:0] | LSB                                                                                                                    | 0000h                     | 00h        | V                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Designation of horizontal cropping position for FID0 on area (8, y)<br>(y = 1 to 8)<br>*Set the value of multiple of 8 |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                        |                           |            |                   |
| 5Ah     | 315Ah            | 7   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 0   |                    | MSB                                                                                                                    | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 5Bh     | 315Bh            | 6   |                    | Fixed to 0                                                                                                             |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 5Ah     | 315Ah            | 0   | FID0_ROIPV8 [12:0] | LSB                                                                                                                    | 0000h                     | 00h        | I                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Designation of vertical cropping position for FID0 on area (x, 8)<br>(x = 1 to 8)<br>*Set the value of multiple of 8   |                           |            |                   |
|         |                  | 6   |                    |                                                                                                                        |                           |            |                   |
| 5Bh     | 315Bh            | 7   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 0   |                    | MSB                                                                                                                    | 0                         | 00h        | —                 |
|         |                  | 1   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 2   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 3   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 4   |                    |                                                                                                                        |                           |            |                   |
|         |                  | 5   |                    | Fixed to 0                                                                                                             |                           |            |                   |
| 5Bh     | 315Bh            | 6   |                    | Fixed to 0                                                                                                             |                           |            |                   |
|         |                  | 7   |                    | Fixed to 0                                                                                                             |                           |            |                   |

| Address |                  | bit | Register Name                                                                                                         | Description | Default value<br>after reset |               | Reflection<br>timing |
|---------|------------------|-----|-----------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|---------------|----------------------|
| 4-wire  | I <sup>2</sup> C |     |                                                                                                                       |             | By<br>register               | By<br>address |                      |
| 5Ch     | 315Ch            | 0   | FID0_ROIWH8<br>[12:0]                                                                                                 | LSB         | 0000h                        | 00h           | V                    |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
|         |                  | 6   |                                                                                                                       |             |                              |               |                      |
| 5Dh     | 315Dh            | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Designation of horizontal cropping<br>size for FID0 on area (8, y)<br>(y = 1 to 8)<br>*Set the value of multiple of 8 | 0000h       | 00h                          | V             |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
| 5Eh     | 315Eh            | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Designation of vertical cropping<br>size for FID0 on area (x, 8)<br>(x = 1 to 8)<br>*Set the value of multiple of 8   | 0000h       | 00h                          | I             |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
| 5Fh     | 315Fh            | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | MSB                                                                                                                   | 0           | 00h                          | —             |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |
|         |                  | 0   | Fixed to 0                                                                                                            | 0           | —                            |               |                      |
|         |                  | 1   |                                                                                                                       |             |                              |               |                      |
|         |                  | 2   |                                                                                                                       |             |                              |               |                      |
|         |                  | 3   |                                                                                                                       |             |                              |               |                      |
|         |                  | 4   |                                                                                                                       |             |                              |               |                      |
|         |                  | 5   |                                                                                                                       |             |                              |               |                      |
| 6       |                  |     |                                                                                                                       |             |                              |               |                      |
|         |                  | 7   |                                                                                                                       |             |                              |               |                      |

Registers corresponding to Chip ID = 04h in Write mode. (I<sup>2</sup>C:32\*\*h)

| Address |                  | bit | Register Name  | Description                                                                                                                          | Default value after reset |            | Reflection timing |   |   |
|---------|------------------|-----|----------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|---|---|
| 4-wire  | I <sup>2</sup> C |     |                |                                                                                                                                      | By register               | By address |                   |   |   |
| 00h     | 3200h            | 0   | AD_SEL [1:0]   | 0h: Normal mode<br>2h: Dual ADC without combination<br>3h: Dual ADC with built-in combination<br>Others: Setting prohibited          | 0h                        | 04h        | S                 |   |   |
|         |                  | 1   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 2   |                |                                                                                                                                      |                           |            | Fixed to 1        | 1 | – |
|         |                  | 3   |                | Fixed to 0                                                                                                                           | 0                         |            | –                 |   |   |
|         |                  | 4   | ADBIT [1:0]    | AD conversion bits setting<br>0h: 10 bit    1h: 12 bit    2h: 8 bit<br>3h: Setting prohibited                                        | 0h                        |            | S                 |   |   |
|         |                  | 5   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 6   |                | Fixed to 0                                                                                                                           |                           |            |                   | 0 | – |
|         |                  | 7   |                | Fixed to 0                                                                                                                           |                           |            |                   | 0 | – |
| 10h     | 3210h            | 0   | EXP_GAIN [2:0] | Gain setting added to high gain side<br>0h: 0dB    1h: +6 dB    2h: +12 dB<br>3h: +18 dB    4h: +24 dB<br>Others: Setting prohibited | 0h                        | 00h        | V                 |   |   |
|         |                  | 1   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 2   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 3   |                |                                                                                                                                      |                           |            | Fixed to 0        | 0 | – |
|         |                  | 4   |                |                                                                                                                                      |                           |            | Fixed to 0        | 0 | – |
|         |                  | 5   |                | Fixed to 0                                                                                                                           | 0                         |            | –                 |   |   |
|         |                  | 6   | CMP_EN_EXP [0] | Gradation compression enable for built-in combination<br>0: Gradation compression OFF<br>1: Gradation compression ON                 | 0                         |            | V                 |   |   |
|         |                  | 7   |                | Fixed to 0                                                                                                                           | 0                         |            | –                 |   |   |
| 12h     | 3212h            | 0   | EXP_TH [12:0]  | LSB                                                                                                                                  | 0FFFh                     | FFh        | I                 |   |   |
|         |                  | 1   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 2   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 3   |                | Threshold for data selection of low gain side or high gain side                                                                      |                           |            |                   |   |   |
|         |                  | 4   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 5   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 6   |                | High gain ≥ EXP_TH:<br>Low gain side data output                                                                                     |                           |            |                   |   |   |
|         |                  | 7   |                | Gigh gain < EXP_TH<br>High gain side data output                                                                                     |                           |            |                   |   |   |
| 13h     | 3213h            | 0   |                | MSB                                                                                                                                  | 0                         | 0Fh        | –                 |   |   |
|         |                  | 1   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 2   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 3   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 4   |                |                                                                                                                                      |                           |            |                   |   |   |
|         |                  | 5   |                | Fixed to 0                                                                                                                           |                           |            |                   | 0 | – |
|         |                  | 6   |                | Fixed to 0                                                                                                                           |                           |            |                   | 0 | – |
|         |                  | 7   |                | Fixed to 0                                                                                                                           |                           |            |                   | 0 | – |

| Address |                  | bit                                                                   | Register Name   | Description                                                          | Default value after reset |            | Reflection timing |  |
|---------|------------------|-----------------------------------------------------------------------|-----------------|----------------------------------------------------------------------|---------------------------|------------|-------------------|--|
| 4-wire  | I <sup>2</sup> C |                                                                       |                 |                                                                      | By register               | By address |                   |  |
| 14h     | 3214h            | 0                                                                     | CCMP1_EXP [3:0] | Set the first start point of gradation compression (data position 1) | 0h                        |            | V                 |  |
|         |                  | 1                                                                     |                 |                                                                      |                           |            |                   |  |
|         |                  | 2                                                                     |                 |                                                                      |                           |            |                   |  |
|         |                  | 3                                                                     |                 | 0000b: Gradation compression off (both of the first and second)      |                           |            |                   |  |
|         |                  |                                                                       |                 | 0001b: 0000h      0010b: 0010h                                       |                           |            |                   |  |
|         |                  |                                                                       |                 | 0011b: 0020h      0100b: 0040h                                       |                           |            |                   |  |
|         |                  |                                                                       |                 | 0101b: 0080h      0110b: 0100h                                       |                           |            |                   |  |
|         |                  |                                                                       |                 | 0111b: 0200h      1000b: 0400h                                       |                           |            |                   |  |
|         |                  |                                                                       |                 | 1001b: 0800h      1010b: 1000h                                       |                           |            |                   |  |
|         |                  |                                                                       |                 | 1011b: 2000h      1100b: 4000h                                       |                           |            |                   |  |
|         |                  |                                                                       |                 | 1101b: 8000h                                                         |                           |            |                   |  |
|         |                  |                                                                       |                 | Others: prohibited                                                   |                           |            |                   |  |
|         |                  |                                                                       |                 | *Set to BLKLEVEL < CCMP1_EXP                                         |                           |            |                   |  |
| 4       | CCMP2_EXP [3:0]  | Set the second start point of gradation compression (data position 2) | 0h              | 00h                                                                  | V                         |            |                   |  |
| 5       |                  |                                                                       |                 |                                                                      |                           |            |                   |  |
| 6       |                  |                                                                       |                 |                                                                      |                           |            |                   |  |
| 7       |                  | 0000b: The second Gradation Compression off                           |                 |                                                                      |                           |            |                   |  |
|         |                  | 0010b: 0010h      0011b: 0020h                                        |                 |                                                                      |                           |            |                   |  |
|         |                  | 0100b: 0040h      0101b: 0080h                                        |                 |                                                                      |                           |            |                   |  |
|         |                  | 0110b: 0100h      0111b: 0200h                                        |                 |                                                                      |                           |            |                   |  |
|         |                  | 1000b: 0400h      1001b: 0800h                                        |                 |                                                                      |                           |            |                   |  |
|         |                  | 1010b: 1000h      1011b: 2000h                                        |                 |                                                                      |                           |            |                   |  |
|         |                  | 1100b: 4000h      1101b: 8000h                                        |                 |                                                                      |                           |            |                   |  |
|         |                  | Others: prohibited                                                    |                 |                                                                      |                           |            |                   |  |
|         |                  | *In case of setting more than 0010, set to CCMP1_EXP < CCMP2_EXP      |                 |                                                                      |                           |            |                   |  |
|         |                  | *Set to BLKLEVEL < CCMP2_EXP                                          |                 |                                                                      |                           |            |                   |  |

| Address |                  | bit | Register Name   | Description                                                                                        | Default value after reset |            | Reflection timing |
|---------|------------------|-----|-----------------|----------------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |                 |                                                                                                    | By register               | By address |                   |
| 15h     | 3215h            | 0   | ACMP1_EXP [3:0] | Set the gain of the first gradation compression                                                    | 0h                        | 00h        | V                 |
|         |                  | 1   |                 |                                                                                                    |                           |            |                   |
|         |                  | 2   |                 |                                                                                                    |                           |            |                   |
|         |                  | 3   |                 |                                                                                                    |                           |            |                   |
|         |                  | 4   |                 |                                                                                                    |                           |            |                   |
|         |                  | 5   |                 |                                                                                                    |                           |            |                   |
|         |                  | 6   |                 |                                                                                                    |                           |            |                   |
|         |                  | 7   |                 |                                                                                                    |                           |            |                   |
|         |                  | 8   |                 |                                                                                                    |                           |            |                   |
|         |                  | 9   |                 |                                                                                                    |                           |            |                   |
| 15h     | 3215h            | 4   | ACMP2_EXP [3:0] | Set the gain of the second gradation compression                                                   | 0h                        | 00h        | V                 |
|         |                  | 5   |                 |                                                                                                    |                           |            |                   |
|         |                  | 6   |                 |                                                                                                    |                           |            |                   |
|         |                  | 7   |                 |                                                                                                    |                           |            |                   |
|         |                  | 8   |                 |                                                                                                    |                           |            |                   |
|         |                  | 9   |                 |                                                                                                    |                           |            |                   |
|         |                  | 10  |                 |                                                                                                    |                           |            |                   |
|         |                  | 11  |                 |                                                                                                    |                           |            |                   |
|         |                  | 12  |                 |                                                                                                    |                           |            |                   |
|         |                  | 13  |                 |                                                                                                    |                           |            |                   |
| 3Ch     | 323Ch            | 0   | LLBLANK [7:0]   | Fixed to 1                                                                                         | 1                         | 31h        | –                 |
|         |                  | 1   |                 |                                                                                                    |                           |            |                   |
|         |                  | 2   |                 |                                                                                                    |                           |            |                   |
|         |                  | 3   |                 |                                                                                                    |                           |            |                   |
|         |                  | 4   |                 |                                                                                                    |                           |            |                   |
|         |                  | 5   |                 |                                                                                                    |                           |            |                   |
|         |                  | 6   |                 |                                                                                                    |                           |            |                   |
|         |                  | 7   |                 |                                                                                                    |                           |            |                   |
| 3Dh     | 323Dh            | 0   |                 | Fixed to 0                                                                                         | 0                         | 00h        | –                 |
|         |                  | 1   |                 |                                                                                                    |                           |            |                   |
|         |                  | 2   |                 |                                                                                                    |                           |            |                   |
|         |                  | 3   |                 |                                                                                                    |                           |            |                   |
|         |                  | 4   |                 |                                                                                                    |                           |            |                   |
|         |                  | 5   |                 |                                                                                                    |                           |            |                   |
|         |                  | 6   |                 |                                                                                                    |                           |            |                   |
|         |                  | 7   |                 |                                                                                                    |                           |            |                   |
| 3Eh     | 323Eh            | 0   | VINT_EN [0]     | Setting of Interrupt mode in Trigger Mode<br>0: V interrupt is disable<br>1: V interrupt is enable | 1                         | 5Bh        | S                 |
|         |                  | 1   | VINT_EN_NOR [0] | Setting of Interrupt mode in Normal Mode<br>0: V interrupt is disable<br>1: V interrupt is enable  | 1                         |            | S                 |
|         |                  | 2   |                 | Dual ADC (Built-in combination): 0Ch<br>Others: Same as the setting value in Normal mode           | 16h                       |            | S                 |
|         |                  | 3   |                 |                                                                                                    |                           |            |                   |
|         |                  | 4   |                 |                                                                                                    |                           |            |                   |
|         |                  | 5   |                 |                                                                                                    |                           |            |                   |
|         |                  | 6   |                 |                                                                                                    |                           |            |                   |
|         |                  | 7   |                 |                                                                                                    |                           |            |                   |

Registers corresponding to Chip ID = 06h in Write mode. (I<sup>2</sup>C:34\*\*h)

| Address |                  | bit | Register Name | Description                                                                               | Default value after reset |            | Reflection timing |
|---------|------------------|-----|---------------|-------------------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                                                           | By register               | By address |                   |
| 30h     | 3430h            | 0   | ODBIT [1:0]   | Number of output bit setting<br>0h: 10 bit 1h: 12 bit 2h: 8 bit<br>3h: Setting prohibited | 0h                        | 00h        | S                 |
|         |                  | 1   |               |                                                                                           |                           |            | —                 |
|         |                  | 2   |               | Fixed to 0                                                                                | 0                         |            | —                 |
|         |                  | 3   |               | Fixed to 0                                                                                | 0                         |            | —                 |
|         |                  | 4   |               | Fixed to 0                                                                                | 0                         |            | —                 |
|         |                  | 5   |               | Fixed to 0                                                                                | 0                         |            | —                 |
|         |                  | 6   |               | Fixed to 0                                                                                | 0                         |            | —                 |
|         |                  | 7   |               | Fixed to 0                                                                                | 0                         |            | —                 |

FRAMOS Technologies Inc.  
Internal use only

Registers corresponding to Chip ID = 07h in Write mode. (I<sup>2</sup>C:35\*\*h)

| Address |                  | bit | Register Name | Description                                                                          | Default value after reset |           |            |           | Reflection timing |
|---------|------------------|-----|---------------|--------------------------------------------------------------------------------------|---------------------------|-----------|------------|-----------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                                                      | By register               |           | By address |           |                   |
| 04h     | 3504h            | 0   | AGAIN [8:0]   | LSB                                                                                  | 0F0h                      | F0h       | V          |           |                   |
|         |                  | 1   |               | Max analog gain setting<br>0 dB to 24 dB (240d)<br>0.1 dB step                       |                           |           |            |           |                   |
|         |                  | 2   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 3   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 4   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 5   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 6   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 7   |               |                                                                                      |                           |           |            |           |                   |
| 05h     | 3505h            | 0   |               | MSB                                                                                  | 0                         | 00h       | —          |           |                   |
|         |                  | 1   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 2   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 3   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 4   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 5   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 6   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 7   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
| 14h     | 3514h            | 0   | GAIN [8:0]    | LSB                                                                                  | 000h                      | 00h       | V          |           |                   |
|         |                  | 1   |               | Gain setting<br>0 dB to "AGAIN + 24 dB"<br>0.1 dB Step<br>Maximum gain: 48 dB (480d) |                           |           |            |           |                   |
|         |                  | 2   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 3   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 4   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 5   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 6   |               |                                                                                      |                           |           |            |           |                   |
|         |                  | 7   |               |                                                                                      |                           |           |            |           |                   |
| 15h     | 3515h            | 0   |               | MSB                                                                                  | 0                         | 00h       | —          |           |                   |
|         |                  | 1   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 2   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 3   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 4   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 5   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 6   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
|         |                  | 7   |               | Fixed to 0                                                                           |                           |           |            |           |                   |
| 21h     | 3521h            | 0   |               | Dual ADC mode (Built-in combination):                                                | 0Dh<br>*1                 | A9h<br>*2 | 0Dh<br>*1  | A9h<br>*2 | S                 |
|         |                  | 1   |               | IMX530: 19h                                                                          |                           |           |            |           |                   |
|         |                  | 2   |               | IMX531: 91h                                                                          |                           |           |            |           |                   |
|         |                  | 3   |               | IMX532: 19h                                                                          |                           |           |            |           |                   |
|         |                  | 4   |               | IMX535: 51h                                                                          |                           |           |            |           |                   |
|         |                  | 5   |               | IMX536: 7Dh                                                                          |                           |           |            |           |                   |
|         |                  | 6   |               | IMX537: 3Dh                                                                          |                           |           |            |           |                   |
|         |                  | 7   |               | Others: Same as the setting value in Normal mode                                     |                           |           |            |           |                   |
| 22h     | 3522h            | 0   |               | Dual ADC mode (Built-in combination):                                                | B1h<br>*1                 | B0h<br>*2 | B1h<br>*1  | B0h<br>*2 | S                 |
|         |                  | 1   |               | IMX530: B2h                                                                          |                           |           |            |           |                   |
|         |                  | 2   |               | IMX531: B1h                                                                          |                           |           |            |           |                   |
|         |                  | 3   |               | IMX532: B2h                                                                          |                           |           |            |           |                   |
|         |                  | 4   |               | IMX535: B1h                                                                          |                           |           |            |           |                   |
|         |                  | 5   |               | IMX536: B0h                                                                          |                           |           |            |           |                   |
|         |                  | 6   |               | IMX537: B0h                                                                          |                           |           |            |           |                   |
|         |                  | 7   |               | Others: Same as the setting value in Normal mode                                     |                           |           |            |           |                   |

| Address |                  | bit | Register Name | Description                                                                                                                                                                                | Default value<br>after reset |           |               |           | Reflection<br>timing |
|---------|------------------|-----|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------|---------------|-----------|----------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                                                                                                                                                            | By<br>register               |           | By<br>address |           |                      |
| 46h     | 3546h            | 0   |               | Dual ADC mode (Built-in combination):<br>IMX530: 86h<br>IMX531: 64h<br>IMX532: 86h<br>IMX535: 54h<br>IMX536: 1Fh<br>IMX537: 0Fh<br><br>Others: Same as the setting value in<br>Normal mode | 43h<br>*1                    | 2Ah<br>*2 | 43h<br>*1     | 2Ah<br>*2 | S                    |
|         |                  | 1   |               |                                                                                                                                                                                            |                              |           |               |           |                      |
|         |                  | 2   |               |                                                                                                                                                                                            |                              |           |               |           |                      |
|         |                  | 3   |               |                                                                                                                                                                                            |                              |           |               |           |                      |
|         |                  | 4   |               |                                                                                                                                                                                            |                              |           |               |           |                      |
|         |                  | 5   |               |                                                                                                                                                                                            |                              |           |               |           |                      |
|         |                  | 6   |               |                                                                                                                                                                                            |                              |           |               |           |                      |
| 7       |                  |     |               |                                                                                                                                                                                            |                              |           |               |           |                      |

\*1 IMX530, IMX531, IMX532

\*2 IMX535, IMX536, IMX537

Registers corresponding to Chip ID = 08h in Write mode. (I<sup>2</sup>C:36\*\*h)

| Address |                  | bit | Register Name | Description                      | Default value after reset |            | Reflection timing |
|---------|------------------|-----|---------------|----------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                  | By register               | By address |                   |
| E8h     | 36E8h            | 0   |               | Refer to the Drive mode setting. | 2Dh                       | 2Dh        | S                 |
|         |                  | 1   |               |                                  |                           |            |                   |
|         |                  | 2   |               |                                  |                           |            |                   |
|         |                  | 3   |               |                                  |                           |            |                   |
|         |                  | 4   |               |                                  |                           |            |                   |
|         |                  | 5   |               |                                  |                           |            |                   |
|         |                  | 6   |               |                                  |                           |            |                   |
|         |                  | 7   |               |                                  |                           |            |                   |

Registers corresponding to Chip ID = 15h in Write mode. (I<sup>2</sup>C:43\*\*h)

| Address |                  | bit | Register Name | Description                                                                     | Default value after reset |            | Reflection timing |
|---------|------------------|-----|---------------|---------------------------------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                                                 | By register               | By address |                   |
| 1Eh     | 431Eh            | 0   |               | IMX535, IMX536, IMX537: 16h<br>Others: Same as the setting value in Normal mode | 02h                       | 02h        | S                 |
|         |                  | 1   |               |                                                                                 |                           |            |                   |
|         |                  | 2   |               |                                                                                 |                           |            |                   |
|         |                  | 3   |               |                                                                                 |                           |            |                   |
|         |                  | 4   |               |                                                                                 |                           |            |                   |
|         |                  | 5   |               |                                                                                 |                           |            |                   |
|         |                  | 6   |               |                                                                                 |                           |            |                   |
|         |                  | 7   |               |                                                                                 |                           |            |                   |
| 1Fh     | 431Fh            | 0   |               | IMX535, IMX536, IMX537: 16h<br>Others: Same as the setting value in Normal mode | 02h                       | 02h        | S                 |
|         |                  | 1   |               |                                                                                 |                           |            |                   |
|         |                  | 2   |               |                                                                                 |                           |            |                   |
|         |                  | 3   |               |                                                                                 |                           |            |                   |
|         |                  | 4   |               |                                                                                 |                           |            |                   |
|         |                  | 5   |               |                                                                                 |                           |            |                   |
|         |                  | 6   |               |                                                                                 |                           |            |                   |
|         |                  | 7   |               |                                                                                 |                           |            |                   |

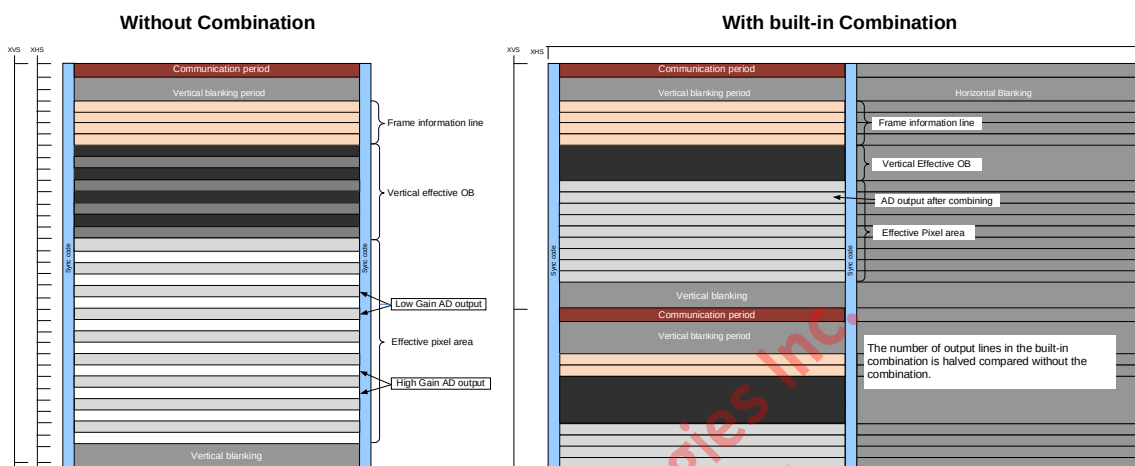
## 5. Dual ADC

### 5.1. Overview

This function outputs 2 different gain data of same pixel. By combining these images, the image which is like high bit image or HDR image can be gotten. Dual ADC has the following two modes.

**Without combination** : The lines for low gain and high gain line are alternately output after the Frame information line (Embedded data line at SLVS-EC) as below.

**With built-in combination**: Low gain lines and high gain lines are combined and output. Therefore, the number of output lines is halved with respect to the combining-less mode.



Dual ADC Output Image

## 6. Readout Drive Mode (without combination)

### 6.1. Readout Drive mode

#### 6.1.1. IMX530

SLVS Output, Data rate: 594 Mbps/ch (IMX530)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 11.0                 | 5320                       | 9200 | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 |                | 5.6                  |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 594                 | 10             | 8.8                  |                            |      | 446.0                | 892.0           | 648.8        |
|            | 4                     | 594                 |                | 4.5                  |                            |      | 870.5                | 1741.0          | 1266.2       |
|            | 8                     | 594                 | 12             | 7.4                  |                            |      | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 |                | 3.7                  |                            |      | 1043.0               | 2086.0          | 1517.1       |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 |                | *2                   |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 446.0                | 892.0           | 648.8        |
|            | 4                     | 594                 |                | *2                   |                            |      | 870.5                | 1741.0          | 1266.2       |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 1043.0               | 2086.0          | 1517.1       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 11.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX530)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 5.5                  | 5320                       | 9200 | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 |                | 2.8                  |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 297                 | 10             | 4.4                  |                            |      | 881.0                | 1762.0          | 1281.5       |
|            | 4                     | 297                 |                | 2.2                  |                            |      | 1726.5               | 3453.0          | 2511.3       |
|            | 8                     | 297                 | 12             | 3.7                  |                            |      | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 |                | 1.9                  |                            |      | 2070.0               | 4140.0          | 3011.0       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 881.0                | 1762.0          | 1281.5       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1726.5               | 3453.0          | 2511.3       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 |                | *2                   |                            |      | 2070.0               | 4140.0          | 3011.0       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 5.5 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX530)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 15.1                 | 5320                       | 9200 | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 |                | 7.9                  |                            |      | 495.0                | 990.0           | 720.0        |
|            | 8                     | 891                 | 10             | 12.2                 |                            |      | 323.0                | 646.0           | 469.9        |
|            | 4                     | 891                 |                | 6.4                  |                            |      | 616.5                | 1233.0          | 896.8        |
|            | 8                     | 891                 | 12             | 11.0                 |                            |      | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 |                | 5.6                  |                            |      | 698.0                | 1396.0          | 1015.3       |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 |                | *2                   |                            |      | 495.0                | 990.0           | 720.0        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 323.0                | 646.0           | 469.9        |
|            | 4                     | 891                 |                | *2                   |                            |      | 616.5                | 1233.0          | 896.8        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 |                | *2                   |                            |      | 698.0                | 1396.0          | 1015.3       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 15.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX530)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 7.7                  | 5320                       | 9200 | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               |                | 4.0                  |                            |      | 972.5                | 1945.0          | 1414.6       |
|            | 8                     | 445.5               | 10             | 6.2                  |                            |      | 633.0                | 1266.0          | 920.8        |
|            | 4                     | 445.5               |                | 3.2                  |                            |      | 1213.5               | 2427.0          | 1765.1       |
|            | 8                     | 445.5               | 12             | 5.5                  |                            |      | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               |                | 2.8                  |                            |      | 1382.5               | 2765.0          | 2011.0       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 972.5                | 1945.0          | 1414.6       |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 633.0                | 1266.0          | 920.8        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1213.5               | 2427.0          | 1765.1       |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1382.5               | 2765.0          | 2011.0       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 7.7 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX530)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 54.1                 | 5320                       | 9200 | 72.5                 | 145.0           | 105.5        |
|            | 4                            | 4752                  |                | 34.0                 |                            |      | 115.5                | 231.0           | 168.0        |
|            | 2                            | 4752                  |                | 17.6                 |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                | 8.9                  |                            |      | 440.0                | 880.0           | 640.0        |
|            | 8                            | 4752                  | 10             | 51.9                 |                            |      | 75.5                 | 151.0           | 109.9        |
|            | 4                            | 4752                  |                | 28.0                 |                            |      | 140.5                | 281.0           | 204.4        |
|            | 2                            | 4752                  |                | 14.3                 |                            |      | 274.5                | 549.0           | 399.3        |
|            | 1                            | 4752                  |                | 7.2                  |                            |      | 546.5                | 1093.0          | 795.0        |
|            | 8                            | 4752                  | 12             | 37.6                 |                            |      | 104.5                | 209.0           | 152.0        |
|            | 4                            | 4752                  |                | 23.5                 |                            |      | 167.5                | 335.0           | 243.7        |
|            | 2                            | 4752                  |                | 12.0                 |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                | 6.0                  |                            |      | 652.5                | 1305.0          | 949.1        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 72.5                 | 145.0           | 105.5        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 115.5                | 231.0           | 168.0        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 440.0                | 880.0           | 640.0        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 75.5                 | 151.0           | 109.9        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 140.5                | 281.0           | 204.4        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 274.5                | 549.0           | 399.3        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 546.5                | 1093.0          | 795.0        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 104.5                | 209.0           | 152.0        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 167.5                | 335.0           | 243.7        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 652.5                | 1305.0          | 949.1        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 54.1 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX530)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 32.3                 | 5320                       | 9200 | 121.5                | 243.0           | 176.8        |
|            | 4                            | 2376                  |                | 17.3                 |                            |      | 227.5                | 455.0           | 331.0        |
|            | 2                            | 2376                  |                | 8.9                  |                            |      | 443.5                | 887.0           | 645.1        |
|            | 1                            | 2376                  |                | 4.4                  |                            |      | 880.0                | 1760.0          | 1280.0       |
|            | 8                            | 2376                  | 10             | 27.4                 |                            |      | 143.5                | 287.0           | 208.8        |
|            | 4                            | 2376                  |                | 14.2                 |                            |      | 277.5                | 555.0           | 403.7        |
|            | 2                            | 2376                  |                | 7.1                  |                            |      | 549.0                | 1098.0          | 798.6        |
|            | 1                            | 2376                  |                | 3.6                  |                            |      | 1093.0               | 2186.0          | 1589.9       |
|            | 8                            | 2376                  | 12             | 22.2                 |                            |      | 177.0                | 354.0           | 257.5        |
|            | 4                            | 2376                  |                | 11.9                 |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                | 6.0                  |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                | 3.0                  |                            |      | 1304.5               | 2609.0          | 1897.5       |
| ROI        | 8                            | 2376                  | 8              | *2                   | *3                         | *3   | 121.5                | 243.0           | 176.8        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 227.5                | 455.0           | 331.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 443.5                | 887.0           | 645.1        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 880.0                | 1760.0          | 1280.0       |
|            | 8                            | 2376                  | 10             | *2                   |                            |      | 143.5                | 287.0           | 208.8        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 277.5                | 555.0           | 403.7        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 549.0                | 1098.0          | 798.6        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 1093.0               | 2186.0          | 1589.9       |
|            | 8                            | 2376                  | 12             | *2                   |                            |      | 177.0                | 354.0           | 257.5        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 1304.5               | 2609.0          | 1897.5       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 32.3 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX530)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 16.4                 | 5320                       | 9200          | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                | 8.7                  |                            |               | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                | 4.4                  |                            |               | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                | 2.2                  |                            |               | 1759.5               | 3519.0          | 2559.3       |
|            | 8                            | 1188                  | 10             | 13.8                 |                            |               | 284.5                | 569.0           | 413.9        |
|            | 4                            | 1188                  |                | 7.1                  |                            |               | 555.0                | 1110.0          | 807.3        |
|            | 2                            | 1188                  |                | 3.6                  |                            |               | 1098.0               | 2196.0          | 1597.1       |
|            | 1                            | 1188                  |                | 1.8                  |                            |               | 2186.0               | 4372.0          | 3179.7       |
|            | 8                            | 1188                  | 12             | 11.2                 |                            |               | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                | 5.9                  |                            |               | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                | 3.0                  |                            |               | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                | 1.5                  |                            |               | 2609.0               | 5218.0          | 3795.0       |
| ROI        | 8                            | 1188                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1759.5               | 3519.0          | 2559.3       |
|            | 8                            | 1188                  | 10             | <sup>*2</sup>        |                            |               | 284.5                | 569.0           | 413.9        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 555.0                | 1110.0          | 807.3        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1098.0               | 2196.0          | 1597.1       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2186.0               | 4372.0          | 3179.7       |
|            | 8                            | 1188                  | 12             | <sup>*2</sup>        |                            |               | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2609.0               | 5218.0          | 3795.0       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 16.4 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

### 6.1.2. IMX531

SLVS Output, Data rate: 594 Mbps/ch (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 13.1                 | 4504                       | 9008 | 307.0                | 614.0           | 446.6        |
|            | 4                     | 594                 |                | 6.7                  |                            |      | 596.0                | 1192.0          | 867.0        |
|            | 8                     | 594                 | 10             | 10.5                 |                            |      | 382.0                | 764.0           | 555.7        |
|            | 4                     | 594                 |                | 5.4                  |                            |      | 743.0                | 1486.0          | 1080.8       |
|            | 8                     | 594                 | 12             | 8.8                  |                            |      | 457.0                | 914.0           | 664.8        |
|            | 4                     | 594                 |                | 4.5                  |                            |      | 890.0                | 1780.0          | 1294.6       |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 307.0                | 614.0           | 446.6        |
|            | 4                     | 594                 |                | *2                   |                            |      | 596.0                | 1192.0          | 867.0        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 382.0                | 764.0           | 555.7        |
|            | 4                     | 594                 |                | *2                   |                            |      | 743.0                | 1486.0          | 1080.8       |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 457.0                | 914.0           | 664.8        |
|            | 4                     | 594                 |                | *2                   |                            |      | 890.0                | 1780.0          | 1294.6       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 13.1 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 6.6                  | 4504                       | 9008 | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 297                 |                | 3.4                  |                            |      | 1178.5               | 2357.0          | 1714.2       |
|            | 8                     | 297                 | 10             | 5.3                  |                            |      | 753.5                | 1507.0          | 1096.0       |
|            | 4                     | 297                 |                | 2.7                  |                            |      | 1471.5               | 2943.0          | 2140.4       |
|            | 8                     | 297                 | 12             | 4.4                  |                            |      | 902.5                | 1805.0          | 1312.8       |
|            | 4                     | 297                 |                | 2.2                  |                            |      | 1764.0               | 3528.0          | 2565.9       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1178.5               | 2357.0          | 1714.2       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 753.5                | 1507.0          | 1096.0       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1471.5               | 2943.0          | 2140.4       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 902.5                | 1805.0          | 1312.8       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1764.0               | 3528.0          | 2565.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 6.6 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 17.8                 | 4504                       | 9008 | 226.0                | 452.0           | 328.8        |
|            | 4                     | 891                 |                | 9.4                  |                            |      | 427.0                | 854.0           | 621.1        |
|            | 8                     | 891                 | 10             | 14.3                 |                            |      | 280.5                | 561.0           | 408.0        |
|            | 4                     | 891                 |                | 7.5                  |                            |      | 531.5                | 1063.0          | 773.1        |
|            | 8                     | 891                 | 12             | 13.1                 |                            |      | 307.0                | 614.0           | 446.6        |
|            | 4                     | 891                 |                | 6.7                  |                            |      | 596.0                | 1192.0          | 867.0        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 226.0                | 452.0           | 328.8        |
|            | 4                     | 891                 |                | *2                   |                            |      | 427.0                | 854.0           | 621.1        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 280.5                | 561.0           | 408.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 531.5                | 1063.0          | 773.1        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 307.0                | 614.0           | 446.6        |
|            | 4                     | 891                 |                | *2                   |                            |      | 596.0                | 1192.0          | 867.0        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 17.8 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 9.1                  | 4504                       | 9008 | 440.0                | 880.0           | 640.0        |
|            | 4                     | 445.5               |                | 4.8                  |                            |      | 836.5                | 1673.0          | 1216.8       |
|            | 8                     | 445.5               | 10             | 7.3                  |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 4                     | 445.5               |                | 3.8                  |                            |      | 1043.5               | 2087.0          | 1517.9       |
|            | 8                     | 445.5               | 12             | 6.6                  |                            |      | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 445.5               |                | 3.4                  |                            |      | 1178.5               | 2357.0          | 1714.2       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 440.0                | 880.0           | 640.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 836.5                | 1673.0          | 1216.8       |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1043.5               | 2087.0          | 1517.9       |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1178.5               | 2357.0          | 1714.2       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 9.1 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 55.2                 | 4504                       | 9008 | 72.5                 | 145.0           | 105.5        |
|            | 4                            | 4752                  |                | 40.5                 |                            |      | 99.0                 | 198.0           | 144.0        |
|            | 2                            | 4752                  |                | 21.4                 |                            |      | 187.5                | 375.0           | 272.8        |
|            | 1                            | 4752                  |                | 10.7                 |                            |      | 373.5                | 747.0           | 543.3        |
|            | 8                            | 4752                  | 10             | 53.0                 |                            |      | 75.5                 | 151.0           | 109.9        |
|            | 4                            | 4752                  |                | 33.3                 |                            |      | 120.5                | 241.0           | 175.3        |
|            | 2                            | 4752                  |                | 17.1                 |                            |      | 234.5                | 469.0           | 341.1        |
|            | 1                            | 4752                  |                | 8.6                  |                            |      | 466.5                | 933.0           | 678.6        |
|            | 8                            | 4752                  | 12             | 38.3                 |                            |      | 104.5                | 209.0           | 152.0        |
|            | 4                            | 4752                  |                | 28.7                 |                            |      | 140.0                | 280.0           | 203.7        |
|            | 2                            | 4752                  |                | 14.4                 |                            |      | 278.5                | 557.0           | 405.1        |
|            | 1                            | 4752                  |                | 7.2                  |                            |      | 554.5                | 1109.0          | 806.6        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 72.5                 | 145.0           | 105.5        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 99.0                 | 198.0           | 144.0        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 187.5                | 375.0           | 272.8        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 373.5                | 747.0           | 543.3        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 75.5                 | 151.0           | 109.9        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 120.5                | 241.0           | 175.3        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 234.5                | 469.0           | 341.1        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 466.5                | 933.0           | 678.6        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 104.5                | 209.0           | 152.0        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 140.0                | 280.0           | 203.7        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 278.5                | 557.0           | 405.1        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 554.5                | 1109.0          | 806.6        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 55.2 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 38.3                 | 4504                       | 9008 | 104.5                | 209.0           | 152.0        |
|            | 4                            | 2376                  |                | 20.6                 |                            |      | 195.0                | 390.0           | 283.7        |
|            | 2                            | 2376                  |                | 10.7                 |                            |      | 374.5                | 749.0           | 544.8        |
|            | 1                            | 2376                  |                | 5.3                  |                            |      | 747.0                | 1494.0          | 1086.6       |
|            | 8                            | 2376                  | 10             | 32.5                 |                            |      | 123.5                | 247.0           | 179.7        |
|            | 4                            | 2376                  |                | 16.9                 |                            |      | 237.5                | 475.0           | 345.5        |
|            | 2                            | 2376                  |                | 8.5                  |                            |      | 469.0                | 938.0           | 682.2        |
|            | 1                            | 2376                  |                | 4.3                  |                            |      | 933.0                | 1866.0          | 1357.1       |
|            | 8                            | 2376                  | 12             | 26.5                 |                            |      | 151.5                | 303.0           | 220.4        |
|            | 4                            | 2376                  |                | 14.4                 |                            |      | 279.5                | 559.0           | 406.6        |
|            | 2                            | 2376                  |                | 7.2                  |                            |      | 557.0                | 1114.0          | 810.2        |
|            | 1                            | 2376                  |                | 3.6                  |                            |      | 1109.0               | 2218.0          | 1613.1       |
| ROI        | 8                            | 2376                  | 8              | *2                   | *3                         | *3   | 104.5                | 209.0           | 152.0        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 195.0                | 390.0           | 283.7        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 374.5                | 749.0           | 544.8        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 747.0                | 1494.0          | 1086.6       |
|            | 8                            | 2376                  | 10             | *2                   |                            |      | 123.5                | 247.0           | 179.7        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 237.5                | 475.0           | 345.5        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 469.0                | 938.0           | 682.2        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 933.0                | 1866.0          | 1357.1       |
|            | 8                            | 2376                  | 12             | *2                   |                            |      | 151.5                | 303.0           | 220.4        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 279.5                | 559.0           | 406.6        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 557.0                | 1114.0          | 810.2        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 1109.0               | 2218.0          | 1613.1       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 38.4 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 19.5                 | 4504                       | 9008          | 206.0                | 412.0           | 299.7        |
|            | 4                            | 1188                  |                | 10.4                 |                            |               | 387.0                | 774.0           | 563.0        |
|            | 2                            | 1188                  |                | 5.3                  |                            |               | 749.0                | 1498.0          | 1089.5       |
|            | 1                            | 1188                  |                | 2.7                  |                            |               | 1493.5               | 2987.0          | 2172.4       |
|            | 8                            | 1188                  | 10             | 16.5                 |                            |               | 244.0                | 488.0           | 355.0        |
|            | 4                            | 1188                  |                | 8.4                  |                            |               | 474.5                | 949.0           | 690.2        |
|            | 2                            | 1188                  |                | 4.3                  |                            |               | 937.5                | 1875.0          | 1363.7       |
|            | 1                            | 1188                  |                | 2.1                  |                            |               | 1865.5               | 3731.0          | 2713.5       |
|            | 8                            | 1188                  | 12             | 13.3                 |                            |               | 301.0                | 602.0           | 437.9        |
|            | 4                            | 1188                  |                | 7.2                  |                            |               | 559.0                | 1118.0          | 813.1        |
|            | 2                            | 1188                  |                | 3.6                  |                            |               | 1114.0               | 2228.0          | 1620.4       |
|            | 1                            | 1188                  |                | 1.8                  |                            |               | 2218.0               | 4436.0          | 3226.2       |
| ROI        | 8                            | 1188                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 206.0                | 412.0           | 299.7        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 387.0                | 774.0           | 563.0        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 749.0                | 1498.0          | 1089.5       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1493.5               | 2987.0          | 2172.4       |
|            | 8                            | 1188                  | 10             | <sup>*2</sup>        |                            |               | 244.0                | 488.0           | 355.0        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 474.5                | 949.0           | 690.2        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 937.5                | 1875.0          | 1363.7       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1865.5               | 3731.0          | 2713.5       |
|            | 8                            | 1188                  | 12             | <sup>*2</sup>        |                            |               | 301.0                | 602.0           | 437.9        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 559.0                | 1118.0          | 813.1        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1114.0               | 2228.0          | 1620.4       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2218.0               | 4436.0          | 3226.2       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 19.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

## 6.1.3. IMX532

SLVS Output, Data rate: 594 Mbps/ch (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 16.5                 | 5320                       | 6064 | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 |                | 8.4                  |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 594                 | 10             | 13.2                 |                            |      | 446.0                | 892.0           | 648.8        |
|            | 4                     | 594                 |                | 6.8                  |                            |      | 870.5                | 1741.0          | 1266.2       |
|            | 8                     | 594                 | 12             | 11.1                 |                            |      | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 |                | 5.6                  |                            |      | 1043.0               | 2086.0          | 1517.1       |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 |                | *2                   |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 446.0                | 892.0           | 648.8        |
|            | 4                     | 594                 |                | *2                   |                            |      | 870.5                | 1741.0          | 1266.2       |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 1043.0               | 2086.0          | 1517.1       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 16.5 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 8.4                  | 5320                       | 6064 | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 |                | 4.2                  |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 297                 | 10             | 6.7                  |                            |      | 881.0                | 1762.0          | 1281.5       |
|            | 4                     | 297                 |                | 3.4                  |                            |      | 1726.5               | 3453.0          | 2511.3       |
|            | 8                     | 297                 | 12             | 5.6                  |                            |      | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 |                | 2.8                  |                            |      | 2070.0               | 4140.0          | 3011.0       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 881.0                | 1762.0          | 1281.5       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1726.5               | 3453.0          | 2511.3       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 |                | *2                   |                            |      | 2070.0               | 4140.0          | 3011.0       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 8.4 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX532)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 22.7                 | 5320                       | 6064 | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 |                | 11.9                 |                            |      | 495.0                | 990.0           | 720.0        |
|            | 8                     | 891                 | 10             | 18.3                 |                            |      | 323.0                | 646.0           | 469.9        |
|            | 4                     | 891                 |                | 9.6                  |                            |      | 616.5                | 1233.0          | 896.8        |
|            | 8                     | 891                 | 12             | 16.5                 |                            |      | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 |                | 8.4                  |                            |      | 698.0                | 1396.0          | 1015.3       |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 |                | *2                   |                            |      | 495.0                | 990.0           | 720.0        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 323.0                | 646.0           | 469.9        |
|            | 4                     | 891                 |                | *2                   |                            |      | 616.5                | 1233.0          | 896.8        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 |                | *2                   |                            |      | 698.0                | 1396.0          | 1015.3       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 22.7 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX532)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 11.6                 | 5320                       | 6064 | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               |                | 6.1                  |                            |      | 972.5                | 1945.0          | 1414.6       |
|            | 8                     | 445.5               | 10             | 9.3                  |                            |      | 633.0                | 1266.0          | 920.8        |
|            | 4                     | 445.5               |                | 4.8                  |                            |      | 1213.5               | 2427.0          | 1765.1       |
|            | 8                     | 445.5               | 12             | 8.4                  |                            |      | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               |                | 4.2                  |                            |      | 1382.5               | 2765.0          | 2011.0       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 972.5                | 1945.0          | 1414.6       |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 633.0                | 1266.0          | 920.8        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1213.5               | 2427.0          | 1765.1       |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1382.5               | 2765.0          | 2011.0       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 11.6 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 80.9                 | 5320                       | 6064 | 72.5                 | 145.0           | 105.5        |
|            | 4                            | 4752                  |                | 51.0                 |                            |      | 115.5                | 231.0           | 168.0        |
|            | 2                            | 4752                  |                | 26.4                 |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                | 13.4                 |                            |      | 440.0                | 880.0           | 640.0        |
|            | 8                            | 4752                  | 10             | 77.7                 |                            |      | 75.5                 | 151.0           | 109.9        |
|            | 4                            | 4752                  |                | 41.9                 |                            |      | 140.5                | 281.0           | 204.4        |
|            | 2                            | 4752                  |                | 21.5                 |                            |      | 274.5                | 549.0           | 399.3        |
|            | 1                            | 4752                  |                | 10.8                 |                            |      | 546.5                | 1093.0          | 795.0        |
|            | 8                            | 4752                  | 12             | 56.3                 |                            |      | 104.5                | 209.0           | 152.0        |
|            | 4                            | 4752                  |                | 35.2                 |                            |      | 167.5                | 335.0           | 243.7        |
|            | 2                            | 4752                  |                | 18.0                 |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                | 9.0                  |                            |      | 652.5                | 1305.0          | 949.1        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 72.5                 | 145.0           | 105.5        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 115.5                | 231.0           | 168.0        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 440.0                | 880.0           | 640.0        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 75.5                 | 151.0           | 109.9        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 140.5                | 281.0           | 204.4        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 274.5                | 549.0           | 399.3        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 546.5                | 1093.0          | 795.0        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 104.5                | 209.0           | 152.0        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 167.5                | 335.0           | 243.7        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 652.5                | 1305.0          | 949.1        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 80.9 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX532)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 48.5                 | 5320                       | 6064 | 121.5                | 243.0           | 176.8        |
|            | 4                            | 2376                  |                | 26.0                 |                            |      | 227.5                | 455.0           | 331.0        |
|            | 2                            | 2376                  |                | 13.3                 |                            |      | 443.5                | 887.0           | 645.1        |
|            | 1                            | 2376                  |                | 6.7                  |                            |      | 880.0                | 1760.0          | 1280.0       |
|            | 8                            | 2376                  | 10             | 41.1                 |                            |      | 143.5                | 287.0           | 208.8        |
|            | 4                            | 2376                  |                | 21.3                 |                            |      | 277.5                | 555.0           | 403.7        |
|            | 2                            | 2376                  |                | 10.8                 |                            |      | 549.0                | 1098.0          | 798.6        |
|            | 1                            | 2376                  |                | 5.4                  |                            |      | 1093.0               | 2186.0          | 1589.9       |
|            | 8                            | 2376                  | 12             | 33.3                 |                            |      | 177.0                | 354.0           | 257.5        |
|            | 4                            | 2376                  |                | 17.8                 |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                | 9.0                  |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                | 4.5                  |                            |      | 1304.5               | 2609.0          | 1897.5       |
| ROI        | 8                            | 2376                  | 8              | *2                   | *3                         | *3   | 121.5                | 243.0           | 176.8        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 227.5                | 455.0           | 331.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 443.5                | 887.0           | 645.1        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 880.0                | 1760.0          | 1280.0       |
|            | 8                            | 2376                  | 10             | *2                   |                            |      | 143.5                | 287.0           | 208.8        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 277.5                | 555.0           | 403.7        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 549.0                | 1098.0          | 798.6        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 1093.0               | 2186.0          | 1589.9       |
|            | 8                            | 2376                  | 12             | *2                   |                            |      | 177.0                | 354.0           | 257.5        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 1304.5               | 2609.0          | 1897.5       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 48.5 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX532)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 24.6                 | 5320                       | 6064          | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                | 13.0                 |                            |               | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                | 6.6                  |                            |               | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                | 3.3                  |                            |               | 1759.5               | 3519.0          | 2559.3       |
|            | 8                            | 1188                  | 10             | 20.8                 |                            |               | 284.5                | 569.0           | 413.9        |
|            | 4                            | 1188                  |                | 10.6                 |                            |               | 555.0                | 1110.0          | 807.3        |
|            | 2                            | 1188                  |                | 5.4                  |                            |               | 1098.0               | 2196.0          | 1597.1       |
|            | 1                            | 1188                  |                | 2.7                  |                            |               | 2186.0               | 4372.0          | 3179.7       |
|            | 8                            | 1188                  | 12             | 16.8                 |                            |               | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                | 8.9                  |                            |               | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                | 4.5                  |                            |               | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                | 2.2                  |                            |               | 2609.0               | 5218.0          | 3795.0       |
| ROI        | 8                            | 1188                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1759.5               | 3519.0          | 2559.3       |
|            | 8                            | 1188                  | 10             | <sup>*2</sup>        |                            |               | 284.5                | 569.0           | 413.9        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 555.0                | 1110.0          | 807.3        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1098.0               | 2196.0          | 1597.1       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2186.0               | 4372.0          | 3179.7       |
|            | 8                            | 1188                  | 12             | <sup>*2</sup>        |                            |               | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2609.0               | 5218.0          | 3795.0       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 24.6 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

#### 6.1.4. IMX535-AAQJ

SLVS Output, Data rate: 594 Mbps/ch (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 21.0                 | 4096                       | 6000 | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 |                | 10.9                 |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 594                 | 10             | 16.9                 |                            |      | 352.0                | 704.0           | 512.0        |
|            | 4                     | 594                 |                | 8.7                  |                            |      | 683.0                | 1366.0          | 993.5        |
|            | 8                     | 594                 | 12             | 14.1                 |                            |      | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 |                | 7.3                  |                            |      | 818.0                | 1636.0          | 1189.9       |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 |                | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 352.0                | 704.0           | 512.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 683.0                | 1366.0          | 993.5        |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 |                | *2                   |                            |      | 818.0                | 1636.0          | 1189.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 10.7                 | 4096                       | 6000 | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 |                | 5.5                  |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 297                 | 10             | 8.6                  |                            |      | 693.5                | 1387.0          | 1008.8       |
|            | 4                     | 297                 |                | 4.4                  |                            |      | 1351.5               | 2703.0          | 1965.9       |
|            | 8                     | 297                 | 12             | 7.2                  |                            |      | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 |                | 3.6                  |                            |      | 1620.0               | 3240.0          | 2356.4       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 693.5                | 1387.0          | 1008.8       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1351.5               | 2703.0          | 1965.9       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1620.0               | 3240.0          | 2356.4       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 10.7 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX535-AAQJ)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 30.8                 | 4096                       | 6000 | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 |                | 16.1                 |                            |      | 369.5                | 739.0           | 537.5        |
|            | 8                     | 891                 | 10             | 24.9                 |                            |      | 238.5                | 477.0           | 347.0        |
|            | 4                     | 891                 |                | 13.0                 |                            |      | 459.5                | 919.0           | 668.4        |
|            | 8                     | 891                 | 12             | 21.0                 |                            |      | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 |                | 10.9                 |                            |      | 548.0                | 1096.0          | 797.1        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 369.5                | 739.0           | 537.5        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 238.5                | 477.0           | 347.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 459.5                | 919.0           | 668.4        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 |                | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 30.8 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX535-AAQJ)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 15.9                 | 4096                       | 6000 | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               |                | 8.2                  |                            |      | 725.5                | 1451.0          | 1055.3       |
|            | 8                     | 445.5               | 10             | 12.8                 |                            |      | 465.5                | 931.0           | 677.1        |
|            | 4                     | 445.5               |                | 6.6                  |                            |      | 904.5                | 1809.0          | 1315.7       |
|            | 8                     | 445.5               | 12             | 10.7                 |                            |      | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               |                | 5.5                  |                            |      | 1082.5               | 2165.0          | 1574.6       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 725.5                | 1451.0          | 1055.3       |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 465.5                | 931.0           | 677.1        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 904.5                | 1809.0          | 1315.7       |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1082.5               | 2165.0          | 1574.6       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 15.9 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 95.6                 | 4096                       | 6000 | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | 64.6                 |                            |      | 91.0                 | 182.0           | 132.4        |
|            | 2                            | 4752                  |                | 34.5                 |                            |      | 172.0                | 344.0           | 250.2        |
|            | 1                            | 4752                  |                | 17.4                 |                            |      | 343.0                | 686.0           | 499.0        |
|            | 8                            | 4752                  | 10             | 90.5                 |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | 53.4                 |                            |      | 110.5                | 221.0           | 160.8        |
|            | 2                            | 4752                  |                | 27.7                 |                            |      | 214.5                | 429.0           | 312.0        |
|            | 1                            | 4752                  |                | 14.0                 |                            |      | 426.5                | 853.0           | 620.4        |
|            | 8                            | 4752                  | 12             | 62.0                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | 46.0                 |                            |      | 128.5                | 257.0           | 187.0        |
|            | 2                            | 4752                  |                | 23.3                 |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                | 11.7                 |                            |      | 508.5                | 1017.0          | 739.7        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 91.0                 | 182.0           | 132.4        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 172.0                | 344.0           | 250.2        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 343.0                | 686.0           | 499.0        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 110.5                | 221.0           | 160.8        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 214.5                | 429.0           | 312.0        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 426.5                | 853.0           | 620.4        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 128.5                | 257.0           | 187.0        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 508.5                | 1017.0          | 739.7        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 95.6 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 61.0                 | 4096                       | 6000          | 96.5                 | 193.0           | 140.4        |
|            | 4                            | 2376                  |                | 33.2                 |                            |               | 178.5                | 357.0           | 259.7        |
|            | 2                            | 2376                  |                | 17.3                 |                            |               | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                | 8.7                  |                            |               | 685.5                | 1371.0          | 997.1        |
|            | 8                            | 2376                  | 10             | 52.0                 |                            |               | 113.5                | 227.0           | 165.1        |
|            | 4                            | 2376                  |                | 27.3                 |                            |               | 217.5                | 435.0           | 316.4        |
|            | 2                            | 2376                  |                | 13.9                 |                            |               | 429.0                | 858.0           | 624.0        |
|            | 1                            | 2376                  |                | 7.0                  |                            |               | 853.0                | 1706.0          | 1240.8       |
|            | 8                            | 2376                  | 12             | 42.4                 |                            |               | 139.5                | 279.0           | 203.0        |
|            | 4                            | 2376                  |                | 23.2                 |                            |               | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                | 11.7                 |                            |               | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                | 5.8                  |                            |               | 1017.0               | 2034.0          | 1479.3       |
| ROI        | 8                            | 2376                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 96.5                 | 193.0           | 140.4        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 178.5                | 357.0           | 259.7        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 685.5                | 1371.0          | 997.1        |
|            | 8                            | 2376                  | 10             | <sup>*2</sup>        |                            |               | 113.5                | 227.0           | 165.1        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 217.5                | 435.0           | 316.4        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 429.0                | 858.0           | 624.0        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 853.0                | 1706.0          | 1240.8       |
|            | 8                            | 2376                  | 12             | <sup>*2</sup>        |                            |               | 139.5                | 279.0           | 203.0        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 1017.0               | 2034.0          | 1479.3       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.0 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 31.2                 | 4096                       | 6000          | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                | 16.8                 |                            |               | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                | 8.7                  |                            |               | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                | 4.3                  |                            |               | 1370.5               | 2741.0          | 1993.5       |
|            | 8                            | 1188                  | 10             | 26.5                 |                            |               | 224.5                | 449.0           | 326.6        |
|            | 4                            | 1188                  |                | 13.7                 |                            |               | 435.0                | 870.0           | 632.8        |
|            | 2                            | 1188                  |                | 6.9                  |                            |               | 858.0                | 1716.0          | 1248.0       |
|            | 1                            | 1188                  |                | 3.5                  |                            |               | 1706.0               | 3412.0          | 2481.5       |
|            | 8                            | 1188                  | 12             | 21.5                 |                            |               | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                | 11.6                 |                            |               | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                | 5.8                  |                            |               | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                | 2.9                  |                            |               | 2034.0               | 4068.0          | 2958.6       |
| ROI        | 8                            | 1188                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1370.5               | 2741.0          | 1993.5       |
|            | 8                            | 1188                  | 10             | <sup>*2</sup>        |                            |               | 224.5                | 449.0           | 326.6        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 435.0                | 870.0           | 632.8        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 858.0                | 1716.0          | 1248.0       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1706.0               | 3412.0          | 2481.5       |
|            | 8                            | 1188                  | 12             | <sup>*2</sup>        |                            |               | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2034.0               | 4068.0          | 2958.6       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 31.2 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

### 6.1.5. IMX535-AAMJ

SLVS Output, Data rate: 594 Mbps/ch (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 21.0                 | 4096                       | 6000 | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 |                | 10.9                 |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 594                 | 10             | 16.9                 |                            |      | 352.0                | 704.0           | 512.0        |
|            | 4                     | 594                 |                | 8.7                  |                            |      | 683.0                | 1366.0          | 993.5        |
|            | 8                     | 594                 | 12             | 14.1                 |                            |      | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 |                | 7.3                  |                            |      | 818.0                | 1636.0          | 1189.9       |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 |                | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 352.0                | 704.0           | 512.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 683.0                | 1366.0          | 993.5        |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 |                | *2                   |                            |      | 818.0                | 1636.0          | 1189.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 10.7                 | 4096                       | 6000 | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 |                | 5.5                  |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 297                 | 10             | 8.6                  |                            |      | 693.5                | 1387.0          | 1008.8       |
|            | 4                     | 297                 |                | 4.4                  |                            |      | 1351.5               | 2703.0          | 1965.9       |
|            | 8                     | 297                 | 12             | 7.2                  |                            |      | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 |                | 3.6                  |                            |      | 1620.0               | 3240.0          | 2356.4       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 693.5                | 1387.0          | 1008.8       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1351.5               | 2703.0          | 1965.9       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 |                | *2                   |                            |      | 1620.0               | 3240.0          | 2356.4       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 10.7 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 30.8                 | 4096                       | 6000 | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 |                | 16.1                 |                            |      | 369.5                | 739.0           | 537.5        |
|            | 8                     | 891                 | 10             | 24.9                 |                            |      | 238.5                | 477.0           | 347.0        |
|            | 4                     | 891                 |                | 13.0                 |                            |      | 459.5                | 919.0           | 668.4        |
|            | 8                     | 891                 | 12             | 21.0                 |                            |      | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 |                | 10.9                 |                            |      | 548.0                | 1096.0          | 797.1        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 369.5                | 739.0           | 537.5        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 238.5                | 477.0           | 347.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 459.5                | 919.0           | 668.4        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 |                | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 30.8 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 15.9                 | 4096                       | 6000 | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               |                | 8.2                  |                            |      | 725.5                | 1451.0          | 1055.3       |
|            | 8                     | 445.5               | 10             | 12.8                 |                            |      | 465.5                | 931.0           | 677.1        |
|            | 4                     | 445.5               |                | 6.6                  |                            |      | 904.5                | 1809.0          | 1315.7       |
|            | 8                     | 445.5               | 12             | 10.7                 |                            |      | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               |                | 5.5                  |                            |      | 1082.5               | 2165.0          | 1574.6       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 725.5                | 1451.0          | 1055.3       |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 465.5                | 931.0           | 677.1        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 904.5                | 1809.0          | 1315.7       |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 1082.5               | 2165.0          | 1574.6       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 15.9 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 95.5                 | 4096                       | 6000 | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | 64.6                 |                            |      | 91.0                 | 182.0           | 132.4        |
|            | 2                            | 4752                  |                | 34.5                 |                            |      | 172.0                | 344.0           | 250.2        |
|            | 1                            | 4752                  |                | 17.3                 |                            |      | 343.0                | 686.0           | 499.0        |
|            | 8                            | 4752                  | 10             | 90.5                 |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | 53.4                 |                            |      | 110.5                | 221.0           | 160.8        |
|            | 2                            | 4752                  |                | 27.7                 |                            |      | 214.5                | 429.0           | 312.0        |
|            | 1                            | 4752                  |                | 14.0                 |                            |      | 426.5                | 853.0           | 620.4        |
|            | 8                            | 4752                  | 12             | 61.9                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | 46.0                 |                            |      | 128.5                | 257.0           | 187.0        |
|            | 2                            | 4752                  |                | 23.3                 |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                | 11.7                 |                            |      | 508.5                | 1017.0          | 739.7        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 91.0                 | 182.0           | 132.4        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 172.0                | 344.0           | 250.2        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 343.0                | 686.0           | 499.0        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 110.5                | 221.0           | 160.8        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 214.5                | 429.0           | 312.0        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 426.5                | 853.0           | 620.4        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 128.5                | 257.0           | 187.0        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 508.5                | 1017.0          | 739.7        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 95.5 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX535-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 61.0                 | 4096                       | 6000          | 96.5                 | 193.0           | 140.4        |
|            | 4                            | 2376                  |                | 33.2                 |                            |               | 178.5                | 357.0           | 259.7        |
|            | 2                            | 2376                  |                | 17.3                 |                            |               | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                | 8.7                  |                            |               | 685.5                | 1371.0          | 997.1        |
|            | 8                            | 2376                  | 10             | 52.0                 |                            |               | 113.5                | 227.0           | 165.1        |
|            | 4                            | 2376                  |                | 27.3                 |                            |               | 217.5                | 435.0           | 316.4        |
|            | 2                            | 2376                  |                | 13.9                 |                            |               | 429.0                | 858.0           | 624.0        |
|            | 1                            | 2376                  |                | 7.0                  |                            |               | 853.0                | 1706.0          | 1240.8       |
|            | 8                            | 2376                  | 12             | 42.4                 |                            |               | 139.5                | 279.0           | 203.0        |
|            | 4                            | 2376                  |                | 23.2                 |                            |               | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                | 11.6                 |                            |               | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                | 5.8                  |                            |               | 1017.0               | 2034.0          | 1479.3       |
| ROI        | 8                            | 2376                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 96.5                 | 193.0           | 140.4        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 178.5                | 357.0           | 259.7        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 685.5                | 1371.0          | 997.1        |
|            | 8                            | 2376                  | 10             | <sup>*2</sup>        |                            |               | 113.5                | 227.0           | 165.1        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 217.5                | 435.0           | 316.4        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 429.0                | 858.0           | 624.0        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 853.0                | 1706.0          | 1240.8       |
|            | 8                            | 2376                  | 12             | <sup>*2</sup>        |                            |               | 139.5                | 279.0           | 203.0        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 1017.0               | 2034.0          | 1479.3       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.0 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 31.2                 | 4096                       | 6000          | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                | 16.8                 |                            |               | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                | 8.7                  |                            |               | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                | 4.3                  |                            |               | 1370.5               | 2741.0          | 1993.5       |
|            | 8                            | 1188                  | 10             | 26.5                 |                            |               | 224.5                | 449.0           | 326.6        |
|            | 4                            | 1188                  |                | 13.7                 |                            |               | 435.0                | 870.0           | 632.8        |
|            | 2                            | 1188                  |                | 6.9                  |                            |               | 858.0                | 1716.0          | 1248.0       |
|            | 1                            | 1188                  |                | 3.5                  |                            |               | 1706.0               | 3412.0          | 2481.5       |
|            | 8                            | 1188                  | 12             | 21.5                 |                            |               | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                | 11.6                 |                            |               | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                | 5.8                  |                            |               | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                | 2.9                  |                            |               | 2034.0               | 4068.0          | 2958.6       |
| ROI        | 8                            | 1188                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1370.5               | 2741.0          | 1993.5       |
|            | 8                            | 1188                  | 10             | <sup>*2</sup>        |                            |               | 224.5                | 449.0           | 326.6        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 435.0                | 870.0           | 632.8        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 858.0                | 1716.0          | 1248.0       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1706.0               | 3412.0          | 2481.5       |
|            | 8                            | 1188                  | 12             | <sup>*2</sup>        |                            |               | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                | <sup>*2</sup>        |                            |               | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                | <sup>*2</sup>        |                            |               | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                | <sup>*2</sup>        |                            |               | 2034.0               | 4068.0          | 2958.6       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 31.2 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

### 6.1.6. IMX536-AAQJ

SLVS Output, Data rate: 594 Mbps/ch (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 30.5                 | 2840                       | 5680 | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 |                | 16.0                 |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 594                 | 10             | 24.6                 |                            |      | 255.0                | 510.0           | 371.0        |
|            | 4                     | 594                 |                | 12.8                 |                            |      | 489.5                | 979.0           | 712.0        |
|            | 8                     | 594                 | 12             | 20.6                 |                            |      | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 |                | 10.7                 |                            |      | 585.5                | 1171.0          | 851.7        |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 255.0                | 510.0           | 371.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 489.5                | 979.0           | 712.0        |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 585.5                | 1171.0          | 851.7        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 30.5 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 15.7                 | 2840                       | 5680 | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 |                | 8.2                  |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 297                 | 10             | 12.6                 |                            |      | 497.0                | 994.0           | 723.0        |
|            | 4                     | 297                 |                | 6.5                  |                            |      | 959.0                | 1918.0          | 1395.0       |
|            | 8                     | 297                 | 12             | 10.6                 |                            |      | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 |                | 5.4                  |                            |      | 1149.0               | 2298.0          | 1671.3       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 |                | *2                   |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 497.0                | 994.0           | 723.0        |
|            | 4                     | 297                 |                | *2                   |                            |      | 959.0                | 1918.0          | 1395.0       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1149.0               | 2298.0          | 1671.3       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 15.7 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX536-AAQJ)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 44.4                 | 2840                       | 5680 | 140.5                | 281.0           | 204.4        |
|            | 4                     | 891                 |                | 23.6                 |                            |      | 266.0                | 532.0           | 387.0        |
|            | 8                     | 891                 | 10             | 36.0                 |                            |      | 173.5                | 347.0           | 252.4        |
|            | 4                     | 891                 |                | 19.0                 |                            |      | 330.0                | 660.0           | 480.0        |
|            | 8                     | 891                 | 12             | 30.5                 |                            |      | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 |                | 16.0                 |                            |      | 393.0                | 786.0           | 571.7        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 140.5                | 281.0           | 204.4        |
|            | 4                     | 891                 |                | *2                   |                            |      | 266.0                | 532.0           | 387.0        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 173.5                | 347.0           | 252.4        |
|            | 4                     | 891                 |                | *2                   |                            |      | 330.0                | 660.0           | 480.0        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 393.0                | 786.0           | 571.7        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 44.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX536-AAQJ)  
 FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 23.3                 | 2840                       | 5680 | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               |                | 12.2                 |                            |      | 516.0                | 1032.0          | 750.6        |
|            | 8                     | 445.5               | 10             | 18.7                 |                            |      | 335.0                | 670.0           | 487.3        |
|            | 4                     | 445.5               |                | 9.8                  |                            |      | 643.0                | 1286.0          | 935.3        |
|            | 8                     | 445.5               | 12             | 15.7                 |                            |      | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               |                | 8.2                  |                            |      | 768.5                | 1537.0          | 1117.9       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 516.0                | 1032.0          | 750.6        |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 335.0                | 670.0           | 487.3        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 643.0                | 1286.0          | 935.3        |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 768.5                | 1537.0          | 1117.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 23.3 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 100.6                | 2840                       | 5680 | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | 94.5                 |                            |      | 65.0                 | 130.0           | 94.6         |
|            | 2                            | 4752                  |                | 50.8                 |                            |      | 122.5                | 245.0           | 178.2        |
|            | 1                            | 4752                  |                | 26.0                 |                            |      | 240.5                | 481.0           | 349.9        |
|            | 8                            | 4752                  | 10             | 95.3                 |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | 78.7                 |                            |      | 78.5                 | 157.0           | 114.2        |
|            | 2                            | 4752                  |                | 41.3                 |                            |      | 151.0                | 302.0           | 219.7        |
|            | 1                            | 4752                  |                | 21.0                 |                            |      | 299.0                | 598.0           | 435.0        |
|            | 8                            | 4752                  | 12             | 65.3                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | 65.3                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | 34.6                 |                            |      | 181.0                | 362.0           | 263.3        |
|            | 1                            | 4752                  |                | 17.5                 |                            |      | 359.0                | 718.0           | 522.2        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 65.0                 | 130.0           | 94.6         |
|            | 2                            | 4752                  |                | *2                   |                            |      | 122.5                | 245.0           | 178.2        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 240.5                | 481.0           | 349.9        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 78.5                 | 157.0           | 114.2        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 151.0                | 302.0           | 219.7        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 299.0                | 598.0           | 435.0        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 181.0                | 362.0           | 263.3        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 359.0                | 718.0           | 522.2        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 100.6 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 88.0                 | 2840                       | 5680          | 70.0                 | 140.0           | 101.9        |
|            | 4                            | 2376                  |                | 49.3                 |                            |               | 126.5                | 253.0           | 184.0        |
|            | 2                            | 2376                  |                | 25.8                 |                            |               | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                | 13.1                 |                            |               | 480.5                | 961.0           | 699.0        |
|            | 8                            | 2376                  | 10             | 75.9                 |                            |               | 81.5                 | 163.0           | 118.6        |
|            | 4                            | 2376                  |                | 40.6                 |                            |               | 154.0                | 308.0           | 224.0        |
|            | 2                            | 2376                  |                | 20.8                 |                            |               | 301.5                | 603.0           | 438.6        |
|            | 1                            | 2376                  |                | 10.5                 |                            |               | 597.5                | 1195.0          | 869.1        |
|            | 8                            | 2376                  | 12             | 62.1                 |                            |               | 100.0                | 200.0           | 145.5        |
|            | 4                            | 2376                  |                | 34.4                 |                            |               | 182.0                | 364.0           | 264.8        |
|            | 2                            | 2376                  |                | 17.4                 |                            |               | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                | 8.7                  |                            |               | 718.0                | 1436.0          | 1044.4       |
| ROI        | 8                            | 2376                  | 8              | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 70.0                 | 140.0           | 101.9        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 126.5                | 253.0           | 184.0        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 480.5                | 961.0           | 699.0        |
|            | 8                            | 2376                  | 10             | <sup>*2</sup>        |                            |               | 81.5                 | 163.0           | 118.6        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 154.0                | 308.0           | 224.0        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 301.5                | 603.0           | 438.6        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 597.5                | 1195.0          | 869.1        |
|            | 8                            | 2376                  | 12             | <sup>*2</sup>        |                            |               | 100.0                | 200.0           | 145.5        |
|            | 4                            | 2376                  |                | <sup>*2</sup>        |                            |               | 182.0                | 364.0           | 264.8        |
|            | 2                            | 2376                  |                | <sup>*2</sup>        |                            |               | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                | <sup>*2</sup>        |                            |               | 718.0                | 1436.0          | 1044.4       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.0 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 45.5                 | 2840                       | 5680 | 137.0                | 274.0           | 199.3        |
|            | 4                            | 1188                  |                | 24.9                 |                            |      | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                | 13.0                 |                            |      | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                | 6.5                  |                            |      | 961.0                | 1922.0          | 1397.9       |
|            | 8                            | 1188                  | 10             | 38.8                 |                            |      | 161.0                | 322.0           | 234.2        |
|            | 4                            | 1188                  |                | 20.4                 |                            |      | 307.5                | 615.0           | 447.3        |
|            | 2                            | 1188                  |                | 10.4                 |                            |      | 602.5                | 1205.0          | 876.4        |
|            | 1                            | 1188                  |                | 5.2                  |                            |      | 1194.5               | 2389.0          | 1737.5       |
|            | 8                            | 1188                  | 12             | 31.6                 |                            |      | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                | 17.3                 |                            |      | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                | 8.7                  |                            |      | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                | 4.4                  |                            |      | 1435.5               | 2871.0          | 2088.0       |
| ROI        | 8                            | 1188                  | 8              | *2                   | *3                         | *3   | 137.0                | 274.0           | 199.3        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 961.0                | 1922.0          | 1397.9       |
|            | 8                            | 1188                  | 10             | *2                   |                            |      | 161.0                | 322.0           | 234.2        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 307.5                | 615.0           | 447.3        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 602.5                | 1205.0          | 876.4        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1194.5               | 2389.0          | 1737.5       |
|            | 8                            | 1188                  | 12             | *2                   |                            |      | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1435.5               | 2871.0          | 2088.0       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 45.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 6.1.7. IMX536-AAMJ

SLVS Output, Data rate: 594 Mbps/ch (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 30.4                 | 2840                       | 5680 | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 |                | 16.0                 |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 594                 | 10             | 24.6                 |                            |      | 255.0                | 510.0           | 371.0        |
|            | 4                     | 594                 |                | 12.8                 |                            |      | 489.5                | 979.0           | 712.0        |
|            | 8                     | 594                 | 12             | 20.6                 |                            |      | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 |                | 10.7                 |                            |      | 585.5                | 1171.0          | 851.7        |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 255.0                | 510.0           | 371.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 489.5                | 979.0           | 712.0        |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 585.5                | 1171.0          | 851.7        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 30.4 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 15.7                 | 2840                       | 5680 | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 |                | 8.2                  |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 297                 | 10             | 12.6                 |                            |      | 497.0                | 994.0           | 723.0        |
|            | 4                     | 297                 |                | 6.5                  |                            |      | 959.0                | 1918.0          | 1395.0       |
|            | 8                     | 297                 | 12             | 10.5                 |                            |      | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 |                | 5.4                  |                            |      | 1149.0               | 2298.0          | 1671.3       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 |                | *2                   |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 497.0                | 994.0           | 723.0        |
|            | 4                     | 297                 |                | *2                   |                            |      | 959.0                | 1918.0          | 1395.0       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1149.0               | 2298.0          | 1671.3       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 15.7 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX536-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 44.4                 | 2840                       | 5680 | 140.5                | 281.0           | 204.4        |
|            | 4                     | 891                 |                | 23.6                 |                            |      | 266.0                | 532.0           | 387.0        |
|            | 8                     | 891                 | 10             | 36.0                 |                            |      | 173.5                | 347.0           | 252.4        |
|            | 4                     | 891                 |                | 19.0                 |                            |      | 330.0                | 660.0           | 480.0        |
|            | 8                     | 891                 | 12             | 30.4                 |                            |      | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 |                | 16.0                 |                            |      | 393.0                | 786.0           | 571.7        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 140.5                | 281.0           | 204.4        |
|            | 4                     | 891                 |                | *2                   |                            |      | 266.0                | 532.0           | 387.0        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 173.5                | 347.0           | 252.4        |
|            | 4                     | 891                 |                | *2                   |                            |      | 330.0                | 660.0           | 480.0        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 393.0                | 786.0           | 571.7        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 44.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX536-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 23.3                 | 2840                       | 5680 | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               |                | 12.2                 |                            |      | 516.0                | 1032.0          | 750.6        |
|            | 8                     | 445.5               | 10             | 18.7                 |                            |      | 335.0                | 670.0           | 487.3        |
|            | 4                     | 445.5               |                | 9.8                  |                            |      | 643.0                | 1286.0          | 935.3        |
|            | 8                     | 445.5               | 12             | 15.7                 |                            |      | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               |                | 8.2                  |                            |      | 768.5                | 1537.0          | 1117.9       |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 516.0                | 1032.0          | 750.6        |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 335.0                | 670.0           | 487.3        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 643.0                | 1286.0          | 935.3        |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 768.5                | 1537.0          | 1117.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 23.3 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 100.6                | 2840                       | 5680 | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | 94.5                 |                            |      | 65.0                 | 130.0           | 94.6         |
|            | 2                            | 4752                  |                | 50.8                 |                            |      | 122.5                | 245.0           | 178.2        |
|            | 1                            | 4752                  |                | 26.0                 |                            |      | 240.5                | 481.0           | 349.9        |
|            | 8                            | 4752                  | 10             | 95.2                 |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | 78.6                 |                            |      | 78.5                 | 157.0           | 114.2        |
|            | 2                            | 4752                  |                | 41.3                 |                            |      | 151.0                | 302.0           | 219.7        |
|            | 1                            | 4752                  |                | 21.0                 |                            |      | 299.0                | 598.0           | 435.0        |
|            | 8                            | 4752                  | 12             | 65.3                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | 65.3                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | 34.5                 |                            |      | 181.0                | 362.0           | 263.3        |
|            | 1                            | 4752                  |                | 17.5                 |                            |      | 359.0                | 718.0           | 522.2        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 65.0                 | 130.0           | 94.6         |
|            | 2                            | 4752                  |                | *2                   |                            |      | 122.5                | 245.0           | 178.2        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 240.5                | 481.0           | 349.9        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 78.5                 | 157.0           | 114.2        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 151.0                | 302.0           | 219.7        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 299.0                | 598.0           | 435.0        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 181.0                | 362.0           | 263.3        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 359.0                | 718.0           | 522.2        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 100.6 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 87.9                 | 2840                       | 5680 | 70.0                 | 140.0           | 101.9        |
|            | 4                            | 2376                  |                | 49.3                 |                            |      | 126.5                | 253.0           | 184.0        |
|            | 2                            | 2376                  |                | 25.8                 |                            |      | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                | 13.1                 |                            |      | 480.5                | 961.0           | 699.0        |
|            | 8                            | 2376                  | 10             | 75.8                 |                            |      | 81.5                 | 163.0           | 118.6        |
|            | 4                            | 2376                  |                | 40.5                 |                            |      | 154.0                | 308.0           | 224.0        |
|            | 2                            | 2376                  |                | 20.8                 |                            |      | 301.5                | 603.0           | 438.6        |
|            | 1                            | 2376                  |                | 10.5                 |                            |      | 597.5                | 1195.0          | 869.1        |
|            | 8                            | 2376                  | 12             | 62.0                 |                            |      | 100.0                | 200.0           | 145.5        |
|            | 4                            | 2376                  |                | 34.3                 |                            |      | 182.0                | 364.0           | 264.8        |
|            | 2                            | 2376                  |                | 17.4                 |                            |      | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                | 8.7                  |                            |      | 718.0                | 1436.0          | 1044.4       |
| ROI        | 8                            | 2376                  | 8              | *2                   | *3                         | *3   | 70.0                 | 140.0           | 101.9        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 126.5                | 253.0           | 184.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 480.5                | 961.0           | 699.0        |
|            | 8                            | 2376                  | 10             | *2                   |                            |      | 81.5                 | 163.0           | 118.6        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 154.0                | 308.0           | 224.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 301.5                | 603.0           | 438.6        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 597.5                | 1195.0          | 869.1        |
|            | 8                            | 2376                  | 12             | *2                   |                            |      | 100.0                | 200.0           | 145.5        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 182.0                | 364.0           | 264.8        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 718.0                | 1436.0          | 1044.4       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 87.9 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX536-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 45.5                 | 2840                       | 5680 | 137.0                | 274.0           | 199.3        |
|            | 4                            | 1188                  |                | 24.9                 |                            |      | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                | 12.9                 |                            |      | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                | 6.5                  |                            |      | 961.0                | 1922.0          | 1397.9       |
|            | 8                            | 1188                  | 10             | 38.8                 |                            |      | 161.0                | 322.0           | 234.2        |
|            | 4                            | 1188                  |                | 20.4                 |                            |      | 307.5                | 615.0           | 447.3        |
|            | 2                            | 1188                  |                | 10.4                 |                            |      | 602.5                | 1205.0          | 876.4        |
|            | 1                            | 1188                  |                | 5.2                  |                            |      | 1194.5               | 2389.0          | 1737.5       |
|            | 8                            | 1188                  | 12             | 31.6                 |                            |      | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                | 17.3                 |                            |      | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                | 8.7                  |                            |      | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                | 4.3                  |                            |      | 1435.5               | 2871.0          | 2088.0       |
| ROI        | 8                            | 1188                  | 8              | *2                   | *3                         | *3   | 137.0                | 274.0           | 199.3        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 961.0                | 1922.0          | 1397.9       |
|            | 8                            | 1188                  | 10             | *2                   |                            |      | 161.0                | 322.0           | 234.2        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 307.5                | 615.0           | 447.3        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 602.5                | 1205.0          | 876.4        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1194.5               | 2389.0          | 1737.5       |
|            | 8                            | 1188                  | 12             | *2                   |                            |      | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1435.5               | 2871.0          | 2088.0       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 45.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

### 6.1.8. IMX537-AAQJ

SLVS Output, Data rate: 594 Mbps/ch (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 46.9                 | 2448                       | 2048 | 181.5                | 363.0           | 264.0        |
|            | 4                     | 594                 |                | 24.8                 |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 594                 | 10             | 37.9                 |                            |      | 225.0                | 450.0           | 327.3        |
|            | 4                     | 594                 |                | 19.9                 |                            |      | 429.5                | 859.0           | 624.8        |
|            | 8                     | 594                 | 12             | 31.8                 |                            |      | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 |                | 16.7                 |                            |      | 513.5                | 1027.0          | 747.0        |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 181.5                | 363.0           | 264.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 225.0                | 450.0           | 327.3        |
|            | 4                     | 594                 |                | *2                   |                            |      | 429.5                | 859.0           | 624.8        |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 |                | *2                   |                            |      | 513.5                | 1027.0          | 747.0        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 46.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 24.4                 | 2448                       | 2048 | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 |                | 12.7                 |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 297                 | 10             | 19.6                 |                            |      | 437.0                | 874.0           | 635.7        |
|            | 4                     | 297                 |                | 10.2                 |                            |      | 839.0                | 1678.0          | 1220.4       |
|            | 8                     | 297                 | 12             | 16.4                 |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 |                | 8.5                  |                            |      | 1005.0               | 2010.0          | 1461.9       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 |                | *2                   |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 437.0                | 874.0           | 635.7        |
|            | 4                     | 297                 |                | *2                   |                            |      | 839.0                | 1678.0          | 1220.4       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1005.0               | 2010.0          | 1461.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 24.4 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 67.9                 | 2448                       | 2048 | 124.5                | 249.0           | 181.1        |
|            | 4                     | 891                 |                | 36.4                 |                            |      | 234.0                | 468.0           | 340.4        |
|            | 8                     | 891                 | 10             | 55.3                 |                            |      | 153.5                | 307.0           | 223.3        |
|            | 4                     | 891                 |                | 29.5                 |                            |      | 290.0                | 580.0           | 421.9        |
|            | 8                     | 891                 | 12             | 46.9                 |                            |      | 181.5                | 363.0           | 264.0        |
|            | 4                     | 891                 |                | 24.8                 |                            |      | 345.0                | 690.0           | 501.9        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 124.5                | 249.0           | 181.1        |
|            | 4                     | 891                 |                | *2                   |                            |      | 234.0                | 468.0           | 340.4        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 153.5                | 307.0           | 223.3        |
|            | 4                     | 891                 |                | *2                   |                            |      | 290.0                | 580.0           | 421.9        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 181.5                | 363.0           | 264.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 345.0                | 690.0           | 501.9        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 67.9 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 35.9                 | 2448                       | 2048 | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               |                | 19.0                 |                            |      | 452.0                | 904.0           | 657.5        |
|            | 8                     | 445.5               | 10             | 29.0                 |                            |      | 295.0                | 590.0           | 429.1        |
|            | 4                     | 445.5               |                | 15.2                 |                            |      | 563.0                | 1126.0          | 819.0        |
|            | 8                     | 445.5               | 12             | 24.4                 |                            |      | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               |                | 12.7                 |                            |      | 672.5                | 1345.0          | 978.2        |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 452.0                | 904.0           | 657.5        |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 295.0                | 590.0           | 429.1        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 563.0                | 1126.0          | 819.0        |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 672.5                | 1345.0          | 978.2        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 35.9 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX537-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 135.9                | 2448                       | 2048 | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | 135.9                |                            |      | 61.0                 | 122.0           | 88.8         |
|            | 2                            | 4752                  |                | 78.7                 |                            |      | 107.0                | 214.0           | 155.7        |
|            | 1                            | 4752                  |                | 40.7                 |                            |      | 209.5                | 419.0           | 304.8        |
|            | 8                            | 4752                  | 10             | 128.8                |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | 122.4                |                            |      | 68.0                 | 136.0           | 99.0         |
|            | 2                            | 4752                  |                | 64.7                 |                            |      | 131.0                | 262.0           | 190.6        |
|            | 1                            | 4752                  |                | 33.0                 |                            |      | 259.0                | 518.0           | 376.8        |
|            | 8                            | 4752                  | 12             | 88.5                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | 88.5                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | 53.7                 |                            |      | 158.0                | 316.0           | 229.9        |
|            | 1                            | 4752                  |                | 27.3                 |                            |      | 313.0                | 626.0           | 455.3        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 61.0                 | 122.0           | 88.8         |
|            | 2                            | 4752                  |                | *2                   |                            |      | 107.0                | 214.0           | 155.7        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 209.5                | 419.0           | 304.8        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 68.0                 | 136.0           | 99.0         |
|            | 2                            | 4752                  |                | *2                   |                            |      | 131.0                | 262.0           | 190.6        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 259.0                | 518.0           | 376.8        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 158.0                | 316.0           | 229.9        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 313.0                | 626.0           | 455.3        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 135.9 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 133.7                | 2448                       | 2048 | 62.0                 | 124.0           | 90.2         |
|            | 4                            | 2376                  |                | 76.7                 |                            |      | 110.0                | 220.0           | 160.0        |
|            | 2                            | 2376                  |                | 40.3                 |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                | 20.4                 |                            |      | 419.0                | 838.0           | 609.5        |
|            | 8                            | 2376                  | 10             | 115.8                |                            |      | 72.0                 | 144.0           | 104.8        |
|            | 4                            | 2376                  |                | 63.2                 |                            |      | 134.0                | 268.0           | 195.0        |
|            | 2                            | 2376                  |                | 32.7                 |                            |      | 261.5                | 523.0           | 380.4        |
|            | 1                            | 2376                  |                | 16.6                 |                            |      | 517.5                | 1035.0          | 752.8        |
|            | 8                            | 2376                  | 12             | 88.5                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 2376                  |                | 53.4                 |                            |      | 159.0                | 318.0           | 231.3        |
|            | 2                            | 2376                  |                | 27.1                 |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                | 13.7                 |                            |      | 626.0                | 1252.0          | 910.6        |
| ROI        | 8                            | 2376                  | 8              | *2                   | *3                         | *3   | 62.0                 | 124.0           | 90.2         |
|            | 4                            | 2376                  |                | *2                   |                            |      | 110.0                | 220.0           | 160.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 419.0                | 838.0           | 609.5        |
|            | 8                            | 2376                  | 10             | *2                   |                            |      | 72.0                 | 144.0           | 104.8        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 134.0                | 268.0           | 195.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 261.5                | 523.0           | 380.4        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 517.5                | 1035.0          | 752.8        |
|            | 8                            | 2376                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 159.0                | 318.0           | 231.3        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 626.0                | 1252.0          | 910.6        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 133.7 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX537-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 69.9                 | 2448                       | 2048 | 121.0                | 242.0           | 176.0        |
|            | 4                            | 1188                  |                | 38.8                 |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                | 20.2                 |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                | 10.2                 |                            |      | 838.0                | 1676.0          | 1219.0       |
|            | 8                            | 1188                  | 10             | 59.9                 |                            |      | 141.5                | 283.0           | 205.9        |
|            | 4                            | 1188                  |                | 31.9                 |                            |      | 268.0                | 536.0           | 389.9        |
|            | 2                            | 1188                  |                | 16.4                 |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 1                            | 1188                  |                | 8.3                  |                            |      | 1035.0               | 2070.0          | 1505.5       |
|            | 8                            | 1188                  | 12             | 48.8                 |                            |      | 174.0                | 348.0           | 253.1        |
|            | 4                            | 1188                  |                | 27.0                 |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                | 13.6                 |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                | 6.8                  |                            |      | 1251.5               | 2503.0          | 1820.4       |
| ROI        | 8                            | 1188                  | 8              | *2                   | *3                         | *3   | 121.0                | 242.0           | 176.0        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 838.0                | 1676.0          | 1219.0       |
|            | 8                            | 1188                  | 10             | *2                   |                            |      | 141.5                | 283.0           | 205.9        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 268.0                | 536.0           | 389.9        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1035.0               | 2070.0          | 1505.5       |
|            | 8                            | 1188                  | 12             | *2                   |                            |      | 174.0                | 348.0           | 253.1        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1251.5               | 2503.0          | 1820.4       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 69.9 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

### 6.1.9. IMX537-AAMJ

SLVS Output, Data rate: 594 Mbps/ch (IMX537-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 8              | 46.8                 | 2448                       | 4096 | 181.5                | 363.0           | 264.0        |
|            | 4                     | 594                 |                | 24.8                 |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 594                 | 10             | 37.9                 |                            |      | 225.0                | 450.0           | 327.3        |
|            | 4                     | 594                 |                | 19.9                 |                            |      | 429.5                | 859.0           | 624.8        |
|            | 8                     | 594                 | 12             | 31.8                 |                            |      | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 |                | 16.7                 |                            |      | 513.5                | 1027.0          | 747.0        |
| ROI        | 8                     | 594                 | 8              | *2                   | *3                         | *3   | 181.5                | 363.0           | 264.0        |
|            | 4                     | 594                 |                | *2                   |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 594                 | 10             | *2                   |                            |      | 225.0                | 450.0           | 327.3        |
|            | 4                     | 594                 |                | *2                   |                            |      | 429.5                | 859.0           | 624.8        |
|            | 8                     | 594                 | 12             | *2                   |                            |      | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 |                | *2                   |                            |      | 513.5                | 1027.0          | 747.0        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 46.8 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX537-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 8              | 24.3                 | 2448                       | 4096 | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 |                | 12.7                 |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 297                 | 10             | 19.6                 |                            |      | 437.0                | 874.0           | 635.7        |
|            | 4                     | 297                 |                | 10.2                 |                            |      | 839.0                | 1678.0          | 1220.4       |
|            | 8                     | 297                 | 12             | 16.4                 |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 |                | 8.5                  |                            |      | 1005.0               | 2010.0          | 1461.9       |
| ROI        | 8                     | 297                 | 8              | *2                   | *3                         | *3   | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 |                | *2                   |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 297                 | 10             | *2                   |                            |      | 437.0                | 874.0           | 635.7        |
|            | 4                     | 297                 |                | *2                   |                            |      | 839.0                | 1678.0          | 1220.4       |
|            | 8                     | 297                 | 12             | *2                   |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 |                | *2                   |                            |      | 1005.0               | 2010.0          | 1461.9       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 24.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 8              | 67.8                 | 2448                       | 4096 | 124.5                | 249.0           | 181.1        |
|            | 4                     | 891                 |                | 36.4                 |                            |      | 234.0                | 468.0           | 340.4        |
|            | 8                     | 891                 | 10             | 55.3                 |                            |      | 153.5                | 307.0           | 223.3        |
|            | 4                     | 891                 |                | 29.4                 |                            |      | 290.0                | 580.0           | 421.9        |
|            | 8                     | 891                 | 12             | 46.8                 |                            |      | 181.5                | 363.0           | 264.0        |
|            | 4                     | 891                 |                | 24.8                 |                            |      | 345.0                | 690.0           | 501.9        |
| ROI        | 8                     | 891                 | 8              | *2                   | *3                         | *3   | 124.5                | 249.0           | 181.1        |
|            | 4                     | 891                 |                | *2                   |                            |      | 234.0                | 468.0           | 340.4        |
|            | 8                     | 891                 | 10             | *2                   |                            |      | 153.5                | 307.0           | 223.3        |
|            | 4                     | 891                 |                | *2                   |                            |      | 290.0                | 580.0           | 421.9        |
|            | 8                     | 891                 | 12             | *2                   |                            |      | 181.5                | 363.0           | 264.0        |
|            | 4                     | 891                 |                | *2                   |                            |      | 345.0                | 690.0           | 501.9        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 67.8 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 8              | 35.9                 | 2448                       | 4096 | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               |                | 18.9                 |                            |      | 452.0                | 904.0           | 657.5        |
|            | 8                     | 445.5               | 10             | 29.0                 |                            |      | 295.0                | 590.0           | 429.1        |
|            | 4                     | 445.5               |                | 15.2                 |                            |      | 563.0                | 1126.0          | 819.0        |
|            | 8                     | 445.5               | 12             | 24.3                 |                            |      | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               |                | 12.7                 |                            |      | 672.5                | 1345.0          | 978.2        |
| ROI        | 8                     | 445.5               | 8              | *2                   | *3                         | *3   | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 452.0                | 904.0           | 657.5        |
|            | 8                     | 445.5               | 10             | *2                   |                            |      | 295.0                | 590.0           | 429.1        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 563.0                | 1126.0          | 819.0        |
|            | 8                     | 445.5               | 12             | *2                   |                            |      | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               |                | *2                   |                            |      | 672.5                | 1345.0          | 978.2        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 35.9 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX537-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 8              | 135.8                | 2448                       | 4096 | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | 135.8                |                            |      | 61.0                 | 122.0           | 88.8         |
|            | 2                            | 4752                  |                | 78.7                 |                            |      | 107.0                | 214.0           | 155.7        |
|            | 1                            | 4752                  |                | 40.6                 |                            |      | 209.5                | 419.0           | 304.8        |
|            | 8                            | 4752                  | 10             | 128.7                |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | 122.3                |                            |      | 68.0                 | 136.0           | 99.0         |
|            | 2                            | 4752                  |                | 64.6                 |                            |      | 131.0                | 262.0           | 190.6        |
|            | 1                            | 4752                  |                | 33.0                 |                            |      | 259.0                | 518.0           | 376.8        |
|            | 8                            | 4752                  | 12             | 88.4                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | 88.4                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | 53.7                 |                            |      | 158.0                | 316.0           | 229.9        |
|            | 1                            | 4752                  |                | 27.3                 |                            |      | 313.0                | 626.0           | 455.3        |
| ROI        | 8                            | 4752                  | 8              | *2                   | *3                         | *3   | 61.0                 | 122.0           | 88.8         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 61.0                 | 122.0           | 88.8         |
|            | 2                            | 4752                  |                | *2                   |                            |      | 107.0                | 214.0           | 155.7        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 209.5                | 419.0           | 304.8        |
|            | 8                            | 4752                  | 10             | *2                   |                            |      | 64.5                 | 129.0           | 93.9         |
|            | 4                            | 4752                  |                | *2                   |                            |      | 68.0                 | 136.0           | 99.0         |
|            | 2                            | 4752                  |                | *2                   |                            |      | 131.0                | 262.0           | 190.6        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 259.0                | 518.0           | 376.8        |
|            | 8                            | 4752                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 4752                  |                | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 2                            | 4752                  |                | *2                   |                            |      | 158.0                | 316.0           | 229.9        |
|            | 1                            | 4752                  |                | *2                   |                            |      | 313.0                | 626.0           | 455.3        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 135.8 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX537-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 8              | 133.6                | 2448                       | 4096 | 62.0                 | 124.0           | 90.2         |
|            | 4                            | 2376                  |                | 76.6                 |                            |      | 110.0                | 220.0           | 160.0        |
|            | 2                            | 2376                  |                | 40.2                 |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                | 20.4                 |                            |      | 419.0                | 838.0           | 609.5        |
|            | 8                            | 2376                  | 10             | 115.7                |                            |      | 72.0                 | 144.0           | 104.8        |
|            | 4                            | 2376                  |                | 63.1                 |                            |      | 134.0                | 268.0           | 195.0        |
|            | 2                            | 2376                  |                | 32.7                 |                            |      | 261.5                | 523.0           | 380.4        |
|            | 1                            | 2376                  |                | 16.5                 |                            |      | 517.5                | 1035.0          | 752.8        |
|            | 8                            | 2376                  | 12             | 88.4                 |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 2376                  |                | 53.4                 |                            |      | 159.0                | 318.0           | 231.3        |
|            | 2                            | 2376                  |                | 27.1                 |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                | 13.7                 |                            |      | 626.0                | 1252.0          | 910.6        |
| ROI        | 8                            | 2376                  | 8              | *2                   | *3                         | *3   | 62.0                 | 124.0           | 90.2         |
|            | 4                            | 2376                  |                | *2                   |                            |      | 110.0                | 220.0           | 160.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 419.0                | 838.0           | 609.5        |
|            | 8                            | 2376                  | 10             | *2                   |                            |      | 72.0                 | 144.0           | 104.8        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 134.0                | 268.0           | 195.0        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 261.5                | 523.0           | 380.4        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 517.5                | 1035.0          | 752.8        |
|            | 8                            | 2376                  | 12             | *2                   |                            |      | 95.0                 | 190.0           | 138.2        |
|            | 4                            | 2376                  |                | *2                   |                            |      | 159.0                | 318.0           | 231.3        |
|            | 2                            | 2376                  |                | *2                   |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                | *2                   |                            |      | 626.0                | 1252.0          | 910.6        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 133.6 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|----------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 8              | 69.8                 | 2448                       | 4096 | 121.0                | 242.0           | 176.0        |
|            | 4                            | 1188                  |                | 38.8                 |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                | 20.2                 |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                | 10.2                 |                            |      | 838.0                | 1676.0          | 1219.0       |
|            | 8                            | 1188                  | 10             | 59.9                 |                            |      | 141.5                | 283.0           | 205.9        |
|            | 4                            | 1188                  |                | 31.9                 |                            |      | 268.0                | 536.0           | 389.9        |
|            | 2                            | 1188                  |                | 16.4                 |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 1                            | 1188                  |                | 8.3                  |                            |      | 1035.0               | 2070.0          | 1505.5       |
|            | 8                            | 1188                  | 12             | 48.8                 |                            |      | 174.0                | 348.0           | 253.1        |
|            | 4                            | 1188                  |                | 27.0                 |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                | 13.6                 |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                | 6.8                  |                            |      | 1251.5               | 2503.0          | 1820.4       |
| ROI        | 8                            | 1188                  | 8              | *2                   | *3                         | *3   | 121.0                | 242.0           | 176.0        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 838.0                | 1676.0          | 1219.0       |
|            | 8                            | 1188                  | 10             | *2                   |                            |      | 141.5                | 283.0           | 205.9        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 268.0                | 536.0           | 389.9        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 523.0                | 1046.0          | 760.8        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1035.0               | 2070.0          | 1505.5       |
|            | 8                            | 1188                  | 12             | *2                   |                            |      | 174.0                | 348.0           | 253.1        |
|            | 4                            | 1188                  |                | *2                   |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                | *2                   |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                | *2                   |                            |      | 1251.5               | 2503.0          | 1820.4       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 69.8 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 7. Drive Mode Setting (without combination)

Please refer to the product specification for register setup other than those lists.

### 7.1. Drive mode setting

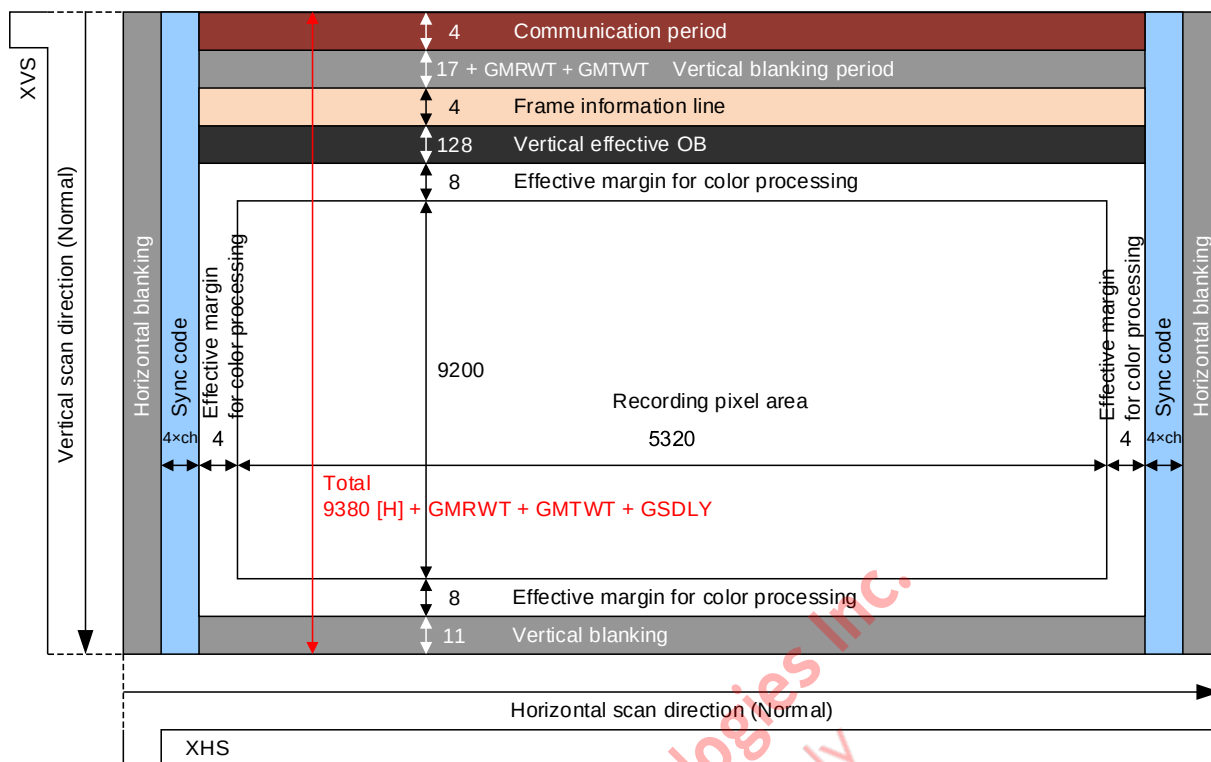
#### 7.1.1. IMX530

##### 7.1.1.1. All-pixel scan

SLVS Output (IMX530)

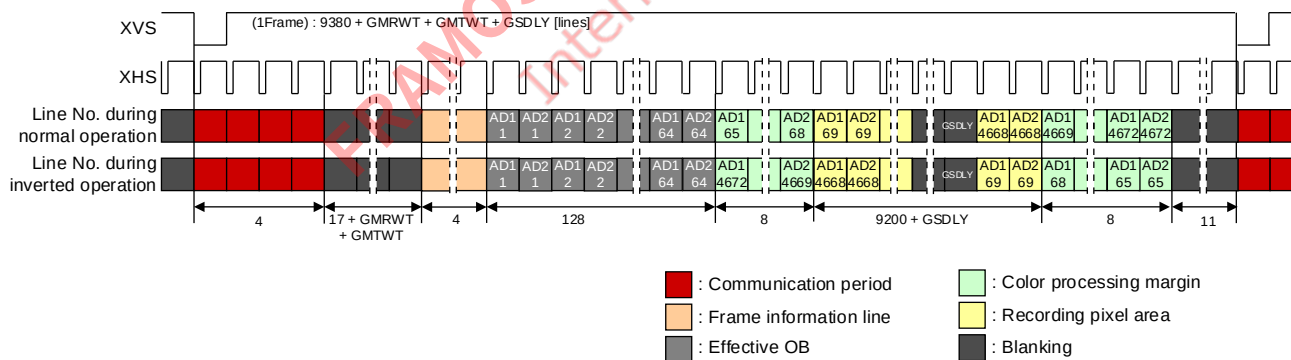
| Address       | bit   | Register name | Initial Value | Setting value     |                  |                   |                  |                   |                  | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|------------------|-------------------|------------------|-------------------|------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                  | AD = 10 bit       |                  | AD = 12 bit       |                  |                                       |           |
|               |       |               |               | 8 ch              | 4 ch             | 8 ch              | 4 ch             | 8 ch              | 4 ch             |                                       |           |
|               |       |               |               | 11.0<br>[frame/s] | 5.6<br>[frame/s] | 8.8<br>[frame/s]  | 4.5<br>[frame/s] | 7.4<br>[frame/s]  | 3.7<br>[frame/s] |                                       | FREQ = 0h |
|               |       |               |               | 5.5<br>[frame/s]  | 2.8<br>[frame/s] | 4.4<br>[frame/s]  | 2.2<br>[frame/s] | 3.7<br>[frame/s]  | 1.9<br>[frame/s] |                                       | FREQ = 1h |
|               |       |               |               | 15.1<br>[frame/s] | 7.9<br>[frame/s] | 12.2<br>[frame/s] | 6.4<br>[frame/s] | 11.0<br>[frame/s] | 5.6<br>[frame/s] |                                       | FREQ = 2h |
|               |       |               |               | 7.7<br>[frame/s]  | 4.0<br>[frame/s] | 6.2<br>[frame/s]  | 3.2<br>[frame/s] | 5.5<br>[frame/s]  | 2.8<br>[frame/s] |                                       | FREQ = 3h |
|               |       |               |               | 594 Mbps/ch       |                  |                   |                  |                   |                  |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch       |                  |                   |                  |                   |                  |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                   |                  | FREQ = 2h         |                  |                   |                  |                                       |           |
| 445.5 Mbps/ch |       |               |               |                   |                  | FREQ = 3h         |                  |                   |                  |                                       |           |
| Chip ID = 02h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |           |
| D4h           | [7:0] | VMAX          | 12ECh         | 24BCh             | 24B4h            | 24B8h             | 24B0h            | 24B4h             | 24B0h            | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 24B4h             | 24B0h            | 24B0h             | 24B0h            | 24B0h             | 24B0h            | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 24BCh             | 24B8h            | 24BCh             | 24B4h            | 24BCh             | 24B4h            | FREQ = 2h                             |           |
|               |       |               |               | 24B4h             | 24B0h            | 24B4h             | 24B0h            | 24B4h             | 24B0h            | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 2CCh              | 574h             | 37Ch              | 6CDh             | 42Bh              | 826h             | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 584h              | ACDh             | 6E2h              | D7Dh             | 83Fh              | 102Ch            | FREQ = 1h                             |           |
|               |       |               |               | 208h              | 3DEh             | 286h              | 4D1h             | 2CCh              | 574h             | FREQ = 2h                             |           |
|               |       |               |               | 3F8h              | 799h             | 4F2h              | 97Bh             | 584h              | ACDh             | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 48h           | 08h               | 04h              | 08h               | 04h              | 04h               | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 08h              | 08h               | 04h              | 08h               | 04h              | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                  |                   |                  |                   |                  |                                       |           |
| E6h           | [7:0] | GSDLY         | 34h           | 0Ch               | 08h              | 08h               | 04h              | 08h               | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 08h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 0Ch               | 08h              | 0Ch               | 08h              | 0Ch               | 08h              | FREQ = 2h                             |           |
|               |       |               |               | 08h               | 04h              | 08h               | 04h              | 08h               | 04h              | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                  |                   |                  |                   |                  |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                  | 0h                |                  | 1h                |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                  |                   |                  |                   |                  |                                       |           |
| 3Dh           | [0]   |               |               |                   |                  |                   |                  |                   |                  |                                       |           |
| Chip ID = 06h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                  | 0h                |                  | 1h                |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 09h               |                  |                   |                  |                   |                  |                                       |           |

(IMX530)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX530)

| Address         | bit   | Register name | Initial Value | Setting value     |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  | Remarks                               |           |
|-----------------|-------|---------------|---------------|-------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|------------------|---------------------------------------|-----------|
|                 |       |               |               | AD = 8 bit        |                   |                   |                  | AD = 10 bit       |                   |                   |                  | AD = 12 bit       |                   |                   |                  |                                       |           |
|                 |       |               |               | 8 lane            | 4 lane            | 2 lane            | 1 lane           | 8 lane            | 4 lane            | 2 lane            | 1 lane           | 8 lane            | 4 lane            | 2 lane            | 1 lane           | FREQ = 0h                             |           |
|                 |       |               |               | 54.1<br>[frame/s] | 34.0<br>[frame/s] | 17.6<br>[frame/s] | 8.9<br>[frame/s] | 51.9<br>[frame/s] | 28.0<br>[frame/s] | 14.3<br>[frame/s] | 7.2<br>[frame/s] | 37.6<br>[frame/s] | 23.5<br>[frame/s] | 12.0<br>[frame/s] | 6.0<br>[frame/s] |                                       |           |
|                 |       |               |               | 32.3<br>[frame/s] | 17.3<br>[frame/s] | 8.9<br>[frame/s]  | 4.4<br>[frame/s] | 27.4<br>[frame/s] | 14.2<br>[frame/s] | 7.1<br>[frame/s]  | 3.6<br>[frame/s] | 22.2<br>[frame/s] | 11.9<br>[frame/s] | 6.0<br>[frame/s]  | 3.0<br>[frame/s] |                                       | FREQ = 1h |
|                 |       |               |               | 16.4<br>[frame/s] | 8.7<br>[frame/s]  | 4.4<br>[frame/s]  | 2.2<br>[frame/s] | 13.8<br>[frame/s] | 7.1<br>[frame/s]  | 3.6<br>[frame/s]  | 1.8<br>[frame/s] | 11.1<br>[frame/s] | 5.9<br>[frame/s]  | 3.0<br>[frame/s]  | 1.5<br>[frame/s] |                                       |           |
|                 |       |               |               | 4752 Mbps/lane    |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  | FREQ = 0h                             |           |
| 2376 Mbps/lane  |       |               |               |                   |                   |                   |                  |                   |                   |                   |                  | FREQ = 1h         |                   |                   |                  |                                       |           |
| 1188 Mbps /lane |       |               |               |                   |                   |                   |                  |                   |                   |                   |                  | FREQ = 2h         |                   |                   |                  |                                       |           |
| Chip ID = 02h   |       |               |               |                   |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| D4h             | [7:0] | VMAX          | 12ECh         | 24F8h             | 24DCh             | 24C4h             | 24B8h            | 24F8h             | 24D4h             | 24BCh             | 24B4h            | 24E4h             | 24D0h             | 24BCh             | 24B4h            | FREQ = 0h                             |           |
| D5h             | [7:0] |               |               | 24DCh             | 24C4h             | 24B8h             | 24B0h            | 24D4h             | 24BCh             | 24B4h             | 24B0h            | 24CCh             | 24BCh             | 24B4h             | 24B0h            | FREQ = 1h                             |           |
| D6h             | [7:0] |               |               | 24C4h             | 24B8h             | 24B0h             | 24B0h            | 24BCh             | 24B4h             | 24B0h             | 24B0h            | 24BCh             | 24B4h             | 24B0h             | 24B0h            | FREQ = 2h                             |           |
| D8h             | [7:0] | HMAX          | 0A6h          | 91h               | E7h               | 1BFh              | 370h             | 97h               | 119h              | 225h              | 445h             | D1h               | 14Fh              | 28Fh              | 519h             | FREQ = 0h                             |           |
| D9h             | [7:0] |               |               | F3h               | 1C7h              | 377h              | 6E0h             | 11Fh              | 22Bh              | 44Ah              | 88Ah             | 162h              | 297h              | 51Eh              | A31h             | FREQ = 1h                             |           |
|                 |       |               |               | 1E0h              | 389h              | 6EEh              | DBFh             | 239h              | 456h              | 894h              | 1114h            | 2C0h              | 52Ah              | A3Bh              | 1462h            | FREQ = 2h                             |           |
| E2h             | [7:0] | GMRWT         | 0Ch           | 0Ch               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | FREQ = 0h                             |           |
|                 |       |               |               | 08h               | 04h               | 04h               | 04h              | 08h               | 04h               | 04h               | 04h              | 08h               | 04h               | 04h               | 04h              | FREQ = 1h                             |           |
|                 |       |               |               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h               | 04h               | 04h              | 04h               | 04h               | 04h               | 04h              | FREQ = 2h                             |           |
| E3h             | [7:0] | GMTWT         | 48h           | 1Ch               | 14h               | 0Ch               | 08h              | 1Ch               | 10h               | 08h               | 04h              | 14h               | 10h               | 08h               | 04h              | FREQ = 0h                             |           |
|                 |       |               |               | 14h               | 0Ch               | 08h               | 04h              | 10h               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | FREQ = 1h                             |           |
|                 |       |               |               | 0Ch               | 08h               | 04h               | 04h              | 08h               | 04h               | 04h               | 04h              | 08h               | 04h               | 04h               | 04h              | FREQ = 2h                             |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| E6h             | [7:0] | GSDLY         | 34h           | 2Ch               | 1Ch               | 10h               | 08h              | 2Ch               | 18h               | 0Ch               | 08h              | 20h               | 14h               | 0Ch               | 08h              | FREQ = 0h                             |           |
|                 |       |               |               | 1Ch               | 10h               | 08h               | 04h              | 18h               | 0Ch               | 08h               | 04h              | 14h               | 0Ch               | 08h               | 04h              | FREQ = 1h                             |           |
|                 |       |               |               | 10h               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | FREQ = 2h                             |           |
| Chip ID = 04h   |       |               |               |                   |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                |                   |                   |                  | 0h                |                   |                   |                  | 1h                |                   |                   |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| 3Dh             | [0]   |               |               |                   |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| Chip ID = 06h   |       |               |               |                   |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                |                   |                   |                  | 0h                |                   |                   |                  | 1h                |                   |                   |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h   |       |               |               |                   |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |
| F8h             | [7:0] |               | 2Dh           | 09h               |                   |                   |                  |                   |                   |                   |                  |                   |                   |                   |                  |                                       |           |

The diagram illustrates the timing of the video signal, showing the vertical scan direction (Normal) and the horizontal scan direction (Normal). The vertical scan direction is indicated by a downward arrow on the left, and the horizontal scan direction is indicated by a rightward arrow at the bottom.

The vertical scan direction is divided into several fields:

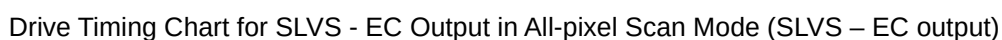
- Start Code**: 4byte x Lane
- Packet Header**: 24byte
- Effective margin for color processing**: 4
- Communication period**: 4
- Vertical blanking period**: 17 + GMRWT + GMTWT
- Embedded Data**: 4
- Vertical effective OB**: 128
- Effective margin for color processing**: 8
- Recording pixel area**: 9200 (vertical) x 5320 (horizontal)
- Effective margin for color processing**: 8
- Vertical blanking**: 11
- End Code**: 4byte x Lane
- Deskew Code**: 4byte x Lane
- Idle Code**

The horizontal scan direction is divided into several fields:

- Start Code**: 4byte x Lane
- Packet Header**: 24byte
- Effective margin for color processing**: 4
- Communication period**: 4
- Vertical blanking period**: 17 + GMRWT + GMTWT
- Embedded Data**: 4
- Vertical effective OB**: 128
- Effective margin for color processing**: 8
- Recording pixel area**: 9200 (vertical) x 5320 (horizontal)
- Effective margin for color processing**: 8
- Vertical blanking**: 11
- End Code**: 4byte x Lane
- Deskew Code**: 4byte x Lane
- Idle Code**

The total horizontal length is indicated by a red arrow and the formula:  $9380 [H] + GMRWT + GMTWT + GSDLY$ .

The output of each AD is output alternately every other line.



### 7.1.1.2. ROI, Overlap ROI

#### SLVS Output (IMX530)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 2CCh                                                                                                          | 574h      | 37Ch        | 6CDh      | 42Bh        | 826h      | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 584h                                                                                                          | ACDh      | 6E2h        | D7Dh      | 83Fh        | 102Ch     | FREQ = 1h                             |           |
|               |       |               |               | 208h                                                                                                          | 3DEh      | 286h        | 4D1h      | 2CCh        | 574h      | FREQ = 2h                             |           |
|               |       | 3F8h          | 799h          | 4F2h                                                                                                          | 97Bh      | 584h        | ACDh      | FREQ = 3h   |           |                                       |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 2h                             |           |
| E3h           | [7:0] | GMTWT         | 48h           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 08h                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
| E6h           | [7:0] | GSDLY         | 34h           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 2h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 3h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 0h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           |                                       |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

## SLVS – EC output (IMX530)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |
|-----------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|
|                 |       |               |               | AD = 8 bit                                                                                                 |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |
|                 |       |               |               | 8 lane                                                                                                     | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |
|                 |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |
|                 |       |               |               | 4752 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 2376 Mbps/lane  |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 1h                             |
| 1188 Mbps /lane |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 2h                             |
| Chip ID = 02h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D4h             | [7:0] | VMAX          | 12ECh         | *1                                                                                                         | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              |                                       |
| D5h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D6h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D8h             | [7:0] | HMAX          | 0A6h          | 91h                                                                                                        | E7h             | 1BFh            | 370h            | 97h             | 119h            | 225h            | 445h            | D1h             | 14Fh             | 28Fh             | 519h             | FREQ = 0h                             |
| D9h             | [7:0] |               |               | F3h                                                                                                        | 1C7h            | 377h            | 6E0h            | 11Fh            | 22Bh            | 44Ah            | 88Ah            | 162h            | 297h             | 51Eh             | A31h             | FREQ = 1h                             |
|                 |       |               |               | 1E0h                                                                                                       | 389h            | 6EEh            | DBFh            | 239h            | 456h            | 894h            | 1114h           | 2C0h            | 52Ah             | A3Bh             | 1462h            | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 0Ch           | 08h                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 0h                             |
|                 |       |               |               | 08h                                                                                                        | 04h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 1h                             |
|                 |       |               |               | 04h                                                                                                        | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h              | 04h              | 04h              | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 48h           | 1Ch                                                                                                        | 14h             | 0Ch             | 08h             | 1Ch             | 10h             | 08h             | 04h             | 14h             | 10h              | 08h              | 04h              | FREQ = 0h                             |
|                 |       |               |               | 14h                                                                                                        | 0Ch             | 08h             | 04h             | 10h             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 1h                             |
|                 |       |               |               | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| E6h             | [7:0] | GSDLY         | 34h           | 2Ch                                                                                                        | 1Ch             | 10h             | 08h             | 2Ch             | 18h             | 0Ch             | 08h             | 20h             | 14h              | 0Ch              | 08h              | FREQ = 0h                             |
|                 |       |               |               | 1Ch                                                                                                        | 10h             | 08h             | 04h             | 18h             | 0Ch             | 08h             | 04h             | 14h             | 0Ch              | 08h              | 04h              | FREQ = 1h                             |
|                 |       |               |               | 10h                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 2h                             |
| Chip ID = 03h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 04h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 3Dh             | [0]   |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 06h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| E8h             | [7:0] |               | 2Dh           | 09h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

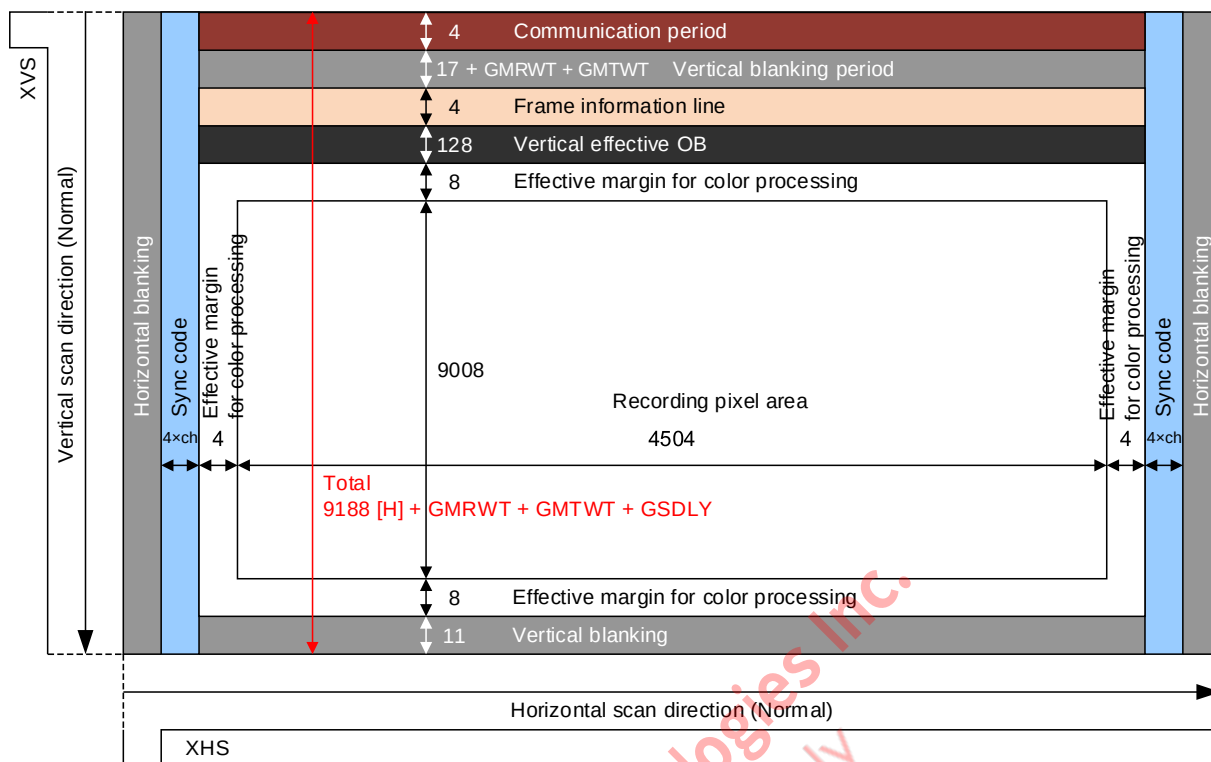
## 7.1.2. IMX531

## 7.1.2.1. All-pixel scan

SLVS Output (IMX531)

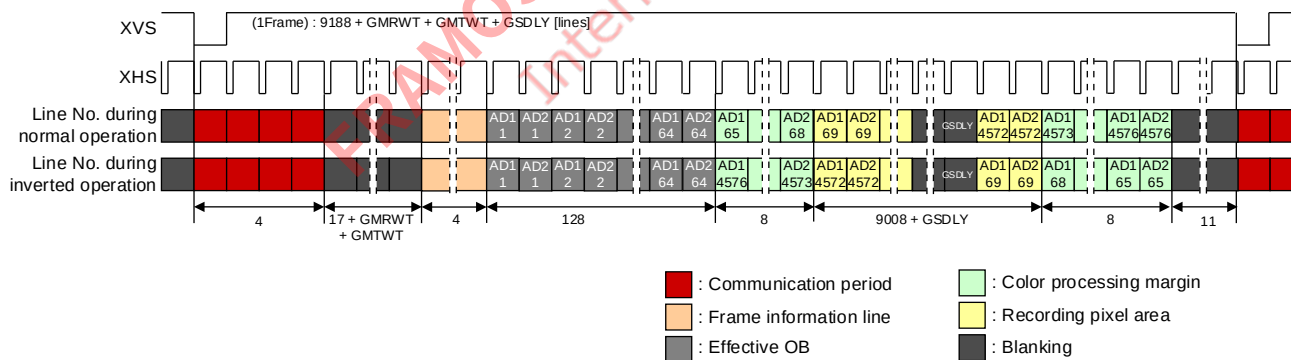
| Address       | bit   | Register name | Initial Value | Setting value     |                  |                   |                  |                   |                  | Remarks                               |
|---------------|-------|---------------|---------------|-------------------|------------------|-------------------|------------------|-------------------|------------------|---------------------------------------|
|               |       |               |               | AD = 8 bit        |                  | AD = 10 bit       |                  | AD = 12 bit       |                  |                                       |
|               |       |               |               | 8 ch              | 4 ch             | 8 ch              | 4 ch             | 8 ch              | 4 ch             |                                       |
|               |       |               |               | 13.1<br>[frame/s] | 6.7<br>[frame/s] | 10.5<br>[frame/s] | 5.4<br>[frame/s] | 8.8<br>[frame/s]  | 4.5<br>[frame/s] | FREQ = 0h                             |
|               |       |               |               | 6.6<br>[frame/s]  | 3.4<br>[frame/s] | 5.3<br>[frame/s]  | 2.7<br>[frame/s] | 4.4<br>[frame/s]  | 2.2<br>[frame/s] | FREQ = 1h                             |
|               |       |               |               | 17.8<br>[frame/s] | 9.4<br>[frame/s] | 14.3<br>[frame/s] | 7.5<br>[frame/s] | 13.1<br>[frame/s] | 6.7<br>[frame/s] | FREQ = 2h                             |
|               |       |               |               | 9.1<br>[frame/s]  | 4.8<br>[frame/s] | 7.3<br>[frame/s]  | 3.8<br>[frame/s] | 6.6<br>[frame/s]  | 3.4<br>[frame/s] | FREQ = 3h                             |
|               |       |               |               | 594 Mbps/ch       |                  |                   |                  |                   |                  | FREQ = 0h                             |
|               |       |               |               | 297 Mbps/ch       |                  |                   |                  |                   |                  | FREQ = 1h                             |
|               |       |               |               | 891 Mbps/ch       |                  |                   |                  |                   |                  | FREQ = 2h                             |
| 445.5 Mbps/ch |       |               |               |                   |                  | FREQ = 3h         |                  |                   |                  |                                       |
| Chip ID = 02h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |
| D4h           | [7:0] | VMAX          | 12ECh         | 23FCh             | 23F4h            | 23F8h             | 23F4h            | 23F8h             | 23F0h            | FREQ = 0h                             |
| D5h           | [7:0] |               |               | 23F4h             | 23F0h            | 23F4h             | 23F0h            | 23F0h             | 23F0h            | FREQ = 1h                             |
| D6h           | [7:0] |               |               | 2404h             | 23F8h            | 23FCh             | 23F4h            | 23FCh             | 23F4h            | FREQ = 2h                             |
|               |       |               |               | 23F8h             | 23F0h            | 23F4h             | 23F0h            | 23F4h             | 23F0h            | FREQ = 3h                             |
| D8h           | [7:0] | HMAX          | 0A6h          | 266h              | 4A8h             | 2FCh              | 5CEh             | 392h              | 6F4h             | FREQ = 0h                             |
| D9h           | [7:0] |               |               | 4B8h              | 935h             | 5E3h              | B7Fh             | 70Dh              | DC8h             | FREQ = 1h                             |
|               |       |               |               | 1C4h              | 356h             | 231h              | 427h             | 266h              | 4A8h             | FREQ = 2h                             |
|               |       |               |               | 370h              | 689h             | 448h              | 827h             | 4B8h              | 935h             | FREQ = 3h                             |
| E2h           | [7:0] | GMRWT         | 0Ch           | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 0h                             |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 2h                             |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 3h                             |
| E3h           | [7:0] | GMTWT         | 48h           | 08h               | 04h              | 08h               | 04h              | 08h               | 04h              | FREQ = 0h                             |
|               |       |               |               | 04h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |
|               |       |               |               | 0Ch               | 08h              | 08h               | 04h              | 08h               | 04h              | FREQ = 2h                             |
|               |       |               |               | 08h               | 04h              | 04h               | 04h              | 04h               | 04h              | FREQ = 3h                             |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                  |                   |                  |                   |                  |                                       |
| E6h           | [7:0] | GSDLY         | 34h           | 0Ch               | 08h              | 08h               | 08h              | 08h               | 04h              | FREQ = 0h                             |
|               |       |               |               | 08h               | 04h              | 08h               | 04h              | 04h               | 04h              | FREQ = 1h                             |
|               |       |               |               | 10h               | 08h              | 0Ch               | 08h              | 0Ch               | 08h              | FREQ = 2h                             |
|               |       |               |               | 08h               | 04h              | 08h               | 04h              | 08h               | 04h              | FREQ = 3h                             |
| Chip ID = 04h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                  |                   |                  |                   |                  |                                       |
|               | [5:4] | ADBIT         | 0h            | 2h                |                  | 0h                |                  | 1h                |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                  |                   |                  |                   |                  |                                       |
| 3Dh           | [0]   |               |               |                   |                  |                   |                  |                   |                  |                                       |
| Chip ID = 06h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                  | 0h                |                  | 1h                |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h |       |               |               |                   |                  |                   |                  |                   |                  |                                       |
| E8h           | [7:0] |               | 2Dh           | 09h               |                  |                   |                  |                   |                  |                                       |

(IMX531)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

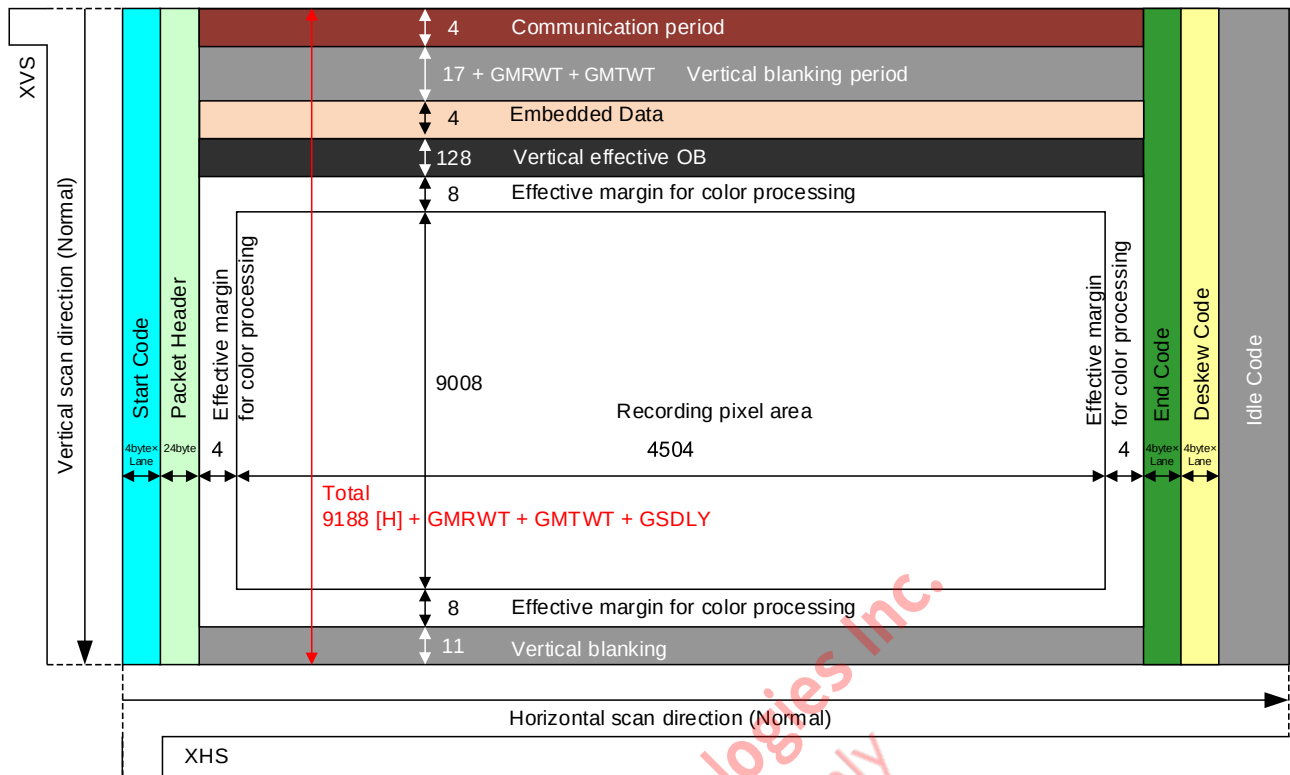


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX531)

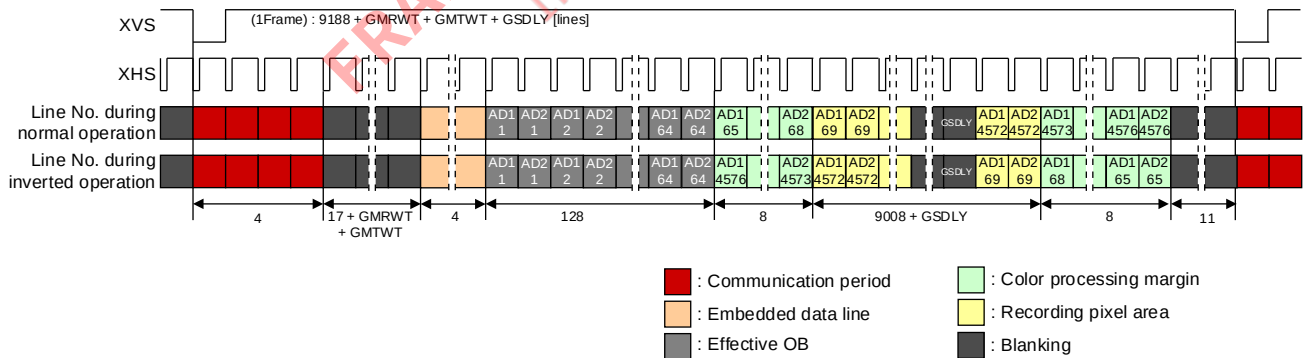
| Address         | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  | Remarks                               |
|-----------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|------------------|---------------------------------------|
|                 |       |               |               | AD = 8 bit        |                   |                   |                   | AD = 10 bit       |                   |                   |                  | AD = 12 bit       |                   |                   |                  |                                       |
|                 |       |               |               | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane           | 8 lane            | 4 lane            | 2 lane            | 1 lane           |                                       |
|                 |       |               |               | 55.2<br>[frame/s] | 40.5<br>[frame/s] | 21.4<br>[frame/s] | 10.7<br>[frame/s] | 53.0<br>[frame/s] | 33.3<br>[frame/s] | 17.1<br>[frame/s] | 8.6<br>[frame/s] | 38.4<br>[frame/s] | 28.7<br>[frame/s] | 14.4<br>[frame/s] | 7.2<br>[frame/s] |                                       |
|                 |       |               |               | 38.3<br>[frame/s] | 20.6<br>[frame/s] | 10.7<br>[frame/s] | 5.3<br>[frame/s]  | 32.5<br>[frame/s] | 16.9<br>[frame/s] | 8.5<br>[frame/s]  | 4.3<br>[frame/s] | 26.5<br>[frame/s] | 14.4<br>[frame/s] | 7.2<br>[frame/s]  | 3.6<br>[frame/s] |                                       |
|                 |       |               |               | 19.5<br>[frame/s] | 10.4<br>[frame/s] | 5.3<br>[frame/s]  | 2.7<br>[frame/s]  | 16.5<br>[frame/s] | 8.4<br>[frame/s]  | 4.3<br>[frame/s]  | 2.1<br>[frame/s] | 13.3<br>[frame/s] | 7.2<br>[frame/s]  | 3.6<br>[frame/s]  | 1.8<br>[frame/s] |                                       |
|                 |       |               |               | 4752 Mbps/lane    |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| 2376 Mbps/lane  |       |               |               |                   |                   |                   |                   |                   |                   |                   |                  | FREQ = 1h         |                   |                   |                  |                                       |
| 1188 Mbps /lane |       |               |               |                   |                   |                   |                   |                   |                   |                   |                  | FREQ = 2h         |                   |                   |                  |                                       |
| Chip ID = 02h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| D4h             | [7:0] | VMAX          | 12ECh         | 2438h             | 2428h             | 240Ch             | 23FCh             | 2438h             | 241Ch             | 2404h             | 23F8h            | 2424h             | 2414h             | 23FCh             | 23F4h            | FREQ = 0h                             |
| D5h             | [7:0] |               |               | 2424h             | 2408h             | 23FCh             | 23F4h             | 241Ch             | 2404h             | 23F8h             | 23F0h            | 2414h             | 23FCh             | 23F4h             | 23F0h            | FREQ = 1h                             |
| D6h             | [7:0] |               |               | 2408h             | 23F8h             | 23F4h             | 23F0h             | 2404h             | 23F8h             | 23F0h             | 23F0h            | 23FCh             | 23F4h             | 23F0h             | 23F0h            | FREQ = 2h                             |
| D8h             | [7:0] | HMAX          | 0A6h          | 91h               | C6h               | 177h              | 2EBh              | 97h               | F1h               | 1D5h              | 3A5h             | D1h               | 118h              | 22Dh              | 455h             | FREQ = 0h                             |
| D9h             | [7:0] |               |               | D1h               | 186h              | 2EDh              | 5D6h              | F7h               | 1DBh              | 3AAh              | 74Ah             | 12Fh              | 22Fh              | 45Ah              | 8AAh             | FREQ = 1h                             |
|                 |       |               |               | 19Ch              | 306h              | 5DAh              | BABh              | 1E8h              | 3B5h              | 753h              | E93h             | 25Ah              | 45Eh              | 8B4h              | 1154h            | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 0Ch           | 0Ch               | 0Ch               | 08h               | 04h               | 0Ch               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | FREQ = 0h                             |
|                 |       |               |               | 0Ch               | 08h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h              | 08h               | 04h               | 04h               | 04h              | FREQ = 1h                             |
|                 |       |               |               | 08h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h               | 04h               | 04h              | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 48h           | 1Ch               | 18h               | 0Ch               | 08h               | 1Ch               | 14h               | 0Ch               | 08h              | 14h               | 10h               | 08h               | 04h              | FREQ = 0h                             |
|                 |       |               |               | 14h               | 0Ch               | 08h               | 04h               | 14h               | 0Ch               | 08h               | 04h              | 10h               | 08h               | 04h               | 04h              | FREQ = 1h                             |
|                 |       |               |               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h              | 08h               | 04h               | 04h               | 04h              | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| E6h             | [7:0] | GSDLY         | 34h           | 2Ch               | 20h               | 14h               | 0Ch               | 2Ch               | 1Ch               | 10h               | 08h              | 20h               | 18h               | 0Ch               | 08h              | FREQ = 0h                             |
|                 |       |               |               | 20h               | 10h               | 0Ch               | 08h               | 1Ch               | 10h               | 08h               | 04h              | 18h               | 0Ch               | 08h               | 04h              | FREQ = 1h                             |
|                 |       |               |               | 10h               | 08h               | 08h               | 04h               | 10h               | 08h               | 04h               | 04h              | 0Ch               | 08h               | 04h               | 04h              | FREQ = 2h                             |
| Chip ID = 04h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                  | 1h                |                   |                   |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| 3Dh             | [0]   |               |               |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| Chip ID = 06h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                  | 1h                |                   |                   |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |
| F8h             | [7:0] |               | 2Dh           | 09h               |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |                                       |

(IMX531)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 7.1.2.2. ROI, Overlap ROI

## SLVS Output (IMX531)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 266h                                                                                                          | 4A8h      | 2FCh        | 5CEh      | 392h        | 6F4h      | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 4B8h                                                                                                          | 935h      | 5E3h        | B7Fh      | 70Dh        | DC8h      | FREQ = 1h                             |           |
|               |       |               |               | 1C4h                                                                                                          | 356h      | 231h        | 427h      | 266h        | 4A8h      | FREQ = 2h                             |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 370h                                                                                                          | 689h      | 448h        | 827h      | 4B8h        | 935h      | FREQ = 3h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | 04h                                   | FREQ = 0h |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | 04h                                   | FREQ = 1h |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | 04h                                   | FREQ = 2h |
| E3h           | [7:0] | GMTWT         | 48h           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | 04h                                   | FREQ = 0h |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | 04h                                   | FREQ = 1h |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | 04h                                   | FREQ = 2h |
| E5h           | [7:0] | GAINDLY       | 04h           | 08h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | 04h                                   | FREQ = 3h |
| E6h           | [7:0] | GSDLY         | 34h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 08h         | 08h       | 08h         | 04h       | 04h                                   | FREQ = 0h |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | 04h                                   | FREQ = 1h |
|               |       |               |               | 10h                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | 08h                                   | FREQ = 2h |
| 08h           | 04h   | 08h           | 04h           | 08h                                                                                                           | 04h       | 08h         | 04h       | FREQ = 3h   |           |                                       |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

## SLVS – EC output (IMX531)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |  |
|-----------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|--|
|                 |       |               |               | AD = 8 bit                                                                                                 |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |  |
|                 |       |               |               | 8 lane                                                                                                     | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |  |
|                 |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |  |
|                 |       |               |               | 4752 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| 2376 Mbps/lane  |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 0h                             |  |
| 1188 Mbps /lane |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 1h                             |  |
|                 |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 2h                             |  |
| Chip ID = 02h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| D4h             | [7:0] | VMAX          | 12ECh         | *1                                                                                                         | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              |                                       |  |
| D5h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| D6h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| D8h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| D9h             | [7:0] | HMAX          | 0A6h          | 91h                                                                                                        | C6h             | 177h            | 2EBh            | 97h             | F1h             | 1D5h            | 3A5h            | D1h             | 118h             | 22Dh             | 455h             | FREQ = 0h                             |  |
| E2h             | [7:0] | GMRWT         | 0Ch           | D1h                                                                                                        | 186h            | 2EDh            | 5D6h            | F7h             | 1DBh            | 3AAh            | 74Ah            | 12Fh            | 22Fh             | 45Ah             | 8AAh             | FREQ = 1h                             |  |
|                 |       |               |               | 19Ch                                                                                                       | 306h            | 5DAh            | BABh            | 1E8h            | 3B5h            | 753h            | E93h            | 25Ah            | 45Eh             | 8B4h             | 1154h            | FREQ = 2h                             |  |
|                 |       |               |               | 0Ch                                                                                                        | 0Ch             | 08h             | 04h             | 0Ch             | 08h             | 04h             | 0Ch             | 08h             | 04h              | 04h              | 04h              | FREQ = 0h                             |  |
|                 |       |               |               | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 1h                             |  |
| E3h             | [7:0] | GMTWT         | 48h           | 08h                                                                                                        | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h              | 04h              | 04h              | FREQ = 2h                             |  |
|                 |       |               |               | 1Ch                                                                                                        | 18h             | 0Ch             | 08h             | 1Ch             | 14h             | 0Ch             | 08h             | 14h             | 10h              | 08h              | 04h              | FREQ = 0h                             |  |
|                 |       |               |               | 14h                                                                                                        | 0Ch             | 08h             | 04h             | 14h             | 0Ch             | 08h             | 04h             | 10h             | 08h              | 04h              | 04h              | FREQ = 1h                             |  |
|                 |       |               |               | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 2h                             |  |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| E6h             | [7:0] | GSDLY         | 34h           | 2Ch                                                                                                        | 20h             | 14h             | 0Ch             | 2Ch             | 1Ch             | 10h             | 08h             | 20h             | 18h              | 0Ch              | 08h              | FREQ = 0h                             |  |
|                 |       |               |               | 20h                                                                                                        | 10h             | 0Ch             | 08h             | 1Ch             | 10h             | 08h             | 04h             | 18h             | 0Ch              | 08h              | 04h              | FREQ = 1h                             |  |
|                 |       |               |               | 10h                                                                                                        | 08h             | 08h             | 04h             | 10h             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 2h                             |  |
| Chip ID = 03h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| Chip ID = 04h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |  |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| 3Dh             | [0]   |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| Chip ID = 06h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |  |
| Chip ID = 08h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |
| E8h             | [7:0] |               | 2Dh           | 09h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |  |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

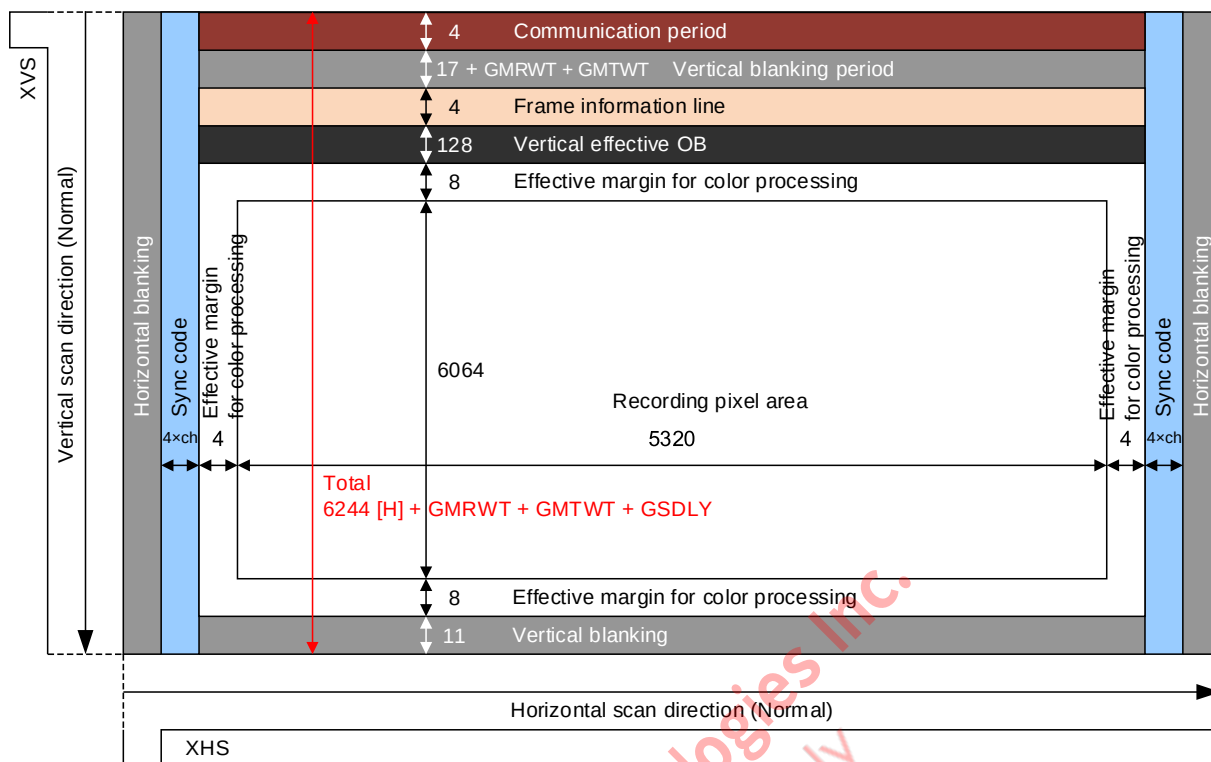
## 7.1.3. IMX532

## 7.1.3.1. All-pixel scan

SLVS Output (IMX532)

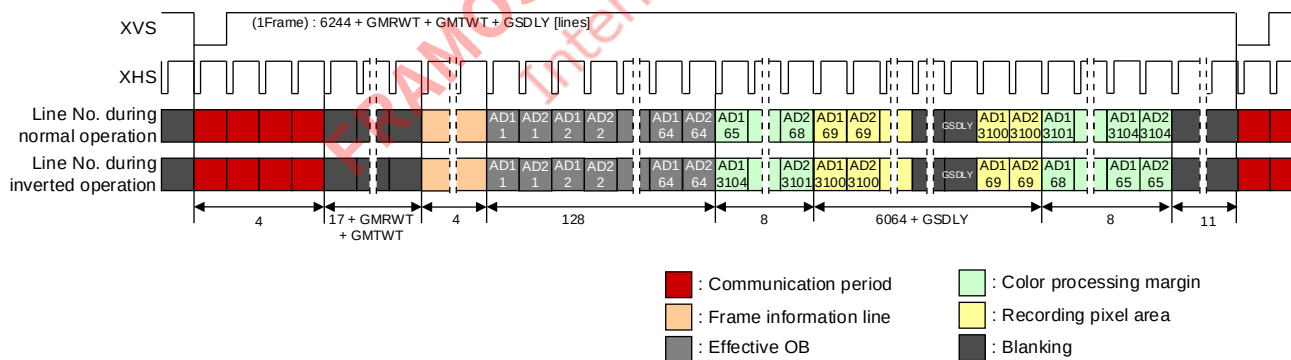
| Address       | bit   | Register name | Initial Value | Setting value     |                   |                   |                  |                   |                  | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|-------------------|-------------------|------------------|-------------------|------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                   | AD = 10 bit       |                  | AD = 12 bit       |                  |                                       |           |
|               |       |               |               | 8 ch              | 4 ch              | 8 ch              | 4 ch             | 8 ch              | 4 ch             |                                       |           |
|               |       |               |               | 16.5<br>[frame/s] | 8.4<br>[frame/s]  | 13.2<br>[frame/s] | 6.8<br>[frame/s] | 11.1<br>[frame/s] | 5.6<br>[frame/s] |                                       | FREQ = 0h |
|               |       |               |               | 8.4<br>[frame/s]  | 4.2<br>[frame/s]  | 6.7<br>[frame/s]  | 3.4<br>[frame/s] | 5.6<br>[frame/s]  | 2.8<br>[frame/s] |                                       | FREQ = 1h |
|               |       |               |               | 22.7<br>[frame/s] | 11.9<br>[frame/s] | 18.3<br>[frame/s] | 9.6<br>[frame/s] | 16.5<br>[frame/s] | 8.4<br>[frame/s] |                                       | FREQ = 2h |
|               |       |               |               | 11.6<br>[frame/s] | 6.1<br>[frame/s]  | 9.3<br>[frame/s]  | 4.8<br>[frame/s] | 8.4<br>[frame/s]  | 4.2<br>[frame/s] |                                       | FREQ = 3h |
|               |       |               |               | 594 Mbps/ch       |                   |                   |                  |                   |                  |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch       |                   |                   |                  |                   |                  |                                       | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch       |                   |                   |                  |                   |                  |                                       | FREQ = 2h |
| 445.5 Mbps/ch |       |               |               |                   |                   | FREQ = 3h         |                  |                   |                  |                                       |           |
| Chip ID = 02h |       |               |               |                   |                   |                   |                  |                   |                  |                                       |           |
| D4h           | [7:0] | VMAX          | 12ECh         | 187Ch             | 1874h             | 1878h             | 1870h            | 1874h             | 1870h            | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 1874h             | 1870h             | 1870h             | 1870h            | 1870h             | 1870h            | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 187Ch             | 1878h             | 187Ch             | 1874h            | 187Ch             | 1874h            | FREQ = 2h                             |           |
|               |       |               |               | 1874h             | 1870h             | 1874h             | 1870h            | 1874h             | 1870h            | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 2CCh              | 574h              | 37Ch              | 6CDh             | 42Bh              | 826h             | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 584h              | ACDh              | 6E2h              | D7Dh             | 83Fh              | 102Ch            | FREQ = 1h                             |           |
|               |       |               |               | 208h              | 3DEh              | 286h              | 4D1h             | 2CCh              | 574h             | FREQ = 2h                             |           |
|               |       |               |               | 3F8h              | 799h              | 4F2h              | 97Bh             | 584h              | ACDh             | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 04h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 48h           | 08h               | 04h               | 08h               | 04h              | 04h               | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 08h               | 08h               | 04h              | 08h               | 04h              | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                  |                   |                  |                                       |           |
| E6h           | [7:0] | GSDLY         | 34h           | 0Ch               | 08h               | 08h               | 04h              | 08h               | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 08h               | 04h               | 04h               | 04h              | 04h               | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 0Ch               | 08h               | 0Ch               | 08h              | 0Ch               | 08h              | FREQ = 2h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h              | 08h               | 04h              | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                   |                   |                  |                   |                  |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                  |                   |                  |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                   | 0h                |                  | 1h                |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                  |                   |                  |                                       |           |
| 3Dh           | [0]   |               |               |                   |                   |                   |                  |                   |                  |                                       |           |
| Chip ID = 06h |       |               |               |                   |                   |                   |                  |                   |                  |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                   | 0h                |                  | 1h                |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                   |                   |                  |                   |                  |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 09h               |                   |                   |                  |                   |                  |                                       |           |

(IMX532)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



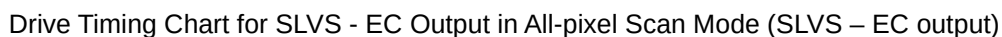
### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX532)

| Address         | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  | Remarks                               |
|-----------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|---------------------------------------|
|                 |       |               |               | AD = 8 bit        |                   |                   |                   | AD = 10 bit       |                   |                   |                   | AD = 12 bit       |                   |                   |                  |                                       |
|                 |       |               |               | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane           |                                       |
|                 |       |               |               | 80.9<br>[frame/s] | 51.0<br>[frame/s] | 26.4<br>[frame/s] | 13.4<br>[frame/s] | 77.7<br>[frame/s] | 41.9<br>[frame/s] | 21.5<br>[frame/s] | 10.8<br>[frame/s] | 56.3<br>[frame/s] | 35.2<br>[frame/s] | 18.0<br>[frame/s] | 9.0<br>[frame/s] | FREQ = 0h                             |
|                 |       |               |               | 48.5<br>[frame/s] | 26.0<br>[frame/s] | 13.3<br>[frame/s] | 6.7<br>[frame/s]  | 41.1<br>[frame/s] | 21.3<br>[frame/s] | 10.8<br>[frame/s] | 5.4<br>[frame/s]  | 33.3<br>[frame/s] | 17.8<br>[frame/s] | 9.0<br>[frame/s]  | 4.5<br>[frame/s] | FREQ = 1h                             |
|                 |       |               |               | 24.6<br>[frame/s] | 13.0<br>[frame/s] | 6.6<br>[frame/s]  | 3.3<br>[frame/s]  | 20.8<br>[frame/s] | 10.6<br>[frame/s] | 5.4<br>[frame/s]  | 2.7<br>[frame/s]  | 16.8<br>[frame/s] | 8.9<br>[frame/s]  | 4.5<br>[frame/s]  | 2.2<br>[frame/s] | FREQ = 2h                             |
|                 |       |               |               | 4752 Mbps/lane    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  | FREQ = 0h                             |
| 2376 Mbps/lane  |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 1h         |                   |                   |                  |                                       |
| 1188 Mbps /lane |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 2h         |                   |                   |                  |                                       |
| Chip ID = 02h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| D4h             | [7:0] | VMAX          | 12ECh         | 18B8h             | 189Ch             | 1884h             | 1878h             | 18B8h             | 1894h             | 187Ch             | 1874h             | 18A4h             | 1890h             | 187Ch             | 1874h            | FREQ = 0h                             |
| D5h             | [7:0] |               |               | 189Ch             | 1884h             | 1878h             | 1870h             | 1894h             | 187Ch             | 1874h             | 1870h             | 188Ch             | 187Ch             | 1874h             | 1870h            | FREQ = 1h                             |
| D6h             | [7:0] |               |               | 1884h             | 1878h             | 1870h             | 1870h             | 187Ch             | 1874h             | 1870h             | 1870h             | 187Ch             | 1874h             | 1870h             | 1870h            | FREQ = 2h                             |
| D8h             | [7:0] | HMAX          | 0A6h          | 91h               | E7h               | 1BFh              | 370h              | 97h               | 119h              | 225h              | 445h              | D1h               | 14Fh              | 28Fh              | 519h             | FREQ = 0h                             |
| D9h             | [7:0] |               |               | F3h               | 1C7h              | 377h              | 6E0h              | 11Fh              | 22Bh              | 44Ah              | 88Ah              | 162h              | 297h              | 51Eh              | A31h             | FREQ = 1h                             |
|                 |       |               |               | 1E0h              | 389h              | 6EEh              | DBFh              | 239h              | 456h              | 894h              | 1114h             | 2C0h              | 52Ah              | A3Bh              | 1462h            | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 0Ch           | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h              | FREQ = 0h                             |
|                 |       |               |               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h              | FREQ = 1h                             |
|                 |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h              | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 48h           | 1Ch               | 14h               | 0Ch               | 08h               | 1Ch               | 10h               | 08h               | 04h               | 14h               | 10h               | 08h               | 04h              | FREQ = 0h                             |
|                 |       |               |               | 14h               | 0Ch               | 08h               | 04h               | 10h               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h              | FREQ = 1h                             |
|                 |       |               |               | 0Ch               | 08h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h              | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| E6h             | [7:0] | GSDLY         | 34h           | 2Ch               | 1Ch               | 10h               | 08h               | 2Ch               | 18h               | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h              | FREQ = 0h                             |
|                 |       |               |               | 1Ch               | 10h               | 08h               | 04h               | 18h               | 0Ch               | 08h               | 04h               | 14h               | 0Ch               | 08h               | 04h              | FREQ = 1h                             |
|                 |       |               |               | 10h               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h              | FREQ = 2h                             |
| Chip ID = 04h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| 3Dh             | [0]   |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| Chip ID = 06h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |
| F8h             | [7:0] |               | 2Dh           | 09h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                                       |

[illegible]

The output of each AD is output alternately every other line.



## 7.1.3.2. ROI, Overlap ROI

## SLVS Output (IMX532)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 2CCh                                                                                                          | 574h      | 37Ch        | 6CDh      | 42Bh        | 826h      | FREQ = 0h                             |           |
|               |       |               |               | 584h                                                                                                          | ACDh      | 6E2h        | D7Dh      | 83Fh        | 102Ch     | FREQ = 1h                             |           |
| D9h           | [7:0] |               |               | 208h                                                                                                          | 3DEh      | 286h        | 4D1h      | 2CCh        | 574h      | FREQ = 2h                             |           |
|               |       |               |               | 3F8h                                                                                                          | 799h      | 4F2h        | 97Bh      | 584h        | ACDh      | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 48h           | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
| E6h           | [7:0] | GSDLY         | 34h           | 0Ch                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 2h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 3h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

## SLVS – EC output (IMX532)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |           |
|-----------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|-----------|
|                 |       |               |               | AD = 8 bit                                                                                                 |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |           |
|                 |       |               |               | 8 lane                                                                                                     | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |           |
|                 |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |           |
|                 |       |               |               | 4752 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
|                 |       |               |               | 2376 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 1188 Mbps /lane |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| Chip ID = 02h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| D4h             | [7:0] | VMAX          | 12ECh         | *1                                                                                                         | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              |                                       |           |
| D5h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| D6h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| D8h             | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| D9h             | [7:0] | HMAX          | 0A6h          | 91h                                                                                                        | E7h             | 1BFh            | 370h            | 97h             | 119h            | 225h            | 445h            | D1h             | 14Fh             | 28Fh             | 519h             | FREQ = 0h                             |           |
| E2h             | [7:0] | GMRWT         | 0Ch           | F3h                                                                                                        | 1C7h            | 377h            | 6E0h            | 11Fh            | 22Bh            | 44Ah            | 88Ah            | 162h            | 297h             | 51Eh             | A31h             | FREQ = 1h                             |           |
|                 |       |               |               | 1E0h                                                                                                       | 389h            | 6EEh            | DBFh            | 239h            | 456h            | 894h            | 1114h           | 2C0h            | 52Ah             | A3Bh             | 1462h            | FREQ = 2h                             |           |
|                 |       |               |               | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | 04h                                   | FREQ = 0h |
|                 |       |               |               | 08h                                                                                                        | 04h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | 04h                                   | FREQ = 1h |
| E3h             | [7:0] | GMTWT         | 48h           | 04h                                                                                                        | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h             | 04h              | 04h              | 04h              | FREQ = 2h                             |           |
|                 |       |               |               | 1Ch                                                                                                        | 14h             | 0Ch             | 08h             | 1Ch             | 10h             | 08h             | 04h             | 14h             | 10h              | 08h              | 04h              | 04h                                   | FREQ = 0h |
|                 |       |               |               | 14h                                                                                                        | 0Ch             | 08h             | 04h             | 10h             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | 04h                                   | FREQ = 1h |
| E5h             | [7:0] | GAINDLY       | 04h           | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 2h                             |           |
| E6h             | [7:0] | GSDLY         | 34h           | 04h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
|                 |       |               |               | 2Ch                                                                                                        | 1Ch             | 10h             | 08h             | 2Ch             | 18h             | 0Ch             | 08h             | 20h             | 14h              | 0Ch              | 08h              |                                       | FREQ = 0h |
|                 |       |               |               | 1Ch                                                                                                        | 10h             | 08h             | 04h             | 18h             | 0Ch             | 08h             | 04h             | 14h             | 0Ch              | 08h              | 04h              |                                       | FREQ = 1h |
|                 |       |               |               | 10h                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 2h                             |           |
| Chip ID = 03h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| Chip ID = 04h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 3Dh             | [0]   |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| Chip ID = 06h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| E8h             | [7:0] |               | 2Dh           | 09h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

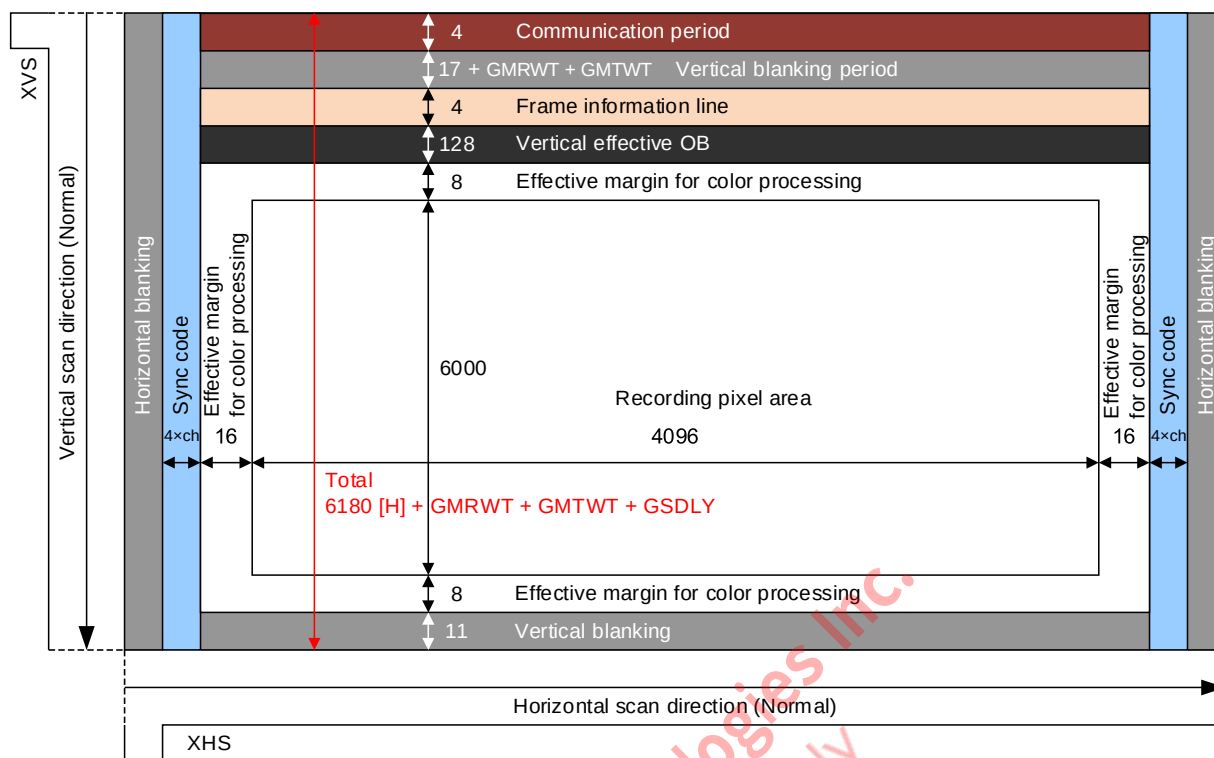
# 7.1.4. IMX535-AAQJ

## 7.1.4.1. All-pixel scan

SLVS Output (IMX535-AAQJ)

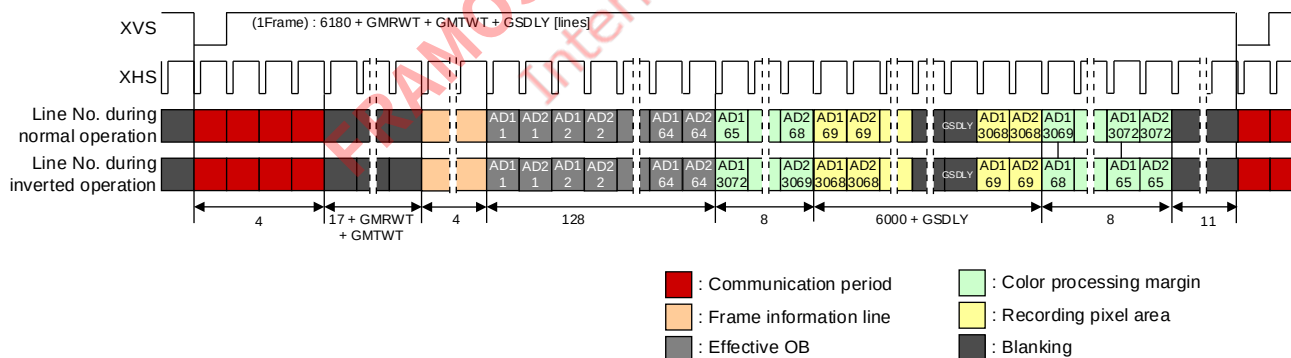
| Address       | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                   | AD = 10 bit       |                   | AD = 12 bit       |                   |                                       |           |
|               |       |               |               | 8 ch              | 4 ch              | 8 ch              | 4 ch              | 8 ch              | 4 ch              | FREQ = 0h                             |           |
|               |       |               |               | 21.0<br>[frame/s] | 10.9<br>[frame/s] | 16.9<br>[frame/s] | 8.7<br>[frame/s]  | 14.1<br>[frame/s] | 7.3<br>[frame/s]  |                                       |           |
|               |       |               |               | 10.7<br>[frame/s] | 5.5<br>[frame/s]  | 8.6<br>[frame/s]  | 4.4<br>[frame/s]  | 7.2<br>[frame/s]  | 3.6<br>[frame/s]  |                                       | FREQ = 1h |
|               |       |               |               | 30.8<br>[frame/s] | 16.1<br>[frame/s] | 24.9<br>[frame/s] | 13.0<br>[frame/s] | 21.0<br>[frame/s] | 10.9<br>[frame/s] |                                       |           |
|               |       |               |               | 15.9<br>[frame/s] | 8.2<br>[frame/s]  | 12.8<br>[frame/s] | 6.6<br>[frame/s]  | 10.7<br>[frame/s] | 5.5<br>[frame/s]  | FREQ = 3h                             |           |
|               |       |               |               | 594 Mbps/ch       |                   |                   |                   |                   |                   | FREQ = 0h                             |           |
|               |       |               |               | 297 Mbps/ch       |                   |                   |                   |                   |                   | FREQ = 1h                             |           |
|               |       |               |               | 891 Mbps/ch       |                   |                   |                   |                   |                   | FREQ = 2h                             |           |
| 445.5 Mbps/ch |       |               |               |                   |                   | FREQ = 3h         |                   |                   |                   |                                       |           |
| Chip ID = 02h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | 1850h             | 1840h             | 184Ch             | 183Ch             | 1844h             | 1838h             | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 1840h             | 1834h             | 183Ch             | 1834h             | 1838h             | 1834h             | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 1864h             | 1848h             | 185Ch             | 1840h             | 1850h             | 1840h             | FREQ = 2h                             |           |
|               |       |               |               | 1848h             | 183Ch             | 1840h             | 1834h             | 1840h             | 1834h             | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 081h          | 236h              | 448h              | 2C0h              | 556h              | 34Ah              | 664h              | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 458h              | 875h              | 56Bh              | A8Fh              | 67Dh              | CA8h              | FREQ = 1h                             |           |
|               |       |               |               | 181h              | 2E3h              | 1DDh              | 397h              | 236h              | 448h              | FREQ = 2h                             |           |
|               |       |               |               | 2ECh              | 5ABh              | 3A3h              | 711h              | 458h              | 875h              | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch               | 10h               | 18h               | 0Ch               | 14h               | 0Ch               | FREQ = 0h                             |           |
|               |       |               |               | 10h               | 08h               | 0Ch               | 08h               | 0Ch               | 08h               | FREQ = 1h                             |           |
|               |       |               |               | 28h               | 14h               | 20h               | 10h               | 1Ch               | 10h               | FREQ = 2h                             |           |
|               |       |               |               | 14h               | 0Ch               | 10h               | 08h               | 10h               | 08h               | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch               | 08h               | 0Ch               | 08h               | 08h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 10h               | 0Ch               | 10h               | 08h               | 0Ch               | 08h               | FREQ = 2h                             |           |
|               |       |               |               | 0Ch               | 08h               | 08h               | 04h               | 08h               | 04h               | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                                       |           |
| 3Dh           | [0]   |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh               |                   | 1Bh               |                   | 13h               |                   |                                       |           |
| Chip ID = 15h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |

(IMX535-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

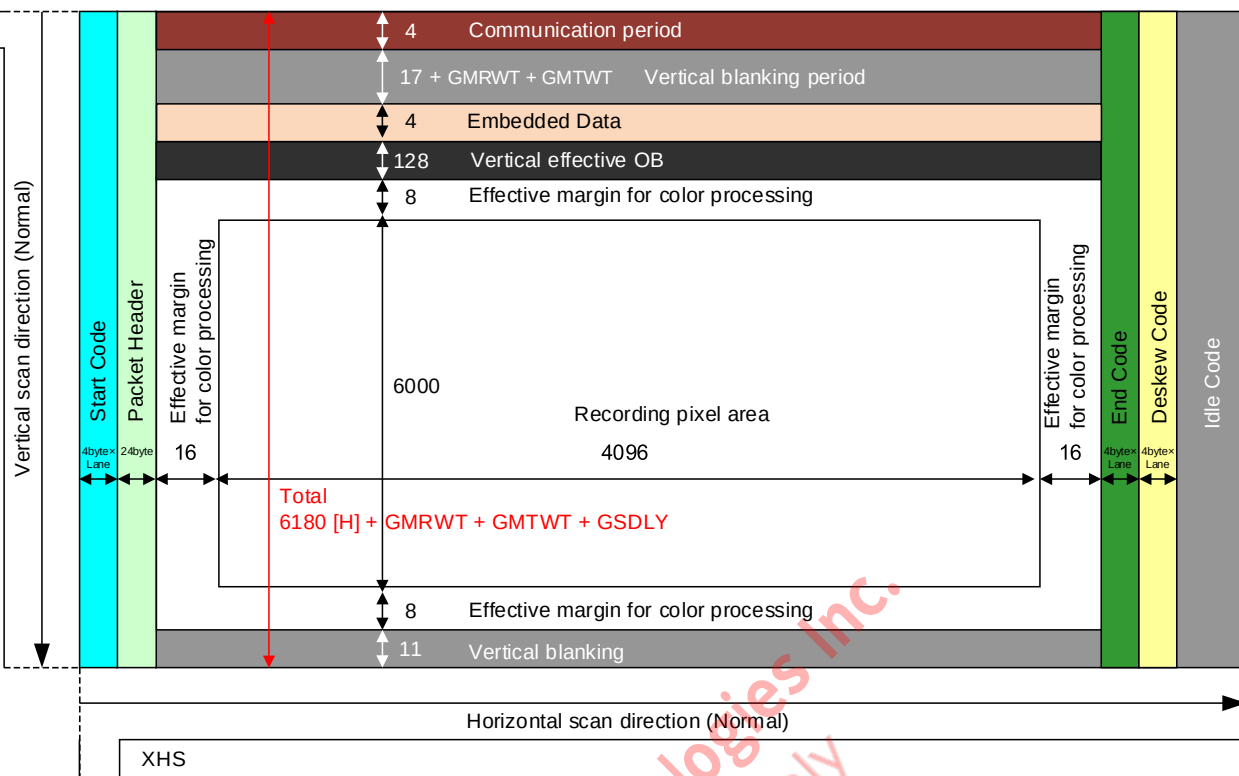


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX535-AAQJ)

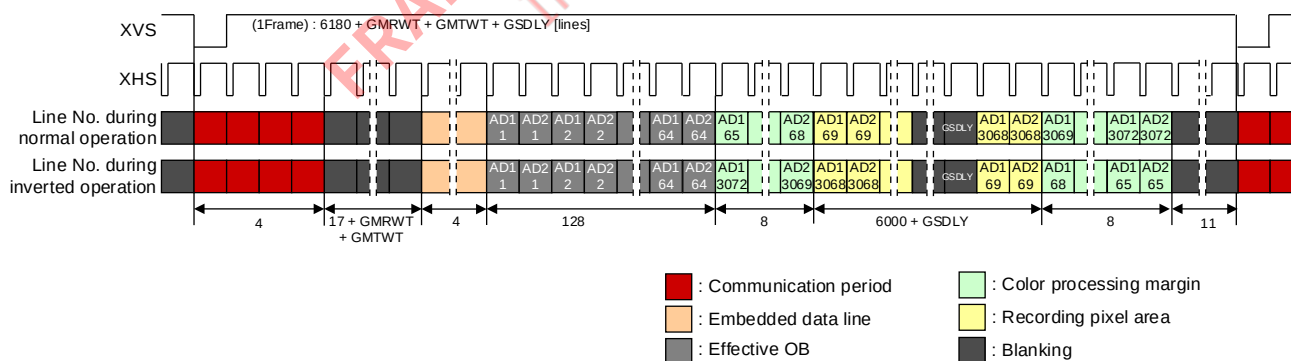
| Address         | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | Remarks                               |           |
|-----------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|                 |       |               |               | AD = 8 bit        |                   |                   |                   | AD = 10 bit       |                   |                   |                   | AD = 12 bit       |                   |                   |                   |                                       |           |
|                 |       |               |               | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            |                                       |           |
|                 |       |               |               | 95.6<br>[frame/s] | 64.6<br>[frame/s] | 34.5<br>[frame/s] | 17.4<br>[frame/s] | 90.5<br>[frame/s] | 53.4<br>[frame/s] | 27.7<br>[frame/s] | 14.0<br>[frame/s] | 62.0<br>[frame/s] | 46.0<br>[frame/s] | 23.3<br>[frame/s] | 11.7<br>[frame/s] |                                       | FREQ = 0h |
|                 |       |               |               | 61.0<br>[frame/s] | 33.2<br>[frame/s] | 17.3<br>[frame/s] | 8.7<br>[frame/s]  | 52.0<br>[frame/s] | 27.3<br>[frame/s] | 13.9<br>[frame/s] | 7.0<br>[frame/s]  | 42.4<br>[frame/s] | 23.2<br>[frame/s] | 11.7<br>[frame/s] | 5.8<br>[frame/s]  |                                       | FREQ = 1h |
|                 |       |               |               | 31.2<br>[frame/s] | 16.8<br>[frame/s] | 8.7<br>[frame/s]  | 4.3<br>[frame/s]  | 26.5<br>[frame/s] | 13.7<br>[frame/s] | 6.9<br>[frame/s]  | 3.5<br>[frame/s]  | 21.5<br>[frame/s] | 11.6<br>[frame/s] | 5.8<br>[frame/s]  | 2.9<br>[frame/s]  |                                       | FREQ = 2h |
|                 |       |               |               | 4752 Mbps/lane    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       | FREQ = 0h |
| 2376 Mbps/lane  |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 1h         |                   |                   |                   |                                       |           |
| 1188 Mbps /lane |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |           |
| Chip ID = 02h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h             | [7:0] | VMAX          | 0CD4h         | 18DCh             | 18A4h             | 186Ch             | 184Ch             | 18D4h             | 188Ch             | 1860h             | 1844h             | 189Ch             | 187Ch             | 1850h             | 1840h             | FREQ = 0h                             |           |
| D5h             | [7:0] |               |               | 189Ch             | 1868h             | 184Ch             | 183Ch             | 188Ch             | 1860h             | 1844h             | 1838h             | 1878h             | 1850h             | 1840h             | 1834h             | FREQ = 1h                             |           |
| D6h             | [7:0] |               |               | 1864h             | 1848h             | 183Ch             | 1834h             | 185Ch             | 1844h             | 1838h             | 1834h             | 1850h             | 1840h             | 1834h             | 1830h             | FREQ = 2h                             |           |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah               | B6h               | 158h              | 2AEh              | 81h               | DDh               | 1ADh              | 355h              | BEh               | 101h              | 1FFh              | 3F9h              | FREQ = 0h                             |           |
| D9h             | [7:0] |               |               | C1h               | 165h              | 2AFh              | 55Bh              | E3h               | 1B3h              | 35Ah              | 6AAh              | 117h              | 201h              | 3FEh              | 7F2h              | FREQ = 1h                             |           |
|                 |       |               |               | 17Ch              | 2C6h              | 55Eh              | AB5h              | 1C1h              | 366h              | 6B4h              | D54h              | 22Ah              | 402h              | 7FCh              | FE4h              | FREQ = 2h                             |           |
| E2h             | [7:0] | GMRWT         | 10h           | 10h               | 0Ch               | 08h               | 04h               | 10h               | 0Ch               | 08h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 0h                             |           |
|                 |       |               |               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|                 |       |               |               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 2h                             |           |
| E3h             | [7:0] | GMTWT         | 70h           | 74h               | 50h               | 2Ch               | 18h               | 70h               | 40h               | 24h               | 14h               | 4Ch               | 38h               | 1Ch               | 10h               | FREQ = 0h                             |           |
|                 |       |               |               | 4Ch               | 28h               | 18h               | 0Ch               | 40h               | 24h               | 14h               | 0Ch               | 34h               | 1Ch               | 10h               | 08h               | FREQ = 1h                             |           |
|                 |       |               |               | 28h               | 14h               | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h               | 1Ch               | 10h               | 08h               | 04h               | FREQ = 2h                             |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| E6h             | [7:0] | GSDLY         | 30h           | 34h               | 24h               | 14h               | 0Ch               | 30h               | 1Ch               | 10h               | 08h               | 20h               | 18h               | 0Ch               | 08h               | FREQ = 0h                             |           |
|                 |       |               |               | 20h               | 14h               | 0Ch               | 08h               | 1Ch               | 10h               | 08h               | 04h               | 18h               | 0Ch               | 08h               | 04h               | FREQ = 1h                             |           |
|                 |       |               |               | 10h               | 0Ch               | 08h               | 04h               | 10h               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 2h                             |           |
| Chip ID = 04h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 3Dh             | [0]   |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h             | [7:0] |               | 2Dh           | 1Bh               |                   |                   |                   | 1Bh               |                   |                   |                   | 13h               |                   |                   |                   |                                       |           |
| Chip ID = 15h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh             | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Fh             | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |

(IMX535-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 7.1.4.2. ROI, Overlap ROI

## SLVS Output (IMX535-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 081h          | 236h                                                                                                          | 448h      | 2C0h        | 556h      | 34Ah        | 664h      | FREQ = 0h                             |           |
|               |       |               |               | 458h                                                                                                          | 875h      | 56Bh        | A8Fh      | 67Dh        | CA8h      | FREQ = 1h                             |           |
| D9h           | [7:0] |               |               | 181h                                                                                                          | 2E3h      | 1DDh        | 397h      | 236h        | 448h      | FREQ = 2h                             |           |
|               |       |               |               | 2ECh                                                                                                          | 5ABh      | 3A3h        | 711h      | 458h        | 875h      | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch                                                                                                           | 10h       | 18h         | 0Ch       | 14h         | 0Ch       | FREQ = 0h                             |           |
|               |       |               |               | 10h                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 1h                             |           |
|               |       |               |               | 28h                                                                                                           | 14h       | 20h         | 10h       | 1Ch         | 10h       | FREQ = 2h                             |           |
|               |       |               |               | 14h                                                                                                           | 0Ch       | 10h         | 08h       | 10h         | 08h       | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch                                                                                                           | 08h       | 0Ch         | 08h       | 08h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 10h                                                                                                           | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 2h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 3h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                           |           | 1Bh         |           | 13h         |           |                                       |           |
| Chip ID = 15h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

## SLVS – EC output (IMX535-AAQJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |           |           |
|-----------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|-----------|-----------|
|                 |       |               |               | AD = 8 bit                                                                                                 |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |           |           |
|                 |       |               |               | 8 lane                                                                                                     | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |           |           |
|                 |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |           |           |
|                 |       |               |               | 4752 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       | FREQ = 0h |           |
| 2376 Mbps/lane  |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 1h                             |           |           |
| 1188 Mbps /lane |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | FREQ = 2h                             |           |           |
| Chip ID = 02h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              |                                       |           |           |
| D5h             | [7:0] |               |               | HMAX                                                                                                       | 081h            | 7Ah             | B6h             | 158h            | 2AEh            | 81h             | DDh             | 1ADh            | 355h             | BEh              | 101h             | 1FFh                                  | 3F9h      | FREQ = 0h |
| D6h             | [7:0] |               |               |                                                                                                            |                 | C1h             | 165h            | 2AFh            | 55Bh            | E3h             | 1B3h            | 35Ah            | 6AAh             | 117h             | 201h             | 3FEh                                  | 7F2h      | FREQ = 1h |
| D8h             | [7:0] | 17Ch          | 2C6h          |                                                                                                            |                 | 55Eh            | AB5h            | 1C1h            | 366h            | 6B4h            | D54h            | 22Ah            | 402h             | 7FCh             | FE4h             | FREQ = 2h                             |           |           |
| D9h             | [7:0] | GMRWT         | 10h           | 10h                                                                                                        | 0Ch             | 08h             | 04h             | 10h             | 0Ch             | 08h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 0h                             |           |           |
|                 |       |               |               | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 1h                             |           |           |
|                 |       |               |               | 08h                                                                                                        | 04h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 04h             | 04h              | 04h              | 04h              | FREQ = 2h                             |           |           |
| E2h             | [7:0] | GMTWT         | 70h           | 74h                                                                                                        | 50h             | 2Ch             | 18h             | 70h             | 40h             | 24h             | 14h             | 4Ch             | 38h              | 1Ch              | 10h              | FREQ = 0h                             |           |           |
|                 |       |               |               | 4Ch                                                                                                        | 28h             | 18h             | 0Ch             | 40h             | 24h             | 14h             | 0Ch             | 34h             | 1Ch              | 10h              | 08h              | FREQ = 1h                             |           |           |
|                 |       |               |               | 28h                                                                                                        | 14h             | 0Ch             | 08h             | 20h             | 14h             | 0Ch             | 08h             | 1Ch             | 10h              | 08h              | 04h              | FREQ = 2h                             |           |           |
| E3h             | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| E5h             | [7:0] | GSDLY         | 30h           | 34h                                                                                                        | 24h             | 14h             | 0Ch             | 30h             | 1Ch             | 10h             | 08h             | 20h             | 18h              | 0Ch              | 08h              | FREQ = 0h                             |           |           |
|                 |       |               |               | 20h                                                                                                        | 14h             | 0Ch             | 08h             | 1Ch             | 10h             | 08h             | 04h             | 18h             | 0Ch              | 08h              | 04h              | FREQ = 1h                             |           |           |
|                 |       |               |               | 10h                                                                                                        | 0Ch             | 08h             | 04h             | 10h             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 2h                             |           |           |
| Chip ID = 03h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| Chip ID = 04h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 3Dh             | [0]   |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| Chip ID = 06h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |           |
| Chip ID = 08h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| E8h             | [7:0] |               | 2Dh           | 1Bh                                                                                                        |                 |                 |                 | 1Bh             |                 |                 |                 | 13h             |                  |                  |                  |                                       |           |           |
| Chip ID = 15h   |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$

Overlap ROI :

$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$

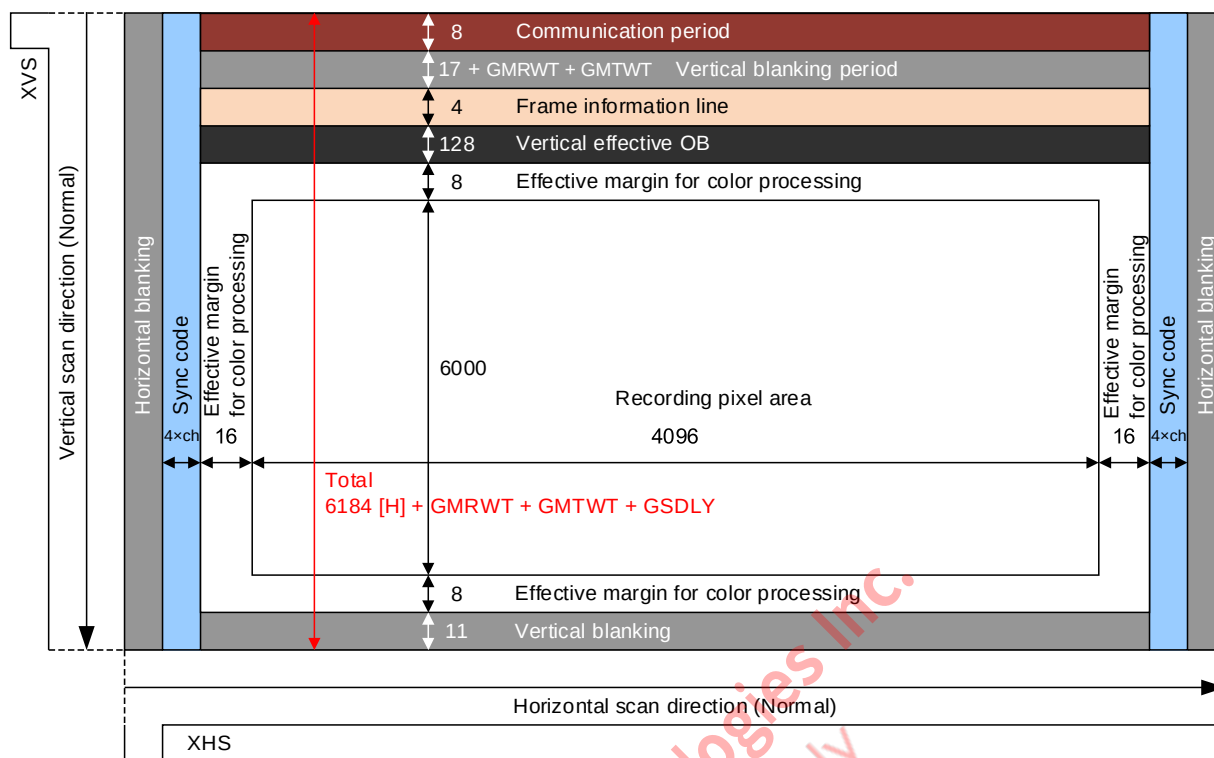
## 7.1.5. IMX535-AAMJ

## 7.1.5.1. All-pixel scan

SLVS Output (IMX535-AAMJ)

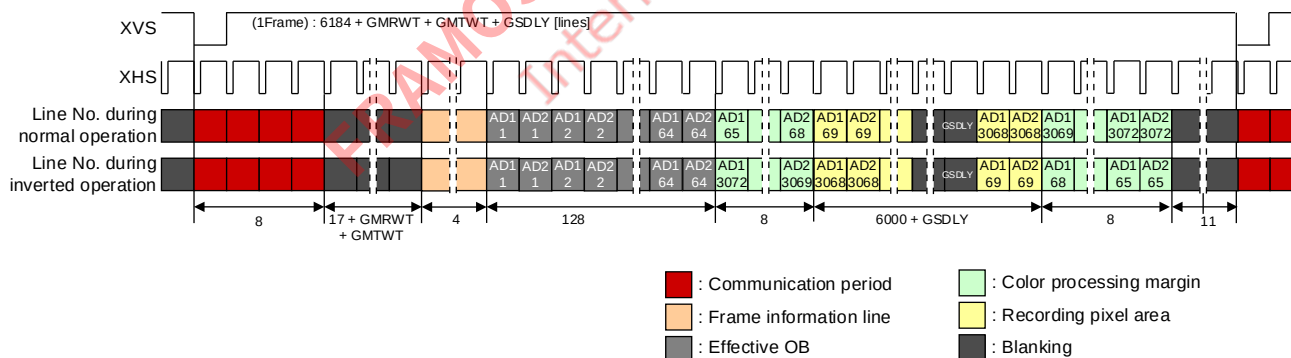
| Address       | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                   | AD = 10 bit       |                   | AD = 12 bit       |                   |                                       |           |
|               |       |               |               | 8 ch              | 4 ch              | 8 ch              | 4 ch              | 8 ch              | 4 ch              |                                       |           |
|               |       |               |               | 21.0<br>[frame/s] | 10.9<br>[frame/s] | 16.9<br>[frame/s] | 8.7<br>[frame/s]  | 14.1<br>[frame/s] | 7.3<br>[frame/s]  |                                       | FREQ = 0h |
|               |       |               |               | 10.7<br>[frame/s] | 5.5<br>[frame/s]  | 8.6<br>[frame/s]  | 4.4<br>[frame/s]  | 7.2<br>[frame/s]  | 3.6<br>[frame/s]  |                                       | FREQ = 1h |
|               |       |               |               | 30.8<br>[frame/s] | 16.1<br>[frame/s] | 24.9<br>[frame/s] | 13.0<br>[frame/s] | 21.0<br>[frame/s] | 10.9<br>[frame/s] |                                       | FREQ = 2h |
|               |       |               |               | 15.9<br>[frame/s] | 8.2<br>[frame/s]  | 12.8<br>[frame/s] | 6.6<br>[frame/s]  | 10.7<br>[frame/s] | 5.5<br>[frame/s]  |                                       | FREQ = 3h |
|               |       |               |               | 594 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 2h |
| 445.5 Mbps/ch |       |               |               |                   |                   | FREQ = 3h         |                   |                   |                   |                                       |           |
| Chip ID = 02h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | 1854h             | 1844h             | 1850h             | 1840h             | 1848h             | 183Ch             | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 1844h             | 1838h             | 1840h             | 1838h             | 183Ch             | 1838h             | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 1868h             | 184Ch             | 1860h             | 1844h             | 1854h             | 1844h             | FREQ = 2h                             |           |
|               |       |               |               | 184Ch             | 1840h             | 1844h             | 1838h             | 1844h             | 1838h             | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 081h          | 236h              | 448h              | 2C0h              | 556h              | 34Ah              | 664h              | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 458h              | 875h              | 56Bh              | A8Fh              | 67Dh              | CA8h              | FREQ = 1h                             |           |
|               |       |               |               | 181h              | 2E3h              | 1DDh              | 397h              | 236h              | 448h              | FREQ = 2h                             |           |
|               |       |               |               | 2ECh              | 5ABh              | 3A3h              | 711h              | 458h              | 875h              | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch               | 10h               | 18h               | 0Ch               | 14h               | 0Ch               | FREQ = 0h                             |           |
|               |       |               |               | 10h               | 08h               | 0Ch               | 08h               | 0Ch               | 08h               | FREQ = 1h                             |           |
|               |       |               |               | 28h               | 14h               | 20h               | 10h               | 1Ch               | 10h               | FREQ = 2h                             |           |
|               |       |               |               | 14h               | 0Ch               | 10h               | 08h               | 10h               | 08h               | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch               | 08h               | 0Ch               | 08h               | 08h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 10h               | 0Ch               | 10h               | 08h               | 0Ch               | 08h               | FREQ = 2h                             |           |
|               |       |               |               | 0Ch               | 08h               | 08h               | 04h               | 08h               | 04h               | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                                       |           |
| 3Dh           | [0]   |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh               |                   | 1Bh               |                   | 13h               |                   |                                       |           |
| Chip ID = 15h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |

(IMX535-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

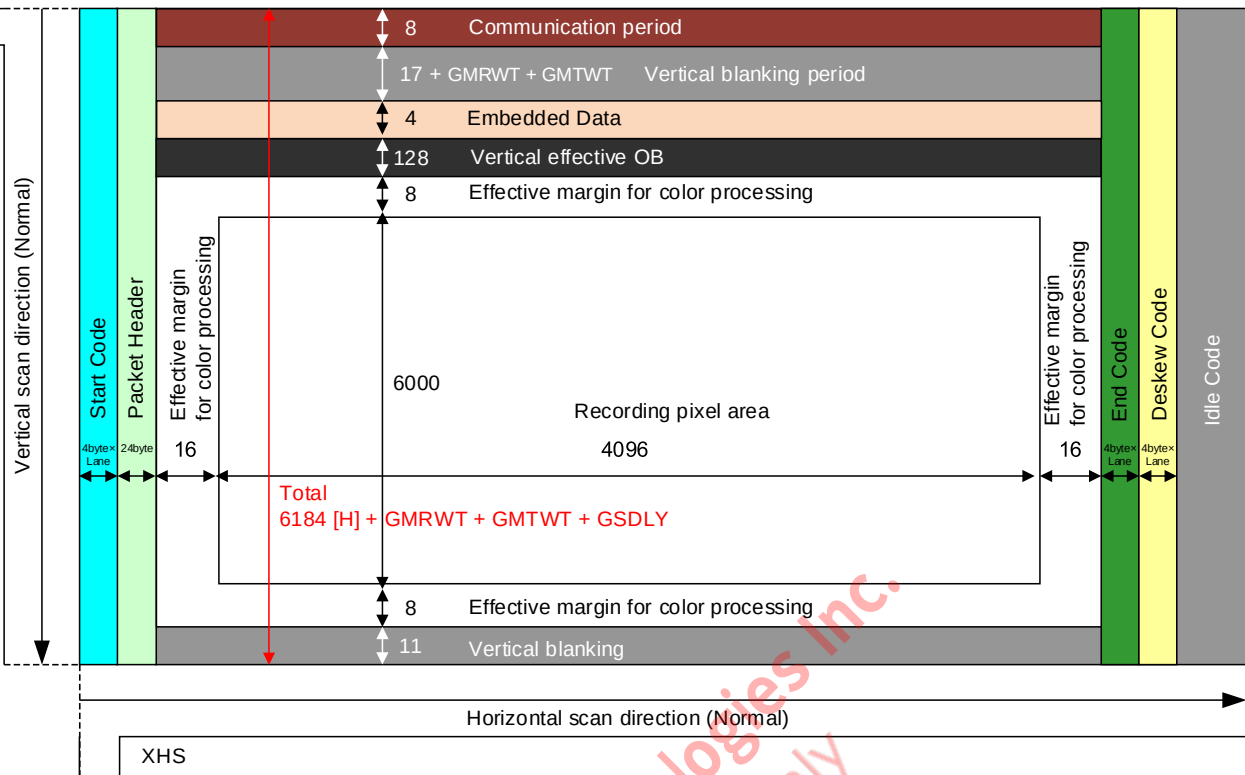


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX535-AAMJ)

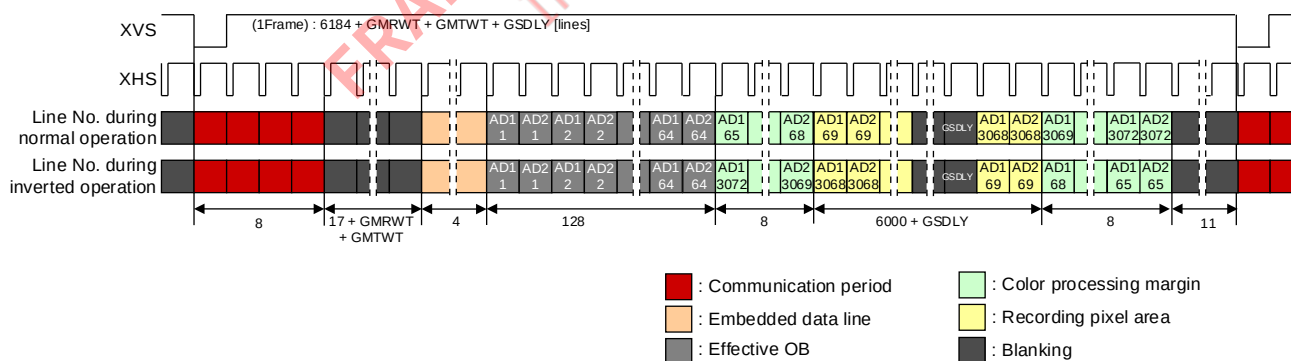
| Address         | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | Remarks                               |           |
|-----------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|                 |       |               |               | AD = 8 bit        |                   |                   |                   | AD = 10 bit       |                   |                   |                   | AD = 12 bit       |                   |                   |                   |                                       |           |
|                 |       |               |               | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | FREQ = 0h                             |           |
|                 |       |               |               | 95.5<br>[frame/s] | 64.6<br>[frame/s] | 34.5<br>[frame/s] | 17.3<br>[frame/s] | 90.5<br>[frame/s] | 53.4<br>[frame/s] | 27.7<br>[frame/s] | 14.0<br>[frame/s] | 61.9<br>[frame/s] | 46.0<br>[frame/s] | 23.3<br>[frame/s] | 11.7<br>[frame/s] |                                       |           |
|                 |       |               |               | 61.0<br>[frame/s] | 33.2<br>[frame/s] | 17.3<br>[frame/s] | 8.7<br>[frame/s]  | 52.0<br>[frame/s] | 27.3<br>[frame/s] | 13.9<br>[frame/s] | 7.0<br>[frame/s]  | 42.4<br>[frame/s] | 23.2<br>[frame/s] | 11.6<br>[frame/s] | 5.8<br>[frame/s]  |                                       | FREQ = 1h |
|                 |       |               |               | 31.2<br>[frame/s] | 16.8<br>[frame/s] | 8.7<br>[frame/s]  | 4.3<br>[frame/s]  | 26.5<br>[frame/s] | 13.7<br>[frame/s] | 6.9<br>[frame/s]  | 3.5<br>[frame/s]  | 21.5<br>[frame/s] | 11.6<br>[frame/s] | 5.8<br>[frame/s]  | 2.9<br>[frame/s]  |                                       |           |
|                 |       |               |               | 4752 Mbps/lane    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       | FREQ = 0h |
| 2376 Mbps/lane  |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 1h         |                   |                   |                   |                                       |           |
| 1188 Mbps /lane |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |           |
| Chip ID = 02h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h             | [7:0] | VMAX          | 0CD4h         | 18E0h             | 18A8h             | 1870h             | 1850h             | 18D8h             | 1890h             | 1864h             | 1848h             | 18A0h             | 1880h             | 1854h             | 1844h             | FREQ = 0h                             |           |
| D5h             | [7:0] |               |               | 18A0h             | 186Ch             | 1850h             | 1840h             | 1890h             | 1864h             | 1848h             | 183Ch             | 187Ch             | 1854h             | 1844h             | 1838h             | FREQ = 1h                             |           |
| D6h             | [7:0] |               |               | 1868h             | 184Ch             | 1840h             | 1838h             | 1860h             | 1848h             | 183Ch             | 1838h             | 1854h             | 1844h             | 1838h             | 1834h             | FREQ = 2h                             |           |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah               | B6h               | 158h              | 2AEh              | 81h               | DDh               | 1ADh              | 355h              | BEh               | 101h              | 1FFh              | 3F9h              | FREQ = 0h                             |           |
| D9h             | [7:0] |               |               | C1h               | 165h              | 2AFh              | 55Bh              | E3h               | 1B3h              | 35Ah              | 6AAh              | 117h              | 201h              | 3FEh              | 7F2h              | FREQ = 1h                             |           |
|                 |       |               |               | 17Ch              | 2C6h              | 55Eh              | AB5h              | 1C1h              | 366h              | 6B4h              | D54h              | 22Ah              | 402h              | 7FCh              | FE4h              | FREQ = 2h                             |           |
| E2h             | [7:0] | GMRWT         | 10h           | 10h               | 0Ch               | 08h               | 04h               | 10h               | 0Ch               | 08h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 0h                             |           |
|                 |       |               |               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|                 |       |               |               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 2h                             |           |
| E3h             | [7:0] | GMTWT         | 70h           | 74h               | 50h               | 2Ch               | 18h               | 70h               | 40h               | 24h               | 14h               | 4Ch               | 38h               | 1Ch               | 10h               | FREQ = 0h                             |           |
|                 |       |               |               | 4Ch               | 28h               | 18h               | 0Ch               | 40h               | 24h               | 14h               | 0Ch               | 34h               | 1Ch               | 10h               | 08h               | FREQ = 1h                             |           |
|                 |       |               |               | 28h               | 14h               | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h               | 1Ch               | 10h               | 08h               | 04h               | FREQ = 2h                             |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| E6h             | [7:0] | GSDLY         | 30h           | 34h               | 24h               | 14h               | 0Ch               | 30h               | 1Ch               | 10h               | 08h               | 20h               | 18h               | 0Ch               | 08h               | FREQ = 0h                             |           |
|                 |       |               |               | 20h               | 14h               | 0Ch               | 08h               | 1Ch               | 10h               | 08h               | 04h               | 18h               | 0Ch               | 08h               | 04h               | FREQ = 1h                             |           |
|                 |       |               |               | 10h               | 0Ch               | 08h               | 04h               | 10h               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 2h                             |           |
| Chip ID = 04h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 3Dh             | [0]   |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h             | [7:0] |               | 2Dh           | 1Bh               |                   |                   |                   | 1Bh               |                   |                   |                   | 13h               |                   |                   |                   |                                       |           |
| Chip ID = 15h   |       |               |               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh             | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Fh             | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |

(IMX535-AAMJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 7.1.5.2. ROI, Overlap ROI

## SLVS Output (IMX535-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 081h          | 236h                                                                                                          | 448h      | 2C0h        | 556h      | 34Ah        | 664h      | FREQ = 0h                             |           |
|               |       |               |               | 458h                                                                                                          | 875h      | 56Bh        | A8Fh      | 67Dh        | CA8h      | FREQ = 1h                             |           |
| D9h           | [7:0] |               |               | 181h                                                                                                          | 2E3h      | 1DDh        | 397h      | 236h        | 448h      | FREQ = 2h                             |           |
|               |       |               |               | 2ECh                                                                                                          | 5ABh      | 3A3h        | 711h      | 458h        | 875h      | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch                                                                                                           | 10h       | 18h         | 0Ch       | 14h         | 0Ch       | FREQ = 0h                             |           |
|               |       |               |               | 10h                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 1h                             |           |
|               |       |               |               | 28h                                                                                                           | 14h       | 20h         | 10h       | 1Ch         | 10h       | FREQ = 2h                             |           |
|               |       |               |               | 14h                                                                                                           | 0Ch       | 10h         | 08h       | 10h         | 08h       | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch                                                                                                           | 08h       | 0Ch         | 08h       | 08h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 10h                                                                                                           | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 2h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 3h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                           |           | 1Bh         |           | 13h         |           |                                       |           |
| Chip ID = 15h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 168$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 168$$

## SLVS – EC output (IMX535-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|
|               |       |               |               | AD = 8 bit                                                                                                 |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |
|               |       |               |               | 8 lane                                                                                                     | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |
|               |       |               |               | 4752 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              |                                       |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D8h           | [7:0] | HMAX          | 081h          | 7Ah                                                                                                        | B6h             | 158h            | 2AEh            | 81h             | DDh             | 1ADh            | 355h            | BEh             | 101h             | 1FFh             | 3F9h             | FREQ = 0h                             |
| D9h           | [7:0] |               |               | C1h                                                                                                        | 165h            | 2AFh            | 55Bh            | E3h             | 1B3h            | 35Ah            | 6AAh            | 117h            | 201h             | 3FEh             | 7F2h             | FREQ = 1h                             |
|               |       |               |               | 17Ch                                                                                                       | 2C6h            | 55Eh            | AB5h            | 1C1h            | 366h            | 6B4h            | D54h            | 22Ah            | 402h             | 7FCh             | FE4h             | FREQ = 2h                             |
| E2h           | [7:0] | GMRWT         | 10h           | 10h                                                                                                        | 0Ch             | 08h             | 04h             | 10h             | 0Ch             | 08h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 0h                             |
|               |       |               |               | 0Ch                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 1h                             |
|               |       |               |               | 08h                                                                                                        | 04h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 04h             | 04h              | 04h              | 04h              | FREQ = 2h                             |
| E3h           | [7:0] | GMTWT         | 70h           | 74h                                                                                                        | 50h             | 2Ch             | 18h             | 70h             | 40h             | 24h             | 14h             | 4Ch             | 38h              | 1Ch              | 10h              | FREQ = 0h                             |
|               |       |               |               | 4Ch                                                                                                        | 28h             | 18h             | 0Ch             | 40h             | 24h             | 14h             | 0Ch             | 34h             | 1Ch              | 10h              | 08h              | FREQ = 1h                             |
|               |       |               |               | 28h                                                                                                        | 14h             | 0Ch             | 08h             | 20h             | 14h             | 0Ch             | 08h             | 1Ch             | 10h              | 08h              | 04h              | FREQ = 2h                             |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| E6h           | [7:0] | GSDLY         | 30h           | 34h                                                                                                        | 24h             | 14h             | 0Ch             | 30h             | 1Ch             | 10h             | 08h             | 20h             | 18h              | 0Ch              | 08h              | FREQ = 0h                             |
|               |       |               |               | 20h                                                                                                        | 14h             | 0Ch             | 08h             | 1Ch             | 10h             | 08h             | 04h             | 18h             | 0Ch              | 08h              | 04h              | FREQ = 1h                             |
|               |       |               |               | 10h                                                                                                        | 0Ch             | 08h             | 04h             | 10h             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 2h                             |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                        |                 |                 |                 | 1Bh             |                 |                 |                 | 13h             |                  |                  |                  |                                       |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 168$

Overlap ROI :

$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 168$

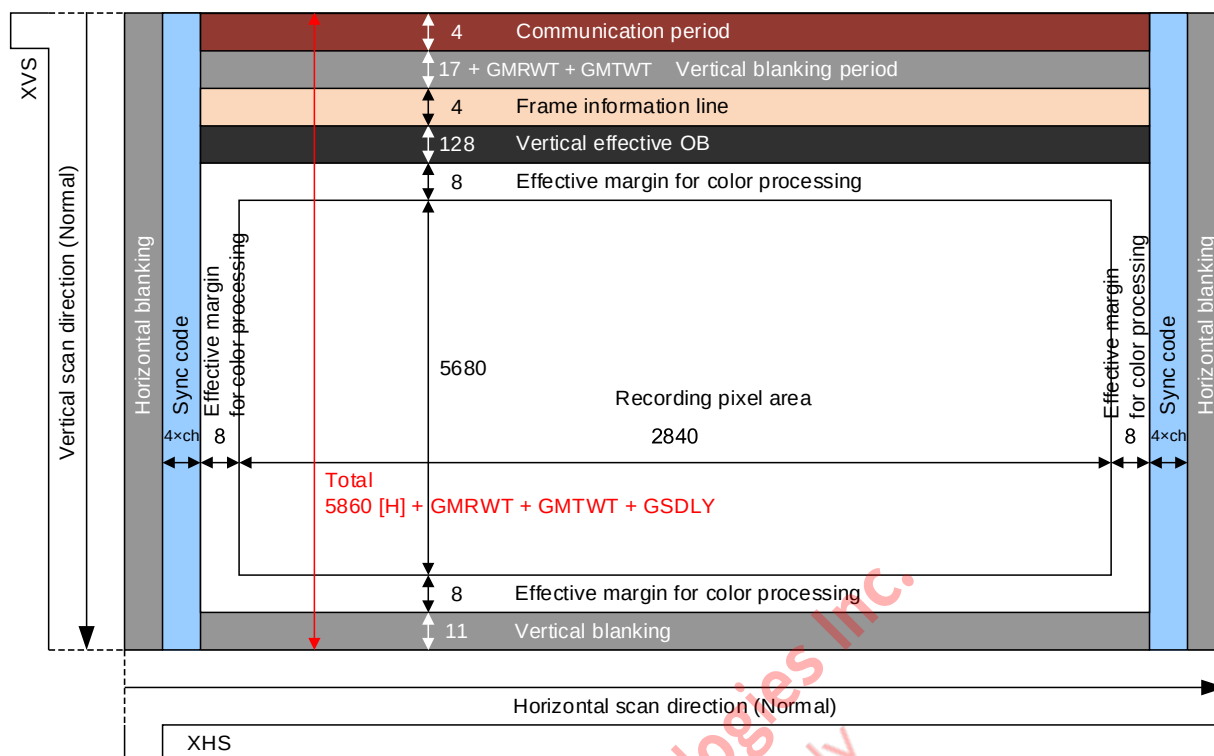
## 7.1.6. IMX536-AAQJ

## 7.1.6.1. All-pixel scan

SLVS Output (IMX536-AAQJ)

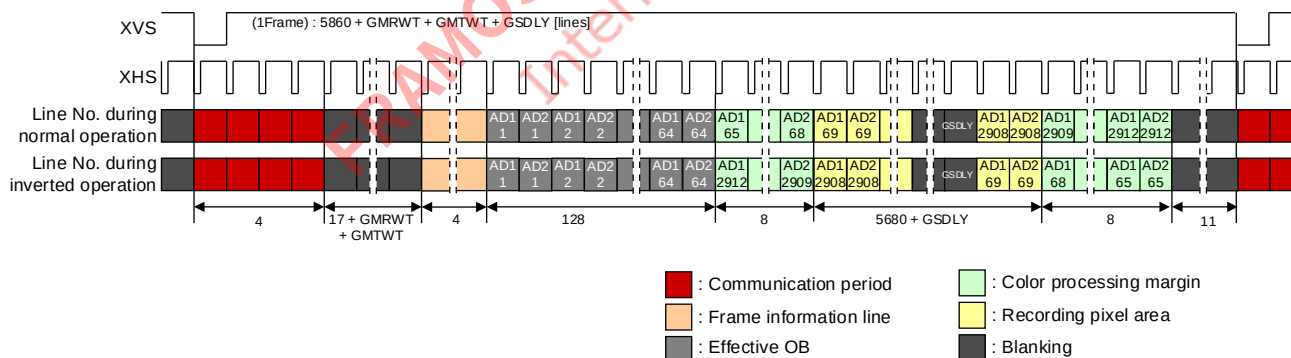
| Address       | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                   | AD = 10 bit       |                   | AD = 12 bit       |                   |                                       |           |
|               |       |               |               | 8 ch              | 4 ch              | 8 ch              | 4 ch              | 8 ch              | 4 ch              |                                       |           |
|               |       |               |               | 30.5<br>[frame/s] | 16.0<br>[frame/s] | 24.6<br>[frame/s] | 12.8<br>[frame/s] | 20.6<br>[frame/s] | 10.7<br>[frame/s] |                                       | FREQ = 0h |
|               |       |               |               | 15.7<br>[frame/s] | 8.2<br>[frame/s]  | 12.6<br>[frame/s] | 6.5<br>[frame/s]  | 10.6<br>[frame/s] | 5.4<br>[frame/s]  |                                       | FREQ = 1h |
|               |       |               |               | 44.4<br>[frame/s] | 23.6<br>[frame/s] | 36.0<br>[frame/s] | 19.0<br>[frame/s] | 30.5<br>[frame/s] | 16.0<br>[frame/s] |                                       | FREQ = 2h |
|               |       |               |               | 23.3<br>[frame/s] | 12.2<br>[frame/s] | 18.7<br>[frame/s] | 9.8<br>[frame/s]  | 15.7<br>[frame/s] | 8.2<br>[frame/s]  |                                       | FREQ = 3h |
|               |       |               |               | 594 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |           |
| 445.5 Mbps/ch |       |               |               |                   |                   | FREQ = 3h         |                   |                   |                   |                                       |           |
| Chip ID = 02h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | 1720h             | 1704h             | 1710h             | 1700h             | 170Ch             | 1700h             | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 1704h             | 16F8h             | 1700h             | 16F4h             | 16FCh             | 16F4h             | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 1738h             | 1710h             | 172Ch             | 170Ch             | 1720h             | 1704h             | FREQ = 2h                             |           |
|               |       |               |               | 1710h             | 1700h             | 170Ch             | 16FCh             | 1704h             | 16F8h             | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 081h          | 19Bh              | 312h              | 1FEh              | 3D3h              | 261h              | 493h              | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 31Eh              | 601h              | 3E2h              | 77Eh              | 4A6h              | 8FAh              | FREQ = 1h                             |           |
|               |       |               |               | 119h              | 214h              | 15Bh              | 294h              | 19Bh              | 312h              | FREQ = 2h                             |           |
|               |       |               |               | 21Bh              | 408h              | 29Eh              | 506h              | 31Eh              | 601h              | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 08h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 08h               | 04h               | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 24h               | 14h               | 1Ch               | 10h               | 18h               | 10h               | FREQ = 0h                             |           |
|               |       |               |               | 14h               | 0Ch               | 10h               | 08h               | 0Ch               | 08h               | FREQ = 1h                             |           |
|               |       |               |               | 34h               | 1Ch               | 2Ch               | 18h               | 24h               | 14h               | FREQ = 2h                             |           |
|               |       |               |               | 1Ch               | 10h               | 18h               | 0Ch               | 14h               | 0Ch               | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 10h               | 08h               | 0Ch               | 08h               | 0Ch               | 08h               | FREQ = 0h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 08h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 18h               | 0Ch               | 14h               | 0Ch               | 10h               | 08h               | FREQ = 2h                             |           |
|               |       |               |               | 0Ch               | 08h               | 0Ch               | 08h               | 08h               | 04h               | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                                       |           |
| 3Dh           | [0]   |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh               |                   | 1Bh               |                   | 13h               |                   |                                       |           |
| Chip ID = 15h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |

(IMX536-AAQJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

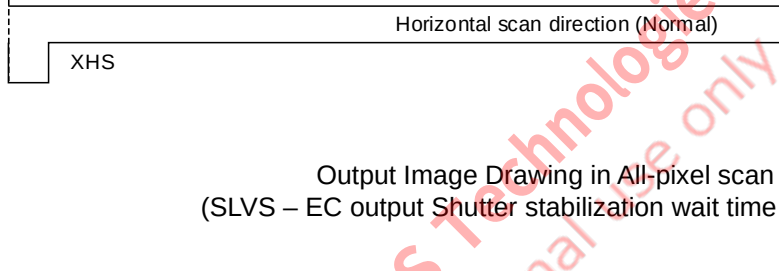


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX536-AAQJ)

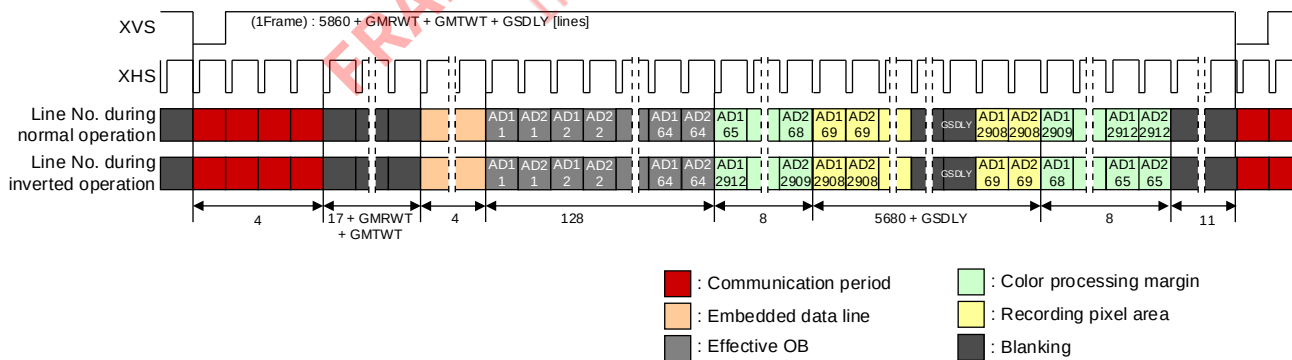
| Address         | bit   | Register name | Initial Value | Setting value      |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | Remarks                               |           |
|-----------------|-------|---------------|---------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|                 |       |               |               | AD = 8 bit         |                   |                   |                   | AD = 10 bit       |                   |                   |                   | AD = 12 bit       |                   |                   |                   |                                       |           |
|                 |       |               |               | 8 lane             | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | FREQ = 0h                             |           |
|                 |       |               |               | 100.6<br>[frame/s] | 94.6<br>[frame/s] | 50.8<br>[frame/s] | 26.1<br>[frame/s] | 95.3<br>[frame/s] | 78.7<br>[frame/s] | 41.3<br>[frame/s] | 21.0<br>[frame/s] | 65.3<br>[frame/s] | 65.3<br>[frame/s] | 34.6<br>[frame/s] | 17.5<br>[frame/s] |                                       |           |
|                 |       |               |               | 88.0<br>[frame/s]  | 49.3<br>[frame/s] | 25.8<br>[frame/s] | 13.1<br>[frame/s] | 75.9<br>[frame/s] | 40.6<br>[frame/s] | 20.8<br>[frame/s] | 10.5<br>[frame/s] | 62.1<br>[frame/s] | 34.4<br>[frame/s] | 17.4<br>[frame/s] | 8.7<br>[frame/s]  |                                       | FREQ = 1h |
|                 |       |               |               | 45.5<br>[frame/s]  | 24.9<br>[frame/s] | 13.0<br>[frame/s] | 6.5<br>[frame/s]  | 38.8<br>[frame/s] | 20.4<br>[frame/s] | 10.4<br>[frame/s] | 5.2<br>[frame/s]  | 31.6<br>[frame/s] | 17.3<br>[frame/s] | 8.7<br>[frame/s]  | 4.4<br>[frame/s]  |                                       |           |
|                 |       |               |               | 4752 Mbps/lane     |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 0h                             |           |
| 2376 Mbps/lane  |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   | FREQ = 1h         |                   |                   |                   |                                       |           |
| 1188 Mbps /lane |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |           |
| Chip ID = 02h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h             | [7:0] | VMAX          | 0CD4h         | 179Ch              | 1794h             | 1744h             | 1718h             | 1794h             | 1778h             | 1734h             | 170Ch             | 175Ch             | 175Ch             | 1728h             | 1708h             | FREQ = 0h                             |           |
| D5h             | [7:0] |               |               | 1788h              | 173Ch             | 1718h             | 1700h             | 1770h             | 1730h             | 170Ch             | 16FCh             | 1758h             | 1728h             | 1708h             | 16FCh             | FREQ = 1h                             |           |
| D6h             | [7:0] |               |               | 1738h              | 1718h             | 1700h             | 16F4h             | 172Ch             | 170Ch             | 16FCh             | 16F4h             | 1720h             | 1708h             | 16FCh             | 16F4h             | FREQ = 2h                             |           |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                | 82h               | F5h               | 1E1h              | 81h               | 9Dh               | 12Eh              | 256h              | BEh               | BEh               | 16Ah              | 2CEh              | FREQ = 0h                             |           |
| D9h             | [7:0] |               |               | 8Ch                | FDh               | 1E5h              | 3C1h              | A3h               | 134h              | 25Bh              | 4ABh              | C8h               | 16Ch              | 2D3h              | 59Ch              | FREQ = 1h                             |           |
|                 |       |               |               | 112h               | 1F7h              | 3CAh              | 782h              | 142h              | 267h              | 4B5h              | 955h              | 18Ch              | 2D6h              | 5A5h              | B37h              | FREQ = 2h                             |           |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                | 10h               | 08h               | 04h               | 10h               | 10h               | 08h               | 04h               | 0Ch               | 0Ch               | 08h               | 04h               | FREQ = 0h                             |           |
|                 |       |               |               | 10h                | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 1h                             |           |
|                 |       |               |               | 08h                | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 2h                             |           |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                | 70h               | 3Ch               | 20h               | 70h               | 5Ch               | 30h               | 18h               | 4Ch               | 4Ch               | 28h               | 14h               | FREQ = 0h                             |           |
|                 |       |               |               | 68h                | 38h               | 20h               | 10h               | 58h               | 30h               | 18h               | 0Ch               | 48h               | 28h               | 14h               | 0Ch               | FREQ = 1h                             |           |
|                 |       |               |               | 34h                | 20h               | 10h               | 08h               | 2Ch               | 18h               | 0Ch               | 08h               | 24h               | 14h               | 0Ch               | 08h               | FREQ = 2h                             |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                | 30h               | 1Ch               | 10h               | 30h               | 28h               | 18h               | 0Ch               | 20h               | 20h               | 14h               | 0Ch               | FREQ = 0h                             |           |
|                 |       |               |               | 2Ch                | 18h               | 10h               | 08h               | 28h               | 14h               | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h               | FREQ = 1h                             |           |
|                 |       |               |               | 18h                | 10h               | 08h               | 04h               | 14h               | 0Ch               | 08h               | 04h               | 10h               | 0Ch               | 08h               | 04h               | FREQ = 2h                             |           |
| Chip ID = 04h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                 |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 3Dh             | [0]   |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                 |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h             | [7:0] |               | 2Dh           | 1Bh                |                   |                   |                   | 1Bh               |                   |                   |                   | 13h               |                   |                   |                   |                                       |           |
| Chip ID = 15h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh             | [7:0] |               | 02h           | 16h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Fh             | [7:0] |               | 02h           | 16h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |           |

(IMX536-AAQJ)



(SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 7.1.6.2. ROI, Overlap ROI

## SLVS Output (IMX536-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 081h          | 19Bh                                                                                                          | 312h      | 1FEh        | 3D3h      | 261h        | 493h      | FREQ = 0h                             |           |
|               |       |               |               | 31Eh                                                                                                          | 601h      | 3E2h        | 77Eh      | 4A6h        | 8FAh      | FREQ = 1h                             |           |
| D9h           | [7:0] |               |               | 119h                                                                                                          | 214h      | 15Bh        | 294h      | 19Bh        | 312h      | FREQ = 2h                             |           |
|               |       |               |               | 21Bh                                                                                                          | 408h      | 29Eh        | 506h      | 31Eh        | 601h      | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 08h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 24h                                                                                                           | 14h       | 1Ch         | 10h       | 18h         | 10h       | FREQ = 0h                             |           |
|               |       |               |               | 14h                                                                                                           | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 1h                             |           |
|               |       |               |               | 34h                                                                                                           | 1Ch       | 2Ch         | 18h       | 24h         | 14h       | FREQ = 2h                             |           |
|               |       |               |               | 1Ch                                                                                                           | 10h       | 18h         | 0Ch       | 14h         | 0Ch       | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 0h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 18h                                                                                                           | 0Ch       | 14h         | 0Ch       | 10h         | 08h       | FREQ = 2h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 0Ch         | 08h       | 08h         | 04h       | FREQ = 3h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                           |           | 1Bh         |           | 13h         |           |                                       |           |
| Chip ID = 15h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

## SLVS – EC output (IMX536-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                 |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |           |
|               |       |               |               | 8 lane                                                                                                     | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |           |
|               |       |               |               | 4752 Mbps/lane                                                                                             |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       | FREQ = 1h |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| D8h           | [7:0] | HMAX          | 081h          | 7Ah                                                                                                        | 82h             | F5h             | 1E1h            | 81h             | 9Dh             | 12Eh            | 256h            | BEh             | BEh              | 16Ah             | 2CEh             | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 8Ch                                                                                                        | FDh             | 1E5h            | 3C1h            | A3h             | 134h            | 25Bh            | 4ABh            | C8h             | 16Ch             | 2D3h             | 59Ch             | FREQ = 1h                             |           |
|               |       |               |               | 112h                                                                                                       | 1F7h            | 3CAh            | 782h            | 142h            | 267h            | 4B5h            | 955h            | 18Ch            | 2D6h             | 5A5h             | B37h             | FREQ = 2h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 10h                                                                                                        | 10h             | 08h             | 04h             | 10h             | 10h             | 08h             | 04h             | 0Ch             | 0Ch              | 08h              | 04h              | FREQ = 0h                             |           |
|               |       |               |               | 10h                                                                                                        | 08h             | 04h             | 04h             | 0Ch             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                        | 04h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 2h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 74h                                                                                                        | 70h             | 3Ch             | 20h             | 70h             | 5Ch             | 30h             | 18h             | 4Ch             | 4Ch              | 28h              | 14h              | FREQ = 0h                             |           |
|               |       |               |               | 68h                                                                                                        | 38h             | 20h             | 10h             | 58h             | 30h             | 18h             | 0Ch             | 48h             | 28h              | 14h              | 0Ch              | FREQ = 1h                             |           |
|               |       |               |               | 34h                                                                                                        | 20h             | 10h             | 08h             | 2Ch             | 18h             | 0Ch             | 08h             | 24h             | 14h              | 0Ch              | 08h              | FREQ = 2h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 34h                                                                                                        | 30h             | 1Ch             | 10h             | 30h             | 28h             | 18h             | 0Ch             | 20h             | 20h              | 14h              | 0Ch              | FREQ = 0h                             |           |
|               |       |               |               | 2Ch                                                                                                        | 18h             | 10h             | 08h             | 28h             | 14h             | 0Ch             | 08h             | 20h             | 14h              | 0Ch              | 08h              | FREQ = 1h                             |           |
|               |       |               |               | 18h                                                                                                        | 10h             | 08h             | 04h             | 14h             | 0Ch             | 08h             | 04h             | 10h             | 0Ch              | 08h              | 04h              | FREQ = 2h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                        |                 |                 |                 | 1Bh             |                 |                 |                 | 13h             |                  |                  |                  |                                       |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$

Overlap ROI :

$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$

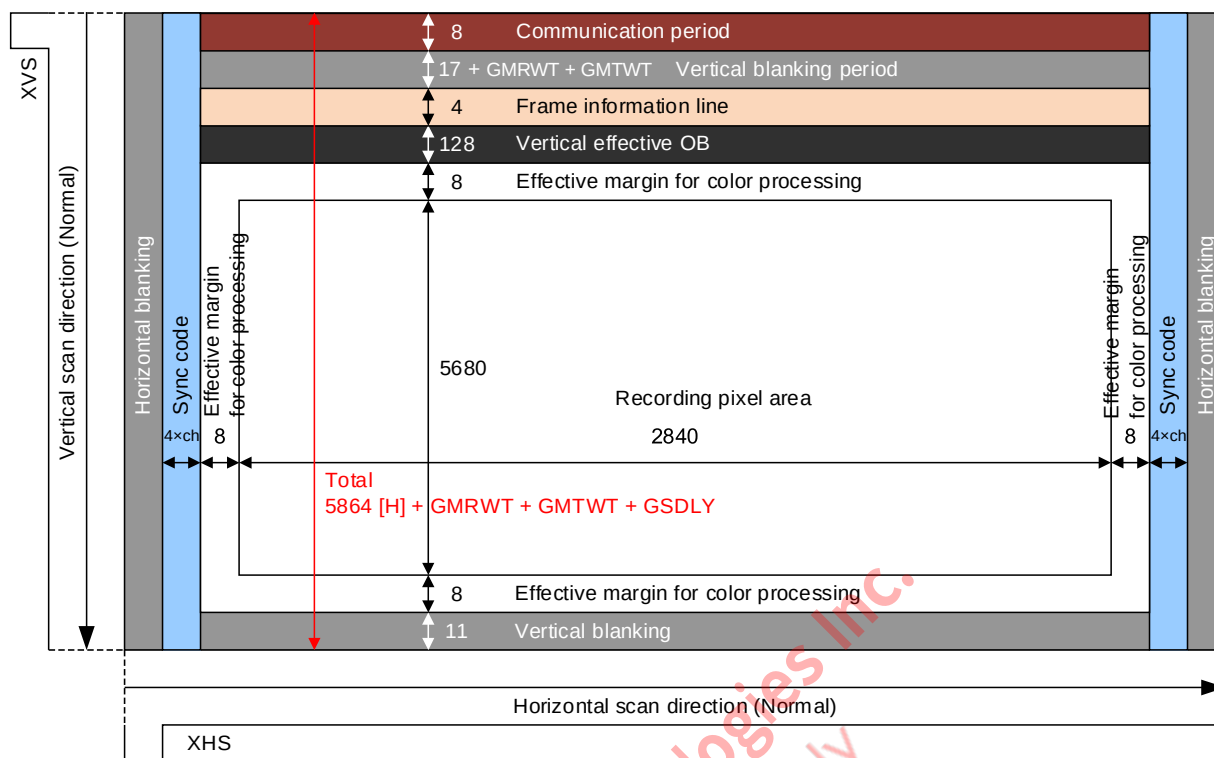
## 7.1.7. IMX536-AAMJ

## 7.1.7.1. All-pixel scan

SLVS Output (IMX536-AAMJ)

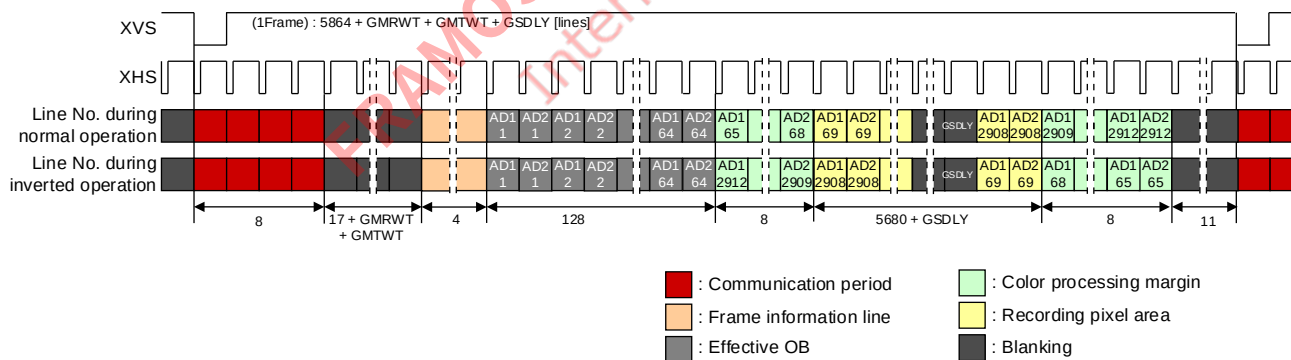
| Address       | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                   | AD = 10 bit       |                   | AD = 12 bit       |                   |                                       |           |
|               |       |               |               | 8 ch              | 4 ch              | 8 ch              | 4 ch              | 8 ch              | 4 ch              |                                       |           |
|               |       |               |               | 30.4<br>[frame/s] | 16.0<br>[frame/s] | 24.6<br>[frame/s] | 12.8<br>[frame/s] | 20.6<br>[frame/s] | 10.7<br>[frame/s] |                                       | FREQ = 0h |
|               |       |               |               | 15.7<br>[frame/s] | 8.2<br>[frame/s]  | 12.6<br>[frame/s] | 6.5<br>[frame/s]  | 10.5<br>[frame/s] | 5.4<br>[frame/s]  |                                       | FREQ = 1h |
|               |       |               |               | 44.4<br>[frame/s] | 23.6<br>[frame/s] | 36.0<br>[frame/s] | 19.0<br>[frame/s] | 30.4<br>[frame/s] | 16.0<br>[frame/s] |                                       | FREQ = 2h |
|               |       |               |               | 23.3<br>[frame/s] | 12.2<br>[frame/s] | 18.7<br>[frame/s] | 9.8<br>[frame/s]  | 15.7<br>[frame/s] | 8.2<br>[frame/s]  |                                       | FREQ = 3h |
|               |       |               |               | 594 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 2h |
| 445.5 Mbps/ch |       |               |               |                   |                   | FREQ = 3h         |                   |                   |                   |                                       |           |
| Chip ID = 02h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | 1724h             | 1708h             | 1714h             | 1704h             | 1710h             | 1704h             | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 1708h             | 16FCh             | 1704h             | 16F8h             | 1700h             | 16F8h             | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 173Ch             | 1714h             | 1730h             | 1710h             | 1724h             | 1708h             | FREQ = 2h                             |           |
|               |       |               |               | 1714h             | 1704h             | 1710h             | 1700h             | 1708h             | 16FCh             | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 081h          | 19Bh              | 312h              | 1FEh              | 3D3h              | 261h              | 493h              | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 31Eh              | 601h              | 3E2h              | 77Eh              | 4A6h              | 8FAh              | FREQ = 1h                             |           |
|               |       |               |               | 119h              | 214h              | 15Bh              | 294h              | 19Bh              | 312h              | FREQ = 2h                             |           |
|               |       |               |               | 21Bh              | 408h              | 29Eh              | 506h              | 31Eh              | 601h              | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 08h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 08h               | 04h               | FREQ = 2h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 24h               | 14h               | 1Ch               | 10h               | 18h               | 10h               | FREQ = 0h                             |           |
|               |       |               |               | 14h               | 0Ch               | 10h               | 08h               | 0Ch               | 08h               | FREQ = 1h                             |           |
|               |       |               |               | 34h               | 1Ch               | 2Ch               | 18h               | 24h               | 14h               | FREQ = 2h                             |           |
|               |       |               |               | 1Ch               | 10h               | 18h               | 0Ch               | 14h               | 0Ch               | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 10h               | 08h               | 0Ch               | 08h               | 0Ch               | 08h               | FREQ = 0h                             |           |
|               |       |               |               | 08h               | 04h               | 08h               | 04h               | 08h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 18h               | 0Ch               | 14h               | 0Ch               | 10h               | 08h               | FREQ = 2h                             |           |
|               |       |               |               | 0Ch               | 08h               | 0Ch               | 08h               | 08h               | 04h               | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                                       |           |
| 3Dh           | [0]   |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh               |                   | 1Bh               |                   | 13h               |                   |                                       |           |
| Chip ID = 15h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |

(IMX536-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

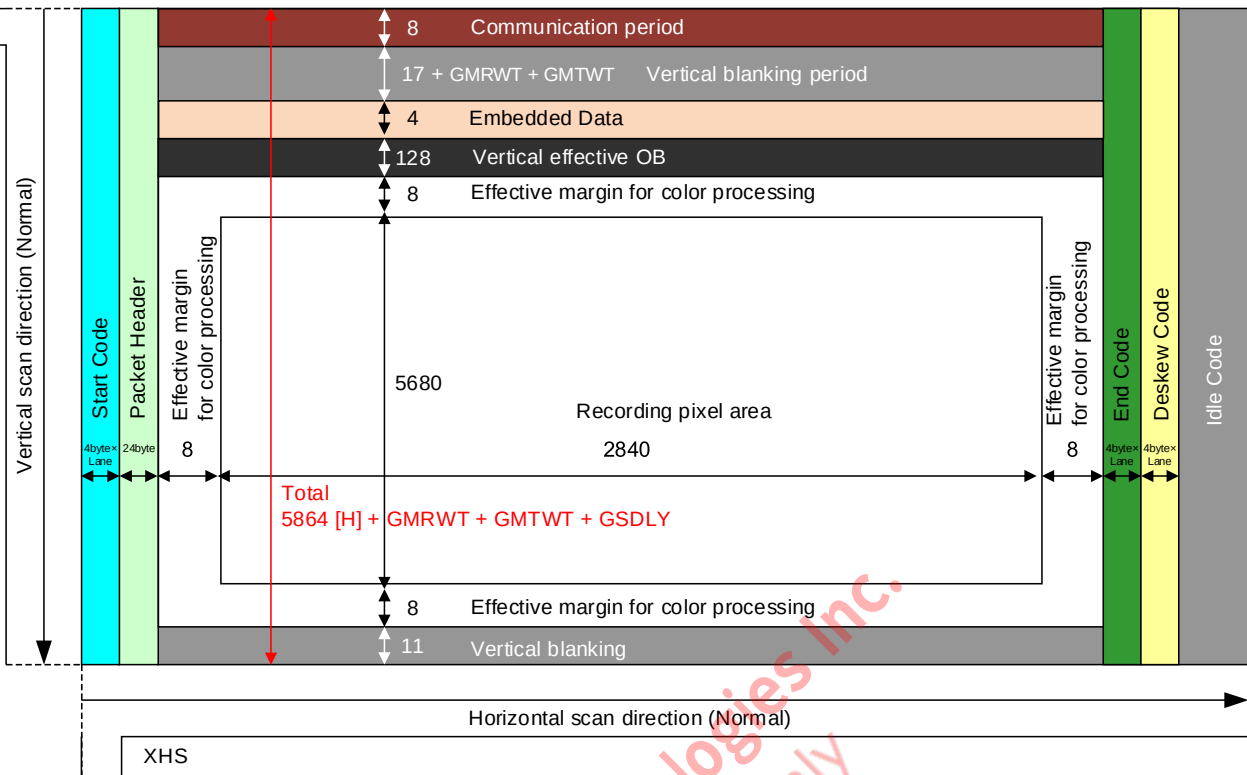


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX536-AAMJ)

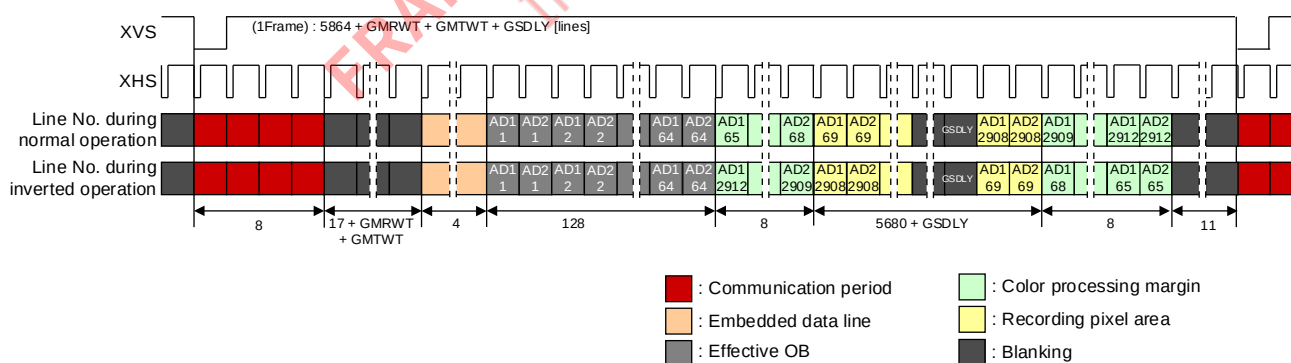
| Address         | bit   | Register name | Initial Value | Setting value      |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | Remarks                               |
|-----------------|-------|---------------|---------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|
|                 |       |               |               | AD = 8 bit         |                   |                   |                   | AD = 10 bit       |                   |                   |                   | AD = 12 bit       |                   |                   |                   |                                       |
|                 |       |               |               | 8 lane             | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            |                                       |
|                 |       |               |               | 100.6<br>[frame/s] | 94.5<br>[frame/s] | 50.8<br>[frame/s] | 26.0<br>[frame/s] | 95.2<br>[frame/s] | 78.6<br>[frame/s] | 41.3<br>[frame/s] | 21.0<br>[frame/s] | 65.3<br>[frame/s] | 65.3<br>[frame/s] | 34.5<br>[frame/s] | 17.5<br>[frame/s] | FREQ = 0h                             |
|                 |       |               |               | 87.9<br>[frame/s]  | 49.3<br>[frame/s] | 25.8<br>[frame/s] | 13.1<br>[frame/s] | 75.8<br>[frame/s] | 40.5<br>[frame/s] | 20.8<br>[frame/s] | 10.5<br>[frame/s] | 62.0<br>[frame/s] | 34.3<br>[frame/s] | 17.4<br>[frame/s] | 8.7<br>[frame/s]  | FREQ = 1h                             |
|                 |       |               |               | 45.5<br>[frame/s]  | 24.9<br>[frame/s] | 12.9<br>[frame/s] | 6.5<br>[frame/s]  | 38.8<br>[frame/s] | 20.4<br>[frame/s] | 10.4<br>[frame/s] | 5.2<br>[frame/s]  | 31.6<br>[frame/s] | 17.3<br>[frame/s] | 8.7<br>[frame/s]  | 4.3<br>[frame/s]  | FREQ = 2h                             |
|                 |       |               |               | 4752 Mbps/lane     |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 0h                             |
|                 |       |               |               | 2376 Mbps/lane     |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   | FREQ = 1h                             |
| 1188 Mbps /lane |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |
| Chip ID = 02h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| D4h             | [7:0] | VMAX          | 0CD4h         | 17A0h              | 1798h             | 1748h             | 171Ch             | 1798h             | 177Ch             | 1738h             | 1710h             | 1760h             | 1760h             | 172Ch             | 170Ch             | FREQ = 0h                             |
| D5h             | [7:0] |               |               | 178Ch              | 1740h             | 171Ch             | 1704h             | 1774h             | 1734h             | 1710h             | 1700h             | 175Ch             | 172Ch             | 170Ch             | 1700h             | FREQ = 1h                             |
| D6h             | [7:0] |               |               | 173Ch              | 171Ch             | 1704h             | 16F8h             | 1730h             | 1710h             | 1700h             | 16F8h             | 1724h             | 170Ch             | 1700h             | 16F8h             | FREQ = 2h                             |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                | 82h               | F5h               | 1E1h              | 81h               | 9Dh               | 12Eh              | 256h              | BEh               | BEh               | 16Ah              | 2CEh              | FREQ = 0h                             |
| D9h             | [7:0] |               |               | 8Ch                | FDh               | 1E5h              | 3C1h              | A3h               | 134h              | 25Bh              | 4ABh              | C8h               | 16Ch              | 2D3h              | 59Ch              | FREQ = 1h                             |
|                 |       |               |               | 112h               | 1F7h              | 3CAh              | 782h              | 142h              | 267h              | 4B5h              | 955h              | 18Ch              | 2D6h              | 5A5h              | B37h              | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                | 10h               | 08h               | 04h               | 10h               | 10h               | 08h               | 04h               | 0Ch               | 0Ch               | 08h               | 04h               | FREQ = 0h                             |
|                 |       |               |               | 10h                | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 1h                             |
|                 |       |               |               | 08h                | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                | 70h               | 3Ch               | 20h               | 70h               | 5Ch               | 30h               | 18h               | 4Ch               | 4Ch               | 28h               | 14h               | FREQ = 0h                             |
|                 |       |               |               | 68h                | 38h               | 20h               | 10h               | 58h               | 30h               | 18h               | 0Ch               | 48h               | 28h               | 14h               | 0Ch               | FREQ = 1h                             |
|                 |       |               |               | 34h                | 20h               | 10h               | 08h               | 2Ch               | 18h               | 0Ch               | 08h               | 24h               | 14h               | 0Ch               | 08h               | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                | 30h               | 1Ch               | 10h               | 30h               | 28h               | 18h               | 0Ch               | 20h               | 20h               | 14h               | 0Ch               | FREQ = 0h                             |
|                 |       |               |               | 2Ch                | 18h               | 10h               | 08h               | 28h               | 14h               | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h               | FREQ = 1h                             |
|                 |       |               |               | 18h                | 10h               | 08h               | 04h               | 14h               | 0Ch               | 08h               | 04h               | 10h               | 0Ch               | 08h               | 04h               | FREQ = 2h                             |
| Chip ID = 04h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                 |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| 3Dh             | [0]   |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| Chip ID = 06h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                 |                   |                   |                   | 0h                |                   |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| E8h             | [7:0] |               | 2Dh           | 1Bh                |                   |                   |                   | 1Bh               |                   |                   |                   | 13h               |                   |                   |                   |                                       |
| Chip ID = 15h   |       |               |               |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| 1Eh             | [7:0] |               | 02h           | 16h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |
| 1Fh             | [7:0] |               | 02h           | 16h                |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                                       |

(IMX536-AAMJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 7.1.7.2. ROI, Overlap ROI

## SLVS Output (IMX536-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 081h          | 19Bh                                                                                                          | 312h      | 1FEh        | 3D3h      | 261h        | 493h      | FREQ = 0h                             |           |
|               |       |               |               | 31Eh                                                                                                          | 601h      | 3E2h        | 77Eh      | 4A6h        | 8FAh      | FREQ = 1h                             |           |
| D9h           | [7:0] |               |               | 119h                                                                                                          | 214h      | 15Bh        | 294h      | 19Bh        | 312h      | FREQ = 2h                             |           |
|               |       |               |               | 21Bh                                                                                                          | 408h      | 29Eh        | 506h      | 31Eh        | 601h      | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 08h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 24h                                                                                                           | 14h       | 1Ch         | 10h       | 18h         | 10h       | FREQ = 0h                             |           |
|               |       |               |               | 14h                                                                                                           | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 1h                             |           |
|               |       |               |               | 34h                                                                                                           | 1Ch       | 2Ch         | 18h       | 24h         | 14h       | FREQ = 2h                             |           |
|               |       |               |               | 1Ch                                                                                                           | 10h       | 18h         | 0Ch       | 14h         | 0Ch       | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 0h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 08h         | 04h       | 08h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 18h                                                                                                           | 0Ch       | 14h         | 0Ch       | 10h         | 08h       | FREQ = 2h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 0Ch         | 08h       | 08h         | 04h       | FREQ = 3h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                           |           | 1Bh         |           | 13h         |           |                                       |           |
| Chip ID = 15h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 168$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 168$$

## SLVS – EC output (IMX536-AAMJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                              |           |           |           |             |           |           |           |             |           |           |           | Remarks                               |
|-----------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-------------|-----------|-----------|-----------|-------------|-----------|-----------|-----------|---------------------------------------|
|                 |       |               |               | AD = 8 bit                                                                                                 |           |           |           | AD = 10 bit |           |           |           | AD = 12 bit |           |           |           |                                       |
|                 |       |               |               | 8 lane                                                                                                     | 4 lane    | 2 lane    | 1 lane    | 8 lane      | 4 lane    | 2 lane    | 1 lane    | 8 lane      | 4 lane    | 2 lane    | 1 lane    |                                       |
|                 |       |               |               | *1                                                                                                         | *2        | *3        | *4        | *5          | *6        | *7        | *8        | *9          | *10       | *11       | *12       |                                       |
|                 |       |               |               | [frame/s]                                                                                                  | [frame/s] | [frame/s] | [frame/s] | [frame/s]   | [frame/s] | [frame/s] | [frame/s] | [frame/s]   | [frame/s] | [frame/s] | [frame/s] |                                       |
|                 |       |               |               | 4752 Mbps/lane                                                                                             |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 2376 Mbps/lane  |       |               |               |                                                                                                            |           |           |           |             |           |           |           | FREQ = 1h   |           |           |           |                                       |
| 1188 Mbps /lane |       |               |               |                                                                                                            |           |           |           |             |           |           |           | FREQ = 2h   |           |           |           |                                       |
| Chip ID = 02h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2        | *3        | *4        | *5          | *6        | *7        | *8        | *9          | *10       | *11       | *12       |                                       |
| D5h             | [7:0] |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| D6h             | [7:0] |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                                                                                                        | 82h       | F5h       | 1E1h      | 81h         | 9Dh       | 12Eh      | 256h      | BEh         | BEh       | 16Ah      | 2CEh      | FREQ = 0h                             |
| D9h             | [7:0] |               |               | 8Ch                                                                                                        | FDh       | 1E5h      | 3C1h      | A3h         | 134h      | 25Bh      | 4ABh      | C8h         | 16Ch      | 2D3h      | 59Ch      | FREQ = 1h                             |
|                 |       |               |               | 112h                                                                                                       | 1F7h      | 3CAh      | 782h      | 142h        | 267h      | 4B5h      | 955h      | 18Ch        | 2D6h      | 5A5h      | B37h      | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                                                                                                        | 10h       | 08h       | 04h       | 10h         | 10h       | 08h       | 04h       | 0Ch         | 0Ch       | 08h       | 04h       | FREQ = 0h                             |
|                 |       |               |               | 10h                                                                                                        | 08h       | 04h       | 04h       | 0Ch         | 08h       | 04h       | 04h       | 0Ch         | 08h       | 04h       | 04h       | FREQ = 1h                             |
|                 |       |               |               | 08h                                                                                                        | 04h       | 04h       | 04h       | 08h         | 04h       | 04h       | 04h       | 08h         | 04h       | 04h       | 04h       | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                                                                                                        | 70h       | 3Ch       | 20h       | 70h         | 5Ch       | 30h       | 18h       | 4Ch         | 4Ch       | 28h       | 14h       | FREQ = 0h                             |
|                 |       |               |               | 68h                                                                                                        | 38h       | 20h       | 10h       | 58h         | 30h       | 18h       | 0Ch       | 48h         | 28h       | 14h       | 0Ch       | FREQ = 1h                             |
|                 |       |               |               | 34h                                                                                                        | 20h       | 10h       | 08h       | 2Ch         | 18h       | 0Ch       | 08h       | 24h         | 14h       | 0Ch       | 08h       | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                                                                                                        | 30h       | 1Ch       | 10h       | 30h         | 28h       | 18h       | 0Ch       | 20h         | 20h       | 14h       | 0Ch       | FREQ = 0h                             |
|                 |       |               |               | 2Ch                                                                                                        | 18h       | 10h       | 08h       | 28h         | 14h       | 0Ch       | 08h       | 20h         | 14h       | 0Ch       | 08h       | FREQ = 1h                             |
|                 |       |               |               | 18h                                                                                                        | 10h       | 08h       | 04h       | 14h         | 0Ch       | 08h       | 04h       | 10h         | 0Ch       | 08h       | 04h       | FREQ = 2h                             |
| Chip ID = 03h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |           |           |           |             |           |           |           |             |           |           |           |                                       |
| Chip ID = 04h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |           |           |           |             |           |           |           |             |           |           |           |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |           |           |           | 0h          |           |           |           | 1h          |           |           |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 3Dh             | [0]   |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| Chip ID = 06h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |           |           |           | 0h          |           |           |           | 1h          |           |           |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| E8h             | [7:0] |               | 2Dh           | 1Bh                                                                                                        |           |           |           | 1Bh         |           |           |           | 13h         |           |           |           |                                       |
| Chip ID = 15h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 168$

Overlap ROI :

$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 168$

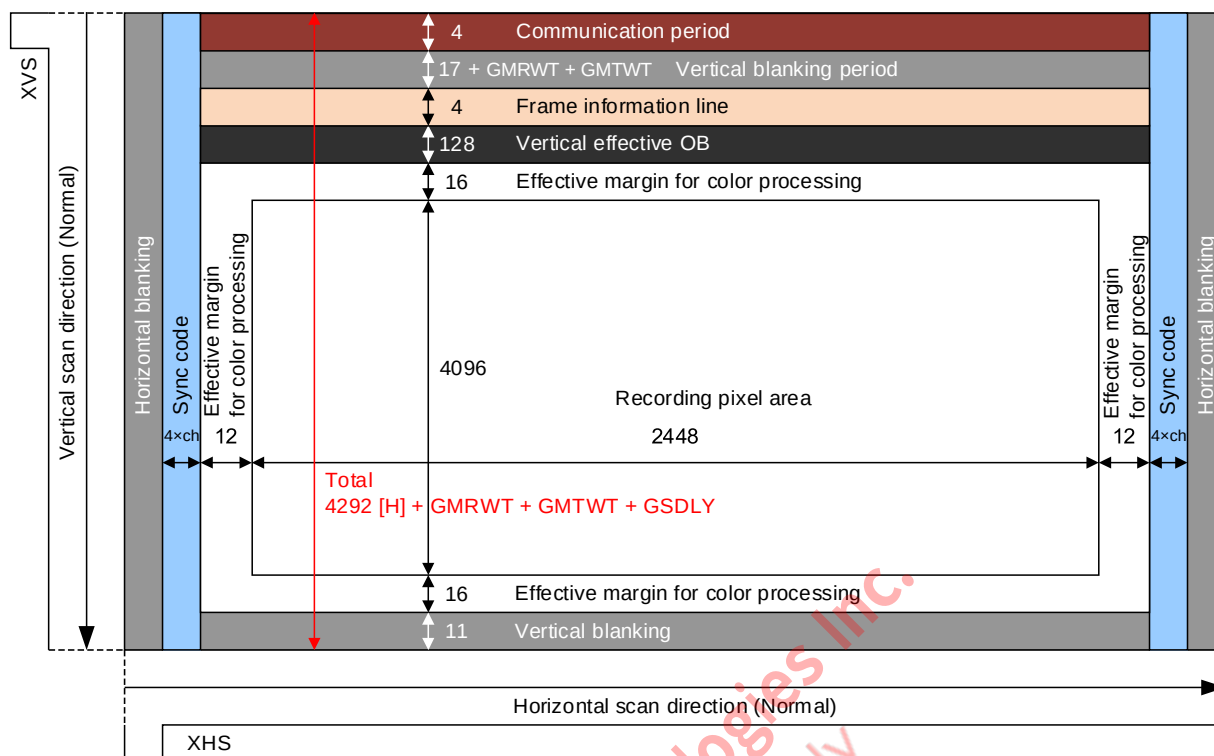
## 7.1.8. IMX537-AAQJ

## 7.1.8.1. All-pixel scan

SLVS Output (IMX537-AAQJ)

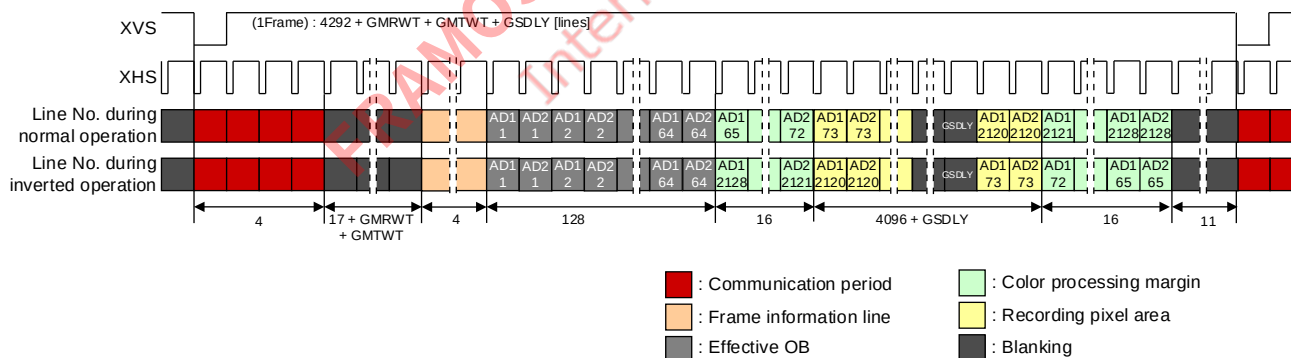
| Address       | bit       | Register name | Initial Value | Setting value |           |             |           |             |           | Remarks                               |
|---------------|-----------|---------------|---------------|---------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|
|               |           |               |               | AD = 8 bit    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |
|               |           |               |               | 8 ch          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |
|               |           |               |               | 46.9          | 24.8      | 37.9        | 19.9      | 31.8        | 16.7      |                                       |
|               |           |               |               | [frame/s]     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |
|               |           |               |               | 24.4          | 12.7      | 19.6        | 10.2      | 16.4        | 8.5       |                                       |
|               |           |               |               | [frame/s]     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |
|               |           |               |               | 67.9          | 36.4      | 55.3        | 29.5      | 46.9        | 24.8      |                                       |
|               |           |               |               | [frame/s]     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |
|               |           |               |               | 35.9          | 19.0      | 29.0        | 15.2      | 24.4        | 12.7      |                                       |
| [frame/s]     | [frame/s] | [frame/s]     | [frame/s]     | [frame/s]     | [frame/s] |             |           |             |           |                                       |
| 594 Mbps/ch   |           |               |               |               |           |             |           |             | FREQ = 0h |                                       |
| 297 Mbps/ch   |           |               |               |               |           |             |           |             | FREQ = 1h |                                       |
| 891 Mbps/ch   |           |               |               |               |           |             |           |             | FREQ = 2h |                                       |
| 445.5 Mbps/ch |           |               |               |               |           |             |           |             | FREQ = 3h |                                       |
| Chip ID = 02h |           |               |               |               |           |             |           |             |           |                                       |
| D4h           | [7:0]     | VMAX          | 0CD4h         | 1108h         | 10ECh     | 10FCh       | 10E4h     | 10F0h       | 10E0h     | FREQ = 0h                             |
| D5h           | [7:0]     |               |               | 10ECh         | 10DCh     | 10E4h       | 10D8h     | 10E0h       | 10D4h     | FREQ = 1h                             |
| D6h           | [7:0]     |               |               | 1124h         | 10FCh     | 1110h       | 10F0h     | 1108h       | 10ECh     | FREQ = 2h                             |
|               |           |               |               | 10FCh         | 10E0h     | 10ECh       | 10E0h     | 10ECh       | 10DCh     | FREQ = 3h                             |
| D8h           | [7:0]     | HMAX          | 081h          | 16Bh          | 2B2h      | 1C2h        | 35Bh      | 219h        | 403h      | FREQ = 0h                             |
| D9h           | [7:0]     |               |               | 2BEh          | 541h      | 36Ah        | 68Eh      | 416h        | 7DAh      | FREQ = 1h                             |
|               |           |               |               | F9h           | 1D4h      | 133h        | 244h      | 16Bh        | 2B2h      | FREQ = 2h                             |
| E2h           | [7:0]     | GMRWT         | 10h           | 1DBh          | 388h      | 24Eh        | 466h      | 2BEh        | 541h      | FREQ = 3h                             |
|               |           |               |               | 08h           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 0h                             |
|               |           |               |               | 04h           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |
|               |           |               |               | 08h           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |
| E3h           | [7:0]     | GMTWT         | 70h           | 08h           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |
|               |           |               |               | 28h           | 18h       | 20h         | 14h       | 1Ch         | 10h       | FREQ = 0h                             |
|               |           |               |               | 18h           | 0Ch       | 14h         | 0Ch       | 10h         | 08h       | FREQ = 1h                             |
|               |           |               |               | 3Ch           | 20h       | 30h         | 1Ch       | 28h         | 18h       | FREQ = 2h                             |
| E5h           | [7:0]     | GAINDLY       | 04h           | 20h           | 10h       | 18h         | 10h       | 18h         | 0Ch       | FREQ = 3h                             |
| E6h           | [7:0]     | GSDLY         | 30h           | 04h           |           |             |           |             |           |                                       |
|               |           |               |               | 14h           | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 0h                             |
|               |           |               |               | 0Ch           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 1h                             |
|               |           |               |               | 1Ch           | 10h       | 14h         | 0Ch       | 14h         | 0Ch       | FREQ = 2h                             |
|               |           |               |               | 10h           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 3h                             |
| Chip ID = 04h |           |               |               |               |           |             |           |             |           |                                       |
| 00h           | [1:0]     | AD_SEL        | 0h            | 2h            |           |             |           |             |           |                                       |
|               | [5:4]     | ADBIT         | 0h            | 2h            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch           | [7:1]     | LLBLANK       | 18h           | 18h           |           |             |           |             |           |                                       |
| 3Dh           | [0]       |               |               |               |           |             |           |             |           |                                       |
| Chip ID = 06h |           |               |               |               |           |             |           |             |           |                                       |
| 30h           | [1:0]     | ODBIT         | 0h            | 2h            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h |           |               |               |               |           |             |           |             |           |                                       |
| E8h           | [7:0]     |               | 2Dh           | 1Bh           |           | 1Bh         |           | 13h         |           |                                       |
| Chip ID = 15h |           |               |               |               |           |             |           |             |           |                                       |
| 1Eh           | [7:0]     |               | 02h           | 16h           |           |             |           |             |           |                                       |
| 1Fh           | [7:0]     |               | 02h           | 16h           |           |             |           |             |           |                                       |

(IMX537-AAQJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

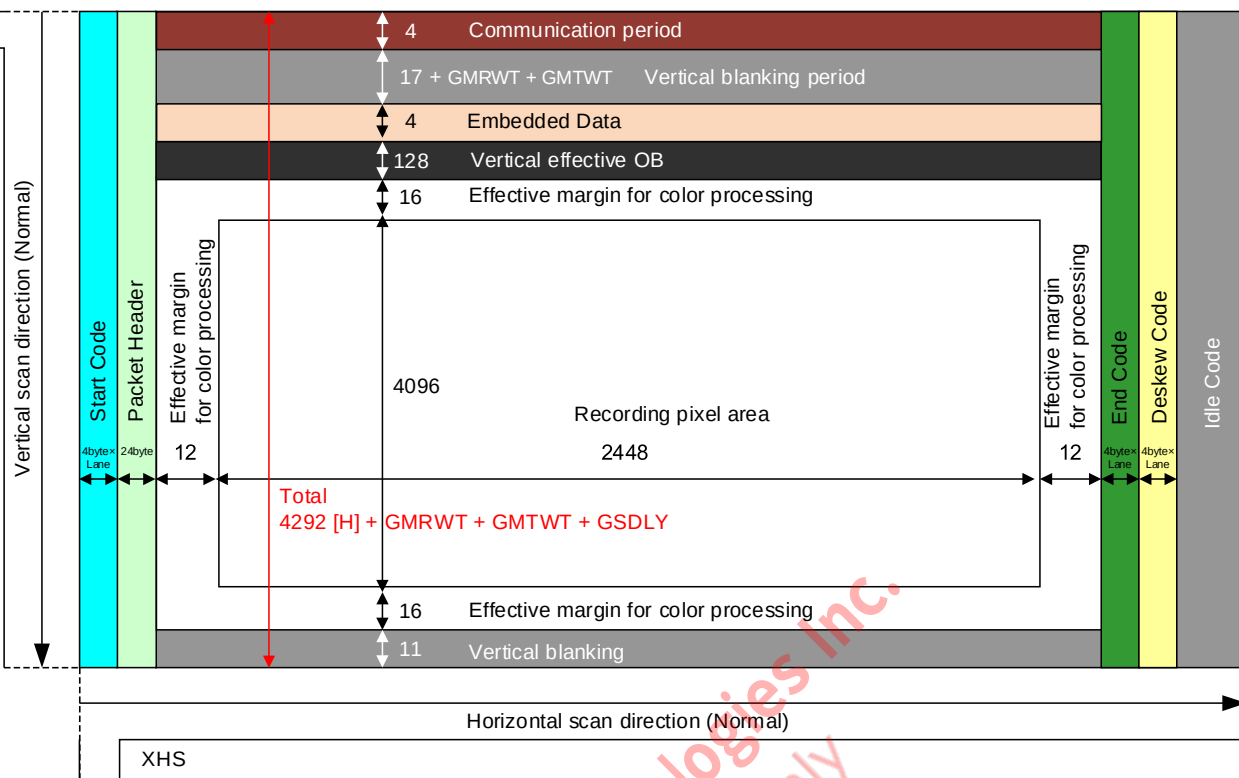


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX537-AAQJ)

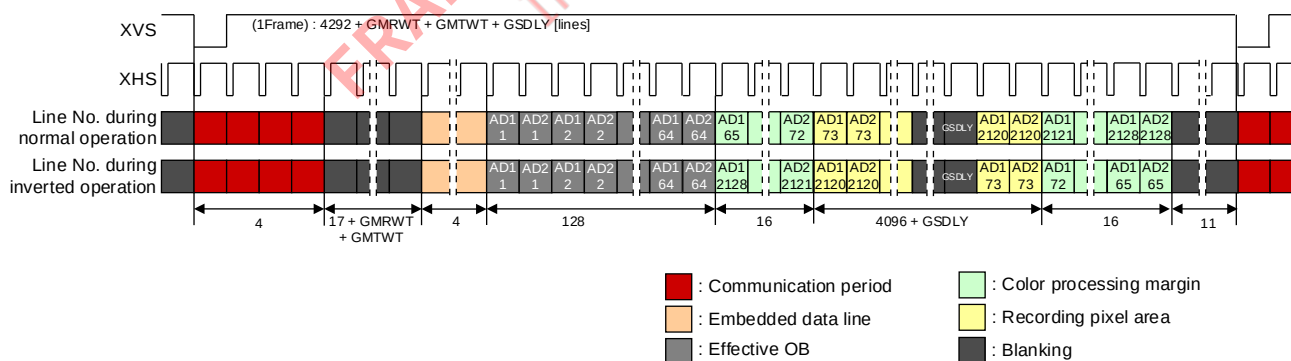
| Address         | bit   | Register name | Initial Value | Setting value      |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   | Remarks                               |
|-----------------|-------|---------------|---------------|--------------------|--------------------|-------------------|-------------------|--------------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|
|                 |       |               |               | AD = 8 bit         |                    |                   |                   | AD = 10 bit        |                    |                   |                   | AD = 12 bit       |                   |                   |                   |                                       |
|                 |       |               |               | 8 lane             | 4 lane             | 2 lane            | 1 lane            | 8 lane             | 4 lane             | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            |                                       |
|                 |       |               |               | 135.9<br>[frame/s] | 135.9<br>[frame/s] | 78.7<br>[frame/s] | 40.7<br>[frame/s] | 128.8<br>[frame/s] | 122.4<br>[frame/s] | 64.7<br>[frame/s] | 33.0<br>[frame/s] | 88.5<br>[frame/s] | 88.5<br>[frame/s] | 53.7<br>[frame/s] | 27.3<br>[frame/s] |                                       |
|                 |       |               |               | 133.7<br>[frame/s] | 76.7<br>[frame/s]  | 40.3<br>[frame/s] | 20.4<br>[frame/s] | 115.8<br>[frame/s] | 63.2<br>[frame/s]  | 32.7<br>[frame/s] | 16.6<br>[frame/s] | 88.5<br>[frame/s] | 53.4<br>[frame/s] | 27.1<br>[frame/s] | 13.7<br>[frame/s] |                                       |
|                 |       |               |               | 69.9<br>[frame/s]  | 38.8<br>[frame/s]  | 20.2<br>[frame/s] | 10.2<br>[frame/s] | 59.9<br>[frame/s]  | 31.9<br>[frame/s]  | 16.4<br>[frame/s] | 8.3<br>[frame/s]  | 48.8<br>[frame/s] | 27.0<br>[frame/s] | 13.6<br>[frame/s] | 6.8<br>[frame/s]  |                                       |
|                 |       |               |               | 4752 Mbps/lane     |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| 2376 Mbps/lane  |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   | FREQ = 1h         |                   |                   |                   |                                       |
| 1188 Mbps /lane |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |
| Chip ID = 02h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| D4h             | [7:0] | VMAX          | 0CD4h         | 117Ch              | 117Ch              | 1134h             | 1100h             | 1174h              | 116Ch              | 111Ch             | 10F0h             | 113Ch             | 113Ch             | 1110h             | 10EC<br>h         | FREQ = 0h                             |
| D5h             | [7:0] |               |               | 117Ch              | 1130h              | 1100h             | 10E4h             | 1164h              | 111Ch              | 10F0h             | 10E0h             | 113Ch             | 1110h             | 10EC<br>h         | 10DC<br>h         | FREQ = 1h                             |
| D6h             | [7:0] |               |               | 1124h              | 1100h              | 10E4h             | 10D8h             | 1118h              | 10F0h              | 10E0h             | 10D4h             | 110Ch             | 10EC<br>h         | 10DC<br>h         | 10D4h             | FREQ = 2h                             |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                | 7Ah                | D6h               | 1A3h              | 81h                | 88h                | 106h              | 206h              | BEh               | BEh               | 13Ch              | 272h              | FREQ = 0h                             |
| D9h             | [7:0] |               |               | 7Ch                | DCh                | 1A7h              | 346h              | 90h                | 10Ch               | 20Bh              | 40Bh              | BEh               | 13Eh              | 277h              | 4E4h              | FREQ = 1h                             |
|                 |       |               |               | F2h                | 1B7h               | 34Eh              | 68Ch              | 11Bh               | 218h               | 416h              | 816h              | 15Ch              | 27Ah              | 4EDh              | 9C7h              | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                | 10h                | 0Ch               | 08h               | 10h                | 10h                | 08h               | 04h               | 0Ch               | 0Ch               | 08h               | 04h               | FREQ = 0h                             |
|                 |       |               |               | 10h                | 0Ch                | 08h               | 04h               | 10h                | 08h                | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 1h                             |
|                 |       |               |               | 08h                | 08h                | 04h               | 04h               | 08h                | 04h                | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                | 74h                | 44h               | 24h               | 70h                | 68h                | 38h               | 1Ch               | 4Ch               | 4Ch               | 30h               | 18h               | FREQ = 0h                             |
|                 |       |               |               | 74h                | 44h                | 24h               | 14h               | 64h                | 38h                | 1Ch               | 10h               | 4Ch               | 30h               | 18h               | 0Ch               | FREQ = 1h                             |
|                 |       |               |               | 3Ch                | 24h                | 14h               | 0Ch               | 34h                | 1Ch                | 10h               | 08h               | 2Ch               | 18h               | 0Ch               | 08h               | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                | 34h                | 20h               | 10h               | 30h                | 30h                | 18h               | 0Ch               | 20h               | 20h               | 14h               | 0Ch               | FREQ = 0h                             |
|                 |       |               |               | 34h                | 1Ch                | 10h               | 08h               | 2Ch                | 18h                | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h               | FREQ = 1h                             |
|                 |       |               |               | 1Ch                | 10h                | 08h               | 04h               | 18h                | 0Ch                | 08h               | 04h               | 14h               | 0Ch               | 08h               | 04h               | FREQ = 2h                             |
| Chip ID = 04h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                 |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                 |                    |                   |                   | 0h                 |                    |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| 3Dh             | [0]   |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| Chip ID = 06h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                 |                    |                   |                   | 0h                 |                    |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| E8h             | [7:0] |               | 2Dh           | 1Bh                |                    |                   |                   | 1Bh                |                    |                   |                   | 13h               |                   |                   |                   |                                       |
| Chip ID = 15h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| 1Eh             | [7:0] |               | 02h           | 16h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |
| 1Fh             | [7:0] |               | 02h           | 16h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |

(IMX537-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

### 7.1.8.2. ROI, Overlap ROI

#### SLVS Output (IMX537-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |           |             |           |             |           | Remarks                               |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|
|               |       |               |               | AD = 8 bit                                                                                                 |           | AD = 10 bit |           | AD = 12 bit |           |                                       |
|               |       |               |               | 8 ch                                                                                                       | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |
|               |       |               |               | *1                                                                                                         | *2        | *3          | *4        | *5          | *6        |                                       |
|               |       |               |               | [frame/s]                                                                                                  | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |
|               |       |               |               | 594 Mbps/ch                                                                                                |           |             |           |             |           |                                       |
| 297 Mbps/ch   |       |               |               |                                                                                                            |           | FREQ = 1h   |           |             |           |                                       |
| 891 Mbps/ch   |       |               |               |                                                                                                            |           | FREQ = 2h   |           |             |           |                                       |
| 445.5 Mbps/ch |       |               |               |                                                                                                            |           | FREQ = 3h   |           |             |           |                                       |
| Chip ID = 02h |       |               |               |                                                                                                            |           |             |           |             |           |                                       |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2        | *3          | *4        | *5          | *6        |                                       |
| D5h           | [7:0] |               |               |                                                                                                            |           |             |           |             |           |                                       |
| D6h           | [7:0] |               |               |                                                                                                            |           |             |           |             |           |                                       |
| D8h           | [7:0] | HMAX          | 081h          | 16Bh                                                                                                       | 2B2h      | 1C2h        | 35Bh      | 219h        | 403h      | FREQ = 0h                             |
| D9h           | [7:0] |               |               | 2BEh                                                                                                       | 541h      | 36Ah        | 68Eh      | 416h        | 7DAh      | FREQ = 1h                             |
|               |       |               |               | F9h                                                                                                        | 1D4h      | 133h        | 244h      | 16Bh        | 2B2h      | FREQ = 2h                             |
| E2h           | [7:0] | GMRWT         | 10h           | 1DBh                                                                                                       | 388h      | 24Eh        | 466h      | 2BEh        | 541h      | FREQ = 3h                             |
|               |       |               |               | 08h                                                                                                        | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 0h                             |
|               |       |               |               | 04h                                                                                                        | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |
|               |       |               |               | 08h                                                                                                        | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |
| E3h           | [7:0] | GMTWT         | 70h           | 08h                                                                                                        | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |
|               |       |               |               | 28h                                                                                                        | 18h       | 20h         | 14h       | 1Ch         | 10h       | FREQ = 0h                             |
|               |       |               |               | 18h                                                                                                        | 0Ch       | 14h         | 0Ch       | 10h         | 08h       | FREQ = 1h                             |
|               |       |               |               | 3Ch                                                                                                        | 20h       | 30h         | 1Ch       | 28h         | 18h       | FREQ = 2h                             |
| E5h           | [7:0] | GAINDLY       | 04h           | 20h                                                                                                        | 10h       | 18h         | 10h       | 18h         | 0Ch       | FREQ = 3h                             |
|               |       |               |               | 04h                                                                                                        |           |             |           |             |           |                                       |
| E6h           | [7:0] | GSDLY         | 30h           | 14h                                                                                                        | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 0h                             |
|               |       |               |               | 0Ch                                                                                                        | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 1h                             |
|               |       |               |               | 1Ch                                                                                                        | 10h       | 14h         | 0Ch       | 14h         | 0Ch       | FREQ = 2h                             |
|               |       |               |               | 10h                                                                                                        | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 3h                             |
| Chip ID = 03h |       |               |               |                                                                                                            |           |             |           |             |           |                                       |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |           |             |           |             |           |                                       |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |           |             |           |             |           |                                       |
| Chip ID = 04h |       |               |               |                                                                                                            |           |             |           |             |           |                                       |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |           |             |           |             |           |                                       |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |           |             |           |             |           |                                       |
| 3Dh           | [0]   |               |               |                                                                                                            |           |             |           |             |           |                                       |
| Chip ID = 06h |       |               |               |                                                                                                            |           |             |           |             |           |                                       |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h |       |               |               |                                                                                                            |           |             |           |             |           |                                       |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                        |           | 1Bh         |           | 13h         |           |                                       |
| Chip ID = 15h |       |               |               |                                                                                                            |           |             |           |             |           |                                       |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |           |             |           |             |           |                                       |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |           |             |           |             |           |                                       |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$$

## SLVS – EC output (IMX537-AAQJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                              |           |           |           |             |           |           |           |             |           |           |           | Remarks                               |
|-----------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-------------|-----------|-----------|-----------|-------------|-----------|-----------|-----------|---------------------------------------|
|                 |       |               |               | AD = 8 bit                                                                                                 |           |           |           | AD = 10 bit |           |           |           | AD = 12 bit |           |           |           |                                       |
|                 |       |               |               | 8 lane                                                                                                     | 4 lane    | 2 lane    | 1 lane    | 8 lane      | 4 lane    | 2 lane    | 1 lane    | 8 lane      | 4 lane    | 2 lane    | 1 lane    |                                       |
|                 |       |               |               | *1                                                                                                         | *2        | *3        | *4        | *5          | *6        | *7        | *8        | *9          | *10       | *11       | *12       |                                       |
|                 |       |               |               | [frame/s]                                                                                                  | [frame/s] | [frame/s] | [frame/s] | [frame/s]   | [frame/s] | [frame/s] | [frame/s] | [frame/s]   | [frame/s] | [frame/s] | [frame/s] |                                       |
|                 |       |               |               | 4752 Mbps/lane                                                                                             |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 2376 Mbps/lane  |       |               |               |                                                                                                            |           |           |           |             |           |           |           | FREQ = 1h   |           |           |           |                                       |
| 1188 Mbps /lane |       |               |               |                                                                                                            |           |           |           |             |           |           |           | FREQ = 2h   |           |           |           |                                       |
| Chip ID = 02h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2        | *3        | *4        | *5          | *6        | *7        | *8        | *9          | *10       | *11       | *12       |                                       |
| D5h             | [7:0] |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| D6h             | [7:0] |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                                                                                                        | 7Ah       | D6h       | 1A3h      | 81h         | 88h       | 106h      | 206h      | BEh         | BEh       | 13Ch      | 272h      | FREQ = 0h                             |
| D9h             | [7:0] |               |               | 7Ch                                                                                                        | DCh       | 1A7h      | 346h      | 90h         | 10Ch      | 20Bh      | 40Bh      | BEh         | 13Eh      | 277h      | 4E4h      | FREQ = 1h                             |
|                 |       |               |               | F2h                                                                                                        | 1B7h      | 34Eh      | 68Ch      | 11Bh        | 218h      | 416h      | 816h      | 15Ch        | 27Ah      | 4EDh      | 9C7h      | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                                                                                                        | 10h       | 0Ch       | 08h       | 10h         | 10h       | 08h       | 04h       | 0Ch         | 0Ch       | 08h       | 04h       | FREQ = 0h                             |
|                 |       |               |               | 10h                                                                                                        | 0Ch       | 08h       | 04h       | 10h         | 08h       | 04h       | 04h       | 0Ch         | 08h       | 04h       | 04h       | FREQ = 1h                             |
|                 |       |               |               | 08h                                                                                                        | 08h       | 04h       | 04h       | 08h         | 04h       | 04h       | 04h       | 08h         | 04h       | 04h       | 04h       | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                                                                                                        | 74h       | 44h       | 24h       | 70h         | 68h       | 38h       | 1Ch       | 4Ch         | 4Ch       | 30h       | 18h       | FREQ = 0h                             |
|                 |       |               |               | 74h                                                                                                        | 44h       | 24h       | 14h       | 64h         | 38h       | 1Ch       | 10h       | 4Ch         | 30h       | 18h       | 0Ch       | FREQ = 1h                             |
|                 |       |               |               | 3Ch                                                                                                        | 24h       | 14h       | 0Ch       | 34h         | 1Ch       | 10h       | 08h       | 2Ch         | 18h       | 0Ch       | 08h       | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                                                                                                        | 34h       | 20h       | 10h       | 30h         | 30h       | 18h       | 0Ch       | 20h         | 20h       | 14h       | 0Ch       | FREQ = 0h                             |
|                 |       |               |               | 34h                                                                                                        | 1Ch       | 10h       | 08h       | 2Ch         | 18h       | 0Ch       | 08h       | 20h         | 14h       | 0Ch       | 08h       | FREQ = 1h                             |
|                 |       |               |               | 1Ch                                                                                                        | 10h       | 08h       | 04h       | 18h         | 0Ch       | 08h       | 04h       | 14h         | 0Ch       | 08h       | 04h       | FREQ = 2h                             |
| Chip ID = 03h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |           |           |           |             |           |           |           |             |           |           |           |                                       |
| Chip ID = 04h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                         |           |           |           |             |           |           |           |             |           |           |           |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                         |           |           |           | 0h          |           |           |           | 1h          |           |           |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 3Dh             | [0]   |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| Chip ID = 06h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                         |           |           |           | 0h          |           |           |           | 1h          |           |           |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| E8h             | [7:0] |               | 2Dh           | 1Bh                                                                                                        |           |           |           | 1Bh         |           |           |           | 13h         |           |           |           |                                       |
| Chip ID = 15h   |       |               |               |                                                                                                            |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                        |           |           |           |             |           |           |           |             |           |           |           |                                       |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 164$

Overlap ROI :

$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 164$

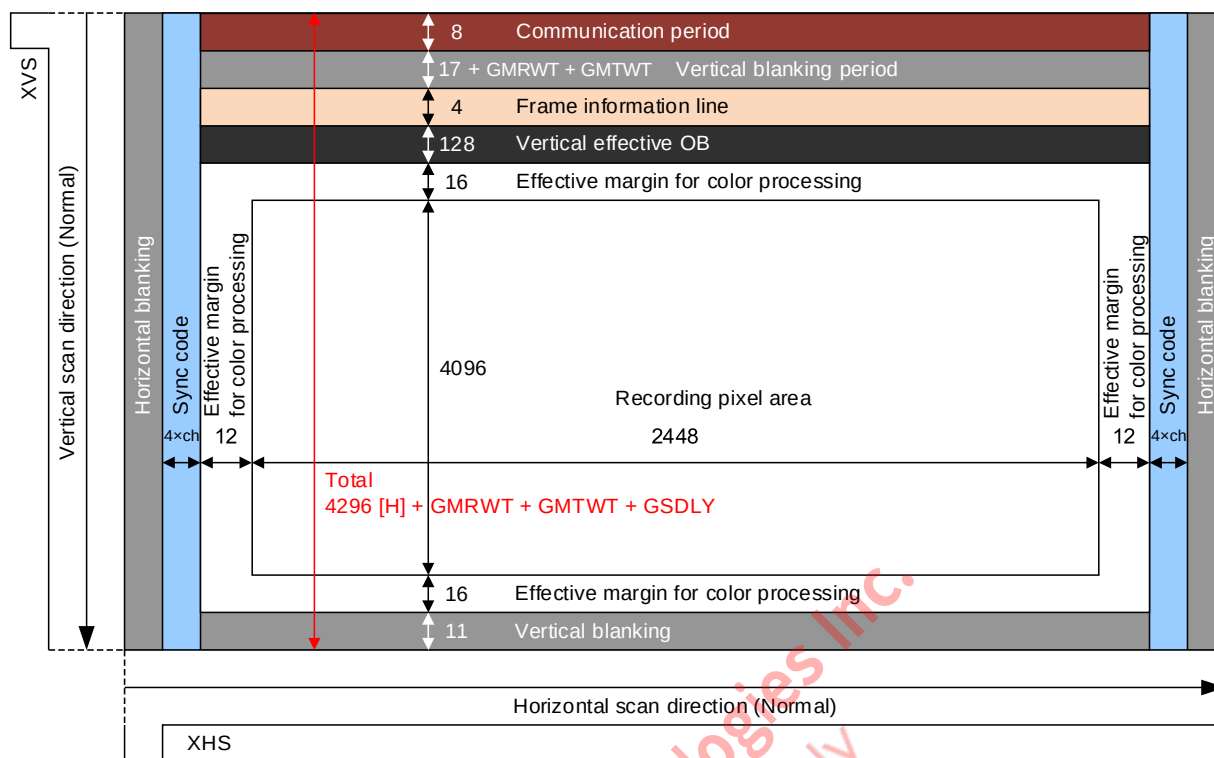
## 7.1.9. IMX537-AAMJ

## 7.1.9.1. All-pixel scan

SLVS Output (IMX537-AAMJ)

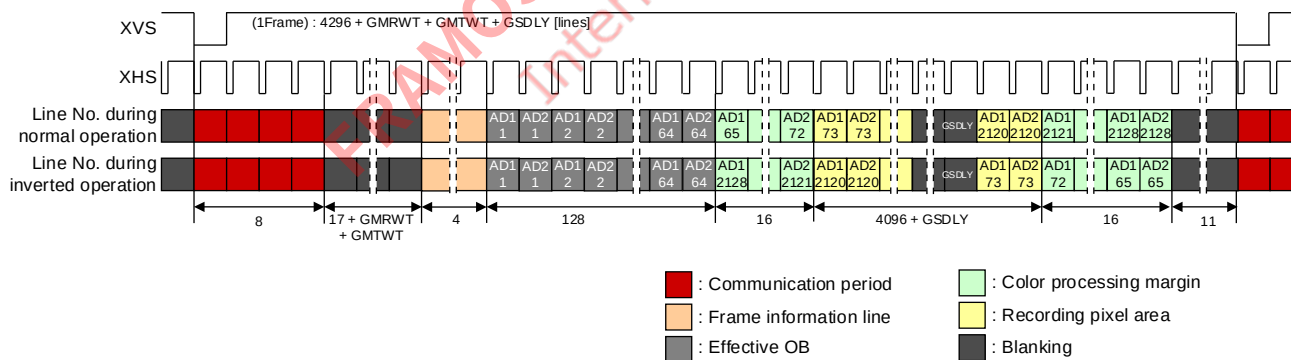
| Address       | bit   | Register name | Initial Value | Setting value     |                   |                   |                   |                   |                   | Remarks                               |           |
|---------------|-------|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit        |                   | AD = 10 bit       |                   | AD = 12 bit       |                   |                                       |           |
|               |       |               |               | 8 ch              | 4 ch              | 8 ch              | 4 ch              | 8 ch              | 4 ch              |                                       |           |
|               |       |               |               | 46.8<br>[frame/s] | 24.8<br>[frame/s] | 37.9<br>[frame/s] | 19.9<br>[frame/s] | 31.8<br>[frame/s] | 16.7<br>[frame/s] |                                       |           |
|               |       |               |               | 24.3<br>[frame/s] | 12.7<br>[frame/s] | 19.6<br>[frame/s] | 10.2<br>[frame/s] | 16.4<br>[frame/s] | 8.5<br>[frame/s]  |                                       |           |
|               |       |               |               | 67.8<br>[frame/s] | 36.4<br>[frame/s] | 55.3<br>[frame/s] | 29.4<br>[frame/s] | 46.8<br>[frame/s] | 24.8<br>[frame/s] |                                       |           |
|               |       |               |               | 35.9<br>[frame/s] | 18.9<br>[frame/s] | 29.0<br>[frame/s] | 15.2<br>[frame/s] | 24.3<br>[frame/s] | 12.7<br>[frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch       |                   |                   |                   |                   |                   |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |           |
| 445.5 Mbps/ch |       |               |               |                   |                   | FREQ = 3h         |                   |                   |                   |                                       |           |
| Chip ID = 02h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | 110Ch             | 10F0h             | 1100h             | 10E8h             | 10F4h             | 10E4h             | FREQ = 0h                             |           |
| D5h           | [7:0] |               |               | 10F0h             | 10E0h             | 10E8h             | 10DCh             | 10E4h             | 10D8h             | FREQ = 1h                             |           |
| D6h           | [7:0] |               |               | 1128h             | 1100h             | 1114h             | 10F4h             | 110Ch             | 10F0h             | FREQ = 2h                             |           |
|               |       |               |               | 1100h             | 10E4h             | 10F0h             | 10E4h             | 10F0h             | 10E0h             | FREQ = 3h                             |           |
| D8h           | [7:0] | HMAX          | 081h          | 16Bh              | 2B2h              | 1C2h              | 35Bh              | 219h              | 403h              | FREQ = 0h                             |           |
| D9h           | [7:0] |               |               | 2BEh              | 541h              | 36Ah              | 68Eh              | 416h              | 7DAh              | FREQ = 1h                             |           |
|               |       |               |               | F9h               | 1D4h              | 133h              | 244h              | 16Bh              | 2B2h              | FREQ = 2h                             |           |
|               |       |               |               | 1DBh              | 388h              | 24Eh              | 466h              | 2BEh              | 541h              | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 08h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 0h                             |           |
|               |       |               |               | 04h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 08h               | 08h               | 08h               | 04h               | 08h               | 04h               | FREQ = 2h                             |           |
|               |       |               |               | 08h               | 04h               | 04h               | 04h               | 04h               | 04h               | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 28h               | 18h               | 20h               | 14h               | 1Ch               | 10h               | FREQ = 0h                             |           |
|               |       |               |               | 18h               | 0Ch               | 14h               | 0Ch               | 10h               | 08h               | FREQ = 1h                             |           |
|               |       |               |               | 3Ch               | 20h               | 30h               | 1Ch               | 28h               | 18h               | FREQ = 2h                             |           |
|               |       |               |               | 20h               | 10h               | 18h               | 10h               | 18h               | 0Ch               | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h               |                   |                   |                   |                   |                   |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 14h               | 0Ch               | 10h               | 08h               | 0Ch               | 08h               | FREQ = 0h                             |           |
|               |       |               |               | 0Ch               | 08h               | 08h               | 04h               | 08h               | 04h               | FREQ = 1h                             |           |
|               |       |               |               | 1Ch               | 10h               | 14h               | 0Ch               | 14h               | 0Ch               | FREQ = 2h                             |           |
|               |       |               |               | 10h               | 08h               | 0Ch               | 08h               | 0Ch               | 08h               | FREQ = 3h                             |           |
| Chip ID = 04h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                |                   |                   |                   |                   |                   |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h               |                   |                   |                   |                   |                   |                                       |           |
| 3Dh           | [0]   |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                |                   | 0h                |                   | 1h                |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh               |                   | 1Bh               |                   | 13h               |                   |                                       |           |
| Chip ID = 15h |       |               |               |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h               |                   |                   |                   |                   |                   |                                       |           |

(IMX537-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.

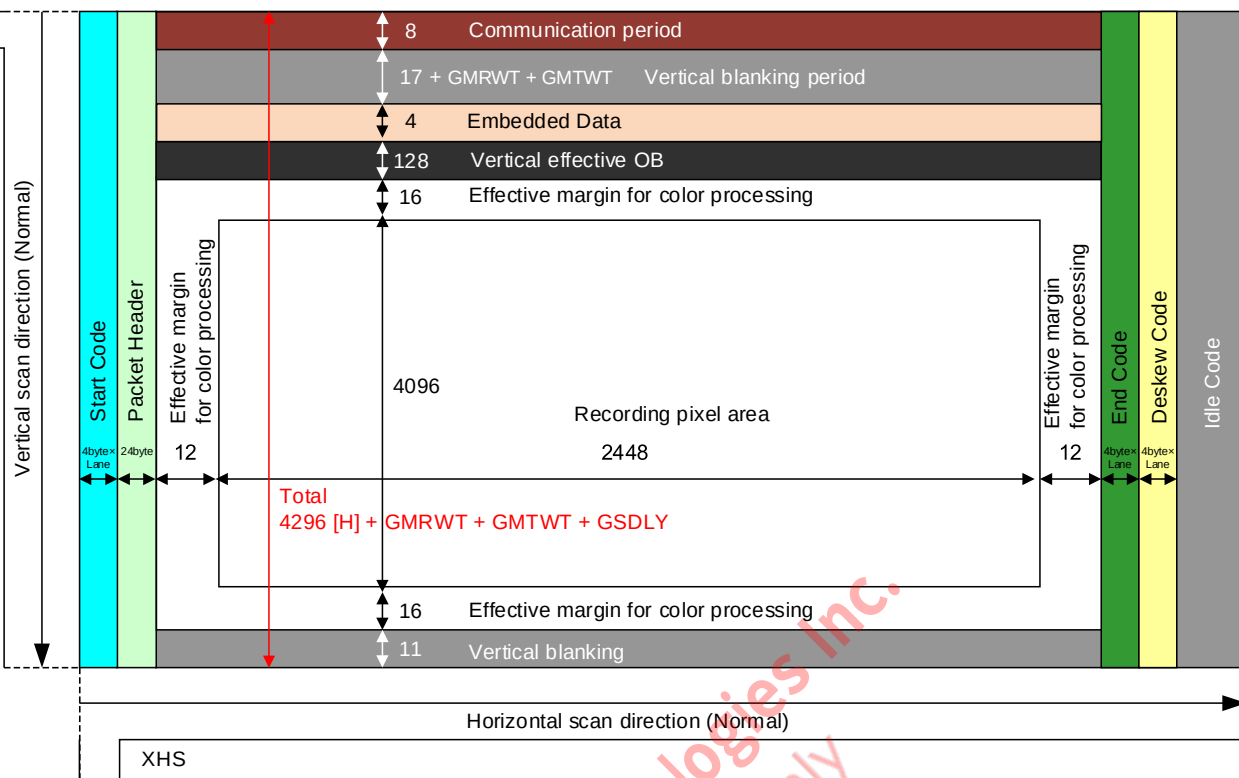


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX537-AAMJ)

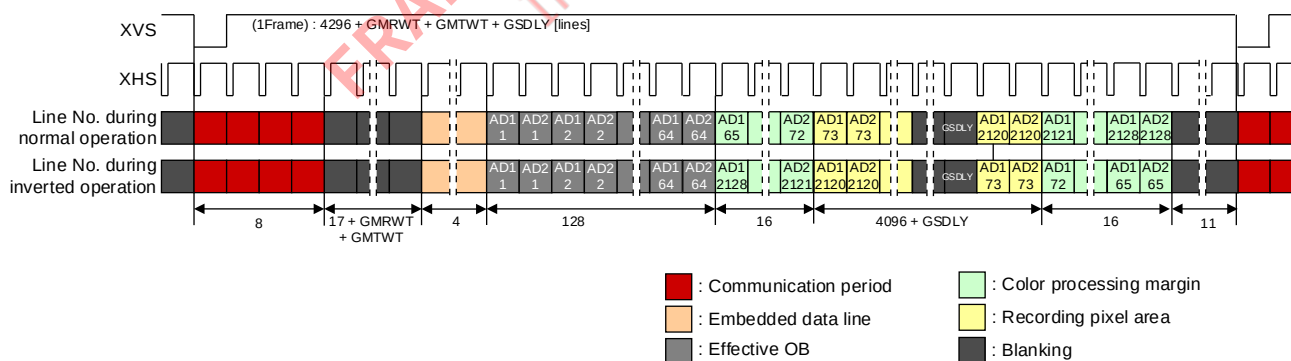
| Address         | bit   | Register name | Initial Value | Setting value      |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   | Remarks                               |           |
|-----------------|-------|---------------|---------------|--------------------|--------------------|-------------------|-------------------|--------------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------------|-----------|
|                 |       |               |               | AD = 8 bit         |                    |                   |                   | AD = 10 bit        |                    |                   |                   | AD = 12 bit       |                   |                   |                   |                                       |           |
|                 |       |               |               | 8 lane             | 4 lane             | 2 lane            | 1 lane            | 8 lane             | 4 lane             | 2 lane            | 1 lane            | 8 lane            | 4 lane            | 2 lane            | 1 lane            |                                       |           |
|                 |       |               |               | 135.8<br>[frame/s] | 135.8<br>[frame/s] | 78.7<br>[frame/s] | 40.6<br>[frame/s] | 128.7<br>[frame/s] | 122.3<br>[frame/s] | 64.6<br>[frame/s] | 33.0<br>[frame/s] | 88.4<br>[frame/s] | 88.4<br>[frame/s] | 53.7<br>[frame/s] | 27.3<br>[frame/s] | FREQ = 0h                             |           |
|                 |       |               |               | 133.6<br>[frame/s] | 76.6<br>[frame/s]  | 40.2<br>[frame/s] | 20.4<br>[frame/s] | 115.7<br>[frame/s] | 63.1<br>[frame/s]  | 32.7<br>[frame/s] | 16.5<br>[frame/s] | 88.4<br>[frame/s] | 53.4<br>[frame/s] | 27.1<br>[frame/s] | 13.7<br>[frame/s] | FREQ = 1h                             |           |
|                 |       |               |               | 69.8<br>[frame/s]  | 38.8<br>[frame/s]  | 20.2<br>[frame/s] | 10.2<br>[frame/s] | 59.9<br>[frame/s]  | 31.9<br>[frame/s]  | 16.4<br>[frame/s] | 8.3<br>[frame/s]  | 48.8<br>[frame/s] | 27.0<br>[frame/s] | 13.6<br>[frame/s] | 6.8<br>[frame/s]  | FREQ = 2h                             |           |
|                 |       |               |               | 4752 Mbps/lane     |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   | FREQ = 0h                             |           |
|                 |       |               |               | 2376 Mbps/lane     |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   | FREQ = 1h                             |           |
| 1188 Mbps /lane |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   | FREQ = 2h         |                   |                   |                   |                                       |           |
| Chip ID = 02h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| D4h             | [7:0] | VMAX          | 0CD4h         | 1180h              | 1180h              | 1138h             | 1104h             | 1178h              | 1170h              | 1120h             | 10F4h             | 1140h             | 1140h             | 1114h             | 10F0h             | FREQ = 0h                             |           |
| D5h             | [7:0] |               |               | 1180h              | 1134h              | 1104h             | 10E8h             | 1168h              | 1120h              | 10F4h             | 10E4h             | 1140h             | 1114h             | 10F0h             | 10E0h             | FREQ = 1h                             |           |
| D6h             | [7:0] |               |               | 1128h              | 1104h              | 10E8h             | 10DCh             | 111Ch              | 10F4h              | 10E4h             | 10D8h             | 1110h             | 10F0h             | 10E0h             | 10D8h             | FREQ = 2h                             |           |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                | 7Ah                | D6h               | 1A3h              | 88h                | 81h                | 88h               | 106h              | 206h              | BEh               | BEh               | 13Ch              | 272h                                  | FREQ = 0h |
| D9h             | [7:0] |               |               | 7Ch                | DCh                | 1A7h              | 346h              | 90h                | 10Ch               | 20Bh              | 40Bh              | BEh               | 13Eh              | 277h              | 4E4h              | FREQ = 1h                             |           |
|                 |       |               |               | F2h                | 1B7h               | 34Eh              | 68Ch              | 11Bh               | 218h               | 416h              | 816h              | 15Ch              | 27Ah              | 4EDh              | 9C7h              | FREQ = 2h                             |           |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                | 10h                | 0Ch               | 08h               | 10h                | 10h                | 08h               | 04h               | 0Ch               | 0Ch               | 08h               | 04h               | FREQ = 0h                             |           |
|                 |       |               |               | 10h                | 0Ch                | 08h               | 04h               | 10h                | 08h                | 04h               | 04h               | 0Ch               | 08h               | 04h               | 04h               | FREQ = 1h                             |           |
|                 |       |               |               | 08h                | 08h                | 04h               | 04h               | 08h                | 04h                | 04h               | 04h               | 08h               | 04h               | 04h               | 04h               | FREQ = 2h                             |           |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                | 74h                | 44h               | 24h               | 70h                | 68h                | 38h               | 1Ch               | 4Ch               | 4Ch               | 30h               | 18h               | FREQ = 0h                             |           |
|                 |       |               |               | 74h                | 44h                | 24h               | 14h               | 64h                | 38h                | 1Ch               | 10h               | 4Ch               | 30h               | 18h               | 0Ch               | FREQ = 1h                             |           |
|                 |       |               |               | 3Ch                | 24h                | 14h               | 0Ch               | 34h                | 1Ch                | 10h               | 08h               | 2Ch               | 18h               | 0Ch               | 08h               | FREQ = 2h                             |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                | 34h                | 20h               | 10h               | 30h                | 30h                | 18h               | 0Ch               | 20h               | 20h               | 14h               | 0Ch               | FREQ = 0h                             |           |
|                 |       |               |               | 34h                | 1Ch                | 10h               | 08h               | 2Ch                | 18h                | 0Ch               | 08h               | 20h               | 14h               | 0Ch               | 08h               | FREQ = 1h                             |           |
|                 |       |               |               | 1Ch                | 10h                | 08h               | 04h               | 18h                | 0Ch                | 08h               | 04h               | 14h               | 0Ch               | 08h               | 04h               | FREQ = 2h                             |           |
| Chip ID = 04h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                 |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
|                 | [5:4] | ADBIT         | 0h            | 2h                 |                    |                   |                   | 0h                 |                    |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| 3Dh             | [0]   |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| Chip ID = 06h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                 |                    |                   |                   | 0h                 |                    |                   |                   | 1h                |                   |                   |                   | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| E8h             | [7:0] |               | 2Dh           | 1Bh                |                    |                   |                   | 1Bh                |                    |                   |                   | 13h               |                   |                   |                   |                                       |           |
| Chip ID = 15h   |       |               |               |                    |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Eh             | [7:0] |               | 02h           | 16h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |
| 1Fh             | [7:0] |               | 02h           | 16h                |                    |                   |                   |                    |                    |                   |                   |                   |                   |                   |                   |                                       |           |

(IMX537-AAMJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The output of each AD is output alternately every other line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 7.1.9.2. ROI, Overlap ROI

## SLVS Output (IMX537-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |             |           |             |           | Remarks                               |           |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-------------|-----------|---------------------------------------|-----------|
|               |       |               |               | AD = 8 bit                                                                                                    |           | AD = 10 bit |           | AD = 12 bit |           |                                       |           |
|               |       |               |               | 8 ch                                                                                                          | 4 ch      | 8 ch        | 4 ch      | 8 ch        | 4 ch      |                                       |           |
|               |       |               |               | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s]   | [frame/s] | [frame/s]   | [frame/s] |                                       |           |
|               |       |               |               | 594 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                   |           |             |           |             |           |                                       | FREQ = 1h |
| 891 Mbps/ch   |       |               |               |                                                                                                               |           | FREQ = 2h   |           |             |           |                                       |           |
| 445.5 Mbps/ch |       |               |               |                                                                                                               |           | FREQ = 3h   |           |             |           |                                       |           |
| Chip ID = 02h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3          | *4        | *5          | *6        |                                       |           |
| D5h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D6h           | [7:0] |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| D8h           | [7:0] | HMAX          | 081h          | 16Bh                                                                                                          | 2B2h      | 1C2h        | 35Bh      | 219h        | 403h      | FREQ = 0h                             |           |
|               |       |               |               | 2BEh                                                                                                          | 541h      | 36Ah        | 68Eh      | 416h        | 7DAh      | FREQ = 1h                             |           |
| D9h           | [7:0] |               |               | F9h                                                                                                           | 1D4h      | 133h        | 244h      | 16Bh        | 2B2h      | FREQ = 2h                             |           |
|               |       |               |               | 1DBh                                                                                                          | 388h      | 24Eh        | 466h      | 2BEh        | 541h      | FREQ = 3h                             |           |
| E2h           | [7:0] | GMRWT         | 10h           | 08h                                                                                                           | 04h       | 08h         | 04h       | 04h         | 04h       | FREQ = 0h                             |           |
|               |       |               |               | 04h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 08h                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 2h                             |           |
|               |       |               |               | 08h                                                                                                           | 04h       | 04h         | 04h       | 04h         | 04h       | FREQ = 3h                             |           |
| E3h           | [7:0] | GMTWT         | 70h           | 28h                                                                                                           | 18h       | 20h         | 14h       | 1Ch         | 10h       | FREQ = 0h                             |           |
|               |       |               |               | 18h                                                                                                           | 0Ch       | 14h         | 0Ch       | 10h         | 08h       | FREQ = 1h                             |           |
|               |       |               |               | 3Ch                                                                                                           | 20h       | 30h         | 1Ch       | 28h         | 18h       | FREQ = 2h                             |           |
|               |       |               |               | 20h                                                                                                           | 10h       | 18h         | 10h       | 18h         | 0Ch       | FREQ = 3h                             |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |           |             |           |             |           |                                       |           |
| E6h           | [7:0] | GSDLY         | 30h           | 14h                                                                                                           | 0Ch       | 10h         | 08h       | 0Ch         | 08h       | FREQ = 0h                             |           |
|               |       |               |               | 0Ch                                                                                                           | 08h       | 08h         | 04h       | 08h         | 04h       | FREQ = 1h                             |           |
|               |       |               |               | 1Ch                                                                                                           | 10h       | 14h         | 0Ch       | 14h         | 0Ch       | FREQ = 2h                             |           |
|               |       |               |               | 10h                                                                                                           | 08h       | 0Ch         | 08h       | 0Ch         | 08h       | FREQ = 3h                             |           |
| Chip ID = 03h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |             |           |             |           |                                       |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |             |           |             |           |                                       |           |
| Chip ID = 04h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |           |             |           |             |           |                                       |           |
|               | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |           |             |           |             |           |                                       |           |
| 3Dh           | [0]   |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| Chip ID = 06h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 30h           | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |           | 0h          |           | 1h          |           | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 08h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| E8h           | [7:0] |               | 2Dh           | 1Bh                                                                                                           |           | 1Bh         |           | 13h         |           |                                       |           |
| Chip ID = 15h |       |               |               |                                                                                                               |           |             |           |             |           |                                       |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |           |             |           |             |           |                                       |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 168$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 168$$

SLVS – EC output (IMX537-AAMJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                                 |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  | Remarks                               |
|-----------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|---------------------------------------|
|                 |       |               |               | AD = 8 bit                                                                                                    |                 |                 |                 | AD = 10 bit     |                 |                 |                 | AD = 12 bit     |                  |                  |                  |                                       |
|                 |       |               |               | 8 lane                                                                                                        | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane          | 2 lane          | 1 lane          | 8 lane          | 4 lane           | 2 lane           | 1 lane           |                                       |
|                 |       |               |               | *1<br>[frame/s]                                                                                               | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s] | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] | *9<br>[frame/s] | *10<br>[frame/s] | *11<br>[frame/s] | *12<br>[frame/s] |                                       |
|                 |       |               |               | 4752 Mbps/lane                                                                                                |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 2376 Mbps/lane  |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 | FREQ = 1h       |                  |                  |                  |                                       |
| 1188 Mbps /lane |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 | FREQ = 2h       |                  |                  |                  |                                       |
| Chip ID = 02h   |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2              | *3              | *4              | *5              | *6              | *7              | *8              | *9              | *10              | *11              | *12              |                                       |
| D5h             | [7:0] |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D6h             | [7:0] |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| D8h             | [7:0] | HMAX          | 081h          | 7Ah                                                                                                           | 7Ah             | D6h             | 1A3h            | 81h             | 88h             | 106h            | 206h            | BEh             | BEh              | 13Ch             | 272h             | FREQ = 0h                             |
| D9h             | [7:0] |               |               | 7Ch                                                                                                           | DCh             | 1A7h            | 346h            | 90h             | 10Ch            | 20Bh            | 40Bh            | BEh             | 13Eh             | 277h             | 4E4h             | FREQ = 1h                             |
|                 |       |               |               | F2h                                                                                                           | 1B7h            | 34Eh            | 68Ch            | 11Bh            | 218h            | 416h            | 816h            | 15Ch            | 27Ah             | 4EDh             | 9C7h             | FREQ = 2h                             |
| E2h             | [7:0] | GMRWT         | 10h           | 10h                                                                                                           | 10h             | 0Ch             | 08h             | 10h             | 10h             | 08h             | 04h             | 0Ch             | 0Ch              | 08h              | 04h              | FREQ = 0h                             |
|                 |       |               |               | 10h                                                                                                           | 0Ch             | 08h             | 04h             | 10h             | 08h             | 04h             | 04h             | 0Ch             | 08h              | 04h              | 04h              | FREQ = 1h                             |
|                 |       |               |               | 08h                                                                                                           | 08h             | 04h             | 04h             | 08h             | 04h             | 04h             | 04h             | 08h             | 04h              | 04h              | 04h              | FREQ = 2h                             |
| E3h             | [7:0] | GMTWT         | 70h           | 74h                                                                                                           | 74h             | 44h             | 24h             | 70h             | 68h             | 38h             | 1Ch             | 4Ch             | 4Ch              | 30h              | 18h              | FREQ = 0h                             |
|                 |       |               |               | 74h                                                                                                           | 44h             | 24h             | 14h             | 64h             | 38h             | 1Ch             | 10h             | 4Ch             | 30h              | 18h              | 0Ch              | FREQ = 1h                             |
|                 |       |               |               | 3Ch                                                                                                           | 24h             | 14h             | 0Ch             | 34h             | 1Ch             | 10h             | 08h             | 2Ch             | 18h              | 0Ch              | 08h              | FREQ = 2h                             |
| E5h             | [7:0] | GAINDLY       | 04h           | 04h                                                                                                           |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| E6h             | [7:0] | GSDLY         | 30h           | 34h                                                                                                           | 34h             | 20h             | 10h             | 30h             | 30h             | 18h             | 0Ch             | 20h             | 20h              | 14h              | 0Ch              | FREQ = 0h                             |
|                 |       |               |               | 34h                                                                                                           | 1Ch             | 10h             | 08h             | 2Ch             | 18h             | 0Ch             | 08h             | 20h             | 14h              | 0Ch              | 08h              | FREQ = 1h                             |
|                 |       |               |               | 1Ch                                                                                                           | 10h             | 08h             | 04h             | 18h             | 0Ch             | 08h             | 04h             | 14h             | 0Ch              | 08h              | 04h              | FREQ = 2h                             |
| Chip ID = 03h   |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 04h   |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 00h             | [1:0] | AD_SEL        | 0h            | 2h                                                                                                            |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
|                 | [5:4] | ADBIT         | 0h            | 2h                                                                                                            |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 18h                                                                                                           |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 3Dh             | [0]   |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| Chip ID = 06h   |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 30h             | [1:0] | ODBIT         | 0h            | 2h                                                                                                            |                 |                 |                 | 0h              |                 |                 |                 | 1h              |                  |                  |                  | 0h: 10 bit<br>1h: 12 bit<br>2h: 8 bit |
| Chip ID = 08h   |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| E8h             | [7:0] |               | 2Dh           | 1Bh                                                                                                           |                 |                 |                 | 1Bh             |                 |                 |                 | 13h             |                  |                  |                  |                                       |
| Chip ID = 15h   |       |               |               |                                                                                                               |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                           |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                           |                 |                 |                 |                 |                 |                 |                 |                 |                  |                  |                  |                                       |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8) × 2 + GMRWT + GMTWT + GSDLY + 168

Overlap ROI :

VMAX= (Maximum vertical output width) × 2 + GMRWT + GMTWT + GSDLY + 168

## 7.2. The register settings for HMAX

When Dual ADC is used, "Register setting" in the datasheet of IMX530/531/532 must be set the registers Initial Value.

| Address       | bit   | Register name | Initial Value | Setting value                                                                    |                                        |                                         |                                          |                       | Remarks                  |
|---------------|-------|---------------|---------------|----------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------|------------------------------------------|-----------------------|--------------------------|
|               |       |               |               | to<br>21Ch<br>(540d)                                                             | 21Dh<br>(541d)<br>to<br>3B4h<br>(948d) | 3B5h<br>(949d)<br>to<br>7F4h<br>(2036d) | 7F5h<br>(2037d)<br>to<br>D28h<br>(3368d) | D29h<br>(3369d)<br>to | Setting range<br>of HMAX |
| Chip ID = 08h |       |               |               |                                                                                  |                                        |                                         |                                          |                       |                          |
| 4Ch           | [7:0] |               | 00h           | When Dual ADC is used, it is necessary to set the registers 00h (Initial Value). |                                        |                                         |                                          |                       |                          |
| 4Eh           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 4Fh           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 74h           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 76h           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 77h           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 9Ch           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 9Eh           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |
| 9Fh           | [7:0] |               | 00h           |                                                                                  |                                        |                                         |                                          |                       |                          |

## 7.3. The register settings for SLVS-EC output 10 bit mode (IMX535, IMX536, IMX537 only)

When SLVS-EC output 10 bit mode is used, it is necessary to set the registers shown in the table below depending on the combination of data rate 4752 Mbps/lane and the number of lanes 8.

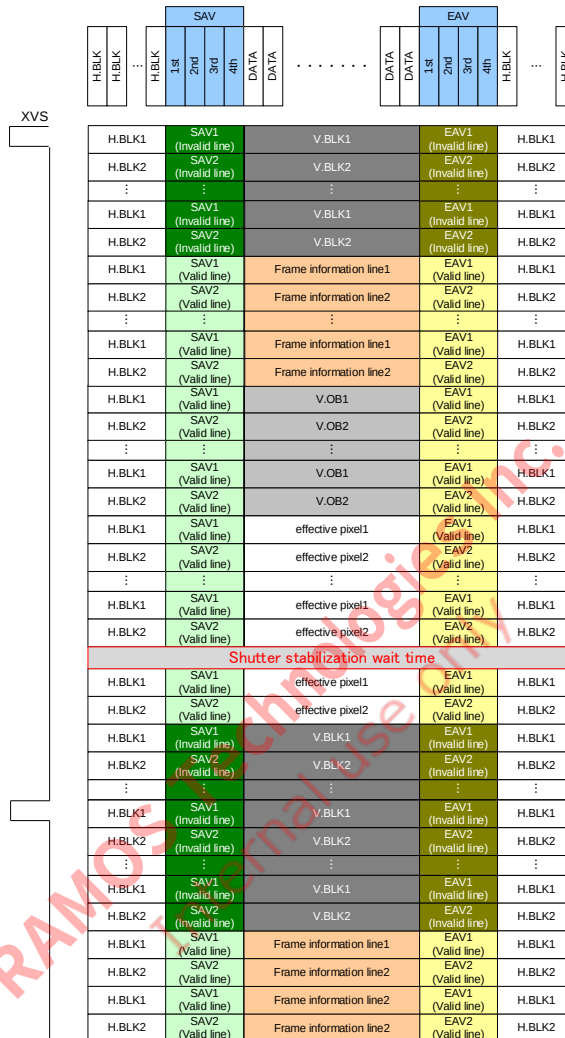
| Address       | bit   | Register name | 4752 Mbps/lane<br>8 lanes |
|---------------|-------|---------------|---------------------------|
| Chip ID = 08h |       |               |                           |
| 2Ch           | [7:0] |               | 64h                       |
| 2Eh           | [7:0] |               | 00h                       |
| 2Fh           | [7:0] |               | 00h                       |
| 54h           | [7:0] |               | 64h                       |
| 56h           | [7:0] |               | 20h                       |
| 57h           | [7:0] |               | 00h                       |
| 7Ch           | [7:0] |               | 64h                       |
| 7Eh           | [7:0] |               | 00h                       |
| 7Fh           | [7:0] |               | 00h                       |
| Chip ID = 12h |       |               |                           |
| 56h           | [7:0] |               | 23h                       |
| 96h           | [7:0] |               | 23h                       |
| Chip ID = 17h |       |               |                           |
| E6h           | [7:0] |               | 3Fh                       |
| F0h           | [7:0] |               | 95h                       |
| F2h           | [7:0] |               | 8Fh                       |
| F8h           | [7:0] |               | 93h                       |
| FAh           | [7:0] |               | 95h                       |
| Chip ID = 18h |       |               |                           |
| 04h           | [7:0] |               | 93h                       |
| 06h           | [7:0] |               | 95h                       |
| 0Ch           | [7:0] |               | 8Fh                       |

| Address       | bit   | Register name | 4752 Mbps/lane<br>8 lanes |
|---------------|-------|---------------|---------------------------|
| Chip ID = 18h |       |               |                           |
| 0Eh           | [7:0] |               | C0h                       |
| 14h           | [7:0] |               | 95h                       |
| 16h           | [7:0] |               | 8Fh                       |
| 34h           | [7:0] |               | 36h                       |
| 36h           | [7:0] |               | 95h                       |
| 3Ch           | [7:0] |               | 38h                       |
| 3Eh           | [7:0] |               | 97h                       |
| 44h           | [7:0] |               | 3Ah                       |
| 46h           | [7:0] |               | 99h                       |
| 4Ch           | [7:0] |               | 33h                       |
| 4Eh           | [7:0] |               | 39h                       |
| 54h           | [7:0] |               | 35h                       |
| 56h           | [7:0] |               | 3Ch                       |
| 5Ch           | [7:0] |               | 37h                       |
| 5Eh           | [7:0] |               | 3Eh                       |
| 6Ah           | [7:0] |               | A3h                       |
| 70h           | [7:0] |               | 9Dh                       |
| 76h           | [7:0] |               | 9Bh                       |
| 78h           | [7:0] |               | BFh                       |
| 98h           | [7:0] |               | 98h                       |
| 9Ah           | [7:0] |               | BEh                       |

## 8. AD Line identification method (without combination)

### 8.1. SLVS Output

In case of SLVS output, by getting 4th code of Sync Code, AD1 (Low gain side) output line and AD2 (High gain side) output line can be distinguished.



Sync Code Output Timing (Dual ADC mode)

### SAV / EAV Sync Code (SLVS output)

#### Sync code-1(AD1)

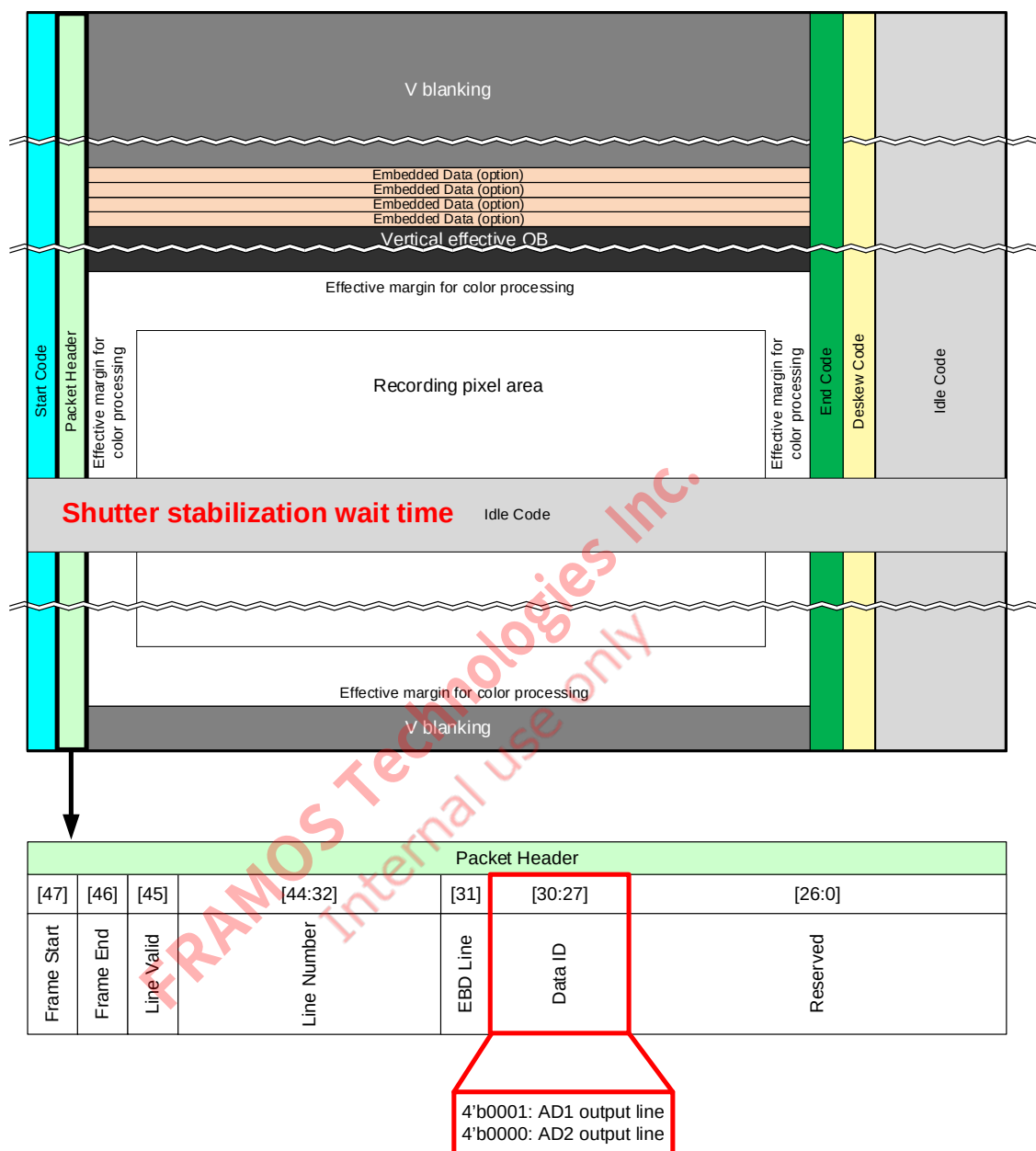
| Sync code           | 1st code |        |        | 2nd code |        |        | 3rd code |        |        | 4th code |        |        |
|---------------------|----------|--------|--------|----------|--------|--------|----------|--------|--------|----------|--------|--------|
|                     | 8 bit    | 10 bit | 12 bit | 8 bit    | 10 bit | 12 bit | 8 bit    | 10 bit | 12 bit | 8 bit    | 10 bit | 12 bit |
| SAV1 (Valid line)   | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | C0h      | 300h   | C00h   |
| EAV1 (Valid line)   | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | DDh      | 374h   | DD0h   |
| SAV1 (Invalid line) | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | EBh      | 3ACh   | EB0h   |
| EAV1 (Invalid line) | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | F6h      | 3D8h   | F60h   |

#### Sync code-2(AD2)

| Sync code           | 1st code |        |        | 2nd code |        |        | 3rd code |        |        | 4th code |        |        |
|---------------------|----------|--------|--------|----------|--------|--------|----------|--------|--------|----------|--------|--------|
|                     | 8 bit    | 10 bit | 12 bit | 8 bit    | 10 bit | 12 bit | 8 bit    | 10 bit | 12 bit | 8 bit    | 10 bit | 12 bit |
| SAV2 (Valid line)   | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | 80h      | 200h   | 800h   |
| EAV2 (Valid line)   | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | 9Dh      | 274h   | 9D0h   |
| SAV2 (Invalid line) | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | ABh      | 2ACh   | AB0h   |
| EAV2 (Invalid line) | FFh      | 3FFh   | FFFh   | 00h      | 000h   | 000h   | 00h      | 000h   | 000h   | B6h      | 2D8h   | B60h   |

## 8.2. SLVS – EC Output

In case of SLVS – EC output, by getting Data ID in Packet Header, AD1 (Low gain side) output line and AD2 (High gain side) output line can be identified.



Data ID in Packet Header (SLVS – EC output)

## 9. Readout Drive Mode (with built-in combination)

When converting from 12-bit A/D to 8-bit output, the gradation compression function different from the gradation compression for built-in combining is used.

Please refer to the separate application note for this gradation compression function.

### 9.1. Readout Drive mode

#### 9.1.1. IMX530

SLVS Output, Data rate: 594 Mbps/ch (IMX530)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 14.7                 | 5320                       | 4600          | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 | (12)                               | 7.5                  |                            |               | 1043.0               | 2086.0          | 1517.1       |
|            | 8                     | 594                 | 12                                 | 21.9                 |                            |               | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 | (8)                                | 11.2                 |                            |               | 698.0                | 1396.0          | 1015.3       |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 1043.0               | 2086.0          | 1517.1       |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 698.0                | 1396.0          | 1015.3       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 14.7 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX530)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 7.4                  | 5320                       | 4600          | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 | (12)                               | 3.8                  |                            |               | 2070.0               | 4140.0          | 3011.0       |
|            | 8                     | 297                 | 12                                 | 11.1                 |                            |               | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 | (8)                                | 5.7                  |                            |               | 1382.5               | 2765.0          | 2011.0       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 2070.0               | 4140.0          | 3011.0       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 1382.5               | 2765.0          | 2011.0       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 7.4 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX530)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 21.9                 | 5320                       | 4600 | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 | (12)                               | 11.2                 |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 891                 | 12                                 | 30.2                 |                            |      | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 | (8)                                | 15.9                 |                            |      | 495.0                | 990.0           | 720.0        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 495.0                | 990.0           | 720.0        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.9 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX530)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 11.1                 | 5320                       | 4600 | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               | (12)                               | 5.7                  |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 445.5               | 12                                 | 15.5                 |                            |      | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               | (8)                                | 8.1                  |                            |      | 972.5                | 1945.0          | 1414.6       |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 972.5                | 1945.0          | 1414.6       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 11.1 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX530)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 37.5                 | 5320                       | 4600 | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | 37.5                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | 24.0                 |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                                    | 12.0                 |                            |      | 652.5                | 1305.0          | 949.1        |
|            | 8                            | 4752                  | 12 (8)                             | 37.5                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | 37.5                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | 35.1                 |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                                    | 17.9                 |                            |      | 440.0                | 880.0           | 640.0        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 652.5                | 1305.0          | 949.1        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 440.0                | 880.0           | 640.0        |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 37.5 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX530)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 37.5                 | 5320                       | 4600 | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | 23.7                 |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                                    | 12.0                 |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | 6.0                  |                            |      | 1304.5               | 2609.0          | 1897.5       |
|            | 8                            | 2376                  | 12 (8)                             | 37.5                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | 23.8                 |                            |      | 330.5                | 661.0           | 480.8        |
|            | 2                            | 2376                  |                                    | 12.0                 |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | 6.0                  |                            |      | 1304.5               | 2609.0          | 1897.5       |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1304.5               | 2609.0          | 1897.5       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 330.5                | 661.0           | 480.8        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1304.5               | 2609.0          | 1897.5       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 37.5 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX530)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 22.3                 | 5320                       | 4600 | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                                    | 11.9                 |                            |      | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                                    | 6.0                  |                            |      | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                                    | 3.0                  |                            |      | 2609.0               | 5218.0          | 3795.0       |
|            | 8                            | 1188                  | 12 (8)                             | 32.7                 |                            |      | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                                    | 17.4                 |                            |      | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                                    | 8.8                  |                            |      | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                                    | 4.4                  |                            |      | 1759.5               | 3519.0          | 2559.3       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 2609.0               | 5218.0          | 3795.0       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1759.5               | 3519.0          | 2559.3       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 22.3 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 9.1.2. IMX531

SLVS Output, Data rate: 594 Mbps/ch (IMX531)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 17.5                 | 4504                       | 4504          | 457.0                | 914.0           | 664.8        |
|            | 4                     | 594                 | (12)                               | 9.0                  |                            |               | 890.0                | 1780.0          | 1294.6       |
|            | 8                     | 594                 | 12                                 | 26.1                 |                            |               | 307.0                | 614.0           | 446.6        |
|            | 4                     | 594                 | (8)                                | 13.5                 |                            |               | 596.0                | 1192.0          | 867.0        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 457.0                | 914.0           | 664.8        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 890.0                | 1780.0          | 1294.6       |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 307.0                | 614.0           | 446.6        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 596.0                | 1192.0          | 867.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 17.5 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX531)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 8.9                  | 4504                       | 4504          | 902.5                | 1805.0          | 1312.8       |
|            | 4                     | 297                 | (12)                               | 4.5                  |                            |               | 1764.0               | 3528.0          | 2565.9       |
|            | 8                     | 297                 | 12                                 | 13.3                 |                            |               | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 297                 | (8)                                | 6.8                  |                            |               | 1178.5               | 2357.0          | 1714.2       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 902.5                | 1805.0          | 1312.8       |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1764.0               | 3528.0          | 2565.9       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 1178.5               | 2357.0          | 1714.2       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 8.9 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 26.1                 | 4504                       | 4504 | 307.0                | 614.0           | 446.6        |
|            | 4                     | 891                 | (12)                               | 13.5                 |                            |      | 596.0                | 1192.0          | 867.0        |
|            | 8                     | 891                 | 12                                 | 35.4                 |                            |      | 226.0                | 452.0           | 328.8        |
|            | 4                     | 891                 | (8)                                | 18.8                 |                            |      | 427.0                | 854.0           | 621.1        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 307.0                | 614.0           | 446.6        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 596.0                | 1192.0          | 867.0        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 226.0                | 452.0           | 328.8        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 427.0                | 854.0           | 621.1        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 26.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 13.3                 | 4504                       | 4504 | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 445.5               | (12)                               | 6.8                  |                            |      | 1178.5               | 2357.0          | 1714.2       |
|            | 8                     | 445.5               | 12                                 | 18.2                 |                            |      | 440.0                | 880.0           | 640.0        |
|            | 4                     | 445.5               | (8)                                | 9.6                  |                            |      | 836.5                | 1673.0          | 1216.8       |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 604.0                | 1208.0          | 878.6        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 1178.5               | 2357.0          | 1714.2       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 440.0                | 880.0           | 640.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 836.5                | 1673.0          | 1216.8       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 13.3 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX531)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 38.3                 | 4504                       | 4504 | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | 38.3                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | 28.8                 |                            |      | 278.5                | 557.0           | 405.1        |
|            | 1                            | 4752                  |                                    | 14.5                 |                            |      | 554.5                | 1109.0          | 806.6        |
|            | 8                            | 4752                  | 12 (8)                             | 38.3                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | 38.3                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | 38.3                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 1                            | 4752                  |                                    | 21.5                 |                            |      | 373.5                | 747.0           | 543.3        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 278.5                | 557.0           | 405.1        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 554.5                | 1109.0          | 806.6        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 373.5                | 747.0           | 543.3        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 38.3 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX531)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 38.3                 | 4504                       | 4504 | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | 28.7                 |                            |      | 279.5                | 559.0           | 406.6        |
|            | 2                            | 2376                  |                                    | 14.4                 |                            |      | 557.0                | 1114.0          | 810.2        |
|            | 1                            | 2376                  |                                    | 7.2                  |                            |      | 1109.0               | 2218.0          | 1613.1       |
|            | 8                            | 2376                  | 12 (8)                             | 38.3                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | 28.7                 |                            |      | 279.5                | 559.0           | 406.6        |
|            | 2                            | 2376                  |                                    | 14.4                 |                            |      | 557.0                | 1114.0          | 810.2        |
|            | 1                            | 2376                  |                                    | 7.2                  |                            |      | 1109.0               | 2218.0          | 1613.1       |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 279.5                | 559.0           | 406.6        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 557.0                | 1114.0          | 810.2        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1109.0               | 2218.0          | 1613.1       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 279.5                | 559.0           | 406.6        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 557.0                | 1114.0          | 810.2        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1109.0               | 2218.0          | 1613.1       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 38.3 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX531)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 26.6                 | 4504                       | 4504          | 301.0                | 602.0           | 437.9        |
|            | 4                            | 1188                  |                                    | 14.4                 |                            |               | 559.0                | 1118.0          | 813.1        |
|            | 2                            | 1188                  |                                    | 7.2                  |                            |               | 1114.0               | 2228.0          | 1620.4       |
|            | 1                            | 1188                  |                                    | 3.6                  |                            |               | 2218.0               | 4436.0          | 3226.2       |
|            | 8                            | 1188                  | 12 (8)                             | 38.3                 |                            |               | 209.0                | 418.0           | 304.0        |
|            | 4                            | 1188                  |                                    | 20.7                 |                            |               | 387.0                | 774.0           | 563.0        |
|            | 2                            | 1188                  |                                    | 10.7                 |                            |               | 749.0                | 1498.0          | 1089.5       |
|            | 1                            | 1188                  |                                    | 5.3                  |                            |               | 1493.5               | 2987.0          | 2172.4       |
| ROI        | 8                            | 1188                  | 12 (12)                            | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 301.0                | 602.0           | 437.9        |
|            | 4                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 559.0                | 1118.0          | 813.1        |
|            | 2                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 1114.0               | 2228.0          | 1620.4       |
|            | 1                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 2218.0               | 4436.0          | 3226.2       |
|            | 8                            | 1188                  | 12 (8)                             | <sup>*2</sup>        |                            |               | 209.0                | 418.0           | 304.0        |
|            | 4                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 387.0                | 774.0           | 563.0        |
|            | 2                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 749.0                | 1498.0          | 1089.5       |
|            | 1                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 1493.5               | 2987.0          | 2172.4       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 26.6 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

### 9.1.3. IMX532

SLVS Output, Data rate: 594 Mbps/ch (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 22.1                 | 5320                       | 3032          | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 | (12)                               | 11.3                 |                            |               | 1043.0               | 2086.0          | 1517.1       |
|            | 8                     | 594                 | 12                                 | 32.9                 |                            |               | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 | (8)                                | 16.9                 |                            |               | 698.0                | 1396.0          | 1015.3       |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 533.5                | 1067.0          | 776.0        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 1043.0               | 2086.0          | 1517.1       |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 358.0                | 716.0           | 520.8        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 698.0                | 1396.0          | 1015.3       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 22.1 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 11.2                 | 5320                       | 3032          | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 | (12)                               | 5.7                  |                            |               | 2070.0               | 4140.0          | 3011.0       |
|            | 8                     | 297                 | 12                                 | 16.7                 |                            |               | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 | (8)                                | 8.5                  |                            |               | 1382.5               | 2765.0          | 2011.0       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 1055.5               | 2111.0          | 1535.3       |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 2070.0               | 4140.0          | 3011.0       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 1382.5               | 2765.0          | 2011.0       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 11.2 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 32.9                 | 5320                       | 3032 | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 | (12)                               | 16.9                 |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 891                 | 12                                 | 45.3                 |                            |      | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 | (8)                                | 23.8                 |                            |      | 495.0                | 990.0           | 720.0        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 358.0                | 716.0           | 520.8        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 698.0                | 1396.0          | 1015.3       |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 260.0                | 520.0           | 378.2        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 495.0                | 990.0           | 720.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.9 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 16.7                 | 5320                       | 3032 | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               | (12)                               | 8.5                  |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 445.5               | 12                                 | 23.2                 |                            |      | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               | (8)                                | 12.1                 |                            |      | 972.5                | 1945.0          | 1414.6       |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 706.0                | 1412.0          | 1027.0       |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 1382.5               | 2765.0          | 2011.0       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 508.0                | 1016.0          | 739.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 972.5                | 1945.0          | 1414.6       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 16.7 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 56.2                 | 5320                       | 3032 | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | 56.2                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | 35.9                 |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                                    | 18.1                 |                            |      | 652.5                | 1305.0          | 949.1        |
|            | 8                            | 4752                  | 12 (8)                             | 56.2                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | 56.2                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | 52.6                 |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                                    | 26.8                 |                            |      | 440.0                | 880.0           | 640.0        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 327.5                | 655.0           | 476.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 652.5                | 1305.0          | 949.1        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 223.5                | 447.0           | 325.1        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 440.0                | 880.0           | 640.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 56.2 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX532)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 56.2                 | 5320                       | 3032 | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | 35.5                 |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                                    | 18.0                 |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | 9.0                  |                            |      | 1304.5               | 2609.0          | 1897.5       |
|            | 8                            | 2376                  | 12 (8)                             | 56.2                 |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | 35.6                 |                            |      | 330.5                | 661.0           | 480.8        |
|            | 2                            | 2376                  |                                    | 18.0                 |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | 9.0                  |                            |      | 1304.5               | 2609.0          | 1897.5       |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 331.5                | 663.0           | 482.2        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1304.5               | 2609.0          | 1897.5       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 209.0                | 418.0           | 304.0        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 330.5                | 661.0           | 480.8        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 655.0                | 1310.0          | 952.8        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1304.5               | 2609.0          | 1897.5       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 56.2 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX532)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 33.5                 | 5320                       | 3032 | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                                    | 17.8                 |                            |      | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                                    | 9.0                  |                            |      | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                                    | 4.5                  |                            |      | 2609.0               | 5218.0          | 3795.0       |
|            | 8                            | 1188                  | 12 (8)                             | 49.0                 |                            |      | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                                    | 26.0                 |                            |      | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                                    | 13.3                 |                            |      | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                                    | 6.7                  |                            |      | 1759.5               | 3519.0          | 2559.3       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 352.0                | 704.0           | 512.0        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 661.0                | 1322.0          | 961.5        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 1309.5               | 2619.0          | 1904.8       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 2609.0               | 5218.0          | 3795.0       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 240.0                | 480.0           | 349.1        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 452.5                | 905.0           | 658.2        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 887.0                | 1774.0          | 1290.2       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1759.5               | 3519.0          | 2559.3       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 33.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 9.1.4. IMX535-AAQJ

SLVS Output, Data rate: 594 Mbps/ch (IMX535-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 28.2                 | 4096                       | 3000          | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 | (12)                               | 14.5                 |                            |               | 818.0                | 1636.0          | 1189.9       |
|            | 8                     | 594                 | 12                                 | 41.8                 |                            |               | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 | (8)                                | 21.7                 |                            |               | 548.0                | 1096.0          | 797.1        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 818.0                | 1636.0          | 1189.9       |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 548.0                | 1096.0          | 797.1        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 28.2 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX535-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 14.3                 | 4096                       | 3000          | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 | (12)                               | 7.3                  |                            |               | 1620.0               | 3240.0          | 2356.4       |
|            | 8                     | 297                 | 12                                 | 21.4                 |                            |               | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 | (8)                                | 11.0                 |                            |               | 1082.5               | 2165.0          | 1574.6       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1620.0               | 3240.0          | 2356.4       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 1082.5               | 2165.0          | 1574.6       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 14.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 41.8                 | 4096                       | 3000 | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 | (12)                               | 21.7                 |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 891                 | 12                                 | 61.1                 |                            |      | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 | (8)                                | 32.1                 |                            |      | 369.5                | 739.0           | 537.5        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 369.5                | 739.0           | 537.5        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.8 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 21.4                 | 4096                       | 3000 | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               | (12)                               | 11.0                 |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 445.5               | 12                                 | 31.7                 |                            |      | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               | (8)                                | 16.4                 |                            |      | 725.5                | 1451.0          | 1055.3       |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 725.5                | 1451.0          | 1055.3       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.4 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX535-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 61.9                 | 4096                       | 3000 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 46.3                 |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                                    | 23.4                 |                            |      | 508.5                | 1017.0          | 739.7        |
|            | 8                            | 4752                  | 12 (8)                             | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 34.5                 |                            |      | 343.0                | 686.0           | 499.0        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 508.5                | 1017.0          | 739.7        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 343.0                | 686.0           | 499.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.9 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX535-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 61.9                 | 4096                       | 3000 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 46.1                 |                            |      | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                                    | 23.3                 |                            |      | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                                    | 11.7                 |                            |      | 1017.0               | 2034.0          | 1479.3       |
|            | 8                            | 2376                  | 12 (8)                             | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 34.5                 |                            |      | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                                    | 17.3                 |                            |      | 685.5                | 1371.0          | 997.1        |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1017.0               | 2034.0          | 1479.3       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 685.5                | 1371.0          | 997.1        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.9 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX535-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 42.7                 | 4096                       | 3000 | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                                    | 23.2                 |                            |      | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                                    | 11.6                 |                            |      | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                                    | 5.8                  |                            |      | 2034.0               | 4068.0          | 2958.6       |
|            | 8                            | 1188                  | 12 (8)                             | 61.9                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 33.4                 |                            |      | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                                    | 17.3                 |                            |      | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                                    | 8.7                  |                            |      | 1370.5               | 2741.0          | 1993.5       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 2034.0               | 4068.0          | 2958.6       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1370.5               | 2741.0          | 1993.5       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 42.7 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 9.1.5. IMX535-AAMJ

SLVS Output, Data rate: 594 Mbps/ch (IMX535-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 28.1                 | 4096                       | 3000          | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 | (12)                               | 14.5                 |                            |               | 818.0                | 1636.0          | 1189.9       |
|            | 8                     | 594                 | 12                                 | 41.7                 |                            |               | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 | (8)                                | 21.7                 |                            |               | 548.0                | 1096.0          | 797.1        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 421.0                | 842.0           | 612.4        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 818.0                | 1636.0          | 1189.9       |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 283.0                | 566.0           | 411.7        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 548.0                | 1096.0          | 797.1        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 28.1 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX535-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 14.3                 | 4096                       | 3000          | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 | (12)                               | 7.3                  |                            |               | 1620.0               | 3240.0          | 2356.4       |
|            | 8                     | 297                 | 12                                 | 21.4                 |                            |               | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 | (8)                                | 11.0                 |                            |               | 1082.5               | 2165.0          | 1574.6       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 830.5                | 1661.0          | 1208.0       |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1620.0               | 3240.0          | 2356.4       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 1082.5               | 2165.0          | 1574.6       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 14.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 41.7                 | 4096                       | 3000 | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 | (12)                               | 21.7                 |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 891                 | 12                                 | 61.0                 |                            |      | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 | (8)                                | 32.0                 |                            |      | 369.5                | 739.0           | 537.5        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 283.0                | 566.0           | 411.7        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 548.0                | 1096.0          | 797.1        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 192.5                | 385.0           | 280.0        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 369.5                | 739.0           | 537.5        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.8 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 21.4                 | 4096                       | 3000 | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               | (12)                               | 11.0                 |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 445.5               | 12                                 | 31.6                 |                            |      | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               | (8)                                | 16.4                 |                            |      | 725.5                | 1451.0          | 1055.3       |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 556.0                | 1112.0          | 808.8        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 1082.5               | 2165.0          | 1574.6       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 374.0                | 748.0           | 544.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 725.5                | 1451.0          | 1055.3       |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.4 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX535-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 61.8                 | 4096                       | 3000 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 46.2                 |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                                    | 23.4                 |                            |      | 508.5                | 1017.0          | 739.7        |
|            | 8                            | 4752                  | 12 (8)                             | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 34.5                 |                            |      | 343.0                | 686.0           | 499.0        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 255.5                | 511.0           | 371.7        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 508.5                | 1017.0          | 739.7        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 343.0                | 686.0           | 499.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.8 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX535-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 61.8                 | 4096                       | 3000 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 46.0                 |                            |      | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                                    | 23.2                 |                            |      | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                                    | 11.7                 |                            |      | 1017.0               | 2034.0          | 1479.3       |
|            | 8                            | 2376                  | 12 (8)                             | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 34.4                 |                            |      | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                                    | 17.3                 |                            |      | 685.5                | 1371.0          | 997.1        |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 256.5                | 513.0           | 373.1        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 511.0                | 1022.0          | 743.3        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 1017.0               | 2034.0          | 1479.3       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 343.5                | 687.0           | 499.7        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 685.5                | 1371.0          | 997.1        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.8 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX535-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 42.6                 | 4096                       | 3000 | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                                    | 23.1                 |                            |      | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                                    | 11.6                 |                            |      | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                                    | 5.8                  |                            |      | 2034.0               | 4068.0          | 2958.6       |
|            | 8                            | 1188                  | 12 (8)                             | 61.8                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 33.3                 |                            |      | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                                    | 17.3                 |                            |      | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                                    | 8.7                  |                            |      | 1370.5               | 2741.0          | 1993.5       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 277.0                | 554.0           | 403.0        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 513.0                | 1026.0          | 746.2        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 1022.0               | 2044.0          | 1486.6       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 2034.0               | 4068.0          | 2958.6       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 355.0                | 710.0           | 516.4        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 687.0                | 1374.0          | 999.3        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1370.5               | 2741.0          | 1993.5       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 42.6 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 9.1.6. IMX536-AAQJ

SLVS Output, Data rate: 594 Mbps/ch (IMX536-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 41.0                 | 2840                       | 2840          | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 | (12)                               | 21.4                 |                            |               | 585.5                | 1171.0          | 851.7        |
|            | 8                     | 594                 | 12                                 | 60.3                 |                            |               | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 | (8)                                | 31.8                 |                            |               | 393.0                | 786.0           | 571.7        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 585.5                | 1171.0          | 851.7        |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 393.0                | 786.0           | 571.7        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX536-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 21.1                 | 2840                       | 2840          | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 | (12)                               | 10.9                 |                            |               | 1149.0               | 2298.0          | 1671.3       |
|            | 8                     | 297                 | 12                                 | 31.3                 |                            |               | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 | (8)                                | 16.3                 |                            |               | 768.5                | 1537.0          | 1117.9       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1149.0               | 2298.0          | 1671.3       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 768.5                | 1537.0          | 1117.9       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.1 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 60.3                 | 2840                       | 2840 | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 | (12)                               | 31.8                 |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 891                 | 12                                 | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | 46.8                 |                            |      | 266.0                | 532.0           | 387.0        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 266.0                | 532.0           | 387.0        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.3 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 31.3                 | 2840                       | 2840 | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               | (12)                               | 16.3                 |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 445.5               | 12                                 | 46.2                 |                            |      | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               | (8)                                | 24.3                 |                            |      | 516.0                | 1032.0          | 750.6        |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 516.0                | 1032.0          | 750.6        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 31.3 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX536-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 65.2                 | 2840                       | 2840 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 34.8                 |                            |      | 359.0                | 718.0           | 522.2        |
|            | 8                            | 4752                  | 12 (8)                             | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 51.7                 |                            |      | 240.5                | 481.0           | 349.9        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 359.0                | 718.0           | 522.2        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 240.5                | 481.0           | 349.9        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 65.2 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX536-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 65.2                 | 2840                       | 2840 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 34.6                 |                            |      | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                                    | 17.5                 |                            |      | 718.0                | 1436.0          | 1044.4       |
|            | 8                            | 2376                  | 12 (8)                             | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 51.3                 |                            |      | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                                    | 26.1                 |                            |      | 480.5                | 961.0           | 699.0        |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 718.0                | 1436.0          | 1044.4       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 480.5                | 961.0           | 699.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 65.2 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX536-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 62.6                 | 2840                       | 2840 | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                                    | 34.4                 |                            |      | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                                    | 17.4                 |                            |      | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                                    | 8.7                  |                            |      | 1435.5               | 2871.0          | 2088.0       |
|            | 8                            | 1188                  | 12 (8)                             | 65.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 49.5                 |                            |      | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                                    | 25.8                 |                            |      | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                                    | 13.1                 |                            |      | 961.0                | 1922.0          | 1397.9       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1435.5               | 2871.0          | 2088.0       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 961.0                | 1922.0          | 1397.9       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 62.6 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 9.1.7. IMX536-AAMJ

SLVS Output, Data rate: 594 Mbps/ch (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 40.9                 | 2840                       | 2840          | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 | (12)                               | 21.4                 |                            |               | 585.5                | 1171.0          | 851.7        |
|            | 8                     | 594                 | 12                                 | 60.2                 |                            |               | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 | (8)                                | 31.8                 |                            |               | 393.0                | 786.0           | 571.7        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 304.5                | 609.0           | 443.0        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 585.5                | 1171.0          | 851.7        |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 205.5                | 411.0           | 299.0        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 393.0                | 786.0           | 571.7        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 40.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 21.0                 | 2840                       | 2840          | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 | (12)                               | 10.9                 |                            |               | 1149.0               | 2298.0          | 1671.3       |
|            | 8                     | 297                 | 12                                 | 31.3                 |                            |               | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 | (8)                                | 16.3                 |                            |               | 768.5                | 1537.0          | 1117.9       |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 595.0                | 1190.0          | 865.5        |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1149.0               | 2298.0          | 1671.3       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 399.0                | 798.0           | 580.4        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 768.5                | 1537.0          | 1117.9       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.0 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX536-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 60.2                 | 2840                       | 2840 | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 | (12)                               | 31.8                 |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 891                 | 12                                 | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | 46.8                 |                            |      | 266.0                | 532.0           | 387.0        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 205.5                | 411.0           | 299.0        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 393.0                | 786.0           | 571.7        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 266.0                | 532.0           | 387.0        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.2 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX536-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 31.3                 | 2840                       | 2840 | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               | (12)                               | 16.3                 |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 445.5               | 12                                 | 46.1                 |                            |      | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               | (8)                                | 24.3                 |                            |      | 516.0                | 1032.0          | 750.6        |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 399.0                | 798.0           | 580.4        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 768.5                | 1537.0          | 1117.9       |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 269.5                | 539.0           | 392.0        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 516.0                | 1032.0          | 750.6        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 31.3 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 65.1                 | 2840                       | 2840 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 34.7                 |                            |      | 359.0                | 718.0           | 522.2        |
|            | 8                            | 4752                  | 12 (8)                             | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 51.6                 |                            |      | 240.5                | 481.0           | 349.9        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 359.0                | 718.0           | 522.2        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 240.5                | 481.0           | 349.9        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 65.1 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 65.1                 | 2840                       | 2840 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 34.5                 |                            |      | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                                    | 17.4                 |                            |      | 718.0                | 1436.0          | 1044.4       |
|            | 8                            | 2376                  | 12 (8)                             | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 65.1                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 51.2                 |                            |      | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                                    | 26.0                 |                            |      | 480.5                | 961.0           | 699.0        |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 361.5                | 723.0           | 525.9        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 718.0                | 1436.0          | 1044.4       |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 242.5                | 485.0           | 352.8        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 480.5                | 961.0           | 699.0        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 65.1 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX536-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 62.5                 | 2840                       | 2840          | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                                    | 34.4                 |                            |               | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                                    | 17.3                 |                            |               | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                                    | 8.7                  |                            |               | 1435.5               | 2871.0          | 2088.0       |
|            | 8                            | 1188                  | 12 (8)                             | 65.1                 |                            |               | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 49.4                 |                            |               | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                                    | 25.8                 |                            |               | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                                    | 13.0                 |                            |               | 961.0                | 1922.0          | 1397.9       |
| ROI        | 8                            | 1188                  | 12 (12)                            | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 198.0                | 396.0           | 288.0        |
|            | 4                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 363.0                | 726.0           | 528.0        |
|            | 2                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 722.5                | 1445.0          | 1051.0       |
|            | 1                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 1435.5               | 2871.0          | 2088.0       |
|            | 8                            | 1188                  | 12 (8)                             | <sup>*2</sup>        |                            |               | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 251.5                | 503.0           | 365.9        |
|            | 2                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 485.0                | 970.0           | 705.5        |
|            | 1                            | 1188                  |                                    | <sup>*2</sup>        |                            |               | 961.0                | 1922.0          | 1397.9       |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 62.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

## 9.1.8. IMX537-AAQJ

SLVS Output, Data rate: 594 Mbps/ch (IMX537-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 63.0                 | 2448                       | 2048          | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 | (12)                               | 33.2                 |                            |               | 513.5                | 1027.0          | 747.0        |
|            | 8                     | 594                 | 12                                 | 88.4                 |                            |               | 190.0                | 380.0           | 276.4        |
|            | 4                     | 594                 | (8)                                | 49.2                 |                            |               | 345.0                | 690.0           | 501.9        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 513.5                | 1027.0          | 747.0        |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 190.0                | 380.0           | 276.4        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 345.0                | 690.0           | 501.9        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 63.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX537-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 32.6                 | 2448                       | 2048          | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 | (12)                               | 17.0                 |                            |               | 1005.0               | 2010.0          | 1461.9       |
|            | 8                     | 297                 | 12                                 | 48.3                 |                            |               | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 | (8)                                | 25.4                 |                            |               | 672.5                | 1345.0          | 978.2        |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1005.0               | 2010.0          | 1461.9       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 672.5                | 1345.0          | 978.2        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.6 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 88.4                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (12)                               | 49.2                 |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 891                 | 12                                 | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | 72.0                 |                            |      | 234.0                | 468.0           | 340.4        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 234.0                | 468.0           | 340.4        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 88.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 48.3                 | 2448                       | 2048 | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               | (12)                               | 25.4                 |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 445.5               | 12                                 | 71.0                 |                            |      | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               | (8)                                | 37.7                 |                            |      | 452.0                | 904.0           | 657.5        |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 452.0                | 904.0           | 657.5        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 48.3 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX537-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 88.4                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 54.2                 |                            |      | 313.0                | 626.0           | 455.3        |
|            | 8                            | 4752                  | 12 (8)                             | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 80.3                 |                            |      | 209.5                | 419.0           | 304.8        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 313.0                | 626.0           | 455.3        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 209.5                | 419.0           | 304.8        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.4 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX537-AAQJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 88.4                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 53.7                 |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                                    | 27.3                 |                            |      | 626.0                | 1252.0          | 910.6        |
|            | 8                            | 2376                  | 12 (8)                             | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 79.5                 |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                                    | 40.6                 |                            |      | 419.0                | 838.0           | 609.5        |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 626.0                | 1252.0          | 910.6        |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 419.0                | 838.0           | 609.5        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.4 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX537-AAQJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 88.4                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 53.5                 |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                                    | 27.1                 |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                                    | 13.7                 |                            |      | 1251.5               | 2503.0          | 1820.4       |
|            | 8                            | 1188                  | 12 (8)                             | 88.4                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 76.7                 |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                                    | 40.2                 |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                                    | 20.4                 |                            |      | 838.0                | 1676.0          | 1219.0       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1251.5               | 2503.0          | 1820.4       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 838.0                | 1676.0          | 1219.0       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.4 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

### 9.1.9. IMX537-AAMJ

SLVS Output, Data rate: 594 Mbps/ch (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 594                 | 12                                 | 62.9                 | 2448                       | 2048          | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 | (12)                               | 33.2                 |                            |               | 513.5                | 1027.0          | 747.0        |
|            | 8                     | 594                 | 12                                 | 88.2                 |                            |               | 190.0                | 380.0           | 276.4        |
|            | 4                     | 594                 | (8)                                | 49.1                 |                            |               | 345.0                | 690.0           | 501.9        |
| ROI        | 8                     | 594                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 268.5                | 537.0           | 390.6        |
|            | 4                     | 594                 | (12)                               | <sup>*2</sup>        |                            |               | 513.5                | 1027.0          | 747.0        |
|            | 8                     | 594                 | 12                                 | <sup>*2</sup>        |                            |               | 190.0                | 380.0           | 276.4        |
|            | 4                     | 594                 | (8)                                | <sup>*2</sup>        |                            |               | 345.0                | 690.0           | 501.9        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 62.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 297 Mbps/ch (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |               | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|---------------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V             | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 297                 | 12                                 | 32.6                 | 2448                       | 2048          | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 | (12)                               | 17.0                 |                            |               | 1005.0               | 2010.0          | 1461.9       |
|            | 8                     | 297                 | 12                                 | 48.2                 |                            |               | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 | (8)                                | 25.3                 |                            |               | 672.5                | 1345.0          | 978.2        |
| ROI        | 8                     | 297                 | 12                                 | <sup>*2</sup>        | <sup>*3</sup>              | <sup>*3</sup> | 523.0                | 1046.0          | 760.8        |
|            | 4                     | 297                 | (12)                               | <sup>*2</sup>        |                            |               | 1005.0               | 2010.0          | 1461.9       |
|            | 8                     | 297                 | 12                                 | <sup>*2</sup>        |                            |               | 351.0                | 702.0           | 510.6        |
|            | 4                     | 297                 | (8)                                | <sup>*2</sup>        |                            |               | 672.5                | 1345.0          | 978.2        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.6 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.

<sup>\*3</sup> Designated cropping area (ROI)

SLVS Output, Data rate: 891 Mbps/ch (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 891                 | 12                                 | 88.2                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (12)                               | 49.1                 |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 891                 | 12                                 | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | 71.9                 |                            |      | 234.0                | 468.0           | 340.4        |
| ROI        | 8                     | 891                 | 12                                 | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (12)                               | *2                   |                            |      | 345.0                | 690.0           | 501.9        |
|            | 8                     | 891                 | 12                                 | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                     | 891                 | (8)                                | *2                   |                            |      | 234.0                | 468.0           | 340.4        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 88.2 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS Output, Data rate: 445.5 Mbps/ch (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 3h

| Drive mode | SLVS ch <sup>*1</sup> | Data rate [Mbps/ch] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|-----------------------|---------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                       |                     |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                     | 445.5               | 12                                 | 48.2                 | 2448                       | 2048 | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               | (12)                               | 25.3                 |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 445.5               | 12                                 | 70.9                 |                            |      | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               | (8)                                | 37.6                 |                            |      | 452.0                | 904.0           | 657.5        |
| ROI        | 8                     | 445.5               | 12                                 | *2                   | *3                         | *3   | 351.0                | 702.0           | 510.6        |
|            | 4                     | 445.5               | (12)                               | *2                   |                            |      | 672.5                | 1345.0          | 978.2        |
|            | 8                     | 445.5               | 12                                 | *2                   |                            |      | 237.5                | 475.0           | 345.5        |
|            | 4                     | 445.5               | (8)                                | *2                   |                            |      | 452.0                | 904.0           | 657.5        |

\*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 48.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

SLVS – EC output, Data rate: 4752 Mbps/Lane (IMX537-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 4752                  | 12 (12)                            | 88.2                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 54.1                 |                            |      | 313.0                | 626.0           | 455.3        |
|            | 8                            | 4752                  | 12 (8)                             | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | 80.1                 |                            |      | 209.5                | 419.0           | 304.8        |
| ROI        | 8                            | 4752                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 313.0                | 626.0           | 455.3        |
|            | 8                            | 4752                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 4752                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 1                            | 4752                  |                                    | *2                   |                            |      | 209.5                | 419.0           | 304.8        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.2 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 2376 Mbps/Lane (IMX537-AAMJ)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 2376                  | 12 (12)                            | 88.2                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 53.6                 |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                                    | 27.2                 |                            |      | 626.0                | 1252.0          | 910.6        |
|            | 8                            | 2376                  | 12 (8)                             | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | 79.4                 |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                                    | 40.5                 |                            |      | 419.0                | 838.0           | 609.5        |
| ROI        | 8                            | 2376                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 315.5                | 631.0           | 459.0        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 626.0                | 1252.0          | 910.6        |
|            | 8                            | 2376                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 2376                  |                                    | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 2                            | 2376                  |                                    | *2                   |                            |      | 211.5                | 423.0           | 307.7        |
|            | 1                            | 2376                  |                                    | *2                   |                            |      | 419.0                | 838.0           | 609.5        |

<sup>\*1</sup> The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.2 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

<sup>\*2</sup> See the section of "ROI mode" in the data sheet.<sup>\*3</sup> Designated cropping area (ROI)

SLVS – EC output, Data rate: 1188 Mbps/Lane (IMX537-AAMJ)  
FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

| Drive mode | SLVS – EC Lane <sup>*1</sup> | Data rate [Mbps/Lane] | A/D conversion (output conversion) | Frame rate [frame/s] | Number of recording pixels |      | Number of INCK in 1H |                 |              |
|------------|------------------------------|-----------------------|------------------------------------|----------------------|----------------------------|------|----------------------|-----------------|--------------|
|            |                              |                       |                                    |                      | H                          | V    | INCK: 37.125 MHz     | INCK: 74.25 MHz | INCK: 54 MHz |
| All pixel  | 8                            | 1188                  | 12 (12)                            | 88.2                 | 2448                       | 2048 | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 53.4                 |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                                    | 27.0                 |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                                    | 13.6                 |                            |      | 1251.5               | 2503.0          | 1820.4       |
|            | 8                            | 1188                  | 12 (8)                             | 88.2                 |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | 76.6                 |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                                    | 40.1                 |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                                    | 20.4                 |                            |      | 838.0                | 1676.0          | 1219.0       |
| ROI        | 8                            | 1188                  | 12 (12)                            | *2                   | *3                         | *3   | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 317.0                | 634.0           | 461.1        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 630.5                | 1261.0          | 917.1        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 1251.5               | 2503.0          | 1820.4       |
|            | 8                            | 1188                  | 12 (8)                             | *2                   |                            |      | 190.0                | 380.0           | 276.4        |
|            | 4                            | 1188                  |                                    | *2                   |                            |      | 219.5                | 439.0           | 319.3        |
|            | 2                            | 1188                  |                                    | *2                   |                            |      | 423.0                | 846.0           | 615.3        |
|            | 1                            | 1188                  |                                    | *2                   |                            |      | 838.0                | 1676.0          | 1219.0       |

\*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 88.2 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

\*2 See the section of "ROI mode" in the data sheet.

\*3 Designated cropping area (ROI)

## 10. Drive Mode Setting (with built-in combination)

Please refer to the product specification for register setup other than those lists.

### 10.1. Drive Mode Setting

#### 10.1.1. IMX530

##### 10.1.1.1. All-pixel scan

SLVS Output (IMX530)

| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |
|               |       |               |               | 8 ch                          | 4 ch              | 8 ch                         | 4 ch              |                         |
|               |       |               |               | 14.7<br>[frame/s]             | 7.5<br>[frame/s]  | 21.9<br>[frame/s]            | 11.2<br>[frame/s] | FREQ = 0h               |
|               |       |               |               | 7.4<br>[frame/s]              | 3.8<br>[frame/s]  | 11.1<br>[frame/s]            | 5.7<br>[frame/s]  | FREQ = 1h               |
|               |       |               |               | 21.9<br>[frame/s]             | 11.2<br>[frame/s] | 30.2<br>[frame/s]            | 15.9<br>[frame/s] | FREQ = 2h               |
|               |       |               |               | 11.1<br>[frame/s]             | 5.7<br>[frame/s]  | 15.5<br>[frame/s]            | 8.1<br>[frame/s]  | FREQ = 3h               |
|               |       |               |               | 594 Mbps/ch                   |                   |                              |                   | FREQ = 0h               |
|               |       |               |               | 297 Mbps/ch                   |                   |                              |                   | FREQ = 1h               |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |
| 445.5 Mbps/ch |       |               |               | FREQ = 3h                     |                   |                              |                   |                         |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |
| D4h           | [7:0] | VMAX          | 12ECh         | 1264h                         | 1260h             | 126Ch                        | 1264h             | FREQ = 0h               |
| D5h           | [7:0] |               |               | 1260h                         | 125Eh             | 1264h                        | 1260h             | FREQ = 1h               |
| D6h           | [7:0] |               |               | 126Ch                         | 1264h             | 1270h                        | 1268h             | FREQ = 2h               |
|               |       |               |               | 1264h                         | 1260h             | 1264h                        | 1262h             | FREQ = 3h               |
| D8h           | [7:0] | HMAX          | 0A6h          | 42Bh                          | 826h              | 2CCh                         | 574h              | FREQ = 0h               |
| D9h           | [7:0] |               |               | 83Fh                          | 102Ch             | 584h                         | ACDh              | FREQ = 1h               |
|               |       |               |               | 2CCh                          | 574h              | 208h                         | 3DEh              | FREQ = 2h               |
|               |       |               |               | 584h                          | ACDh              | 3F8h                         | 799h              | FREQ = 3h               |
| E2h           | [7:0] | GMRWT         | 0Ch           | 02h                           | 02h               | 04h                          | 02h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 04h                           | 02h               | 04h                          | 02h               | FREQ = 2h               |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 3h               |
| E3h           | [7:0] | GMTWT         | 48h           | 04h                           | 02h               | 06h                          | 04h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 04h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 06h                           | 04h               | 08h                          | 06h               | FREQ = 2h               |
|               |       |               |               | 04h                           | 02h               | 04h                          | 04h               | FREQ = 3h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |
| E6h           | [7:0] | GSDLY         | 34h           | 06h                           | 04h               | 0Ah                          | 06h               | FREQ = 0h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 1h               |
|               |       |               |               | 0Ah                           | 06h               | 0Ch                          | 08h               | FREQ = 2h               |
|               |       |               |               | 06h                           | 04h               | 06h                          | 04h               | FREQ = 3h               |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |                   |                              |                   |                         |
| 3Dh           | [0]   |               |               |                               |                   |                              |                   |                         |
| 3Eh           | [7:2] |               |               | 16h                           | 0Ch               |                              |                   |                         |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |
| 21h           | [7:0] |               | 0Dh           | 19h                           |                   |                              |                   |                         |
| 22h           | [7:0] |               | B1h           | B2h                           |                   |                              |                   |                         |
| 46h           | [7:0] |               | 43h           | 86h                           |                   |                              |                   |                         |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |
| E8h           | [7:0] |               | 2Dh           | 09h                           |                   |                              |                   |                         |

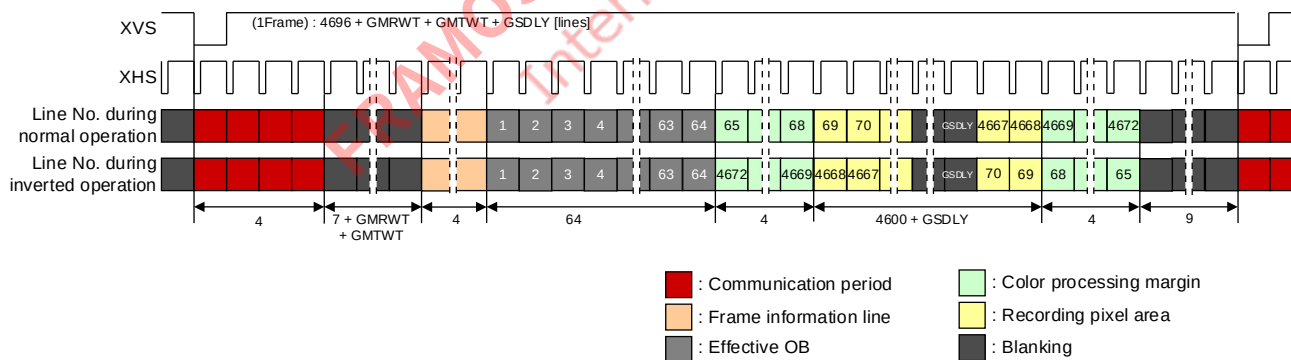
Diagram illustrating the recording area and vertical scan direction (Normal). The diagram shows a vertical sequence of fields:

- Communication period: 4
- Vertical blanking period: 7 + GMRWT + GMTWT
- Frame information line: 4
- Vertical effective OB: 64
- Effective margin for color processing: 4
- Recording pixel area: 5320
- Effective margin for color processing: 4
- Vertical blanking: 9

The total height is indicated as  $4696 [H] + GMRWT + GMTWT + GSDLY$ .

The diagram also shows the horizontal scan direction (Normal) and the vertical scan direction (Normal).

The AD lines for low gain side and high gain side are synthesized and output as one line.

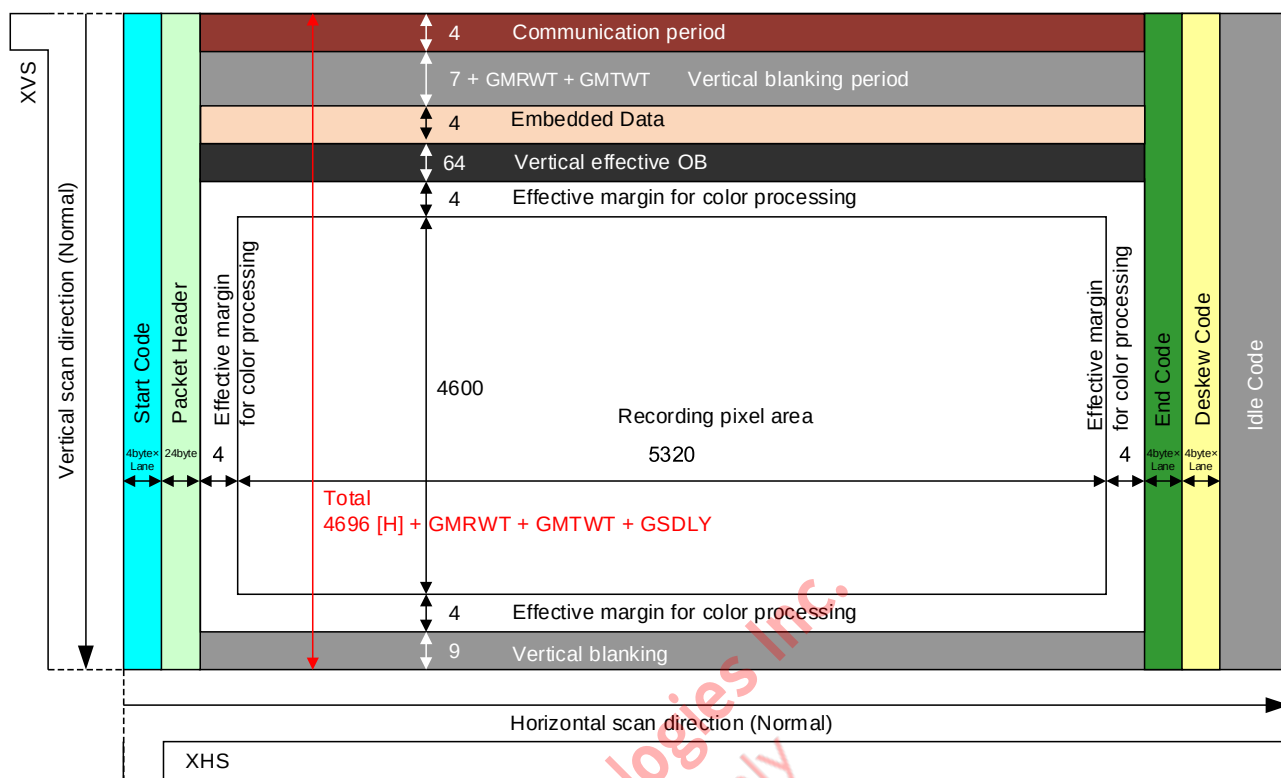


## 169

## SLVS – EC output (IMX530)

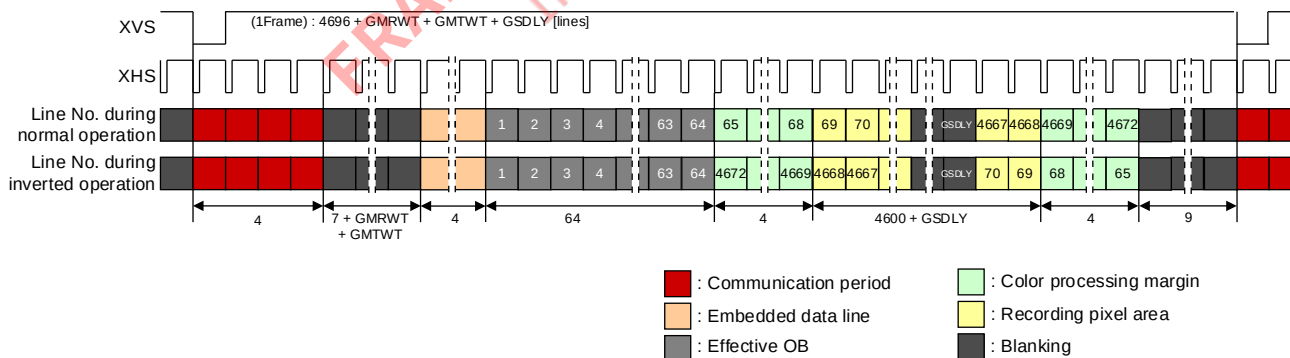
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                   |                   |                              |                   |                   |                   | Remarks                 |           |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|-------------------|-------------------|------------------------------|-------------------|-------------------|-------------------|-------------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   |                   |                   | AD = 12 bit / Output = 8 bit |                   |                   |                   |                         |           |
|               |       |               |               | 8 lane                        | 4 lane            | 2 lane            | 1 lane            | 8 lane                       | 4 lane            | 2 lane            | 1 lane            |                         |           |
|               |       |               |               | 37.5<br>[frame/s]             | 37.5<br>[frame/s] | 24.0<br>[frame/s] | 12.0<br>[frame/s] | 37.5<br>[frame/s]            | 37.5<br>[frame/s] | 35.1<br>[frame/s] | 17.9<br>[frame/s] |                         | FREQ = 0h |
|               |       |               |               | 37.5<br>[frame/s]             | 23.7<br>[frame/s] | 12.0<br>[frame/s] | 6.0<br>[frame/s]  | 37.5<br>[frame/s]            | 23.8<br>[frame/s] | 12.0<br>[frame/s] | 6.0<br>[frame/s]  |                         | FREQ = 1h |
|               |       |               |               | 22.3<br>[frame/s]             | 11.9<br>[frame/s] | 6.0<br>[frame/s]  | 3.0<br>[frame/s]  | 32.7<br>[frame/s]            | 17.4<br>[frame/s] | 8.8<br>[frame/s]  | 4.4<br>[frame/s]  |                         | FREQ = 2h |
|               |       |               |               | 4752 Mbps/lane                |                   |                   |                   |                              |                   |                   |                   |                         | FREQ = 0h |
|               |       |               |               | 2376 Mbps/lane                |                   |                   |                   |                              |                   |                   |                   |                         | FREQ = 1h |
|               |       |               |               | 1188 Mbps /lane               |                   |                   |                   |                              |                   |                   |                   |                         | FREQ = 2h |
| Chip ID = 02h |       |               |               |                               |                   |                   |                   |                              |                   |                   |                   |                         |           |
| D4h           | [7:0] | VMAX          | 12ECh         | 1278h                         | 1278h             | 126Eh             | 1264h             | 1278h                        | 1278h             | 1274h             | 1268h             | FREQ = 0h               |           |
| D5h           | [7:0] |               |               | 1278h                         | 126Eh             | 1264h             | 1260h             | 1278h                        | 126Eh             | 1264h             | 1260h             | FREQ = 1h               |           |
| D6h           | [7:0] |               |               | 126Ch                         | 1264h             | 1260h             | 125Eh             | 1274h                        | 1268h             | 1262h             | 125Eh             | FREQ = 2h               |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 1A2h                          | 1A2h              | 28Fh              | 519h              | 1A2h                         | 1A2h              | 1BFh              | 370h              | FREQ = 0h               |           |
| D9h           | [7:0] |               |               | 1A2h                          | 297h              | 51Eh              | A31h              | 1A2h                         | 295h              | 51Eh              | A31h              | FREQ = 1h               |           |
|               |       |               |               | 2C0h                          | 52Ah              | A3Bh              | 1462h             | 1E0h                         | 389h              | 6EEh              | DBFh              | FREQ = 2h               |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 06h                           | 06h               | 04h               | 02h               | 06h                          | 06h               | 04h               | 02h               | FREQ = 0h               |           |
|               |       |               |               | 06h                           | 04h               | 02h               | 02h               | 06h                          | 04h               | 02h               | 02h               | FREQ = 1h               |           |
|               |       |               |               | 04h                           | 02h               | 02h               | 02h               | 04h                          | 02h               | 02h               | 02h               | FREQ = 2h               |           |
| E3h           | [7:0] | GMTWT         | 48h           | 0Ah                           | 0Ah               | 08h               | 04h               | 0Ah                          | 0Ah               | 0Ah               | 06h               | FREQ = 0h               |           |
|               |       |               |               | 0Ah                           | 08h               | 04h               | 02h               | 0Ah                          | 08h               | 04h               | 02h               | FREQ = 1h               |           |
|               |       |               |               | 06h                           | 04h               | 02h               | 02h               | 0Ah                          | 06h               | 04h               | 02h               | FREQ = 2h               |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                   |                   |                              |                   |                   |                   |                         |           |
| E6h           | [7:0] | GSDLY         | 34h           | 10h                           | 10h               | 0Ah               | 06h               | 10h                          | 10h               | 0Eh               | 08h               | FREQ = 0h               |           |
|               |       |               |               | 10h                           | 0Ah               | 06h               | 04h               | 10h                          | 0Ah               | 06h               | 04h               | FREQ = 1h               |           |
|               |       |               |               | 0Ah                           | 06h               | 04h               | 02h               | 0Eh                          | 08h               | 04h               | 02h               | FREQ = 2h               |           |
| Chip ID = 04h |       |               |               |                               |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                   |                   |                              |                   |                   |                   |                         |           |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                   |                   |                              |                   |                   |                   | 1h: 12 bit              |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 3Dh           | [0]   |               |               |                               |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 3Eh           | [7:2] |               |               | 0Ch                           |                   |                   |                   |                              |                   |                   |                   |                         |           |
| Chip ID = 06h |       |               |               |                               |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   |                   |                   | 2h                           |                   |                   |                   | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h |       |               |               |                               |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 21h           | [7:0] |               | 0Dh           | 19h                           |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 22h           | [7:0] |               | B1h           | B2h                           |                   |                   |                   |                              |                   |                   |                   |                         |           |
| 46h           | [7:0] |               | 43h           | 86h                           |                   |                   |                   |                              |                   |                   |                   |                         |           |
| Chip ID = 08h |       |               |               |                               |                   |                   |                   |                              |                   |                   |                   |                         |           |
| E8h           | [7:0] |               | 2Dh           | 09h                           |                   |                   |                   |                              |                   |                   |                   |                         |           |

(IMX530)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.1.2.ROI, Overlap ROI

## SLVS Output (IMX530)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |  |  |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|--|--|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |  |  |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |  |  |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |  |  |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         |  |  | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                         | *2              | *3                           | *4              |                         |  |  |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 42Bh                                                                                                       | 826h            | 2CCh                         | 574h            | FREQ = 0h               |  |  |           |
| D9h           | [7:0] |               |               | 83Fh                                                                                                       | 102Ch           | 584h                         | ACDh            | FREQ = 1h               |  |  |           |
|               |       |               |               | 2CCh                                                                                                       | 574h            | 208h                         | 3DEh            | FREQ = 2h               |  |  |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 584h                                                                                                       | ACDh            | 3F8h                         | 799h            | FREQ = 3h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 2h               |  |  |           |
| E3h           | [7:0] | GMTWT         | 48h           | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 3h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 08h                          | 06h             | FREQ = 2h               |  |  |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        | 04h             | 04h                          | 04h             | FREQ = 3h               |  |  |           |
| E6h           | [7:0] | GSDLY         | 34h           | 06h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Ch                          | 08h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 06h                          | 04h             | FREQ = 3h               |  |  |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |  |  |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |  |  |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |  |  |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |  |  |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |  |  |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 21h           | [7:0] |               | 0Dh           | 19h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 22h           | [7:0] |               | B1h           | B2h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 46h           | [7:0] |               | 43h           | 86h                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                        |                 |                              |                 |                         |  |  |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

## SLVS – EC output (IMX530)

| Address<br>s     | bit   | Register name | Initial<br>Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|------------------|-------|---------------|------------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                  |       |               |                  | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                  |       |               |                  | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                  |       |               |                  | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|                  |       |               |                  | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                  |       |               |                  | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           |                         |
| 2376 Mbps/lane   |       |               |                  |                                                                                                               |           |           |           |                              |           |           | FREQ = 1h |                         |
| 1188 Mbps /lane  |       |               |                  |                                                                                                               |           |           |           |                              |           |           | FREQ = 2h |                         |
| Chip ID = 02h    |       |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h              | [7:0] | VMAX          | 12ECh            | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h              | [7:0] |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h              | [7:0] |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h              | [7:0] | HMAX          | 0A6h             | 1A2h                                                                                                          | 1A2h      | 28Fh      | 519h      | 1A2h                         | 1A2h      | 1BFh      | 370h      | FREQ = 0h               |
| D9h              | [7:0] |               |                  | 1A2h                                                                                                          | 297h      | 51Eh      | A31h      | 1A2h                         | 295h      | 51Eh      | A31h      | FREQ = 1h               |
|                  |       |               |                  | 2C0h                                                                                                          | 52Ah      | A3Bh      | 1462h     | 1E0h                         | 389h      | 6EEh      | DBFh      | FREQ = 2h               |
| E2h              | [7:0] | GMRWT         | 0CCh             | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 0h               |
|                  |       |               |                  | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 1h               |
|                  |       |               |                  | 04h                                                                                                           | 02h       | 02h       | 02h       | 04h                          | 02h       | 02h       | 02h       | FREQ = 2h               |
| E3h              | [7:0] | GMTWT         | 48h              | 0Ah                                                                                                           | 0Ah       | 08h       | 04h       | 0Ah                          | 0Ah       | 0Ah       | 06h       | FREQ = 0h               |
|                  |       |               |                  | 0Ah                                                                                                           | 08h       | 04h       | 02h       | 0Ah                          | 08h       | 04h       | 02h       | FREQ = 1h               |
|                  |       |               |                  | 06h                                                                                                           | 04h       | 02h       | 02h       | 0Ah                          | 06h       | 04h       | 02h       | FREQ = 2h               |
| E5h              | [7:0] | GAINDLY       | 04h              | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h              | [7:0] | GSDLY         | 34h              | 10h                                                                                                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 0Eh       | 08h       | FREQ = 0h               |
|                  |       |               |                  | 10h                                                                                                           | 0Ah       | 06h       | 04h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 1h               |
|                  |       |               |                  | 0Ah                                                                                                           | 06h       | 04h       | 02h       | 0Eh                          | 08h       | 04h       | 02h       | FREQ = 2h               |
| Chip ID = 03h    |       |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h<br>to<br>05h |       | ROI setting   |                  | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h<br>to<br>5Fh |       | ROI setting   |                  | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h    |       |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h              | [1:0] | AD_SEL        | 0h               | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|                  | [5:4] | ADBIT         | 0h               | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch              | [7:1] | LLBLANK       | 18h              | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh              | [0]   |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh              | [7:2] |               |                  | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h    |       |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h              | [1:0] | ODBIT         | 0h               | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h    |       |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h              | [7:0] |               | 0Dh              | 19h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h              | [7:0] |               | B1h              | B2h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h              | [7:0] |               | 43h              | 86h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h    |       |               |                  |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h              | [7:0] |               | 2Dh              | 09h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

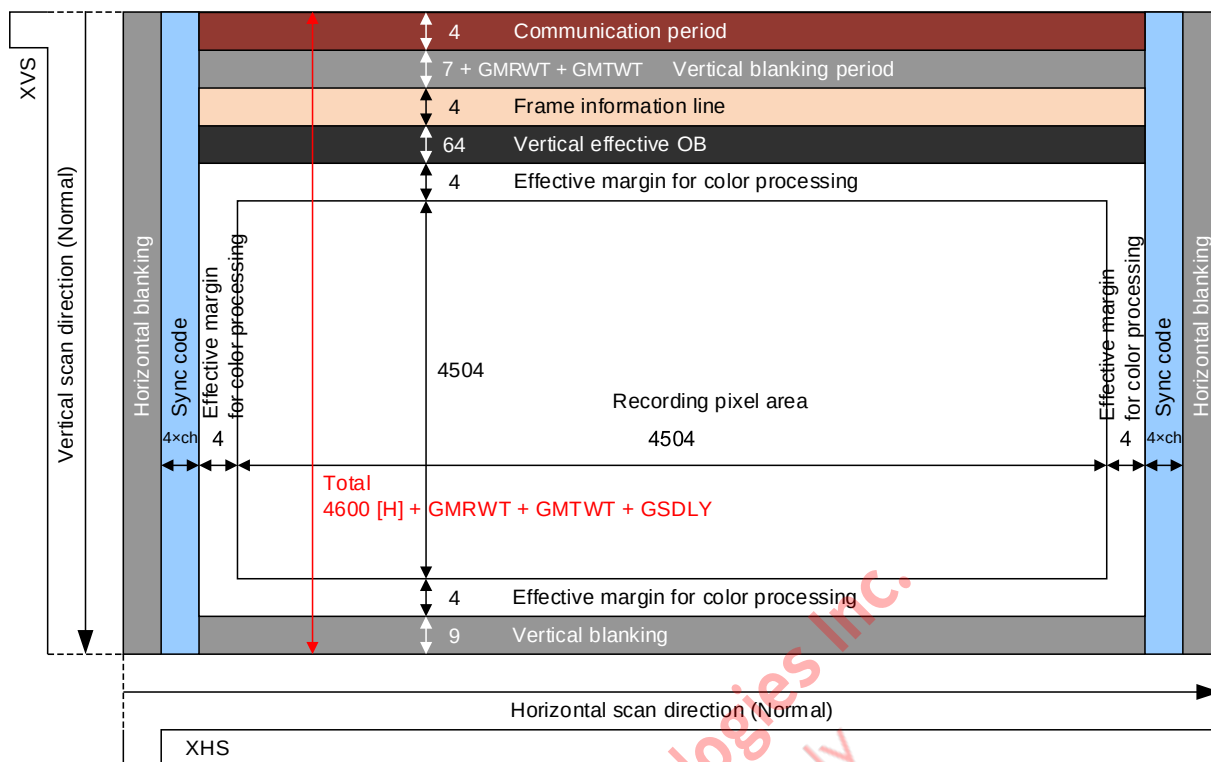
## 10.1.2. IMX531

## 10.1.2.1. All-pixel scan

SLVS Output (IMX531)

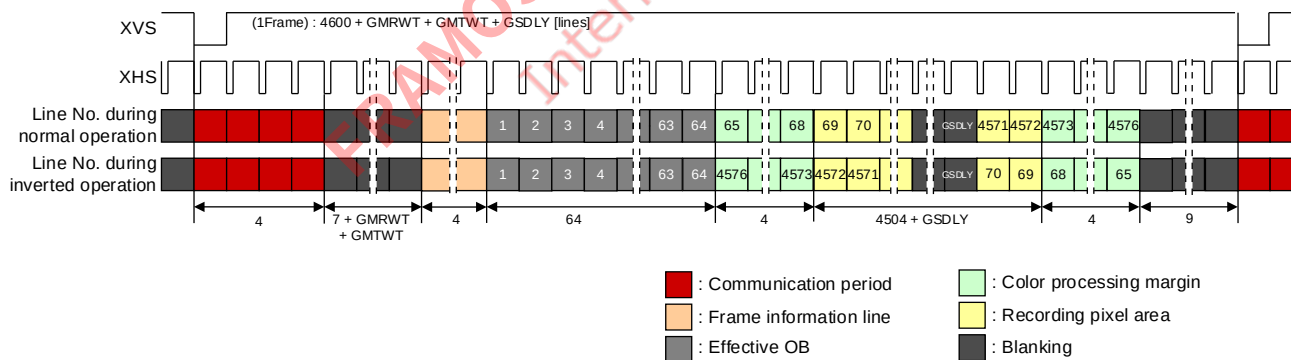
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |
|               |       |               |               | 8 ch                          | 8 ch              | 4 ch                         | 4 ch              |                         |
|               |       |               |               | 17.5<br>[frame/s]             | 9.0<br>[frame/s]  | 26.1<br>[frame/s]            | 13.5<br>[frame/s] | FREQ = 0h               |
|               |       |               |               | 8.9<br>[frame/s]              | 4.5<br>[frame/s]  | 13.3<br>[frame/s]            | 6.8<br>[frame/s]  | FREQ = 1h               |
|               |       |               |               | 26.1<br>[frame/s]             | 13.5<br>[frame/s] | 35.4<br>[frame/s]            | 18.8<br>[frame/s] | FREQ = 2h               |
|               |       |               |               | 13.3<br>[frame/s]             | 6.8<br>[frame/s]  | 18.2<br>[frame/s]            | 9.6<br>[frame/s]  | FREQ = 3h               |
|               |       |               |               | 594 Mbps/ch                   |                   |                              |                   | FREQ = 0h               |
|               |       |               |               | 297 Mbps/ch                   |                   |                              |                   | FREQ = 1h               |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |
| 445.5 Mbps/ch |       |               |               | FREQ = 3h                     |                   |                              |                   |                         |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |
| D4h           | [7:0] | VMAX          | 12ECh         | 1208h                         | 1202h             | 120Eh                        | 1204h             | FREQ = 0h               |
| D5h           | [7:0] |               |               | 1202h                         | 11FEh             | 1204h                        | 1200h             | FREQ = 1h               |
| D6h           | [7:0] |               |               | 120Eh                         | 1204h             | 1214h                        | 1208h             | FREQ = 2h               |
|               |       |               |               | 1204h                         | 1200h             | 1208h                        | 1202h             | FREQ = 3h               |
| D8h           | [7:0] | HMAX          | 0A6h          | 392h                          | 6F4h              | 266h                         | 4A8h              | FREQ = 0h               |
| D9h           | [7:0] |               |               | 70Dh                          | DC8h              | 4B8h                         | 935h              | FREQ = 1h               |
|               |       |               |               | 266h                          | 4A8h              | 1C4h                         | 356h              | FREQ = 2h               |
|               |       |               |               | 4B8h                          | 935h              | 370h                         | 689h              | FREQ = 3h               |
| E2h           | [7:0] | GMRWT         | 0Ch           | 02h                           | 02h               | 04h                          | 02h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 04h                           | 02h               | 04h                          | 02h               | FREQ = 2h               |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 3h               |
| E3h           | [7:0] | GMTWT         | 48h           | 06h                           | 04h               | 08h                          | 04h               | FREQ = 0h               |
|               |       |               |               | 04h                           | 02h               | 04h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 08h                           | 04h               | 0Ah                          | 06h               | FREQ = 2h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 3h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |
| E6h           | [7:0] | GSDLY         | 34h           | 08h                           | 04h               | 0Ah                          | 06h               | FREQ = 0h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 1h               |
|               |       |               |               | 0Ah                           | 06h               | 0Eh                          | 08h               | FREQ = 2h               |
|               |       |               |               | 06h                           | 04h               | 08h                          | 04h               | FREQ = 3h               |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |                   |                              |                   |                         |
| 3Dh           | [0]   |               |               |                               |                   |                              |                   |                         |
| 3Eh           | [7:2] |               |               | 0Ch                           |                   |                              |                   |                         |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |
| 21h           | [7:0] |               | 0Dh           | 91h                           |                   |                              |                   |                         |
| 22h           | [7:0] |               | B1h           | B1h                           |                   |                              |                   |                         |
| 46h           | [7:0] |               | 43h           | 64h                           |                   |                              |                   |                         |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |
| E8h           | [7:0] |               | 2Dh           | 09h                           |                   |                              |                   |                         |

(IMX531)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX531)

| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |           | Remarks                 |           |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|-----------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |           |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |           |
|                 |       |               |               | 38.3                          | 38.3      | 28.8      | 14.5      | 38.3                         | 38.3      | 38.3      | 21.5      |                         | FREQ = 0h |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |           |
|                 |       |               |               | 38.3                          | 28.7      | 14.4      | 7.2       | 38.3                         | 28.7      | 14.4      | 7.2       |                         |           |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |           |
|                 |       |               |               | 26.6                          | 14.4      | 7.2       | 3.6       | 38.3                         | 20.7      | 10.7      | 5.3       |                         | FREQ = 2h |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |           |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |           |                         |           |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |           |                         |           |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |           |                         |           |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| D4h             | [7:0] | VMAX          | 12ECh         | 1218h                         | 1218h     | 1210h     | 1204h     | 1218h                        | 1218h     | 1218h     | 120Ch     | FREQ = 0h               |           |
| D5h             | [7:0] |               |               | 1218h                         | 1210h     | 1204h     | 1200h     | 1218h                        | 1210h     | 1204h     | 1200h     | FREQ = 1h               |           |
| D6h             | [7:0] |               |               | 1210h                         | 1204h     | 1200h     | 11FEh     | 1218h                        | 120Ah     | 1204h     | 1200h     | FREQ = 2h               |           |
| D8h             | [7:0] | HMAX          | 0A6h          | 1A2h                          | 1A2h      | 22Dh      | 455h      | 1A2h                         | 1A2h      | 1A2h      | 2EBh      | FREQ = 0h               |           |
| D9h             | [7:0] |               |               | 1A2h                          | 22Fh      | 45Ah      | 8AAh      | 1A2h                         | 22Fh      | 45Ah      | 8AAh      | FREQ = 1h               |           |
|                 |       |               |               | 25Ah                          | 45Eh      | 8B4h      | 1154h     | 1A2h                         | 306h      | 5DAh      | BABh      | FREQ = 2h               |           |
| E2h             | [7:0] | GMRWT         | 0Ch           | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |           |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 1h               |           |
|                 |       |               |               | 04h                           | 02h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |           |
| E3h             | [7:0] | GMTWT         | 48h           | 0Ah                           | 0Ah       | 08h       | 04h       | 0Ah                          | 0Ah       | 0Ah       | 06h       | FREQ = 0h               |           |
|                 |       |               |               | 0Ah                           | 08h       | 04h       | 02h       | 0Ah                          | 08h       | 04h       | 02h       | FREQ = 1h               |           |
|                 |       |               |               | 08h                           | 04h       | 02h       | 02h       | 0Ah                          | 06h       | 04h       | 02h       | FREQ = 2h               |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |           |                         |           |
| E6h             | [7:0] | GSDLY         | 34h           | 10h                           | 10h       | 0Ch       | 06h       | 10h                          | 10h       | 10h       | 0Ah       | FREQ = 0h               |           |
|                 |       |               |               | 10h                           | 0Ch       | 06h       | 04h       | 10h                          | 0Ch       | 06h       | 04h       | FREQ = 1h               |           |
|                 |       |               |               | 0Ch                           | 06h       | 04h       | 02h       | 10h                          | 08h       | 06h       | 04h       | FREQ = 2h               |           |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |           |                         |           |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           |           | 1h: 12 bit              |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |           |                         |           |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 3Eh             | [7:2] |               |               | 16h                           | 0Ch       |           |           |                              |           |           |           |                         |           |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 21h             | [7:0] |               | 0Dh           | 91h                           |           |           |           |                              |           |           |           |                         |           |
| 22h             | [7:0] |               | B1h           | B1h                           |           |           |           |                              |           |           |           |                         |           |
| 46h             | [7:0] |               | 43h           | 64h                           |           |           |           |                              |           |           |           |                         |           |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| E8h             | [7:0] |               | 2Dh           | 09h                           |           |           |           |                              |           |           |           |                         |           |

The diagram illustrates the timing of a video frame transmission, showing both vertical and horizontal scan directions.

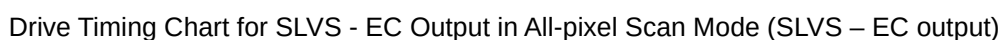
**Vertical Scan Direction (Normal):**

- Start Code:** 4byte x Lane
- Packet Header:** 24byte
- Effective margin for color processing:** 4
- Recording pixel area:** 4504 (height)
- Effective margin for color processing:** 4
- Vertical blanking:** 9
- Embedded Data:** 4
- Vertical effective OB:** 64
- Vertical blanking period:** 7 + GMRWT + GMTWT
- End Code:** 4byte x Lane
- Deskew Code:** 4byte x Lane
- Idle Code:**

**Horizontal Scan Direction (Normal):**

- Total Duration:** Total 4600 [H] + GMRWT + GMTWT + GSDLY

The AD lines for low gain side and high gain side are synthesized and output as one line.



## 10.1.2.2.ROI, Overlap ROI

## SLVS Output (IMX531)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |  |  |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|--|--|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |  |  |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |  |  |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |  |  |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         |  |  | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                         | *2              | *3                           | *4              |                         |  |  |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 392h                                                                                                       | 6F4h            | 266h                         | 4A8h            | FREQ = 0h               |  |  |           |
| D9h           | [7:0] |               |               | 70Dh                                                                                                       | DC8h            | 4B8h                         | 935h            | FREQ = 1h               |  |  |           |
|               |       |               |               | 266h                                                                                                       | 4A8h            | 1C4h                         | 356h            | FREQ = 2h               |  |  |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 4B8h                                                                                                       | 935h            | 370h                         | 689h            | FREQ = 3h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 2h               |  |  |           |
| E3h           | [7:0] | GMTWT         | 48h           | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 3h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 08h                          | 04h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 08h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 2h               |  |  |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 04h                                                                                                        | 06h             | 04h                          | 04h             | FREQ = 3h               |  |  |           |
| E6h           | [7:0] | GSDLY         | 34h           | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Eh                          | 08h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 08h                          | 04h             | FREQ = 3h               |  |  |           |
|               |       |               |               | Chip ID = 03h                                                                                              |                 |                              |                 |                         |  |  |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |  |  |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |  |  |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |  |  |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |  |  |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |  |  |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 21h           | [7:0] |               | 0Dh           | 91h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 22h           | [7:0] |               | B1h           | B1h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 46h           | [7:0] |               | 43h           | 64h                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                        |                 |                              |                 |                         |  |  |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

## SLVS – EC output (IMX531)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|                 |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                                                                                                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                                                                                                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 12ECh         | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h             | [7:0] | HMAX          | 0A6h          | 1A2h                                                                                                          | 1A2h      | 22Dh      | 455h      | 1A2h                         | 1A2h      | 1A2h      | 2EBh      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 1A2h                                                                                                          | 22Fh      | 45Ah      | 8AAh      | 1A2h                         | 22Fh      | 45Ah      | 8AAh      | FREQ = 1h               |
|                 |       |               |               | 25Ah                                                                                                          | 45Eh      | 8B4h      | 1154h     | 1A2h                         | 306h      | 5DAh      | BABh      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 0Ch           | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 04h                                                                                                           | 02h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 48h           | 0Ah                                                                                                           | 0Ah       | 08h       | 04h       | 0Ah                          | 0Ah       | 0Ah       | 06h       | FREQ = 0h               |
|                 |       |               |               | 0Ah                                                                                                           | 08h       | 04h       | 02h       | 0Ah                          | 08h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 08h                                                                                                           | 04h       | 02h       | 02h       | 0Ah                          | 06h       | 04h       | 02h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 34h           | 10h                                                                                                           | 10h       | 0Ch       | 06h       | 10h                          | 10h       | 10h       | 0Ah       | FREQ = 0h               |
|                 |       |               |               | 10h                                                                                                           | 0Ch       | 06h       | 04h       | 10h                          | 0Ch       | 06h       | 04h       | FREQ = 1h               |
|                 |       |               |               | 0Ch                                                                                                           | 06h       | 04h       | 02h       | 10h                          | 08h       | 06h       | 04h       | FREQ = 2h               |
| Chip ID = 03h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | 0Dh           | 91h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B1h           | B1h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 43h           | 64h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 09h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

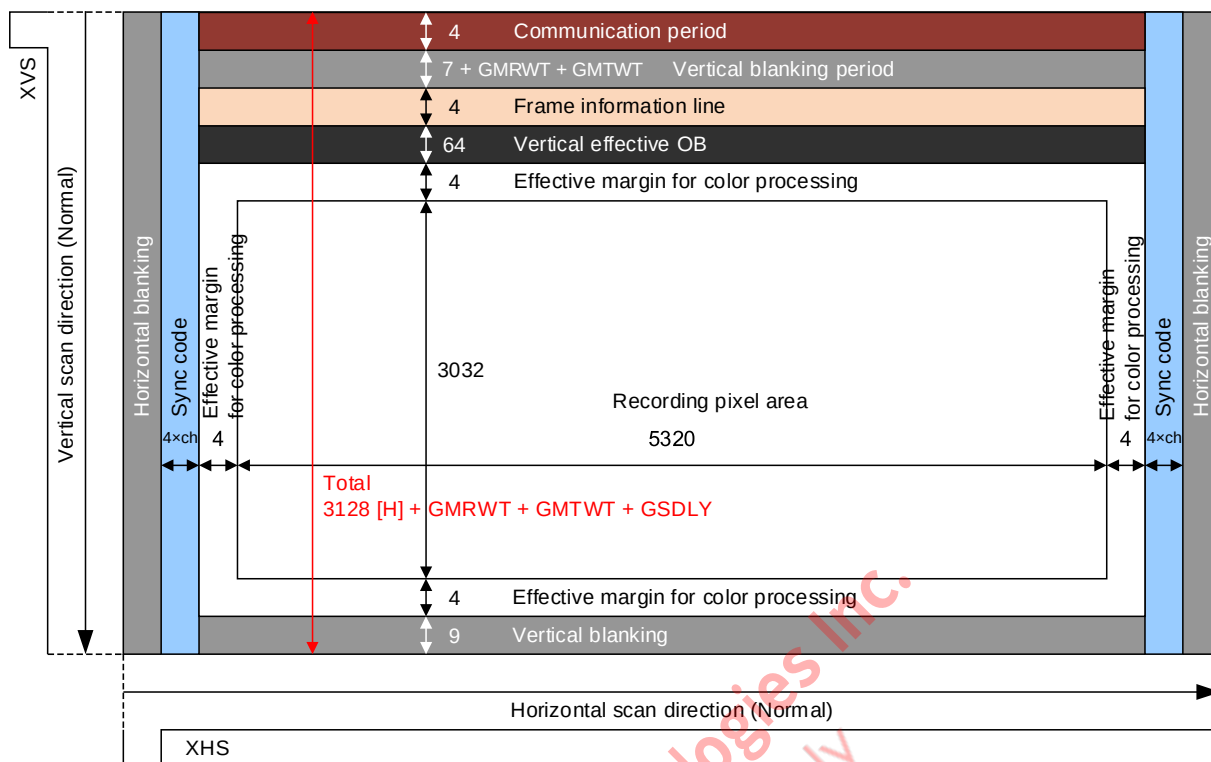
## 10.1.3. IMX532

## 10.1.3.1. All-pixel scan

SLVS Output (IMX532)

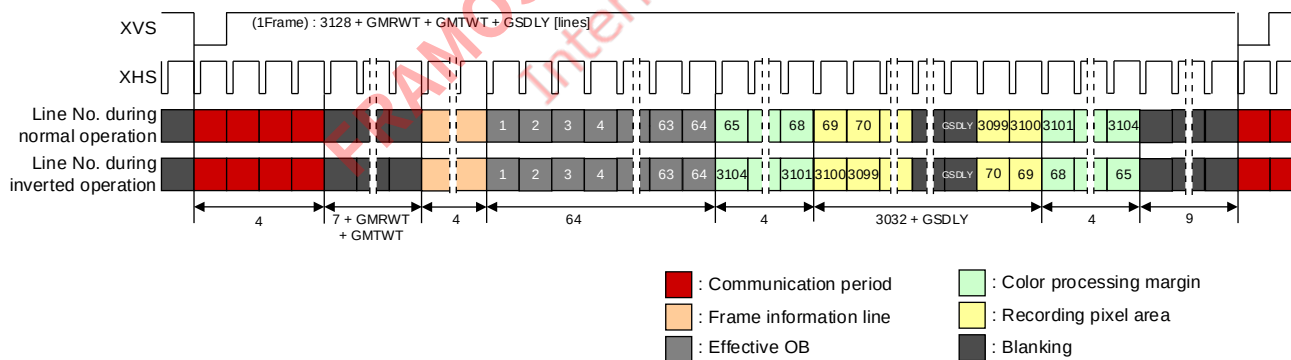
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |  |  |  |           |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|--|--|--|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |  |  |  |           |
|               |       |               |               | 8 ch                          | 8 ch              | 4 ch                         | 4 ch              |                         |  |  |  |           |
|               |       |               |               | 22.1<br>[frame/s]             | 11.3<br>[frame/s] | 32.9<br>[frame/s]            | 16.9<br>[frame/s] |                         |  |  |  |           |
|               |       |               |               |                               |                   |                              |                   |                         |  |  |  | FREQ = 0h |
|               |       |               |               |                               |                   |                              |                   |                         |  |  |  | FREQ = 1h |
|               |       |               |               |                               |                   |                              |                   |                         |  |  |  | FREQ = 2h |
|               |       |               |               |                               |                   |                              |                   |                         |  |  |  | FREQ = 3h |
|               |       |               |               |                               |                   |                              |                   | 594 Mbps/ch             |  |  |  | FREQ = 0h |
|               |       |               |               |                               |                   |                              |                   | 297 Mbps/ch             |  |  |  | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |  |  |  |           |
|               |       |               |               | 445.5 Mbps/ch                 |                   |                              |                   | FREQ = 3h               |  |  |  |           |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |  |  |  |           |
| D4h           | [7:0] | VMAX          | 12ECh         | C44h                          | C40h              | C4Ch                         | C44h              | FREQ = 0h               |  |  |  |           |
| D5h           | [7:0] |               |               | C40h                          | C3Eh              | C44h                         | C40h              | FREQ = 1h               |  |  |  |           |
| D6h           | [7:0] |               |               | C4Ch                          | C44h              | C50h                         | C48h              | FREQ = 2h               |  |  |  |           |
|               |       |               |               | C44h                          | C40h              | C44h                         | C42h              | FREQ = 3h               |  |  |  |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 42Bh                          | 826h              | 2CCh                         | 574h              | FREQ = 0h               |  |  |  |           |
| D9h           | [7:0] |               |               | 83Fh                          | 102Ch             | 584h                         | ACDh              | FREQ = 1h               |  |  |  |           |
|               |       |               |               | 2CCh                          | 574h              | 208h                         | 3DEh              | FREQ = 2h               |  |  |  |           |
|               |       |               |               | 584h                          | ACDh              | 3F8h                         | 799h              | FREQ = 3h               |  |  |  |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 02h                           | 02h               | 04h                          | 02h               | FREQ = 0h               |  |  |  |           |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 1h               |  |  |  |           |
|               |       |               |               | 04h                           | 02h               | 04h                          | 02h               | FREQ = 2h               |  |  |  |           |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 3h               |  |  |  |           |
| E3h           | [7:0] | GMTWT         | 48h           | 04h                           | 02h               | 06h                          | 04h               | FREQ = 0h               |  |  |  |           |
|               |       |               |               | 02h                           | 02h               | 04h                          | 02h               | FREQ = 1h               |  |  |  |           |
|               |       |               |               | 06h                           | 04h               | 08h                          | 06h               | FREQ = 2h               |  |  |  |           |
|               |       |               |               | 04h                           | 02h               | 04h                          | 04h               | FREQ = 3h               |  |  |  |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |  |  |  |           |
| E6h           | [7:0] | GSDLY         | 34h           | 06h                           | 04h               | 0Ah                          | 06h               | FREQ = 0h               |  |  |  |           |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 1h               |  |  |  |           |
|               |       |               |               | 0Ah                           | 06h               | 0Ch                          | 08h               | FREQ = 2h               |  |  |  |           |
|               |       |               |               | 06h                           | 04h               | 06h                          | 04h               | FREQ = 3h               |  |  |  |           |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |  |  |  |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |  |  |  |           |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |  |  |  |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |                   |                              |                   |                         |  |  |  |           |
| 3Dh           | [0]   |               |               |                               |                   |                              |                   |                         |  |  |  |           |
| 3Eh           | [7:2] |               | 16h           | 0Ch                           |                   |                              |                   |                         |  |  |  |           |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |  |  |  |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |  |  |  |           |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |  |  |  |           |
| 21h           | [7:0] |               | 0Dh           | 19h                           |                   |                              |                   |                         |  |  |  |           |
| 22h           | [7:0] |               | B1h           | B2h                           |                   |                              |                   |                         |  |  |  |           |
| 46h           | [7:0] |               | 43h           | 86h                           |                   |                              |                   |                         |  |  |  |           |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |  |  |  |           |
| E8h           | [7:0] |               | 2Dh           | 09h                           |                   |                              |                   |                         |  |  |  |           |

(IMX532)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

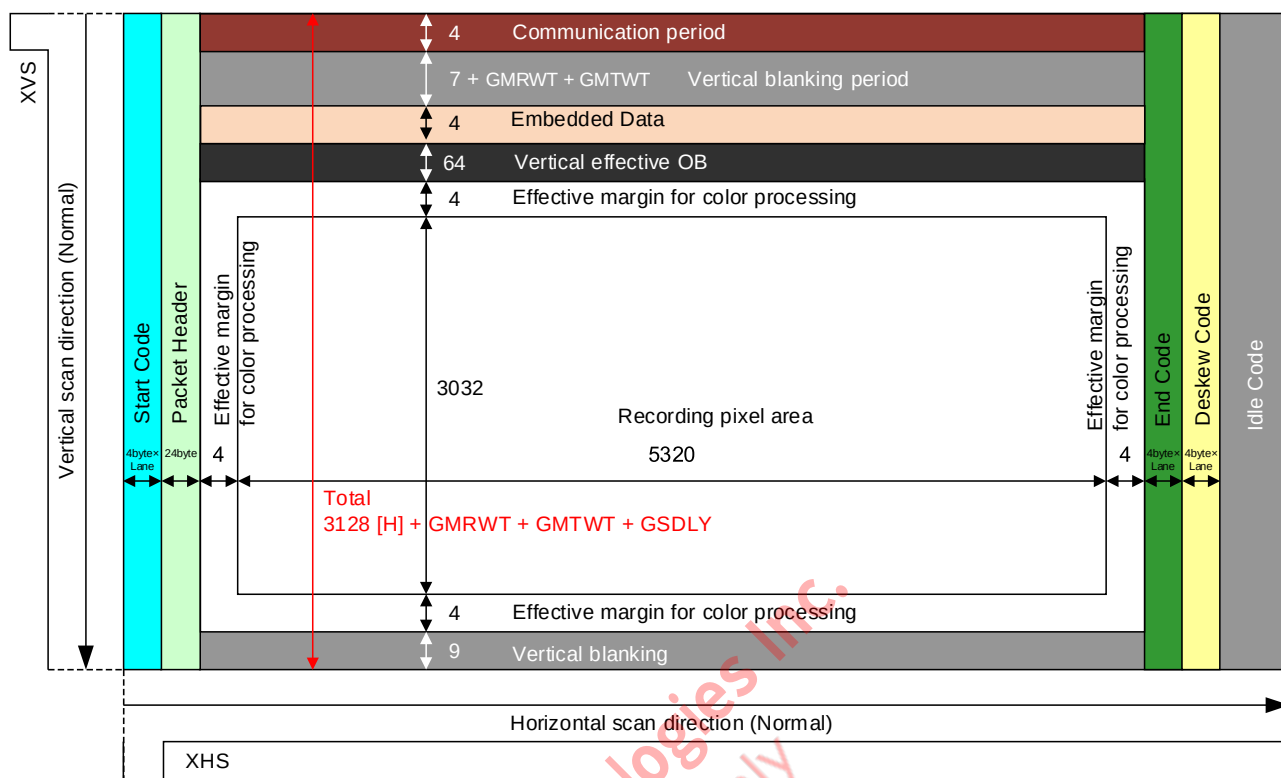


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX532)

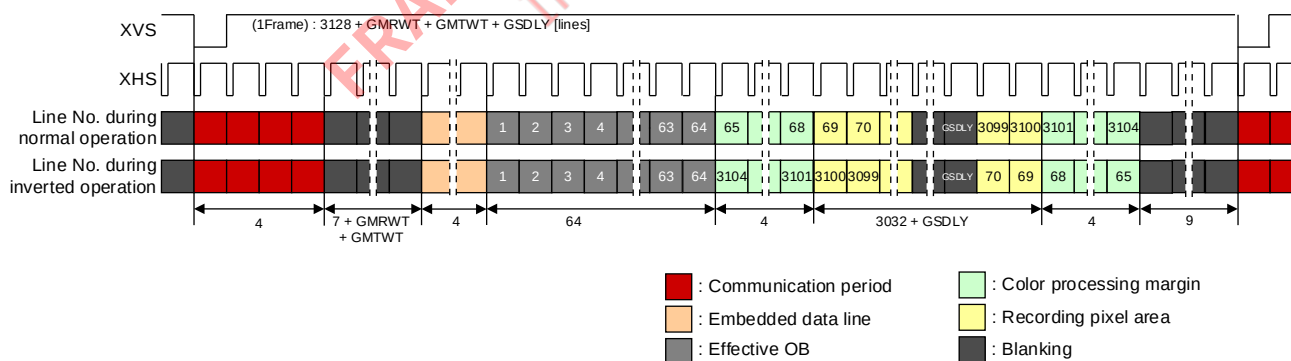
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |           | Remarks                 |           |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|-----------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |           |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |           |
|                 |       |               |               | 56.2                          | 56.2      | 35.9      | 18.1      | 56.2                         | 56.2      | 52.6      | 26.8      |                         | FREQ = 0h |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |           |
|                 |       |               |               | 56.2                          | 35.5      | 18.0      | 9.0       | 56.2                         | 35.6      | 18.0      | 9.0       |                         |           |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |           |
|                 |       |               |               | 33.5                          | 17.8      | 9.0       | 4.5       | 49.0                         | 26.0      | 13.3      | 6.7       |                         | FREQ = 2h |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |           |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |           |                         |           |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |           |                         |           |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |           |                         |           |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| D4h             | [7:0] | VMAX          | 12ECh         | C58h                          | C58h      | C4Eh      | C44h      | C58h                         | C58h      | C54h      | C48h      | FREQ = 0h               |           |
| D5h             | [7:0] |               |               | C58h                          | C4Eh      | C44h      | C40h      | C58h                         | C4Eh      | C44h      | C40h      | FREQ = 1h               |           |
| D6h             | [7:0] |               |               | C4Ch                          | C44h      | C40h      | C3Eh      | C54h                         | C48h      | C42h      | C3Eh      | FREQ = 2h               |           |
| D8h             | [7:0] | HMAX          | 0A6h          | 1A2h                          | 1A2h      | 28Fh      | 519h      | 1A2h                         | 1A2h      | 1BFh      | 370h      | FREQ = 0h               |           |
| D9h             | [7:0] |               |               | 1A2h                          | 297h      | 51Eh      | A31h      | 1A2h                         | 295h      | 51Eh      | A31h      | FREQ = 1h               |           |
|                 |       |               |               | 2C0h                          | 52Ah      | A3Bh      | 1462h     | 1E0h                         | 389h      | 6EEh      | DBFh      | FREQ = 2h               |           |
| E2h             | [7:0] | GMRWT         | 0Ch           | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 0h               |           |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 1h               |           |
|                 |       |               |               | 04h                           | 02h       | 02h       | 02h       | 04h                          | 02h       | 02h       | 02h       | FREQ = 2h               |           |
| E3h             | [7:0] | GMTWT         | 48h           | 0Ah                           | 0Ah       | 08h       | 04h       | 0Ah                          | 0Ah       | 0Ah       | 06h       | FREQ = 0h               |           |
|                 |       |               |               | 0Ah                           | 08h       | 04h       | 02h       | 0Ah                          | 08h       | 04h       | 02h       | FREQ = 1h               |           |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 0Ah                          | 06h       | 04h       | 02h       | FREQ = 2h               |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |           |                         |           |
| E6h             | [7:0] | GSDLY         | 34h           | 10h                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 0Eh       | 08h       | FREQ = 0h               |           |
|                 |       |               |               | 10h                           | 0Ah       | 06h       | 04h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 1h               |           |
|                 |       |               |               | 0Ah                           | 06h       | 04h       | 02h       | 0Eh                          | 08h       | 04h       | 02h       | FREQ = 2h               |           |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |           |                         |           |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           |           | 1h: 12 bit              |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |           |                         |           |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |           |                         |           |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| 21h             | [7:0] |               | 0Dh           | 19h                           |           |           |           |                              |           |           |           |                         |           |
| 22h             | [7:0] |               | B1h           | B2h                           |           |           |           |                              |           |           |           |                         |           |
| 46h             | [7:0] |               | 43h           | 86h                           |           |           |           |                              |           |           |           |                         |           |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |           |
| E8h             | [7:0] |               | 2Dh           | 09h                           |           |           |           |                              |           |           |           |                         |           |

(IMX532)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.3.2. ROI, Overlap ROI

## SLVS Output (IMX532)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |  |  |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|--|--|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |  |  |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |  |  |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |  |  |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         |  |  | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                         | *2              | *3                           | *4              |                         |  |  |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D8h           | [7:0] | HMAX          | 0A6h          | 42Bh                                                                                                       | 826h            | 2CCh                         | 574h            | FREQ = 0h               |  |  |           |
| D9h           | [7:0] |               |               | 83Fh                                                                                                       | 102Ch           | 584h                         | ACDh            | FREQ = 1h               |  |  |           |
|               |       |               |               | 2CCh                                                                                                       | 574h            | 208h                         | 3DEh            | FREQ = 2h               |  |  |           |
|               |       |               |               | 584h                                                                                                       | ACDh            | 3F8h                         | 799h            | FREQ = 3h               |  |  |           |
| E2h           | [7:0] | GMRWT         | 0Ch           | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 3h               |  |  |           |
| E3h           | [7:0] | GMTWT         | 48h           | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 08h                          | 06h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 04h             | FREQ = 3h               |  |  |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |  |  |           |
| E6h           | [7:0] | GSDLY         | 34h           | 06h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Ch                          | 08h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 06h                          | 04h             | FREQ = 3h               |  |  |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |  |  |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |  |  |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 | 1h: 12 bit              |  |  |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 |                         |  |  |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |  |  |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 21h           | [7:0] |               | 0Dh           | 19h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 22h           | [7:0] |               | B1h           | B2h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 46h           | [7:0] |               | 43h           | 86h                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                        |                 |                              |                 |                         |  |  |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

## SLVS – EC output (IMX532)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|               |       |               |               | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|               |       |               |               | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|               |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|               |       |               |               | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           | FREQ = 0h               |
|               |       |               |               | 2376 Mbps/lane                                                                                                |           |           |           |                              |           |           |           | FREQ = 1h               |
|               |       |               |               | 1188 Mbps /lane                                                                                               |           |           |           |                              |           |           |           | FREQ = 2h               |
| Chip ID = 02h |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h           | [7:0] | VMAX          | 12ECh         | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h           | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h           | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h           | [7:0] | HMAX          | 0A6h          | 1A2h                                                                                                          | 1A2h      | 28Fh      | 519h      | 1A2h                         | 1A2h      | 1BFh      | 370h      | FREQ = 0h               |
| D9h           | [7:0] |               |               | 1A2h                                                                                                          | 297h      | 51Eh      | A31h      | 1A2h                         | 295h      | 51Eh      | A31h      | FREQ = 1h               |
|               |       |               |               | 2C0h                                                                                                          | 52Ah      | A3Bh      | 1462h     | 1E0h                         | 389h      | 6EEh      | DBFh      | FREQ = 2h               |
| E2h           | [7:0] | GMRWT         | 0Ch           | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 0h               |
|               |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 1h               |
|               |       |               |               | 04h                                                                                                           | 02h       | 02h       | 02h       | 04h                          | 02h       | 02h       | 02h       | FREQ = 2h               |
| E3h           | [7:0] | GMTWT         | 48h           | 0Ah                                                                                                           | 0Ah       | 08h       | 04h       | 0Ah                          | 0Ah       | 0Ah       | 06h       | FREQ = 0h               |
|               |       |               |               | 0Ah                                                                                                           | 08h       | 04h       | 02h       | 0Ah                          | 08h       | 04h       | 02h       | FREQ = 1h               |
|               |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 0Ah                          | 06h       | 04h       | 02h       | FREQ = 2h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h           | [7:0] | GSDLY         | 34h           | 10h                                                                                                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 0Eh       | 08h       | FREQ = 0h               |
|               |       |               |               | 10h                                                                                                           | 0Ah       | 06h       | 04h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 1h               |
|               |       |               |               | 0Ah                                                                                                           | 06h       | 04h       | 02h       | 0Eh                          | 08h       | 04h       | 02h       | FREQ = 2h               |
| Chip ID = 03h |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh           | [0]   |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh           | [7:2] |               | 16h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h           | [7:0] |               | 0Dh           | 19h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h           | [7:0] |               | B1h           | B2h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h           | [7:0] |               | 43h           | 86h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h           | [7:0] |               | 2Dh           | 09h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 88$$

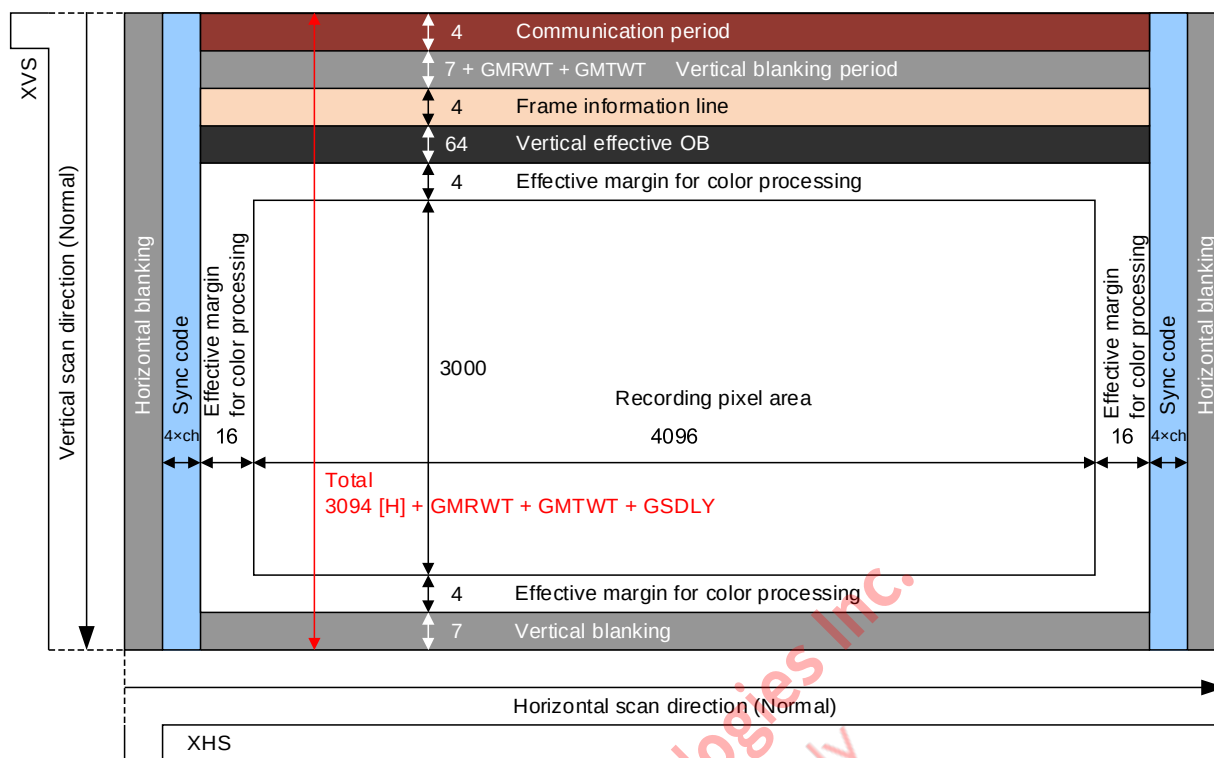
## 10.1.4. IMX535-AAQJ

## 10.1.4.1. All-pixel scan

SLVS Output (IMX535-AAQJ)

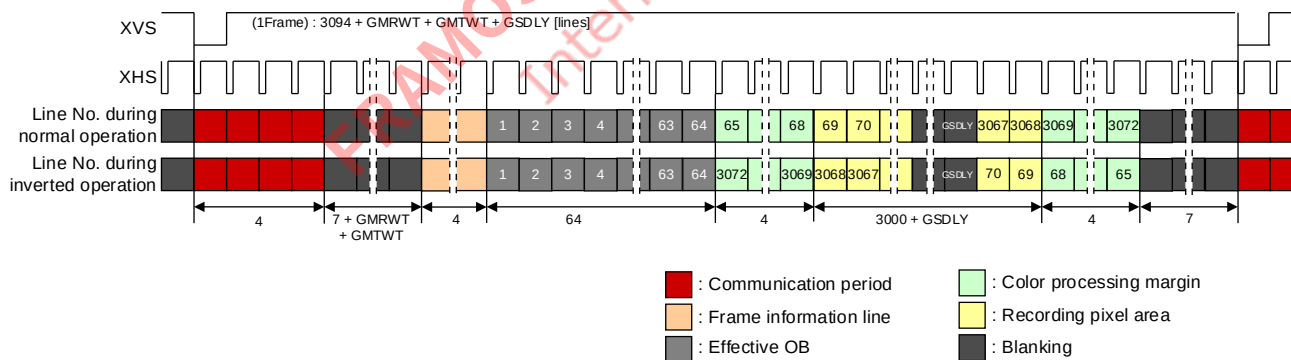
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |
|               |       |               |               | 8 ch                          | 4 ch              | 8 ch                         | 4 ch              |                         |
|               |       |               |               | 28.2<br>[frame/s]             | 14.5<br>[frame/s] | 41.8<br>[frame/s]            | 21.7<br>[frame/s] | FREQ = 0h               |
|               |       |               |               | 14.3<br>[frame/s]             | 7.3<br>[frame/s]  | 21.4<br>[frame/s]            | 11.0<br>[frame/s] | FREQ = 1h               |
|               |       |               |               | 41.8<br>[frame/s]             | 21.7<br>[frame/s] | 61.1<br>[frame/s]            | 32.1<br>[frame/s] | FREQ = 2h               |
|               |       |               |               | 21.4<br>[frame/s]             | 11.0<br>[frame/s] | 31.7<br>[frame/s]            | 16.4<br>[frame/s] | FREQ = 3h               |
|               |       |               |               | 594 Mbps/ch                   |                   |                              |                   | FREQ = 0h               |
|               |       |               |               | 297 Mbps/ch                   |                   |                              |                   | FREQ = 1h               |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |
| 445.5 Mbps/ch |       |               |               | FREQ = 3h                     |                   |                              |                   |                         |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |
| D4h           | [7:0] | VMAX          | 0CD4h         | C34h                          | C26h              | C40h                         | C2Ch              | FREQ = 0h               |
| D5h           | [7:0] |               |               | C26h                          | C20h              | C2Ch                         | C24h              | FREQ = 1h               |
| D6h           | [7:0] |               |               | C40h                          | C2Ch              | C52h                         | C38h              | FREQ = 2h               |
|               |       |               |               | C2Ch                          | C24h              | C38h                         | C28h              | FREQ = 3h               |
| D8h           | [7:0] | HMAX          | 081h          | 34Ah                          | 664h              | 236h                         | 448h              | FREQ = 0h               |
| D9h           | [7:0] |               |               | 67Dh                          | CA8h              | 458h                         | 875h              | FREQ = 1h               |
|               |       |               |               | 236h                          | 448h              | 181h                         | 2E3h              | FREQ = 2h               |
|               |       |               |               | 458h                          | 875h              | 2ECh                         | 5ABh              | FREQ = 3h               |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                           | 02h               | 04h                          | 02h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 02h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 2h               |
|               |       |               |               | 02h                           | 02h               | 04h                          | 02h               | FREQ = 3h               |
| E3h           | [7:0] | GMTWT         | 70h           | 12h                           | 0Ah               | 1Ah                          | 0Eh               | FREQ = 0h               |
|               |       |               |               | 0Ah                           | 06h               | 0Eh                          | 08h               | FREQ = 1h               |
|               |       |               |               | 1Ah                           | 0Eh               | 26h                          | 14h               | FREQ = 2h               |
|               |       |               |               | 0Eh                           | 08h               | 14h                          | 0Ah               | FREQ = 3h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |
| E6h           | [7:0] | GSDLY         | 30h           | 08h                           | 04h               | 0Ch                          | 06h               | FREQ = 0h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 1h               |
|               |       |               |               | 0Ch                           | 06h               | 10h                          | 0Ah               | FREQ = 2h               |
|               |       |               |               | 06h                           | 04h               | 0Ah                          | 06h               | FREQ = 3h               |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |                   |                              |                   |                         |
| 3Dh           | [0]   |               |               |                               |                   |                              |                   |                         |
| 3Eh           | [7:2] |               |               | 0Ch                           |                   |                              |                   |                         |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |
| 21h           | [7:0] |               | A9h           | 51h                           |                   |                              |                   |                         |
| 22h           | [7:0] |               | B0h           | B1h                           |                   |                              |                   |                         |
| 46h           | [7:0] |               | 2Ah           | 54h                           |                   |                              |                   |                         |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |
| E8h           | [7:0] |               | 2Dh           | 13h                           |                   |                              |                   |                         |
| Chip ID = 15h |       |               |               |                               |                   |                              |                   |                         |
| 1Eh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |
| 1Fh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |

(IMX535-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

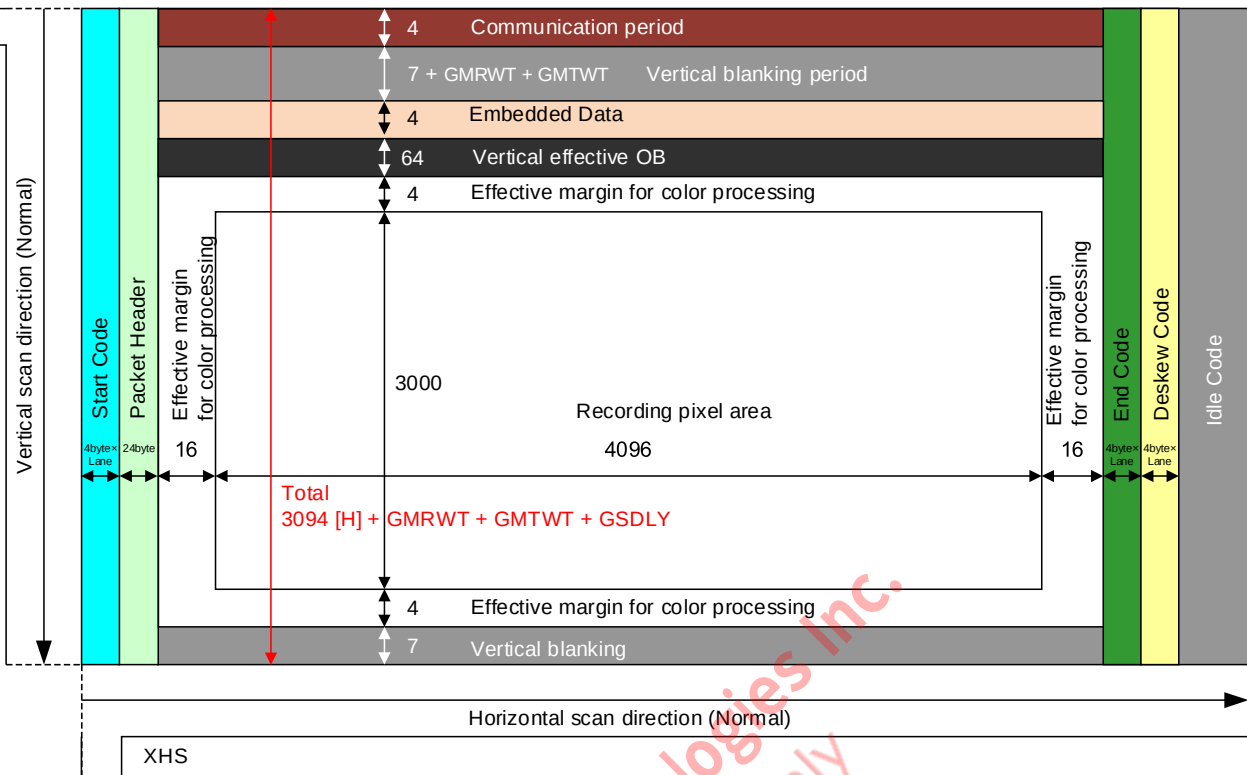


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX535-AAQJ)

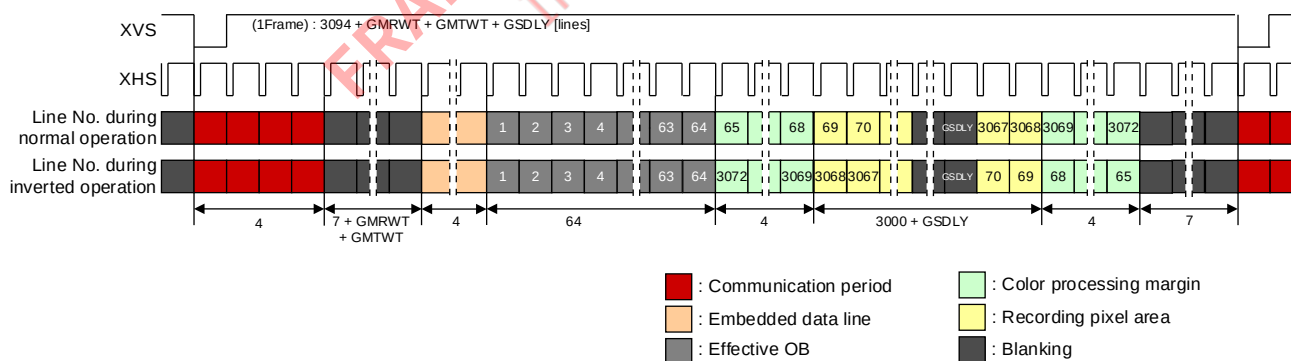
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | 61.9                          | 61.9      | 46.3      | 23.4      | 61.9                         | 61.9      | 61.9      | 34.5      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 61.9                          | 46.1      | 23.3      | 11.7      | 61.9                         | 34.5      | 17.3      |           |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 42.7                          | 23.2      | 11.6      | 5.8       | 61.9                         | 33.4      | 17.3      | 8.7       |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | C52h                          | C52h      | C42h      | C2Ch      | C52h                         | C52h      | C52h      | C3Ah      | FREQ = 0h               |
| D5h             | [7:0] |               |               | C52h                          | C42h      | C2Ch      | C24h      | C52h                         | C52h      | C3Ah      | C2Ah      | FREQ = 1h               |
| D6h             | [7:0] |               |               | C40h                          | C2Ch      | C24h      | C1Eh      | C52h                         | C38h      | C2Ah      | C22h      | FREQ = 2h               |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                          | 17Ch      | 1FFh      | 3F9h      | 17Ch                         | 17Ch      | 17Ch      | 2AEh      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                          | 201h      | 3FEh      | 7F2h      | 17Ch                         | 17Ch      | 2AFh      | 55Bh      | FREQ = 1h               |
|                 |       |               |               | 22Ah                          | 402h      | 7FCh      | FE4h      | 17Ch                         | 2C6h      | 55Eh      | AB5h      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 04h                           | 02h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                           | 26h       | 1Ch       | 0Eh       | 26h                          | 26h       | 26h       | 16h       | FREQ = 0h               |
|                 |       |               |               | 26h                           | 1Ch       | 0Eh       | 08h       | 26h                          | 26h       | 16h       | 0Ch       | FREQ = 1h               |
|                 |       |               |               | 1Ah                           | 0Eh       | 08h       | 04h       | 26h                          | 14h       | 0Ch       | 06h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                           | 10h       | 0Ch       | 06h       | 10h                          | 10h       | 10h       | 0Ah       | FREQ = 0h               |
|                 |       |               |               | 10h                           | 0Ch       | 06h       | 04h       | 10h                          | 10h       | 0Ah       | 06h       | FREQ = 1h               |
|                 |       |               |               | 0Ch                           | 06h       | 04h       | 02h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 2h               |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 51h                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B1h                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 54h                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |

(IMX535-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.4.2. ROI, Overlap ROI

## SLVS Output (IMX535-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3                           | *4              |                         |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D8h           | [7:0] | HMAX          | 081h          | 34Ah                                                                                                       | 664h            | 236h                         | 448h            | FREQ = 0h               |           |
| D9h           | [7:0] |               |               | 67Dh                                                                                                       | CA8h            | 458h                         | 875h            | FREQ = 1h               |           |
|               |       |               |               | 236h                                                                                                       | 448h            | 181h                         | 2E3h            | FREQ = 2h               |           |
|               |       |               |               | 458h                                                                                                       | 875h            | 2ECh                         | 5ABh            | FREQ = 3h               |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 0h               |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 1h               |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 2h               |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 3h               |           |
| E3h           | [7:0] | GMTWT         | 70h           | 12h                                                                                                        | 0Ah             | 1Ah                          | 0Eh             | FREQ = 0h               |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Eh                          | 08h             | FREQ = 1h               |           |
|               |       |               |               | 1Ah                                                                                                        | 0Eh             | 26h                          | 14h             | FREQ = 2h               |           |
|               |       |               |               | 0Eh                                                                                                        | 08h             | 14h                          | 0Ah             | FREQ = 3h               |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |           |
| E6h           | [7:0] | GSDLY         | 30h           | 08h                                                                                                        | 04h             | 0Ch                          | 06h             | FREQ = 0h               |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 1h               |           |
|               |       |               |               | 0Ch                                                                                                        | 06h             | 10h                          | 0Ah             | FREQ = 2h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 3h               |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 21h           | [7:0] |               | A9h           | 51h                                                                                                        |                 |                              |                 |                         |           |
| 22h           | [7:0] |               | B0h           | B1h                                                                                                        |                 |                              |                 |                         |           |
| 46h           | [7:0] |               | 2Ah           | 54h                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

## SLVS – EC output (IMX535-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |                 |                 |                 |                              |                 |                 |                 | Remarks                 |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|------------------------------|-----------------|-----------------|-----------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |                 |                 |                 | AD = 12 bit / Output = 8 bit |                 |                 |                 |                         |
|               |       |               |               | 8 lane                                                                                                        | 4 lane          | 2 lane          | 1 lane          | 8 lane                       | 4 lane          | 2 lane          | 1 lane          |                         |
|               |       |               |               | *1<br>[frame/s]                                                                                               | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s]              | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] |                         |
|               |       |               |               | 4752 Mbps/lane                                                                                                |                 |                 |                 |                              |                 |                 |                 | FREQ = 0h               |
|               |       |               |               | 2376 Mbps/lane                                                                                                |                 |                 |                 |                              |                 |                 |                 | FREQ = 1h               |
|               |       |               |               | 1188 Mbps /lane                                                                                               |                 |                 |                 |                              |                 |                 |                 | FREQ = 2h               |
| Chip ID = 02h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2              | *3              | *4              | *5                           | *6              | *7              | *8              |                         |
| D5h           | [7:0] |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| D6h           | [7:0] |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| D8h           | [7:0] | HMAX          | 081h          | 17Ch                                                                                                          | 17Ch            | 1FFh            | 3F9h            | 17Ch                         | 17Ch            | 17Ch            | 2AEh            | FREQ = 0h               |
| D9h           | [7:0] |               |               | 17Ch                                                                                                          | 201h            | 3FEh            | 7F2h            | 17Ch                         | 17Ch            | 2AFh            | 55Bh            | FREQ = 1h               |
|               |       |               |               | 22Ah                                                                                                          | 402h            | 7FCh            | FE4h            | 17Ch                         | 2C6h            | 55Eh            | AB5h            | FREQ = 2h               |
| E2h           | [7:0] | GMRWT         | 10h           | 06h                                                                                                           | 06h             | 04h             | 02h             | 06h                          | 06h             | 06h             | 04h             | FREQ = 0h               |
|               |       |               |               | 06h                                                                                                           | 04h             | 02h             | 02h             | 06h                          | 06h             | 04h             | 02h             | FREQ = 1h               |
|               |       |               |               | 04h                                                                                                           | 02h             | 02h             | 02h             | 06h                          | 04h             | 02h             | 02h             | FREQ = 2h               |
| E3h           | [7:0] | GMTWT         | 70h           | 26h                                                                                                           | 26h             | 1Ch             | 0Eh             | 26h                          | 26h             | 26h             | 16h             | FREQ = 0h               |
|               |       |               |               | 26h                                                                                                           | 1Ch             | 0Eh             | 08h             | 26h                          | 26h             | 16h             | 0Ch             | FREQ = 1h               |
|               |       |               |               | 1Ah                                                                                                           | 0Eh             | 08h             | 04h             | 26h                          | 14h             | 0Ch             | 06h             | FREQ = 2h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| E6h           | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 10h             | 0Ch             | 06h             | 10h                          | 10h             | 10h             | 0Ah             | FREQ = 0h               |
|               |       |               |               | 10h                                                                                                           | 0Ch             | 06h             | 04h             | 10h                          | 10h             | 0Ah             | 06h             | FREQ = 1h               |
|               |       |               |               | 0Ch                                                                                                           | 06h             | 04h             | 02h             | 10h                          | 0Ah             | 06h             | 04h             | FREQ = 2h               |
| Chip ID = 03h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |                 |                 |                 |                              |                 |                 |                 |                         |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 04h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |                 |                 |                 |                              |                 |                 |                 |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |                 |                 |                 |                              |                 |                 |                 | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 3Dh           | [0]   |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 06h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |                 |                 |                 | 2h                           |                 |                 |                 | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 21h           | [7:0] |               | A9h           | 51h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 22h           | [7:0] |               | B0h           | B1h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 46h           | [7:0] |               | 2Ah           | 54h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 08h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 15h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

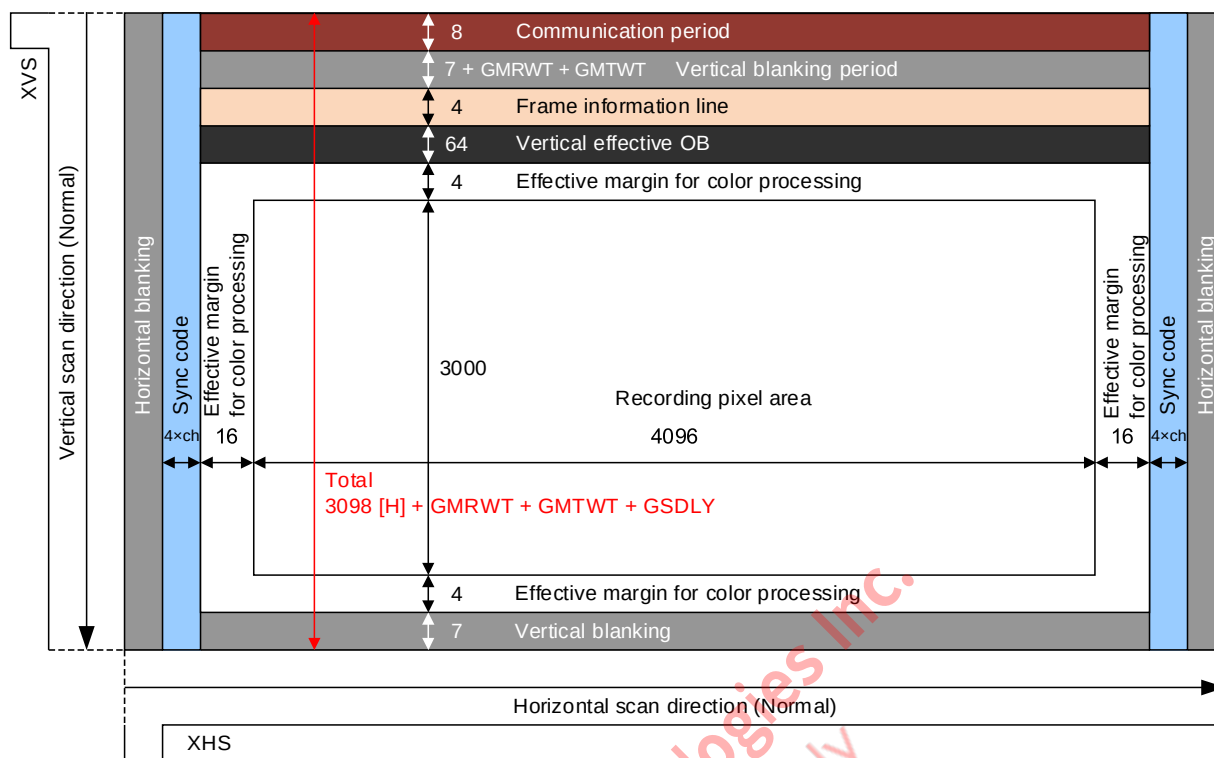
## 10.1.5. IMX535-AAMJ

## 10.1.5.1. All-pixel scan

SLVS Output (IMX535-AAMJ)

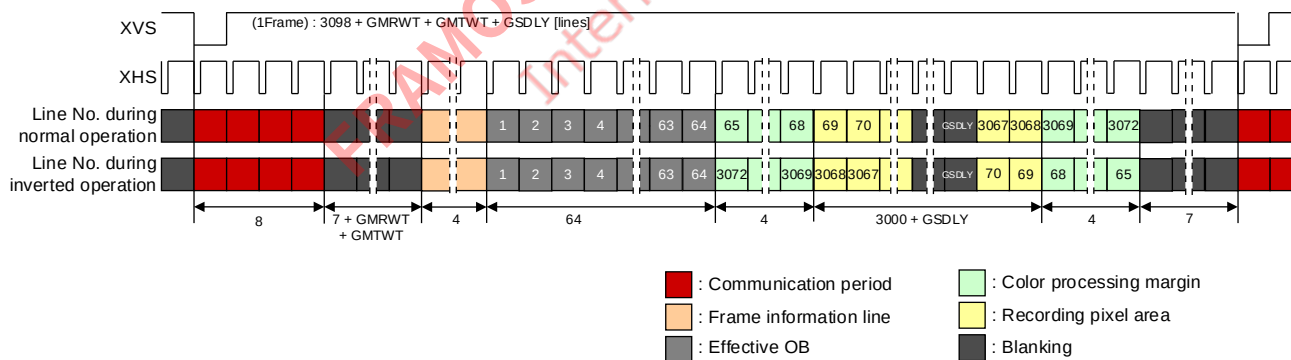
| Address       | bit   | Register name | Initial Value | Setting value                 |      |                              |                   | Remarks                 |                   |           |
|---------------|-------|---------------|---------------|-------------------------------|------|------------------------------|-------------------|-------------------------|-------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |      | AD = 12 bit / Output = 8 bit |                   |                         |                   |           |
|               |       |               |               |                               |      | 8 ch                         | 4 ch              | 8 ch                    | 4 ch              |           |
|               |       |               |               |                               |      | 28.1<br>[frame/s]            | 14.5<br>[frame/s] | 41.7<br>[frame/s]       | 21.7<br>[frame/s] | FREQ = 0h |
|               |       |               |               |                               |      | 14.3<br>[frame/s]            | 7.3<br>[frame/s]  | 21.4<br>[frame/s]       | 11.0<br>[frame/s] | FREQ = 1h |
|               |       |               |               |                               |      | 41.7<br>[frame/s]            | 21.7<br>[frame/s] | 61.0<br>[frame/s]       | 32.0<br>[frame/s] | FREQ = 2h |
|               |       |               |               |                               |      | 21.4<br>[frame/s]            | 11.0<br>[frame/s] | 31.6<br>[frame/s]       | 16.4<br>[frame/s] | FREQ = 3h |
|               |       |               |               |                               |      | 594 Mbps/ch                  |                   |                         |                   | FREQ = 0h |
|               |       |               |               |                               |      | 297 Mbps/ch                  |                   |                         |                   | FREQ = 1h |
|               |       |               |               |                               |      | 891 Mbps/ch                  |                   |                         |                   | FREQ = 2h |
|               |       | 445.5 Mbps/ch |               |                               |      | FREQ = 3h                    |                   |                         |                   |           |
| Chip ID = 02h |       |               |               |                               |      |                              |                   |                         |                   |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | C38h                          | C2Ah | C44h                         | C30h              | FREQ = 0h               |                   |           |
| D5h           | [7:0] |               |               | C2Ah                          | C24h | C30h                         | C28h              | FREQ = 1h               |                   |           |
| D6h           | [7:0] |               |               | C44h                          | C30h | C56h                         | C3Ch              | FREQ = 2h               |                   |           |
|               |       |               |               | C30h                          | C28h | C3Ch                         | C2Ch              | FREQ = 3h               |                   |           |
| D8h           | [7:0] | HMAX          | 081h          | 34Ah                          | 664h | 236h                         | 448h              | FREQ = 0h               |                   |           |
| D9h           | [7:0] |               |               | 67Dh                          | CA8h | 458h                         | 875h              | FREQ = 1h               |                   |           |
|               |       |               |               | 236h                          | 448h | 181h                         | 2E3h              | FREQ = 2h               |                   |           |
|               |       |               |               | 458h                          | 875h | 2ECh                         | 5ABh              | FREQ = 3h               |                   |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                           | 02h  | 04h                          | 02h               | FREQ = 0h               |                   |           |
|               |       |               |               | 02h                           | 02h  | 02h                          | 02h               | FREQ = 1h               |                   |           |
|               |       |               |               | 04h                           | 02h  | 06h                          | 04h               | FREQ = 2h               |                   |           |
|               |       |               |               | 02h                           | 02h  | 04h                          | 02h               | FREQ = 3h               |                   |           |
| E3h           | [7:0] | GMTWT         | 70h           | 12h                           | 0Ah  | 1Ah                          | 0Eh               | FREQ = 0h               |                   |           |
|               |       |               |               | 0Ah                           | 06h  | 0Eh                          | 08h               | FREQ = 1h               |                   |           |
|               |       |               |               | 1Ah                           | 0Eh  | 26h                          | 14h               | FREQ = 2h               |                   |           |
|               |       |               |               | 0Eh                           | 08h  | 14h                          | 0Ah               | FREQ = 3h               |                   |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |      |                              |                   |                         |                   |           |
| E6h           | [7:0] | GSDLY         | 30h           | 08h                           | 04h  | 0Ch                          | 06h               | FREQ = 0h               |                   |           |
|               |       |               |               | 04h                           | 02h  | 06h                          | 04h               | FREQ = 1h               |                   |           |
|               |       |               |               | 0Ch                           | 06h  | 10h                          | 0Ah               | FREQ = 2h               |                   |           |
|               |       |               |               | 06h                           | 04h  | 0Ah                          | 06h               | FREQ = 3h               |                   |           |
| Chip ID = 04h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |      |                              |                   |                         |                   |           |
|               | [5:4] | ADBIT         | 0h            | 1h                            |      |                              |                   | 1h: 12 bit              |                   |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           |                               |      |                              |                   |                         |                   |           |
| 3Dh           | [0]   |               |               | 0Ch                           |      |                              |                   |                         |                   |           |
| 3Eh           | [7:2] |               |               | 0Ch                           |      |                              |                   |                         |                   |           |
| Chip ID = 06h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |      | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |                   |           |
| Chip ID = 07h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 21h           | [7:0] |               | A9h           | 51h                           |      |                              |                   |                         |                   |           |
| 22h           | [7:0] |               | B0h           | B1h                           |      |                              |                   |                         |                   |           |
| 46h           | [7:0] |               | 2Ah           | 54h                           |      |                              |                   |                         |                   |           |
| Chip ID = 08h |       |               |               |                               |      |                              |                   |                         |                   |           |
| E8h           | [7:0] |               | 2Dh           | 13h                           |      |                              |                   |                         |                   |           |
| Chip ID = 15h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 1Eh           | [7:0] |               | 02h           | 16h                           |      |                              |                   |                         |                   |           |
| 1Fh           | [7:0] |               | 02h           | 16h                           |      |                              |                   |                         |                   |           |

(IMX535-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

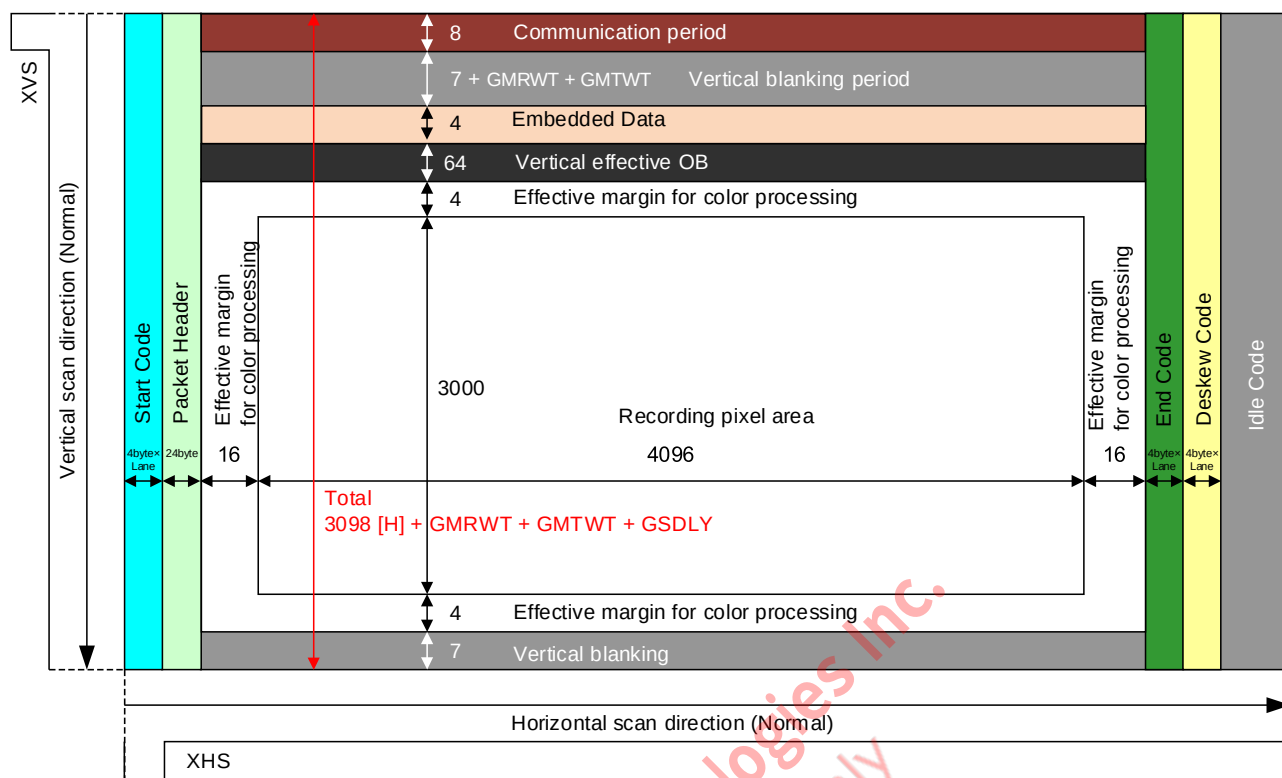


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX535-AAMJ)

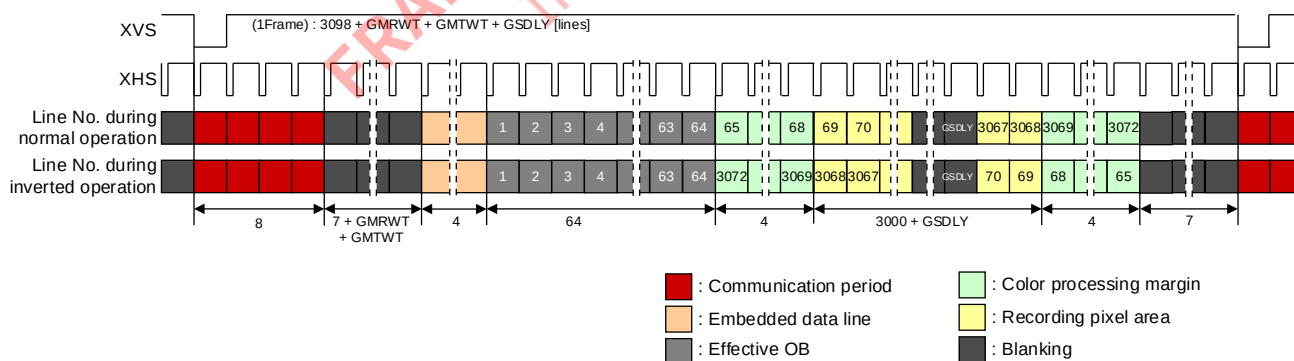
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | 61.8                          | 61.8      | 46.2      | 23.4      | 61.8                         | 61.8      | 61.8      | 34.5      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 61.8                          | 46.0      | 23.2      | 11.7      | 61.8                         | 61.8      | 34.4      | 17.3      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 42.6                          | 23.1      | 11.6      | 5.8       | 61.8                         | 33.3      | 17.3      | 8.7       |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | C56h                          | C56h      | C46h      | C30h      | C56h                         | C56h      | C56h      | C3Eh      | FREQ = 0h               |
| D5h             | [7:0] |               |               | C56h                          | C46h      | C30h      | C28h      | C56h                         | C56h      | C3Eh      | C2Eh      | FREQ = 1h               |
| D6h             | [7:0] |               |               | C44h                          | C30h      | C28h      | C22h      | C56h                         | C3Ch      | C2Eh      | C26h      | FREQ = 2h               |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                          | 17Ch      | 1FFh      | 3F9h      | 17Ch                         | 17Ch      | 17Ch      | 2AEh      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                          | 201h      | 3FEh      | 7F2h      | 17Ch                         | 17Ch      | 2AFh      | 55Bh      | FREQ = 1h               |
|                 |       |               |               | 22Ah                          | 402h      | 7FCh      | FE4h      | 17Ch                         | 2C6h      | 55Eh      | AB5h      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 04h                           | 02h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                           | 26h       | 1Ch       | 0Eh       | 26h                          | 26h       | 26h       | 16h       | FREQ = 0h               |
|                 |       |               |               | 26h                           | 1Ch       | 0Eh       | 08h       | 26h                          | 26h       | 16h       | 0Ch       | FREQ = 1h               |
|                 |       |               |               | 1Ah                           | 0Eh       | 08h       | 04h       | 26h                          | 14h       | 0Ch       | 06h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                           | 10h       | 0Ch       | 06h       | 10h                          | 10h       | 10h       | 0Ah       | FREQ = 0h               |
|                 |       |               |               | 10h                           | 0Ch       | 06h       | 04h       | 10h                          | 10h       | 0Ah       | 06h       | FREQ = 1h               |
|                 |       |               |               | 0Ch                           | 06h       | 04h       | 02h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 2h               |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 51h                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B1h                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 54h                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |

(IMX535-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.5.2.ROI, Overlap ROI

## SLVS Output (IMX535-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |  |  |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|--|--|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |  |  |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |  |  |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |  |  |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         |  |  | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3                           | *4              |                         |  |  |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D8h           | [7:0] | HMAX          | 081h          | 34Ah                                                                                                       | 664h            | 236h                         | 448h            | FREQ = 0h               |  |  |           |
| D9h           | [7:0] |               |               | 67Dh                                                                                                       | CA8h            | 458h                         | 875h            | FREQ = 1h               |  |  |           |
|               |       |               |               | 236h                                                                                                       | 448h            | 181h                         | 2E3h            | FREQ = 2h               |  |  |           |
|               |       |               |               | 458h                                                                                                       | 875h            | 2ECh                         | 5ABh            | FREQ = 3h               |  |  |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 02h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 3h               |  |  |           |
| E3h           | [7:0] | GMTWT         | 70h           | 12h                                                                                                        | 0Ah             | 1Ah                          | 0Eh             | FREQ = 0h               |  |  |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Eh                          | 08h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 1Ah                                                                                                        | 0Eh             | 26h                          | 14h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 0Eh                                                                                                        | 08h             | 14h                          | 0Ah             | FREQ = 3h               |  |  |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |  |  |           |
| E6h           | [7:0] | GSDLY         | 30h           | 08h                                                                                                        | 04h             | 0Ch                          | 06h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 0Ch                                                                                                        | 06h             | 10h                          | 0Ah             | FREQ = 2h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 3h               |  |  |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |  |  |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |  |  |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 | 1h: 12 bit              |  |  |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 |                         |  |  |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |  |  |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 21h           | [7:0] |               | A9h           | 51h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 22h           | [7:0] |               | B0h           | B1h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 46h           | [7:0] |               | 2Ah           | 54h                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |  |  |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

## SLVS – EC output (IMX535-AAMJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|                 |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                                                                                                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                                                                                                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                                                                                                          | 17Ch      | 1FFh      | 3F9h      | 17Ch                         | 17Ch      | 17Ch      | 2AEh      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                                                                                                          | 201h      | 3FEh      | 7F2h      | 17Ch                         | 17Ch      | 2AFh      | 55Bh      | FREQ = 1h               |
|                 |       |               |               | 22Ah                                                                                                          | 402h      | 7FCh      | FE4h      | 17Ch                         | 2C6h      | 55Eh      | AB5h      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 04h                                                                                                           | 02h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                                                                                                           | 26h       | 1Ch       | 0Eh       | 26h                          | 26h       | 26h       | 16h       | FREQ = 0h               |
|                 |       |               |               | 26h                                                                                                           | 1Ch       | 0Eh       | 08h       | 26h                          | 26h       | 16h       | 0Ch       | FREQ = 1h               |
|                 |       |               |               | 1Ah                                                                                                           | 0Eh       | 08h       | 04h       | 26h                          | 14h       | 0Ch       | 06h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 10h       | 0Ch       | 06h       | 10h                          | 10h       | 10h       | 0Ah       | FREQ = 0h               |
|                 |       |               |               | 10h                                                                                                           | 0Ch       | 06h       | 04h       | 10h                          | 10h       | 0Ah       | 06h       | FREQ = 1h               |
|                 |       |               |               | 0Ch                                                                                                           | 06h       | 04h       | 02h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 2h               |
| Chip ID = 03h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 16h                                                                                                           | 0Ch       |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 51h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B1h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 54h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

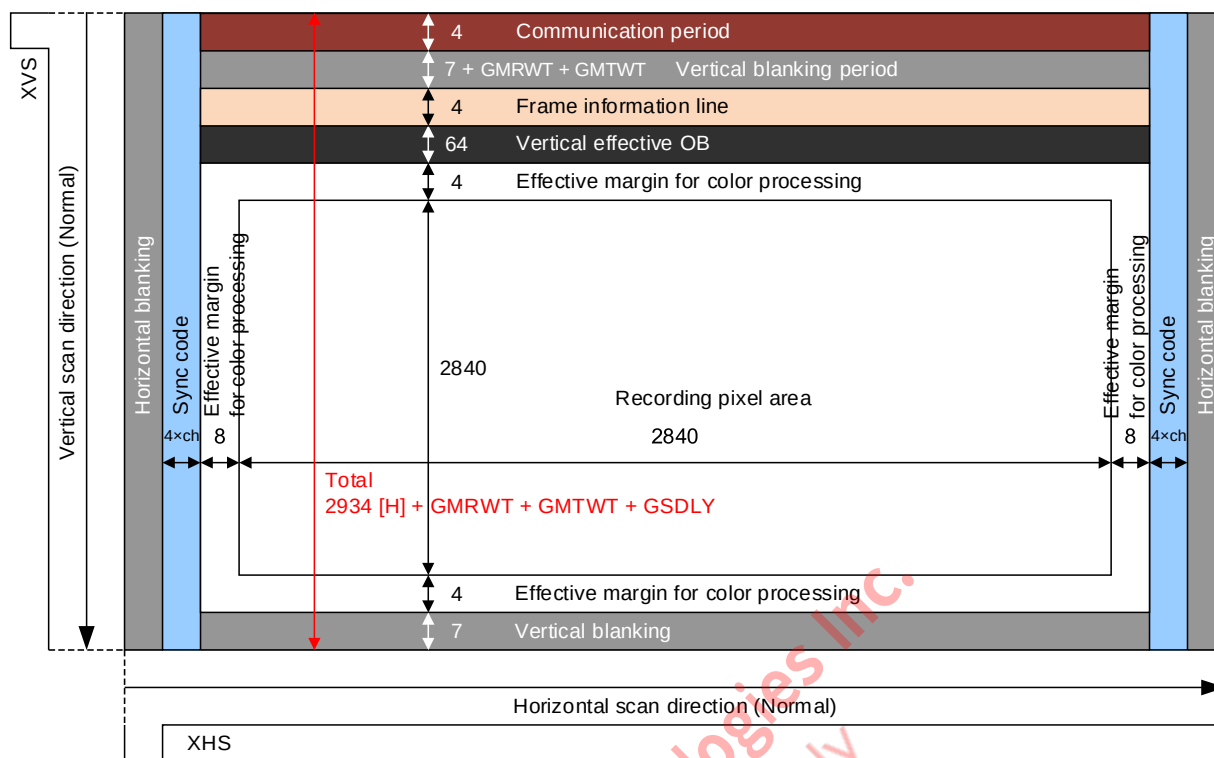
## 10.1.6. IMX536-AAQJ

## 10.1.6.1. All-pixel scan

SLVS Output (IMX536-AAQJ)

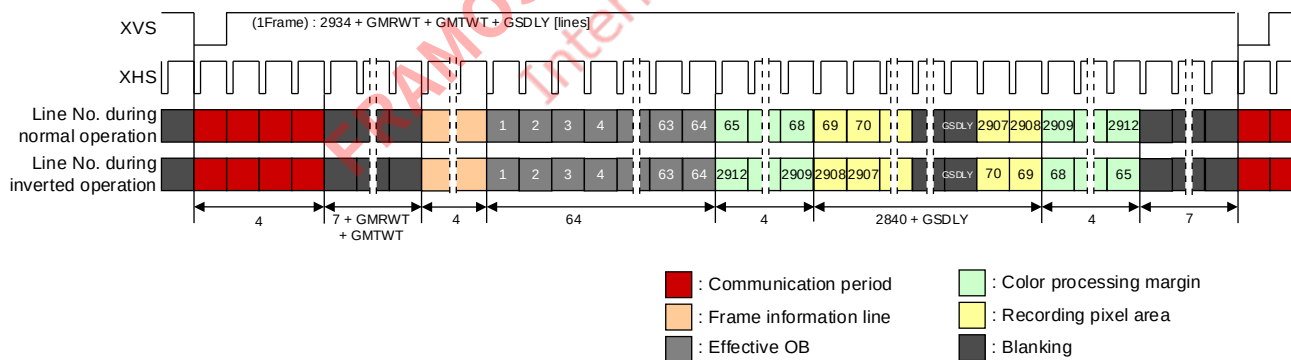
| Address       | bit   | Register name | Initial Value | Setting value                 |      |                              |                   | Remarks                 |                   |           |
|---------------|-------|---------------|---------------|-------------------------------|------|------------------------------|-------------------|-------------------------|-------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |      | AD = 12 bit / Output = 8 bit |                   |                         |                   |           |
|               |       |               |               |                               |      | 8 ch                         | 8 ch              | 4 ch                    | 4 ch              |           |
|               |       |               |               |                               |      | 41.0<br>[frame/s]            | 21.4<br>[frame/s] | 60.3<br>[frame/s]       | 31.8<br>[frame/s] | FREQ = 0h |
|               |       |               |               |                               |      | 21.1<br>[frame/s]            | 10.9<br>[frame/s] | 31.3<br>[frame/s]       | 16.3<br>[frame/s] | FREQ = 1h |
|               |       |               |               |                               |      | 60.3<br>[frame/s]            | 31.8<br>[frame/s] | 65.2<br>[frame/s]       | 46.8<br>[frame/s] | FREQ = 2h |
|               |       |               |               |                               |      | 31.3<br>[frame/s]            | 16.3<br>[frame/s] | 46.2<br>[frame/s]       | 24.3<br>[frame/s] | FREQ = 3h |
|               |       |               |               | 594 Mbps/ch                   |      |                              |                   |                         |                   | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                   |      |                              |                   |                         |                   | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                   |      |                              |                   |                         |                   | FREQ = 2h |
| 445.5 Mbps/ch |       |               |               |                               |      | FREQ = 3h                    |                   |                         |                   |           |
| Chip ID = 02h |       |               |               |                               |      |                              |                   |                         |                   |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | B9Ch                          | B8Ch | BB0h                         | B94h              | FREQ = 0h               |                   |           |
| D5h           | [7:0] |               |               | B8Ah                          | B84h | B94h                         | B86h              | FREQ = 1h               |                   |           |
| D6h           | [7:0] |               |               | BB0h                          | B94h | BB2h                         | BA2h              | FREQ = 2h               |                   |           |
|               |       |               |               | B94h                          | B86h | BA2h                         | B8Ch              | FREQ = 3h               |                   |           |
| D8h           | [7:0] | HMAX          | 081h          | 261h                          | 493h | 19Bh                         | 312h              | FREQ = 0h               |                   |           |
| D9h           | [7:0] |               |               | 4A6h                          | 8FAh | 31Eh                         | 601h              | FREQ = 1h               |                   |           |
|               |       |               |               | 19Bh                          | 312h | 17Ch                         | 214h              | FREQ = 2h               |                   |           |
|               |       |               |               | 31Eh                          | 601h | 21Bh                         | 408h              | FREQ = 3h               |                   |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                           | 02h  | 06h                          | 04h               | FREQ = 0h               |                   |           |
|               |       |               |               | 02h                           | 02h  | 04h                          | 02h               | FREQ = 1h               |                   |           |
|               |       |               |               | 06h                           | 04h  | 06h                          | 04h               | FREQ = 2h               |                   |           |
|               |       |               |               | 04h                           | 02h  | 04h                          | 02h               | FREQ = 3h               |                   |           |
| E3h           | [7:0] | GMTWT         | 70h           | 18h                           | 0Eh  | 24h                          | 12h               | FREQ = 0h               |                   |           |
|               |       |               |               | 0Ch                           | 08h  | 12h                          | 0Ah               | FREQ = 1h               |                   |           |
|               |       |               |               | 24h                           | 12h  | 26h                          | 1Ch               | FREQ = 2h               |                   |           |
|               |       |               |               | 12h                           | 0Ah  | 1Ch                          | 0Eh               | FREQ = 3h               |                   |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |      |                              |                   |                         |                   |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ah                           | 06h  | 10h                          | 08h               | FREQ = 0h               |                   |           |
|               |       |               |               | 06h                           | 04h  | 08h                          | 04h               | FREQ = 1h               |                   |           |
|               |       |               |               | 10h                           | 08h  | 10h                          | 0Ch               | FREQ = 2h               |                   |           |
|               |       |               |               | 08h                           | 04h  | 0Ch                          | 06h               | FREQ = 3h               |                   |           |
| Chip ID = 04h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |      |                              |                   |                         |                   |           |
|               | [5:4] | ADBIT         | 0h            | 1h                            |      |                              |                   | 1h: 12 bit              |                   |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |      |                              |                   |                         |                   |           |
| 3Dh           | [0]   |               |               |                               |      |                              |                   |                         |                   |           |
| 3Eh           | [7:2] |               | 16h           | 0Ch                           |      |                              |                   |                         |                   |           |
| Chip ID = 06h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |      | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |                   |           |
| Chip ID = 07h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 21h           | [7:0] |               | A9h           | 7Dh                           |      |                              |                   |                         |                   |           |
| 22h           | [7:0] |               | B0h           | B0h                           |      |                              |                   |                         |                   |           |
| 46h           | [7:0] |               | 2Ah           | 1Fh                           |      |                              |                   |                         |                   |           |
| Chip ID = 08h |       |               |               |                               |      |                              |                   |                         |                   |           |
| E8h           | [7:0] |               | 2Dh           | 13h                           |      |                              |                   |                         |                   |           |
| Chip ID = 15h |       |               |               |                               |      |                              |                   |                         |                   |           |
| 1Eh           | [7:0] |               | 02h           | 16h                           |      |                              |                   |                         |                   |           |
| 1Fh           | [7:0] |               | 02h           | 16h                           |      |                              |                   |                         |                   |           |

(IMX536-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

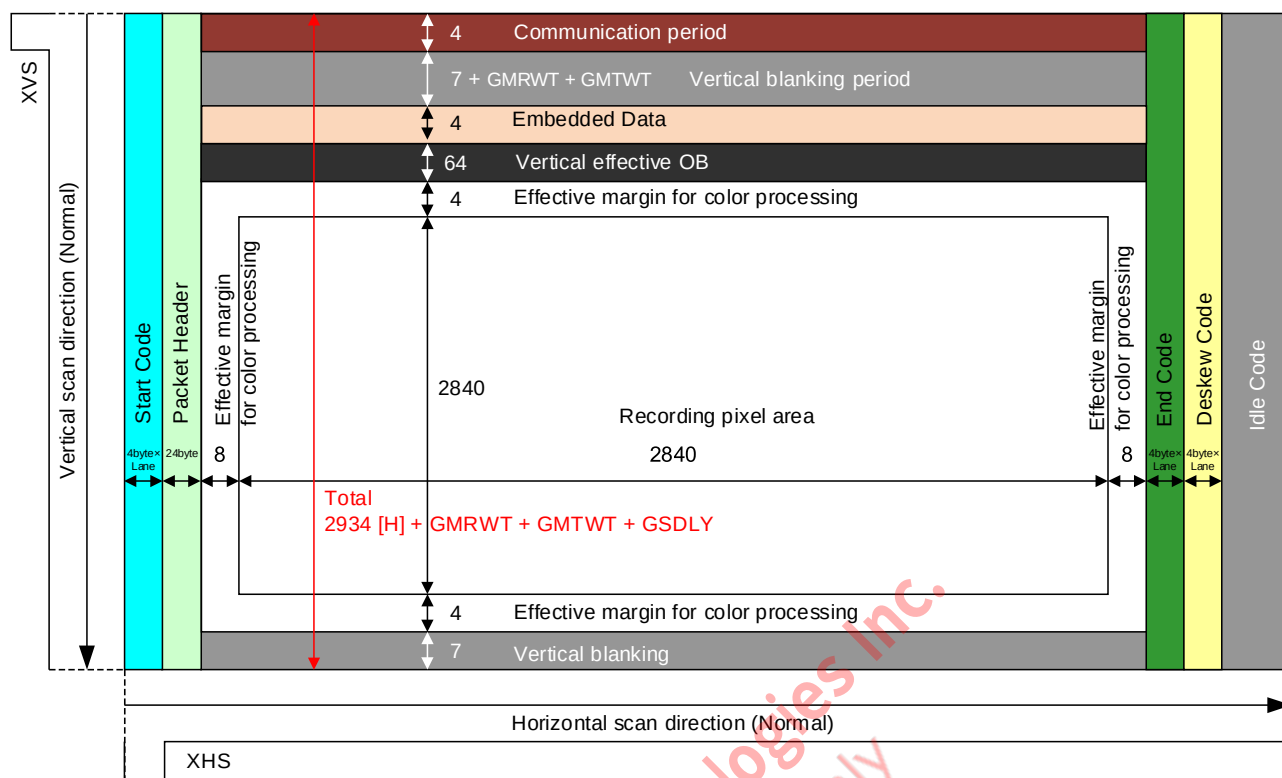


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX536-AAQJ)

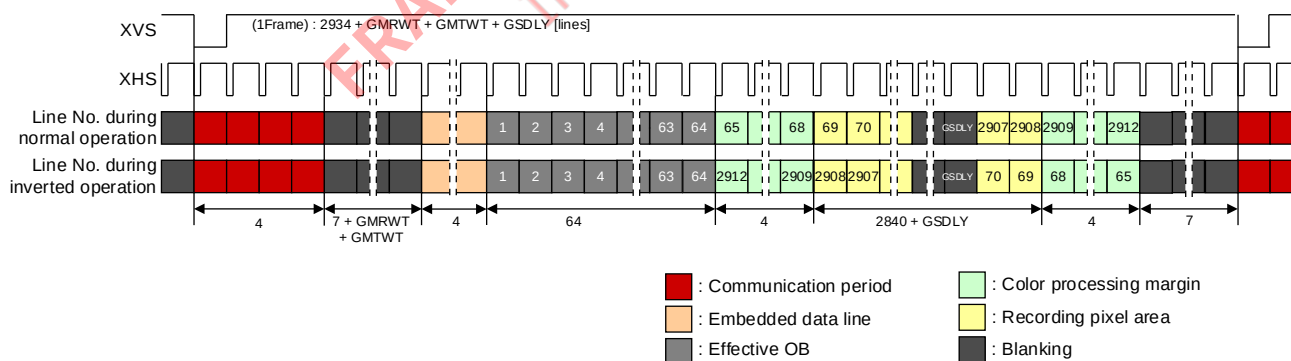
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |            | Remarks                 |           |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|------------|-------------------------|-----------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |            |                         |           |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane     |                         |           |
|                 |       |               |               | 65.2                          | 65.2      | 65.2      | 34.8      | 65.2                         | 65.2      | 65.2      | 51.7       |                         | FREQ = 0h |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s]  |                         |           |
|                 |       |               |               | 65.2                          | 65.2      | 34.6      | 17.5      | 65.2                         | 51.3      | 26.1      | FREQ = 1h  |                         |           |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] |            |                         | [frame/s] |
|                 |       |               |               | 62.6                          | 34.4      | 17.4      | 8.7       | 65.2                         | 49.5      | 25.8      | 13.1       |                         | FREQ = 2h |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s]  |                         |           |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |            |                         |           |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |            |                         |           |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |            |                         |           |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| D4h             | [7:0] | VMAX          | 0CD4h         | BB2h                          | BB2h      | BB2h      | B98h      | BB2h                         | BB2h      | BB2h      | BA6h       | FREQ = 0h               |           |
| D5h             | [7:0] |               |               | BB2h                          | BB2h      | B98h      | B88h      | BB2h                         | BB2h      | BA6h      | B90h       | FREQ = 1h               |           |
| D6h             | [7:0] |               |               | BB0h                          | B98h      | B88h      | B82h      | BB2h                         | BA6h      | B90h      | B84h       | FREQ = 2h               |           |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                          | 17Ch      | 17Ch      | 2CEh      | 17Ch                         | 17Ch      | 17Ch      | 1E1h       | FREQ = 0h               |           |
| D9h             | [7:0] |               |               | 17Ch                          | 17Ch      | 2D3h      | 59Ch      | 17Ch                         | 17Ch      | 1E5h      | 3C1h       | FREQ = 1h               |           |
|                 |       |               |               | 18Ch                          | 2D6h      | 5A5h      | B37h      | 17Ch                         | 1F7h      | 3CAh      | 782h       | FREQ = 2h               |           |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                           | 06h       | 06h       | 04h       | 06h                          | 06h       | 06h       | 04h        | FREQ = 0h               |           |
|                 |       |               |               | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 04h       | 02h        | FREQ = 1h               |           |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h        | FREQ = 2h               |           |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                           | 26h       | 26h       | 14h       | 26h                          | 26h       | 26h       | 1Eh        | FREQ = 0h               |           |
|                 |       |               |               | 26h                           | 26h       | 14h       | 0Ah       | 26h                          | 26h       | 1Eh       | 10h        | FREQ = 1h               |           |
|                 |       |               |               | 24h                           | 14h       | 0Ah       | 06h       | 26h                          | 1Eh       | 10h       | 08h        | FREQ = 2h               |           |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |            |                         |           |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                           | 10h       | 10h       | 0Ah       | 10h                          | 10h       | 10h       | 0Eh        | FREQ = 0h               |           |
|                 |       |               |               | 10h                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 0Eh       | 08h        | FREQ = 1h               |           |
|                 |       |               |               | 10h                           | 0Ah       | 06h       | 04h       | 10h                          | 0Eh       | 08h       | 04h        | FREQ = 2h               |           |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |            |                         |           |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           | 1h: 12 bit |                         |           |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |            |                         |           |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |            |                         |           |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |            | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| 21h             | [7:0] |               | A9h           | 7Dh                           |           |           |           |                              |           |           |            |                         |           |
| 22h             | [7:0] |               | B0h           | B0h                           |           |           |           |                              |           |           |            |                         |           |
| 46h             | [7:0] |               | 2Ah           | 1Fh                           |           |           |           |                              |           |           |            |                         |           |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| E8h             | [7:0] |               | 2Dh           | 13h                           |           |           |           |                              |           |           |            |                         |           |
| Chip ID = 15h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |           |
| 1Eh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |            |                         |           |
| 1Fh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |            |                         |           |

(IMX536-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.6.2.ROI, Overlap ROI

## SLVS Output (IMX536-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3                           | *4              |                         |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D8h           | [7:0] | HMAX          | 081h          | 261h                                                                                                       | 493h            | 19Bh                         | 312h            | FREQ = 0h               |           |
|               |       |               |               | 4A6h                                                                                                       | 8FAh            | 31Eh                         | 601h            | FREQ = 1h               |           |
|               |       |               |               | 19Bh                                                                                                       | 312h            | 17Ch                         | 214h            | FREQ = 2h               |           |
| D9h           | [7:0] |               |               | 31Eh                                                                                                       | 601h            | 21Bh                         | 408h            | FREQ = 3h               |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 0h               |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 06h                          | 04h             | FREQ = 2h               |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 3h               |           |
| E3h           | [7:0] | GMTWT         | 70h           | 18h                                                                                                        | 0Eh             | 24h                          | 12h             | FREQ = 0h               |           |
|               |       |               |               | 0Ch                                                                                                        | 08h             | 12h                          | 0Ah             | FREQ = 1h               |           |
|               |       |               |               | 24h                                                                                                        | 12h             | 26h                          | 1Ch             | FREQ = 2h               |           |
|               |       |               |               | 12h                                                                                                        | 0Ah             | 1Ch                          | 0Eh             | FREQ = 3h               |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ah                                                                                                        | 06h             | 10h                          | 08h             | FREQ = 0h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 08h                          | 04h             | FREQ = 1h               |           |
|               |       |               |               | 10h                                                                                                        | 08h             | 10h                          | 0Ch             | FREQ = 2h               |           |
|               |       |               |               | 08h                                                                                                        | 04h             | 0Ch                          | 06h             | FREQ = 3h               |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         | 2h              |                              |                 | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 21h           | [7:0] |               | A9h           | 7Dh                                                                                                        |                 |                              |                 |                         |           |
| 22h           | [7:0] |               | B0h           | B0h                                                                                                        |                 |                              |                 |                         |           |
| 46h           | [7:0] |               | 2Ah           | 1Fh                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

## SLVS – EC output (IMX536-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                                 |                 |                 |                 |                              |                 |                 |                 | Remarks                 |
|---------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|------------------------------|-----------------|-----------------|-----------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |                 |                 |                 | AD = 12 bit / Output = 8 bit |                 |                 |                 |                         |
|               |       |               |               | 8 lane                                                                                                        | 4 lane          | 2 lane          | 1 lane          | 8 lane                       | 4 lane          | 2 lane          | 1 lane          |                         |
|               |       |               |               | *1<br>[frame/s]                                                                                               | *2<br>[frame/s] | *3<br>[frame/s] | *4<br>[frame/s] | *5<br>[frame/s]              | *6<br>[frame/s] | *7<br>[frame/s] | *8<br>[frame/s] |                         |
|               |       |               |               | 4752 Mbps/lane                                                                                                |                 |                 |                 |                              |                 |                 |                 | FREQ = 0h               |
|               |       |               |               | 2376 Mbps/lane                                                                                                |                 |                 |                 |                              |                 |                 |                 | FREQ = 1h               |
|               |       |               |               | 1188 Mbps /lane                                                                                               |                 |                 |                 |                              |                 |                 |                 | FREQ = 2h               |
| Chip ID = 02h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2              | *3              | *4              | *5                           | *6              | *7              | *8              |                         |
| D5h           | [7:0] |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| D6h           | [7:0] |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| D8h           | [7:0] | HMAX          | 081h          | 17Ch                                                                                                          | 17Ch            | 17Ch            | 2CEh            | 17Ch                         | 17Ch            | 17Ch            | 1E1h            | FREQ = 0h               |
| D9h           | [7:0] |               |               | 17Ch                                                                                                          | 17Ch            | 2D3h            | 59Ch            | 17Ch                         | 17Ch            | 1E5h            | 3C1h            | FREQ = 1h               |
|               |       |               |               | 18Ch                                                                                                          | 2D6h            | 5A5h            | B37h            | 17Ch                         | 1F7h            | 3CAh            | 782h            | FREQ = 2h               |
| E2h           | [7:0] | GMRWT         | 10h           | 06h                                                                                                           | 06h             | 06h             | 04h             | 06h                          | 06h             | 06h             | 04h             | FREQ = 0h               |
|               |       |               |               | 06h                                                                                                           | 06h             | 04h             | 02h             | 06h                          | 06h             | 04h             | 02h             | FREQ = 1h               |
|               |       |               |               | 06h                                                                                                           | 04h             | 02h             | 02h             | 06h                          | 04h             | 02h             | 02h             | FREQ = 2h               |
| E3h           | [7:0] | GMTWT         | 70h           | 26h                                                                                                           | 26h             | 26h             | 14h             | 26h                          | 26h             | 26h             | 1Eh             | FREQ = 0h               |
|               |       |               |               | 26h                                                                                                           | 26h             | 14h             | 0Ah             | 26h                          | 26h             | 1Eh             | 10h             | FREQ = 1h               |
|               |       |               |               | 24h                                                                                                           | 14h             | 0Ah             | 06h             | 26h                          | 26h             | 1Eh             | 08h             | FREQ = 2h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| E6h           | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 10h             | 10h             | 0Ah             | 10h                          | 10h             | 10h             | 0Eh             | FREQ = 0h               |
|               |       |               |               | 10h                                                                                                           | 10h             | 0Ah             | 06h             | 10h                          | 10h             | 0Eh             | 08h             | FREQ = 1h               |
|               |       |               |               | 10h                                                                                                           | 0Ah             | 06h             | 04h             | 10h                          | 0Eh             | 08h             | 04h             | FREQ = 2h               |
| Chip ID = 03h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |                 |                 |                 |                              |                 |                 |                 |                         |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 04h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |                 |                 |                 |                              |                 |                 |                 |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |                 |                 |                 |                              |                 |                 |                 | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 3Dh           | [0]   |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 06h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |                 |                 |                 | 2h                           |                 |                 |                 | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 21h           | [7:0] |               | A9h           | 7Dh                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 22h           | [7:0] |               | B0h           | B0h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 46h           | [7:0] |               | 2Ah           | 1Fh                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 08h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| Chip ID = 15h |       |               |               |                                                                                                               |                 |                 |                 |                              |                 |                 |                 |                         |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                           |                 |                 |                 |                              |                 |                 |                 |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

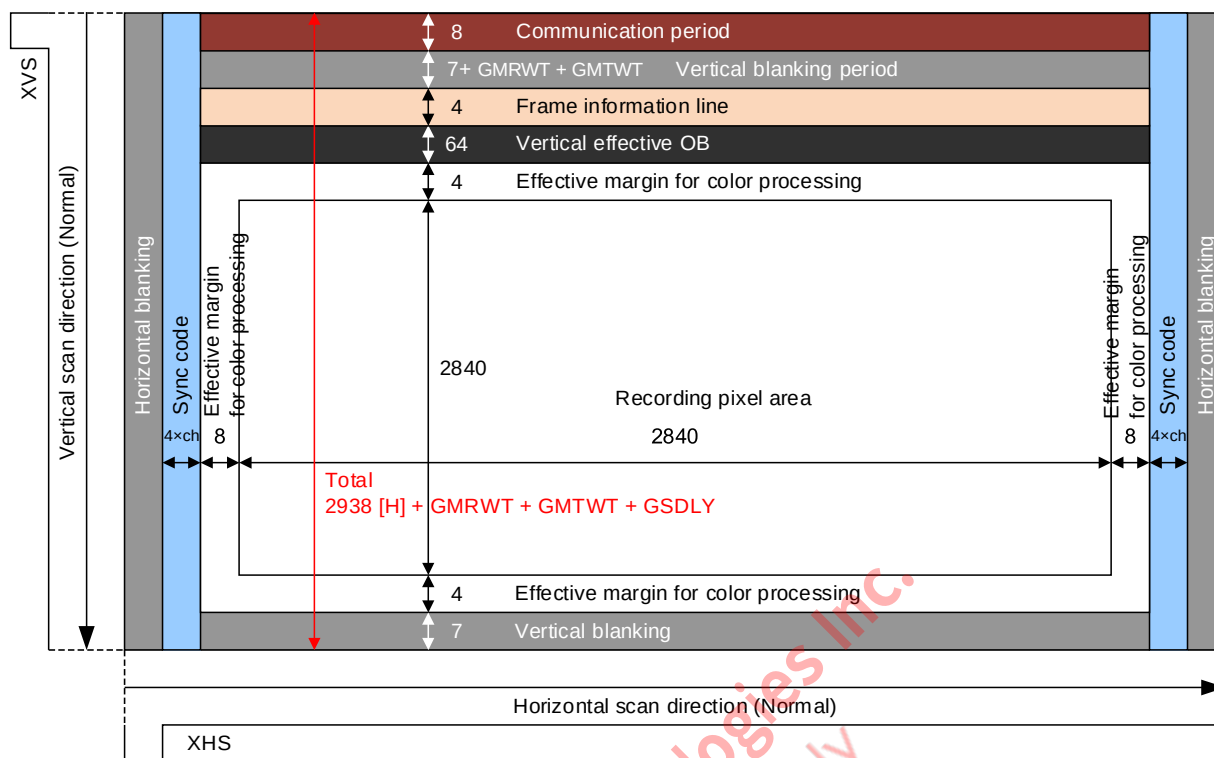
## 10.1.7. IMX536-AAMJ

## 10.1.7.1. All-pixel scan

SLVS Output (IMX536-AAMJ)

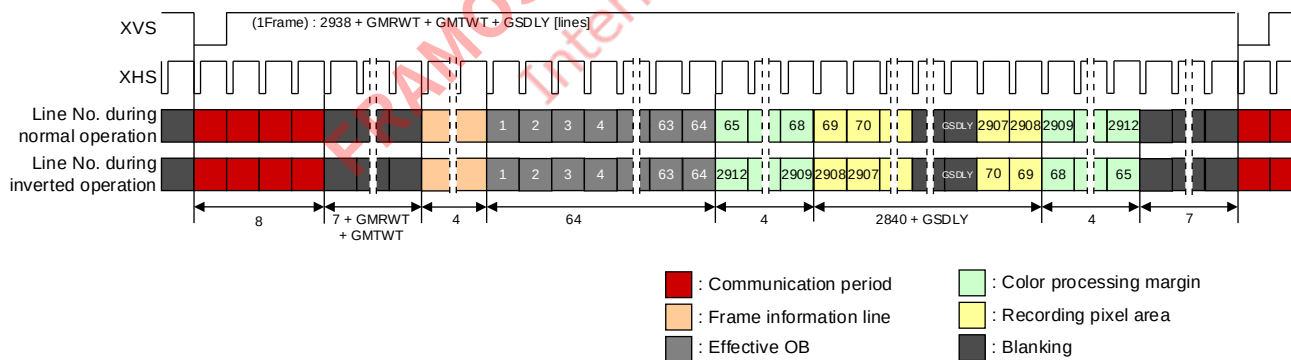
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |
|               |       |               |               | 8 ch                          | 8 ch              | 4 ch                         | 4 ch              |                         |
|               |       |               |               | 40.9<br>[frame/s]             | 21.4<br>[frame/s] | 60.2<br>[frame/s]            | 31.8<br>[frame/s] | FREQ = 0h               |
|               |       |               |               | 21.0<br>[frame/s]             | 10.9<br>[frame/s] | 31.3<br>[frame/s]            | 16.3<br>[frame/s] | FREQ = 1h               |
|               |       |               |               | 60.2<br>[frame/s]             | 31.8<br>[frame/s] | 65.1<br>[frame/s]            | 46.8<br>[frame/s] | FREQ = 2h               |
|               |       |               |               | 31.3<br>[frame/s]             | 16.3<br>[frame/s] | 46.1<br>[frame/s]            | 24.3<br>[frame/s] | FREQ = 3h               |
|               |       |               |               | 594 Mbps/ch                   |                   |                              |                   | FREQ = 0h               |
|               |       |               |               | 297 Mbps/ch                   |                   |                              |                   | FREQ = 1h               |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |
| 445.5 Mbps/ch |       |               |               | FREQ = 3h                     |                   |                              |                   |                         |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |
| D4h           | [7:0] | VMAX          | 0CD4h         | BA0h                          | B90h              | BB4h                         | B98h              | FREQ = 0h               |
| D5h           | [7:0] |               |               | B8Eh                          | B88h              | B98h                         | B8Ah              | FREQ = 1h               |
| D6h           | [7:0] |               |               | BB4h                          | B98h              | BB6h                         | BA6h              | FREQ = 2h               |
|               |       |               |               | B98h                          | B8Ah              | BA6h                         | B90h              | FREQ = 3h               |
| D8h           | [7:0] | HMAX          | 081h          | 261h                          | 493h              | 19Bh                         | 312h              | FREQ = 0h               |
| D9h           | [7:0] |               |               | 4A6h                          | 8FAh              | 31Eh                         | 601h              | FREQ = 1h               |
|               |       |               |               | 19Bh                          | 312h              | 17Ch                         | 214h              | FREQ = 2h               |
|               |       |               |               | 31Eh                          | 601h              | 21Bh                         | 408h              | FREQ = 3h               |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                           | 02h               | 06h                          | 04h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 04h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 06h                           | 04h               | 06h                          | 04h               | FREQ = 2h               |
|               |       |               |               | 04h                           | 02h               | 04h                          | 02h               | FREQ = 3h               |
| E3h           | [7:0] | GMTWT         | 70h           | 18h                           | 0Eh               | 24h                          | 12h               | FREQ = 0h               |
|               |       |               |               | 0Ch                           | 08h               | 12h                          | 0Ah               | FREQ = 1h               |
|               |       |               |               | 24h                           | 12h               | 26h                          | 1Ch               | FREQ = 2h               |
|               |       |               |               | 12h                           | 0Ah               | 1Ch                          | 0Eh               | FREQ = 3h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ah                           | 06h               | 10h                          | 08h               | FREQ = 0h               |
|               |       |               |               | 06h                           | 04h               | 08h                          | 04h               | FREQ = 1h               |
|               |       |               |               | 10h                           | 08h               | 10h                          | 0Ch               | FREQ = 2h               |
|               |       |               |               | 08h                           | 04h               | 0Ch                          | 06h               | FREQ = 3h               |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                           |                   |                              |                   |                         |
| 3Dh           | [0]   |               |               |                               |                   |                              |                   |                         |
| 3Eh           | [7:2] |               | 16h           | 0Ch                           |                   |                              |                   |                         |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |
| 21h           | [7:0] |               | A9h           | 7Dh                           |                   |                              |                   |                         |
| 22h           | [7:0] |               | B0h           | B0h                           |                   |                              |                   |                         |
| 46h           | [7:0] |               | 2Ah           | 1Fh                           |                   |                              |                   |                         |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |
| E8h           | [7:0] |               | 2Dh           | 13h                           |                   |                              |                   |                         |
| Chip ID = 15h |       |               |               |                               |                   |                              |                   |                         |
| 1Eh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |
| 1Fh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |

(IMX536-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

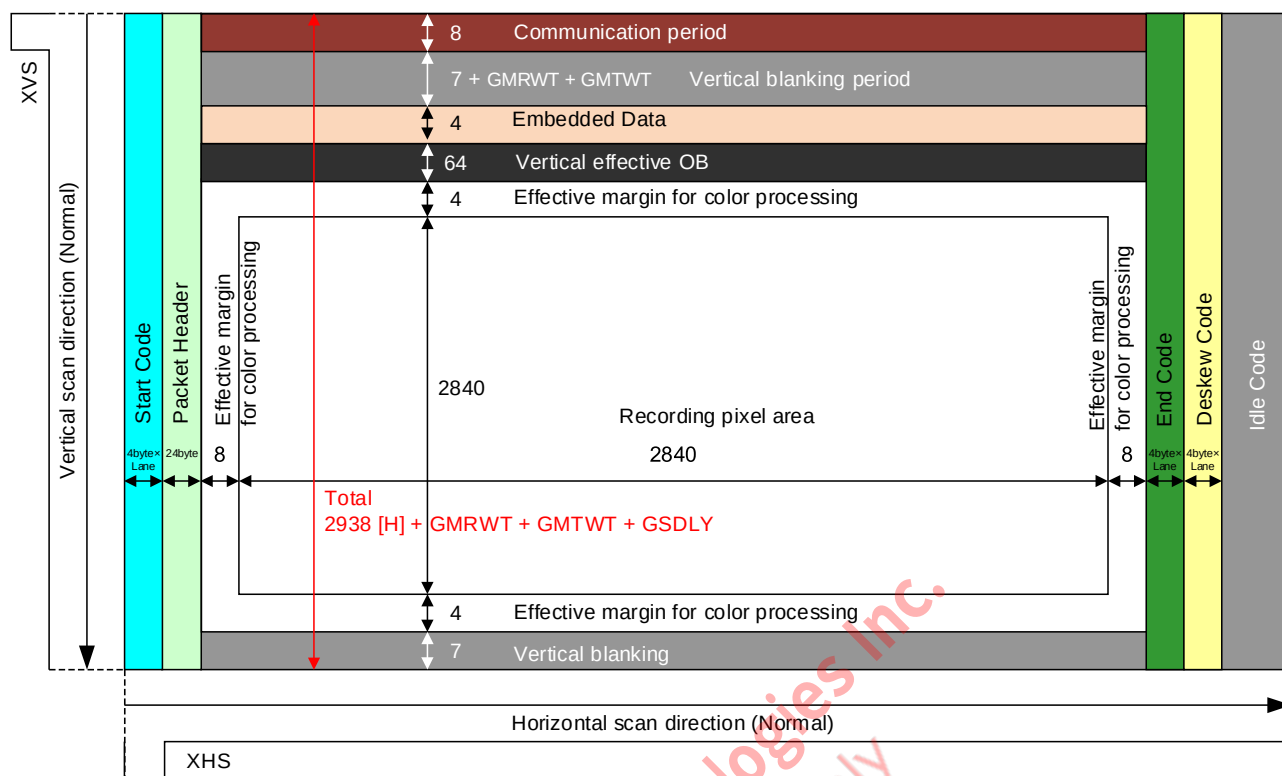


### Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX536-AAMJ)

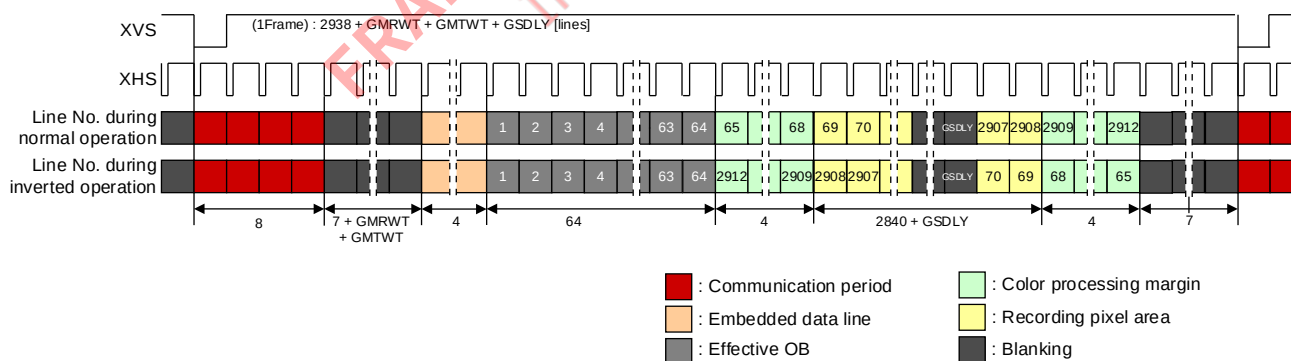
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | 65.1                          | 65.1      | 65.1      | 34.7      | 65.1                         | 65.1      | 65.1      | 51.6      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 65.1                          | 65.1      | 34.5      | 17.4      | 65.1                         | 51.2      | 26.0      |           |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 62.5                          | 34.4      | 17.3      | 8.7       | 65.1                         | 49.4      | 25.8      | 13.0      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | BB6h                          | BB6h      | BB6h      | B9Ch      | BB6h                         | BB6h      | BB6h      | BAAh      | FREQ = 0h               |
| D5h             | [7:0] |               |               | BB6h                          | BB6h      | B9Ch      | B8Ch      | BB6h                         | BB6h      | BAAh      | B94h      | FREQ = 1h               |
| D6h             | [7:0] |               |               | BB4h                          | B9Ch      | B8Ch      | B86h      | BB6h                         | BAAh      | B94h      | B88h      | FREQ = 2h               |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                          | 17Ch      | 17Ch      | 2CEh      | 17Ch                         | 17Ch      | 17Ch      | 1E1h      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                          | 17Ch      | 2D3h      | 59Ch      | 17Ch                         | 17Ch      | 1E5h      | 3C1h      | FREQ = 1h               |
|                 |       |               |               | 18Ch                          | 2D6h      | 5A5h      | B37h      | 17Ch                         | 1F7h      | 3CAh      | 782h      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                           | 06h       | 06h       | 04h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                           | 26h       | 26h       | 14h       | 26h                          | 26h       | 26h       | 1Eh       | FREQ = 0h               |
|                 |       |               |               | 26h                           | 26h       | 14h       | 0Ah       | 26h                          | 26h       | 1Eh       | 10h       | FREQ = 1h               |
|                 |       |               |               | 24h                           | 14h       | 0Ah       | 06h       | 26h                          | 1Eh       | 10h       | 08h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                           | 10h       | 10h       | 0Ah       | 10h                          | 10h       | 10h       | 0Eh       | FREQ = 0h               |
|                 |       |               |               | 10h                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 0Eh       | 08h       | FREQ = 1h               |
|                 |       |               |               | 10h                           | 0Ah       | 06h       | 04h       | 10h                          | 0Eh       | 08h       | 04h       | FREQ = 2h               |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 7Dh                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B0h                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 1Fh                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |

(IMX536-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.7.2.ROI, Overlap ROI

## SLVS Output (IMX536-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3                           | *4              |                         |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D8h           | [7:0] | HMAX          | 081h          | 261h                                                                                                       | 493h            | 19Bh                         | 312h            | FREQ = 0h               |           |
| D9h           | [7:0] |               |               | 4A6h                                                                                                       | 8FAh            | 31Eh                         | 601h            | FREQ = 1h               |           |
|               |       |               |               | 19Bh                                                                                                       | 312h            | 17Ch                         | 214h            | FREQ = 2h               |           |
|               |       |               |               | 31Eh                                                                                                       | 601h            | 21Bh                         | 408h            | FREQ = 3h               |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 0h               |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 06h                          | 04h             | FREQ = 2h               |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 3h               |           |
| E3h           | [7:0] | GMTWT         | 70h           | 18h                                                                                                        | 0Eh             | 24h                          | 12h             | FREQ = 0h               |           |
|               |       |               |               | 0Ch                                                                                                        | 08h             | 12h                          | 0Ah             | FREQ = 1h               |           |
|               |       |               |               | 24h                                                                                                        | 12h             | 26h                          | 1Ch             | FREQ = 2h               |           |
|               |       |               |               | 12h                                                                                                        | 0Ah             | 1Ch                          | 0Eh             | FREQ = 3h               |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ah                                                                                                        | 06h             | 10h                          | 08h             | FREQ = 0h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 08h                          | 04h             | FREQ = 1h               |           |
|               |       |               |               | 10h                                                                                                        | 08h             | 10h                          | 0Ch             | FREQ = 2h               |           |
|               |       |               |               | 08h                                                                                                        | 04h             | 0Ch                          | 06h             | FREQ = 3h               |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         | 2h              |                              |                 | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 21h           | [7:0] |               | A9h           | 7Dh                                                                                                        |                 |                              |                 |                         |           |
| 22h           | [7:0] |               | B0h           | B0h                                                                                                        |                 |                              |                 |                         |           |
| 46h           | [7:0] |               | 2Ah           | 1Fh                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

## SLVS – EC output (IMX536-AAMJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|                 |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                                                                                                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                                                                                                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                                                                                                          | 17Ch      | 17Ch      | 2CEh      | 17Ch                         | 17Ch      | 17Ch      | 1E1h      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                                                                                                          | 17Ch      | 2D3h      | 59Ch      | 17Ch                         | 17Ch      | 1E5h      | 3C1h      | FREQ = 1h               |
|                 |       |               |               | 18Ch                                                                                                          | 2D6h      | 5A5h      | B37h      | 17Ch                         | 1F7h      | 3CAh      | 782h      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                                                                                                           | 06h       | 06h       | 04h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                                                                                                           | 26h       | 26h       | 14h       | 26h                          | 26h       | 26h       | 1Eh       | FREQ = 0h               |
|                 |       |               |               | 26h                                                                                                           | 26h       | 14h       | 0Ah       | 26h                          | 26h       | 1Eh       | 10h       | FREQ = 1h               |
|                 |       |               |               | 24h                                                                                                           | 14h       | 0Ah       | 06h       | 26h                          | 1Eh       | 10h       | 08h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 10h       | 10h       | 0Ah       | 10h                          | 10h       | 10h       | 0Eh       | FREQ = 0h               |
|                 |       |               |               | 10h                                                                                                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 0Eh       | 08h       | FREQ = 1h               |
|                 |       |               |               | 10h                                                                                                           | 0Ah       | 06h       | 04h       | 10h                          | 0Eh       | 08h       | 04h       | FREQ = 2h               |
| Chip ID = 03h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 16h                                                                                                           | 0Ch       |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 7Dh                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B0h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 1Fh                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

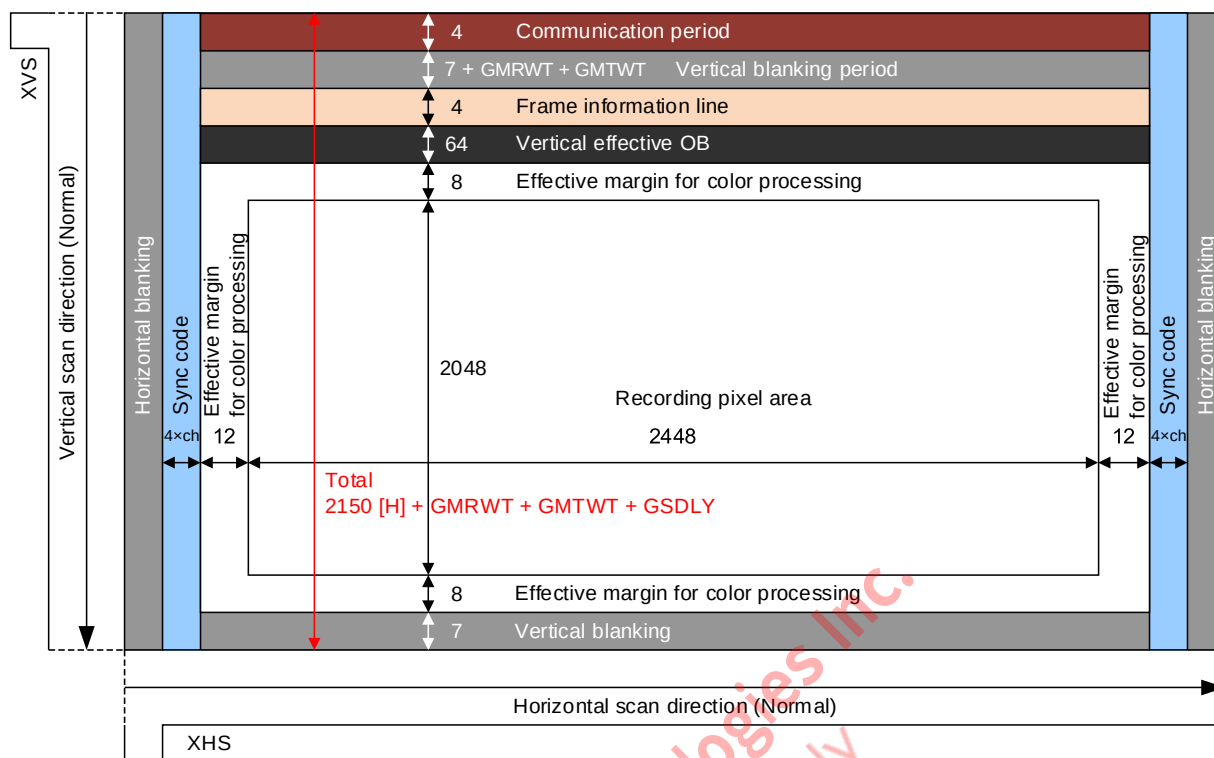
## 10.1.8. IMX537-AAQJ

## 10.1.8.1. All-pixel scan

SLVS Output (IMX537-AAQJ)

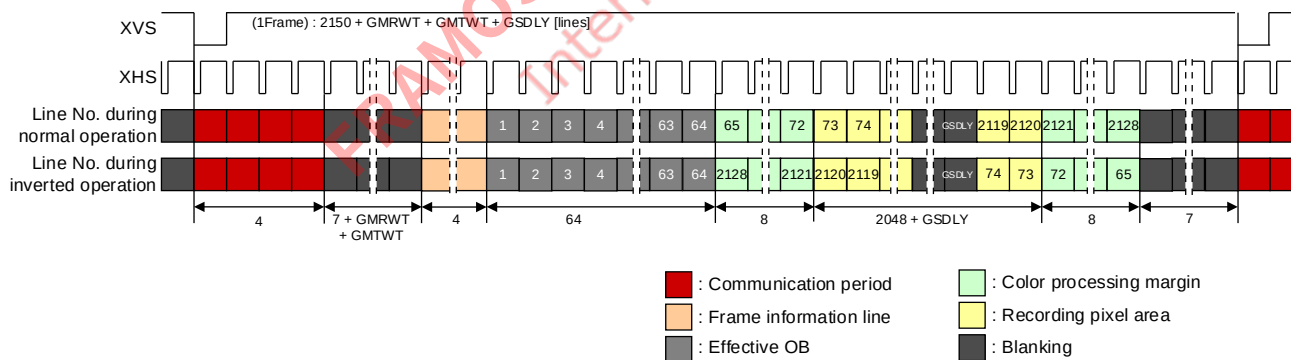
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |
|               |       |               |               | 8 ch                          | 8 ch              | 4 ch                         | 4 ch              |                         |
|               |       |               |               | 63.0<br>[frame/s]             | 33.2<br>[frame/s] | 88.4<br>[frame/s]            | 49.2<br>[frame/s] | FREQ = 0h               |
|               |       |               |               | 32.6<br>[frame/s]             | 17.0<br>[frame/s] | 48.3<br>[frame/s]            | 25.4<br>[frame/s] | FREQ = 1h               |
|               |       |               |               | 88.4<br>[frame/s]             | 49.2<br>[frame/s] | 88.4<br>[frame/s]            | 72.0<br>[frame/s] | FREQ = 2h               |
|               |       |               |               | 48.3<br>[frame/s]             | 25.4<br>[frame/s] | 71.0<br>[frame/s]            | 37.7<br>[frame/s] | FREQ = 3h               |
|               |       |               |               | 594 Mbps/ch                   |                   |                              |                   | FREQ = 0h               |
|               |       |               |               | 297 Mbps/ch                   |                   |                              |                   | FREQ = 1h               |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |
| 445.5 Mbps/ch |       |               |               | FREQ = 3h                     |                   |                              |                   |                         |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |
| D4h           | [7:0] | VMAX          | 0CD4h         | 892h                          | 87Ch              | 8A2h                         | 88Ah              | FREQ = 0h               |
| D5h           | [7:0] |               |               | 87Ch                          | 874h              | 88Ah                         | 87Ah              | FREQ = 1h               |
| D6h           | [7:0] |               |               | 8A2h                          | 88Ah              | 8A2h                         | 89Ah              | FREQ = 2h               |
|               |       |               |               | 88Ah                          | 87Ah              | 898h                         | 882h              | FREQ = 3h               |
| D8h           | [7:0] | HMAX          | 081h          | 219h                          | 403h              | 17Ch                         | 2B2h              | FREQ = 0h               |
| D9h           | [7:0] |               |               | 416h                          | 7DAh              | 2BEh                         | 541h              | FREQ = 1h               |
|               |       |               |               | 17Ch                          | 2B2h              | 17Ch                         | 1D4h              | FREQ = 2h               |
|               |       |               |               | 2BEh                          | 541h              | 1DBh                         | 388h              | FREQ = 3h               |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                           | 02h               | 06h                          | 04h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 04h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 06h                           | 04h               | 06h                          | 06h               | FREQ = 2h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 3h               |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch                           | 0Eh               | 26h                          | 16h               | FREQ = 0h               |
|               |       |               |               | 0Eh                           | 08h               | 16h                          | 0Ch               | FREQ = 1h               |
|               |       |               |               | 26h                           | 16h               | 26h                          | 20h               | FREQ = 2h               |
|               |       |               |               | 16h                           | 0Ch               | 1Eh                          | 10h               | FREQ = 3h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch                           | 06h               | 10h                          | 0Ah               | FREQ = 0h               |
|               |       |               |               | 06h                           | 04h               | 0Ah                          | 06h               | FREQ = 1h               |
|               |       |               |               | 10h                           | 0Ah               | 10h                          | 0Eh               | FREQ = 2h               |
|               |       |               |               | 0Ah                           | 06h               | 0Eh                          | 08h               | FREQ = 3h               |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           |                               |                   |                              |                   |                         |
| 3Dh           | [0]   |               |               | 0Ch                           |                   |                              |                   |                         |
| 3Eh           | [7:2] |               |               | 0Ch                           |                   |                              |                   |                         |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |
| 21h           | [7:0] |               | A9h           | 3Dh                           |                   |                              |                   |                         |
| 22h           | [7:0] |               | B0h           | B0h                           |                   |                              |                   |                         |
| 46h           | [7:0] |               | 2Ah           | 0Fh                           |                   |                              |                   |                         |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |
| E8h           | [7:0] |               | 2Dh           | 13h                           |                   |                              |                   |                         |
| Chip ID = 15h |       |               |               |                               |                   |                              |                   |                         |
| 1Eh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |
| 1Fh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |

(IMX537-AAQJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

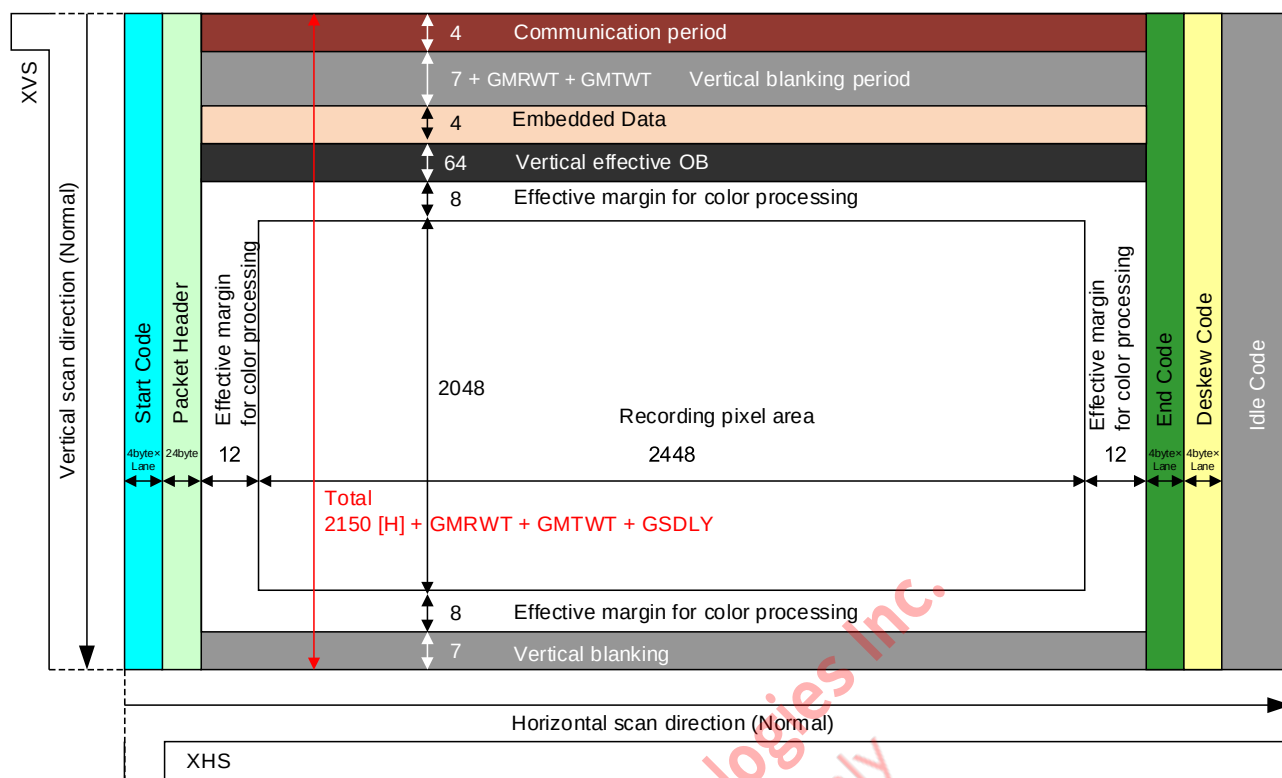


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX537-AAQJ)

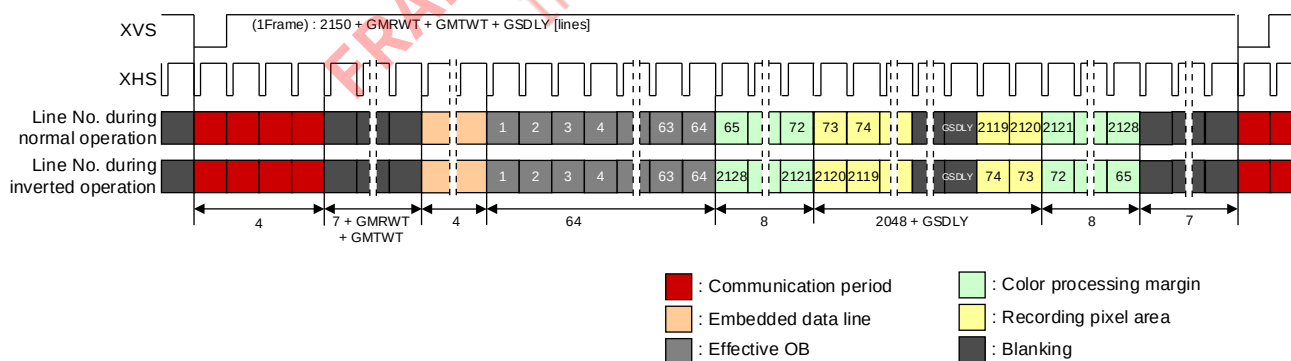
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | 88.4                          | 88.4      | 88.4      | 54.2      | 88.4                         | 88.4      | 88.4      | 80.3      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 88.4                          | 88.4      | 53.7      | 27.3      | 88.4                         | 88.4      | 79.5      | 40.6      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 88.4                          | 53.5      | 27.1      | 13.7      | 88.4                         | 76.7      | 40.2      | 20.4      |                         |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | 8A2h                          | 8A2h      | 8A2h      | 88Ch      | 8A2h                         | 8A2h      | 8A2h      | 89Eh      | FREQ = 0h               |
| D5h             | [7:0] |               |               | 8A2h                          | 8A2h      | 88Ch      | 87Ah      | 8A2h                         | 8A2h      | 89Eh      | 884h      | FREQ = 1h               |
| D6h             | [7:0] |               |               | 8A2h                          | 88Ch      | 87Ah      | 872h      | 8A2h                         | 89Ch      | 884h      | 876h      | FREQ = 2h               |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                          | 17Ch      | 17Ch      | 272h      | 17Ch                         | 17Ch      | 17Ch      | 1A3h      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                          | 17Ch      | 277h      | 4E4h      | 17Ch                         | 17Ch      | 1A7h      | 346h      | FREQ = 1h               |
|                 |       |               |               | 17Ch                          | 27Ah      | 4EDh      | 9C7h      | 17Ch                         | 1B7h      | 34Eh      | 68Ch      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                           | 06h       | 06h       | 04h       | 06h                          | 06h       | 06h       | 06h       | FREQ = 0h               |
|                 |       |               |               | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 1h               |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                           | 26h       | 26h       | 18h       | 26h                          | 26h       | 26h       | 22h       | FREQ = 0h               |
|                 |       |               |               | 26h                           | 26h       | 18h       | 0Ch       | 26h                          | 26h       | 22h       | 12h       | FREQ = 1h               |
|                 |       |               |               | 26h                           | 18h       | 0Ch       | 06h       | 26h                          | 22h       | 12h       | 0Ah       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                           | 10h       | 10h       | 0Ah       | 10h                          | 10h       | 10h       | 10h       | FREQ = 0h               |
|                 |       |               |               | 10h                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 10h       | 08h       | FREQ = 1h               |
|                 |       |               |               | 10h                           | 0Ah       | 06h       | 04h       | 10h                          | 0Eh       | 08h       | 04h       | FREQ = 2h               |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 3Dh                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B0h                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 0Fh                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |           |                         |

(IMX537-AAQJ)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.8.2.ROI, Overlap ROI

## SLVS Output (IMX537-AAQJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |  |  |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|--|--|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |  |  |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |  |  |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |  |  |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         |  |  | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         |  |  | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3                           | *4              |                         |  |  |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| D8h           | [7:0] | HMAX          | 081h          | 219h                                                                                                       | 403h            | 17Ch                         | 2B2h            | FREQ = 0h               |  |  |           |
| D9h           | [7:0] |               |               | 416h                                                                                                       | 7DAh            | 2BEh                         | 541h            | FREQ = 1h               |  |  |           |
|               |       |               |               | 17Ch                                                                                                       | 2B2h            | 17Ch                         | 1D4h            | FREQ = 2h               |  |  |           |
|               |       |               |               | 2BEh                                                                                                       | 541h            | 1DBh                         | 388h            | FREQ = 3h               |  |  |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 06h                          | 06h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 3h               |  |  |           |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch                                                                                                        | 0Eh             | 26h                          | 16h             | FREQ = 0h               |  |  |           |
|               |       |               |               | 0Eh                                                                                                        | 08h             | 16h                          | 0Ch             | FREQ = 1h               |  |  |           |
|               |       |               |               | 26h                                                                                                        | 16h             | 26h                          | 20h             | FREQ = 2h               |  |  |           |
|               |       |               |               | 16h                                                                                                        | 0Ch             | 1Eh                          | 10h             | FREQ = 3h               |  |  |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |  |  |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch                                                                                                        | 06h             | 10h                          | 0Ah             | FREQ = 0h               |  |  |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 1h               |  |  |           |
|               |       |               |               | 10h                                                                                                        | 0Ah             | 10h                          | 0Eh             | FREQ = 2h               |  |  |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Eh                          | 08h             | FREQ = 3h               |  |  |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |  |  |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |  |  |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |  |  |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |  |  |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |  |  |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 21h           | [7:0] |               | A9h           | 3Dh                                                                                                        |                 |                              |                 |                         |  |  |           |
| 22h           | [7:0] |               | B0h           | B0h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 46h           | [7:0] |               | 2Ah           | 0Fh                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                        |                 |                              |                 |                         |  |  |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                              |                 |                         |  |  |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |  |  |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |  |  |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

## SLVS – EC output (IMX537-AAQJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|                 |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                                                                                                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                                                                                                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                                                                                                          | 17Ch      | 1FFh      | 3F9h      | 17Ch                         | 17Ch      | 17Ch      | 2AEh      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                                                                                                          | 201h      | 3FEh      | 7F2h      | 17Ch                         | 17Ch      | 2AFh      | 55Bh      | FREQ = 1h               |
|                 |       |               |               | 22Ah                                                                                                          | 402h      | 7FCh      | FE4h      | 17Ch                         | 2C6h      | 55Eh      | AB5h      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 0h               |
|                 |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 1h               |
|                 |       |               |               | 04h                                                                                                           | 02h       | 02h       | 02h       | 06h                          | 04h       | 02h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                                                                                                           | 26h       | 1Ch       | 0Eh       | 26h                          | 26h       | 26h       | 16h       | FREQ = 0h               |
|                 |       |               |               | 26h                                                                                                           | 1Ch       | 0Eh       | 08h       | 26h                          | 26h       | 16h       | 0Ch       | FREQ = 1h               |
|                 |       |               |               | 1Ah                                                                                                           | 0Eh       | 08h       | 04h       | 26h                          | 14h       | 0Ch       | 06h       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 10h       | 0Ch       | 06h       | 10h                          | 10h       | 10h       | 0Ah       | FREQ = 0h               |
|                 |       |               |               | 10h                                                                                                           | 0Ch       | 06h       | 04h       | 10h                          | 10h       | 0Ah       | 06h       | FREQ = 1h               |
|                 |       |               |               | 0Ch                                                                                                           | 06h       | 04h       | 02h       | 10h                          | 0Ah       | 06h       | 04h       | FREQ = 2h               |
| Chip ID = 03h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 3Dh                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B0h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 0Fh                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 86$$

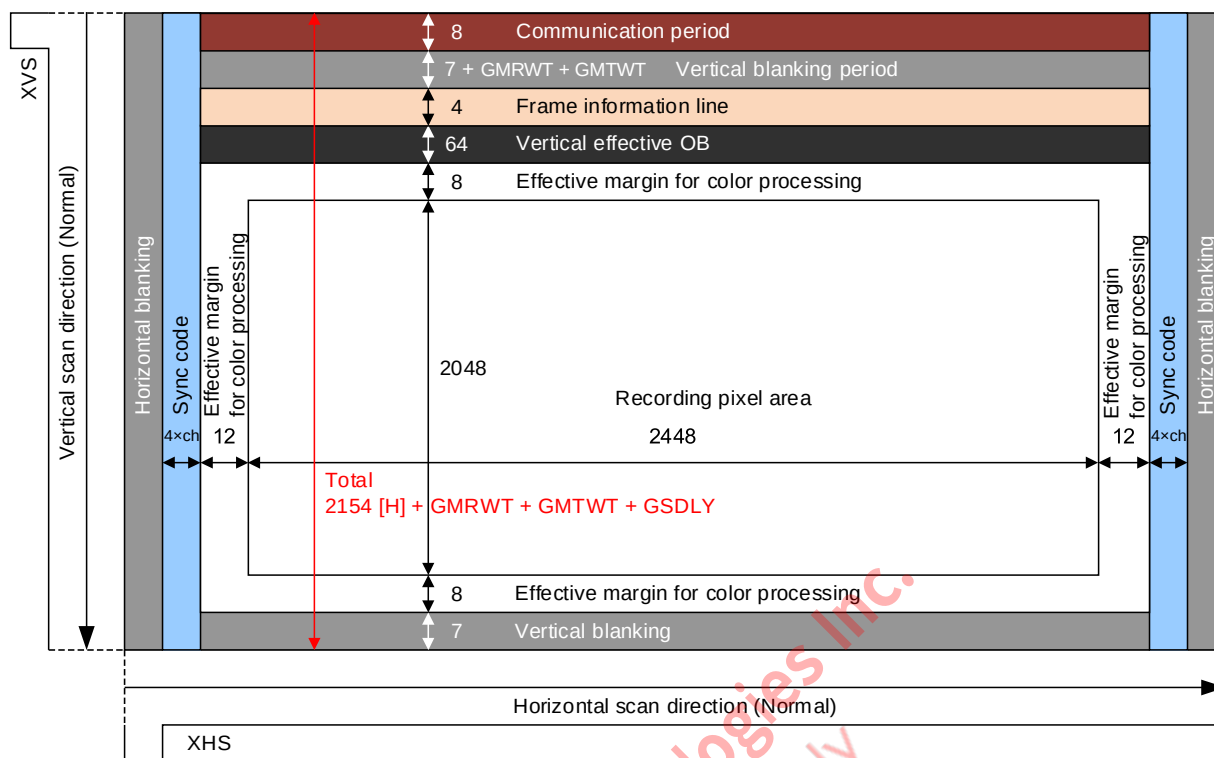
## 10.1.9. IMX537-AAMJ

## 10.1.9.1. All-pixel scan

SLVS Output (IMX537-AAMJ)

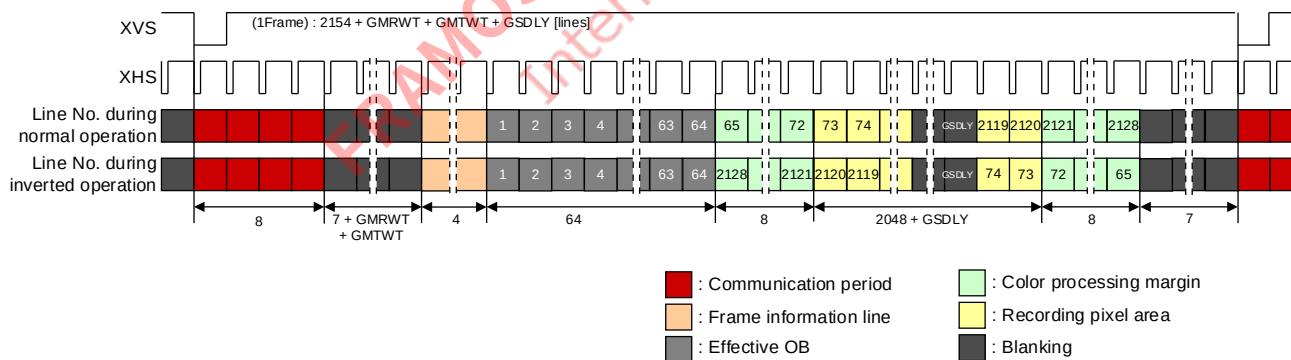
| Address       | bit   | Register name | Initial Value | Setting value                 |                   |                              |                   | Remarks                 |
|---------------|-------|---------------|---------------|-------------------------------|-------------------|------------------------------|-------------------|-------------------------|
|               |       |               |               | AD = 12 bit / Output = 12 bit |                   | AD = 12 bit / Output = 8 bit |                   |                         |
|               |       |               |               | 8 ch                          | 8 ch              | 4 ch                         | 4 ch              |                         |
|               |       |               |               | 62.9<br>[frame/s]             | 33.2<br>[frame/s] | 88.2<br>[frame/s]            | 49.1<br>[frame/s] | FREQ = 0h               |
|               |       |               |               | 32.6<br>[frame/s]             | 17.0<br>[frame/s] | 48.2<br>[frame/s]            | 25.3<br>[frame/s] | FREQ = 1h               |
|               |       |               |               | 88.2<br>[frame/s]             | 49.1<br>[frame/s] | 88.2<br>[frame/s]            | 71.9<br>[frame/s] | FREQ = 2h               |
|               |       |               |               | 48.2<br>[frame/s]             | 25.3<br>[frame/s] | 70.9<br>[frame/s]            | 37.6<br>[frame/s] | FREQ = 3h               |
|               |       |               |               | 594 Mbps/ch                   |                   |                              |                   | FREQ = 0h               |
|               |       |               |               | 297 Mbps/ch                   |                   |                              |                   | FREQ = 1h               |
|               |       |               |               | 891 Mbps/ch                   |                   |                              |                   | FREQ = 2h               |
| 445.5 Mbps/ch |       |               |               | FREQ = 3h                     |                   |                              |                   |                         |
| Chip ID = 02h |       |               |               |                               |                   |                              |                   |                         |
| D4h           | [7:0] | VMAX          | 0CD4h         | 896h                          | 880h              | 8A6h                         | 88Eh              | FREQ = 0h               |
| D5h           | [7:0] |               |               | 880h                          | 878h              | 88Eh                         | 87Eh              | FREQ = 1h               |
| D6h           | [7:0] |               |               | 8A6h                          | 88Eh              | 8A6h                         | 89Eh              | FREQ = 2h               |
|               |       |               |               | 88Eh                          | 87Eh              | 89Ch                         | 886h              | FREQ = 3h               |
| D8h           | [7:0] | HMAX          | 081h          | 219h                          | 403h              | 17Ch                         | 2B2h              | FREQ = 0h               |
| D9h           | [7:0] |               |               | 416h                          | 7DAh              | 2BEh                         | 541h              | FREQ = 1h               |
|               |       |               |               | 17Ch                          | 2B2h              | 17Ch                         | 1D4h              | FREQ = 2h               |
|               |       |               |               | 2BEh                          | 541h              | 1DBh                         | 388h              | FREQ = 3h               |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                           | 02h               | 06h                          | 04h               | FREQ = 0h               |
|               |       |               |               | 02h                           | 02h               | 04h                          | 02h               | FREQ = 1h               |
|               |       |               |               | 06h                           | 04h               | 06h                          | 06h               | FREQ = 2h               |
|               |       |               |               | 04h                           | 02h               | 06h                          | 04h               | FREQ = 3h               |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch                           | 0Eh               | 26h                          | 16h               | FREQ = 0h               |
|               |       |               |               | 0Eh                           | 08h               | 16h                          | 0Ch               | FREQ = 1h               |
|               |       |               |               | 26h                           | 16h               | 26h                          | 20h               | FREQ = 2h               |
|               |       |               |               | 16h                           | 0Ch               | 1Eh                          | 10h               | FREQ = 3h               |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                           |                   |                              |                   |                         |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch                           | 06h               | 10h                          | 0Ah               | FREQ = 0h               |
|               |       |               |               | 06h                           | 04h               | 0Ah                          | 06h               | FREQ = 1h               |
|               |       |               |               | 10h                           | 0Ah               | 10h                          | 0Eh               | FREQ = 2h               |
|               |       |               |               | 0Ah                           | 06h               | 0Eh                          | 08h               | FREQ = 3h               |
| Chip ID = 04h |       |               |               |                               |                   |                              |                   |                         |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                            |                   |                              |                   |                         |
|               | [5:4] | ADBIT         | 0h            | 1h                            |                   |                              |                   | 1h: 12 bit              |
| 3Ch           | [7:1] | LLBLANK       | 18h           |                               |                   |                              |                   |                         |
| 3Dh           | [0]   |               |               | 0Ch                           |                   |                              |                   |                         |
| 3Eh           | [7:2] |               |               | 0Ch                           |                   |                              |                   |                         |
| Chip ID = 06h |       |               |               |                               |                   |                              |                   |                         |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                            |                   | 2h                           |                   | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h |       |               |               |                               |                   |                              |                   |                         |
| 21h           | [7:0] |               | A9h           | 3Dh                           |                   |                              |                   |                         |
| 22h           | [7:0] |               | B0h           | B0h                           |                   |                              |                   |                         |
| 46h           | [7:0] |               | 2Ah           | 0Fh                           |                   |                              |                   |                         |
| Chip ID = 08h |       |               |               |                               |                   |                              |                   |                         |
| E8h           | [7:0] |               | 2Dh           | 13h                           |                   |                              |                   |                         |
| Chip ID = 15h |       |               |               |                               |                   |                              |                   |                         |
| 1Eh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |
| 1Fh           | [7:0] |               | 02h           | 16h                           |                   |                              |                   |                         |

(IMX537-AAMJ)



Output Image Drawing in All-pixel scan Mode  
(SLVS output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.

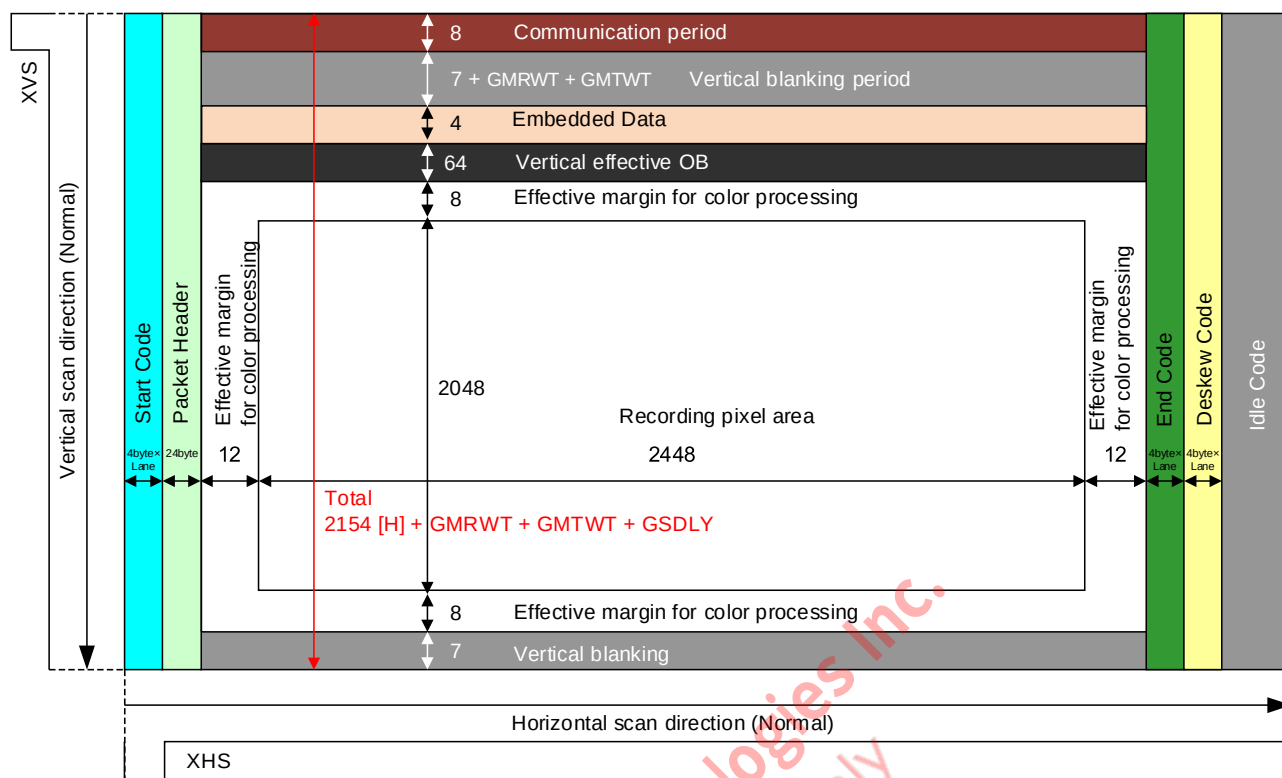


Drive Timing Chart for Serial Output in All-pixel Scan Mode (SLVS output)

## SLVS – EC output (IMX537-AAMJ)

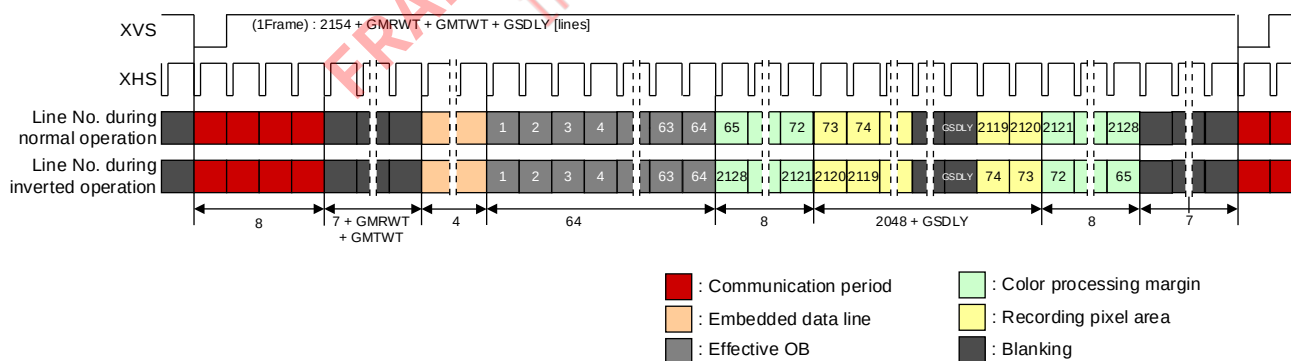
| Address         | bit   | Register name | Initial Value | Setting value                 |           |           |           |                              |           |           |            | Remarks                 |
|-----------------|-------|---------------|---------------|-------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|------------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit |           |           |           | AD = 12 bit / Output = 8 bit |           |           |            |                         |
|                 |       |               |               | 8 lane                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane     |                         |
|                 |       |               |               | 88.2                          | 88.2      | 88.2      | 54.1      | 88.2                         | 88.2      | 88.2      | 80.1       | FREQ = 0h               |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s]  |                         |
|                 |       |               |               | 88.2                          | 88.2      | 53.6      | 27.2      | 88.2                         | 88.2      | 79.4      | 40.5       | FREQ = 1h               |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s]  |                         |
|                 |       |               |               | 88.2                          | 53.4      | 27.0      | 13.6      | 88.2                         | 76.6      | 40.1      | 20.4       | FREQ = 2h               |
|                 |       |               |               | [frame/s]                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s]  |                         |
| 4752 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 0h                    |           |           |            |                         |
| 2376 Mbps/lane  |       |               |               |                               |           |           |           | FREQ = 1h                    |           |           |            |                         |
| 1188 Mbps /lane |       |               |               |                               |           |           |           | FREQ = 2h                    |           |           |            |                         |
| Chip ID = 02h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | 8A6h                          | 8A6h      | 8A6h      | 890h      | 8A6h                         | 8A6h      | 8A6h      | 8A2h       | FREQ = 0h               |
| D5h             | [7:0] |               |               | 8A6h                          | 8A6h      | 890h      | 87Eh      | 8A6h                         | 8A6h      | 8A2h      | 888h       | FREQ = 1h               |
| D6h             | [7:0] |               |               | 8A6h                          | 890h      | 87Eh      | 876h      | 8A6h                         | 8A0h      | 888h      | 87Ah       | FREQ = 2h               |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                          | 17Ch      | 17Ch      | 272h      | 17Ch                         | 17Ch      | 17Ch      | 1A3h       | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                          | 17Ch      | 277h      | 4E4h      | 17Ch                         | 17Ch      | 1A7h      | 346h       | FREQ = 1h               |
|                 |       |               |               | 17Ch                          | 27Ah      | 4EDh      | 9C7h      | 17Ch                         | 1B7h      | 34Eh      | 68Ch       | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                           | 06h       | 06h       | 04h       | 06h                          | 06h       | 06h       | 06h        | FREQ = 0h               |
|                 |       |               |               | 06h                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h        | FREQ = 1h               |
|                 |       |               |               | 06h                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h        | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                           | 26h       | 26h       | 18h       | 26h                          | 26h       | 26h       | 22h        | FREQ = 0h               |
|                 |       |               |               | 26h                           | 26h       | 18h       | 0Ch       | 26h                          | 26h       | 22h       | 12h        | FREQ = 1h               |
|                 |       |               |               | 26h                           | 18h       | 0Ch       | 06h       | 26h                          | 22h       | 12h       | 0Ah        | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                           |           |           |           |                              |           |           |            |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                           | 10h       | 10h       | 0Ah       | 10h                          | 10h       | 10h       | 10h        | FREQ = 0h               |
|                 |       |               |               | 10h                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 10h       | 08h        | FREQ = 1h               |
|                 |       |               |               | 10h                           | 0Ah       | 06h       | 04h       | 10h                          | 0Eh       | 08h       | 04h        | FREQ = 2h               |
| Chip ID = 04h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                            |           |           |           |                              |           |           |            |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                            |           |           |           |                              |           |           | 1h: 12 bit |                         |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                           |           |           |           |                              |           |           |            |                         |
| 3Dh             | [0]   |               |               |                               |           |           |           |                              |           |           |            |                         |
| 3Eh             | [7:2] |               |               | 0Ch                           |           |           |           |                              |           |           |            |                         |
| Chip ID = 06h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                            |           |           |           | 2h                           |           |           |            | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |
| 21h             | [7:0] |               | A9h           | 3Dh                           |           |           |           |                              |           |           |            |                         |
| 22h             | [7:0] |               | B0h           | B0h                           |           |           |           |                              |           |           |            |                         |
| 46h             | [7:0] |               | 2Ah           | 0Fh                           |           |           |           |                              |           |           |            |                         |
| Chip ID = 08h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                           |           |           |           |                              |           |           |            |                         |
| Chip ID = 15h   |       |               |               |                               |           |           |           |                              |           |           |            |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |            |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                           |           |           |           |                              |           |           |            |                         |

(IMX537-AAMJ)



### Output Image Drawing in All-pixel scan Mode (SLVS – EC output Shutter stabilization wait time is not included)

The AD lines for low gain side and high gain side are synthesized and output as one line.



### Drive Timing Chart for SLVS - EC Output in All-pixel Scan Mode (SLVS – EC output)

## 10.1.9.2. ROI, Overlap ROI

## SLVS Output (IMX537-AAMJ)

| Address       | bit   | Register name | Initial Value | Setting value                                                                                              |                 |                              |                 | Remarks                 |           |
|---------------|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|-----------------|-------------------------|-----------|
|               |       |               |               | AD = 12 bit / Output = 12 bit                                                                              |                 | AD = 12 bit / Output = 8 bit |                 |                         |           |
|               |       |               |               | 8 ch                                                                                                       | 4 ch            | 8 ch                         | 4 ch            |                         |           |
|               |       |               |               | *1<br>[frame/s]                                                                                            | *2<br>[frame/s] | *3<br>[frame/s]              | *4<br>[frame/s] |                         |           |
|               |       |               |               | 594 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 0h |
|               |       |               |               | 297 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 1h |
|               |       |               |               | 891 Mbps/ch                                                                                                |                 |                              |                 |                         | FREQ = 2h |
|               |       |               |               | 445.5 Mbps/ch                                                                                              |                 |                              |                 |                         | FREQ = 3h |
| Chip ID = 02h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D4h           | [7:0] | VMAX          | 0CD4h         | *1                                                                                                         | *2              | *3                           | *4              |                         |           |
| D5h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D6h           | [7:0] |               |               |                                                                                                            |                 |                              |                 |                         |           |
| D8h           | [7:0] | HMAX          | 081h          | 219h                                                                                                       | 403h            | 17Ch                         | 2B2h            | FREQ = 0h               |           |
|               |       |               |               | 416h                                                                                                       | 7DAh            | 2BEh                         | 541h            | FREQ = 1h               |           |
|               |       |               |               | 17Ch                                                                                                       | 2B2h            | 17Ch                         | 1D4h            | FREQ = 2h               |           |
| D9h           | [7:0] |               |               | 2BEh                                                                                                       | 541h            | 1DBh                         | 388h            | FREQ = 3h               |           |
| E2h           | [7:0] | GMRWT         | 10h           | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 0h               |           |
|               |       |               |               | 02h                                                                                                        | 02h             | 04h                          | 02h             | FREQ = 1h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 06h                          | 06h             | FREQ = 2h               |           |
|               |       |               |               | 04h                                                                                                        | 02h             | 06h                          | 04h             | FREQ = 3h               |           |
| E3h           | [7:0] | GMTWT         | 70h           | 1Ch                                                                                                        | 0Eh             | 26h                          | 16h             | FREQ = 0h               |           |
|               |       |               |               | 0Eh                                                                                                        | 08h             | 16h                          | 0Ch             | FREQ = 1h               |           |
|               |       |               |               | 26h                                                                                                        | 16h             | 26h                          | 20h             | FREQ = 2h               |           |
|               |       |               |               | 16h                                                                                                        | 0Ch             | 1Eh                          | 10h             | FREQ = 3h               |           |
| E5h           | [7:0] | GAINDLY       | 04h           | 02h                                                                                                        |                 |                              |                 |                         |           |
| E6h           | [7:0] | GSDLY         | 30h           | 0Ch                                                                                                        | 06h             | 10h                          | 0Ah             | FREQ = 0h               |           |
|               |       |               |               | 06h                                                                                                        | 04h             | 0Ah                          | 06h             | FREQ = 1h               |           |
|               |       |               |               | 10h                                                                                                        | 0Ah             | 10h                          | 0Eh             | FREQ = 2h               |           |
|               |       |               |               | 0Ah                                                                                                        | 06h             | 0Eh                          | 08h             | FREQ = 3h               |           |
| Chip ID = 03h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 04h to 05h    |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                     |                 |                              |                 |                         |           |
| 20h to 5Fh    |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size. Other than that, refer to the data sheet. |                 |                              |                 |                         |           |
| Chip ID = 04h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 00h           | [1:0] | AD_SEL        | 0h            | 3h                                                                                                         |                 |                              |                 |                         |           |
|               | [5:4] | ADBIT         | 0h            | 1h                                                                                                         |                 |                              |                 | 1h: 12 bit              |           |
| 3Ch           | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| 3Dh           | [0]   |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 3Eh           | [7:2] |               |               | 0Ch                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 06h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 30h           | [1:0] | ODBIT         | 0h            | 1h                                                                                                         |                 | 2h                           |                 | 1h: 12 bit<br>2h: 8 bit |           |
| Chip ID = 07h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 21h           | [7:0] |               | A9h           | 3Dh                                                                                                        |                 |                              |                 |                         |           |
| 22h           | [7:0] |               | B0h           | B0h                                                                                                        |                 |                              |                 |                         |           |
| 46h           | [7:0] |               | 2Ah           | 0Fh                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 08h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| E8h           | [7:0] |               | 2Dh           | 13h                                                                                                        |                 |                              |                 |                         |           |
| Chip ID = 15h |       |               |               |                                                                                                            |                 |                              |                 |                         |           |
| 1Eh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |
| 1Fh           | [7:0] |               | 02h           | 16h                                                                                                        |                 |                              |                 |                         |           |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

## SLVS – EC output (IMX537-AAMJ)

| Address         | bit   | Register name | Initial Value | Setting value                                                                                                 |           |           |           |                              |           |           |           | Remarks                 |
|-----------------|-------|---------------|---------------|---------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|-------------------------|
|                 |       |               |               | AD = 12 bit / Output = 12 bit                                                                                 |           |           |           | AD = 12 bit / Output = 8 bit |           |           |           |                         |
|                 |       |               |               | 8 lane                                                                                                        | 4 lane    | 2 lane    | 1 lane    | 8 lane                       | 4 lane    | 2 lane    | 1 lane    |                         |
|                 |       |               |               | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
|                 |       |               |               | [frame/s]                                                                                                     | [frame/s] | [frame/s] | [frame/s] | [frame/s]                    | [frame/s] | [frame/s] | [frame/s] |                         |
|                 |       |               |               | 4752 Mbps/lane                                                                                                |           |           |           |                              |           |           |           |                         |
| 2376 Mbps/lane  |       |               |               |                                                                                                               |           |           |           | FREQ = 1h                    |           |           |           |                         |
| 1188 Mbps /lane |       |               |               |                                                                                                               |           |           |           | FREQ = 2h                    |           |           |           |                         |
| Chip ID = 02h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D4h             | [7:0] | VMAX          | 0CD4h         | *1                                                                                                            | *2        | *3        | *4        | *5                           | *6        | *7        | *8        |                         |
| D5h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D6h             | [7:0] |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| D8h             | [7:0] | HMAX          | 081h          | 17Ch                                                                                                          | 17Ch      | 17Ch      | 272h      | 17Ch                         | 17Ch      | 17Ch      | 1A3h      | FREQ = 0h               |
| D9h             | [7:0] |               |               | 17Ch                                                                                                          | 17Ch      | 277h      | 4E4h      | 17Ch                         | 17Ch      | 1A7h      | 346h      | FREQ = 1h               |
|                 |       |               |               | 17Ch                                                                                                          | 27Ah      | 4EDh      | 9C7h      | 17Ch                         | 1B7h      | 34Eh      | 68Ch      | FREQ = 2h               |
| E2h             | [7:0] | GMRWT         | 10h           | 06h                                                                                                           | 06h       | 06h       | 04h       | 06h                          | 06h       | 06h       | 06h       | FREQ = 0h               |
|                 |       |               |               | 06h                                                                                                           | 06h       | 04h       | 02h       | 06h                          | 06h       | 06h       | 04h       | FREQ = 1h               |
|                 |       |               |               | 06h                                                                                                           | 04h       | 02h       | 02h       | 06h                          | 06h       | 04h       | 02h       | FREQ = 2h               |
| E3h             | [7:0] | GMTWT         | 70h           | 26h                                                                                                           | 26h       | 26h       | 18h       | 26h                          | 26h       | 26h       | 22h       | FREQ = 0h               |
|                 |       |               |               | 26h                                                                                                           | 26h       | 18h       | 0Ch       | 26h                          | 26h       | 22h       | 12h       | FREQ = 1h               |
|                 |       |               |               | 26h                                                                                                           | 18h       | 0Ch       | 06h       | 26h                          | 22h       | 12h       | 0Ah       | FREQ = 2h               |
| E5h             | [7:0] | GAINDLY       | 04h           | 02h                                                                                                           |           |           |           |                              |           |           |           |                         |
| E6h             | [7:0] | GSDLY         | 30h           | 10h                                                                                                           | 10h       | 10h       | 0Ah       | 10h                          | 10h       | 10h       | 10h       | FREQ = 0h               |
|                 |       |               |               | 10h                                                                                                           | 10h       | 0Ah       | 06h       | 10h                          | 10h       | 10h       | 08h       | FREQ = 1h               |
|                 |       |               |               | 10h                                                                                                           | 0Ah       | 06h       | 04h       | 10h                          | 0Eh       | 08h       | 04h       | FREQ = 2h               |
| Chip ID = 03h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 04h to 05h      |       | ROI setting   |               | About setting of enable or disable, refer to datasheet                                                        |           |           |           |                              |           |           |           |                         |
| 20h to 5Fh      |       | ROI setting   |               | Set the value of multiple of 8 about cropping position and size.<br>Other than that, refer to the data sheet. |           |           |           |                              |           |           |           |                         |
| Chip ID = 04h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 00h             | [1:0] | AD_SEL        | 0h            | 3h                                                                                                            |           |           |           |                              |           |           |           |                         |
|                 | [5:4] | ADBIT         | 0h            | 1h                                                                                                            |           |           |           |                              |           |           |           | 1h: 12 bit              |
| 3Ch             | [7:1] | LLBLANK       | 18h           | 0Ch                                                                                                           |           |           |           |                              |           |           |           |                         |
| 3Dh             | [0]   |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 3Eh             | [7:2] |               |               | 16h                                                                                                           | 0Ch       |           |           |                              |           |           |           |                         |
| Chip ID = 06h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 30h             | [1:0] | ODBIT         | 0h            | 1h                                                                                                            |           |           |           | 2h                           |           |           |           | 1h: 12 bit<br>2h: 8 bit |
| Chip ID = 07h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 21h             | [7:0] |               | A9h           | 3Dh                                                                                                           |           |           |           |                              |           |           |           |                         |
| 22h             | [7:0] |               | B0h           | B0h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 46h             | [7:0] |               | 2Ah           | 0Fh                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 08h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| E8h             | [7:0] |               | 2Dh           | 13h                                                                                                           |           |           |           |                              |           |           |           |                         |
| Chip ID = 15h   |       |               |               |                                                                                                               |           |           |           |                              |           |           |           |                         |
| 1Eh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |
| 1Fh             | [7:0] |               | 02h           | 16h                                                                                                           |           |           |           |                              |           |           |           |                         |

\*About frame rate, refer to the data sheet.

\*VMAX is as shown below.

ROI :

$$VMAX = (ROIWV1 + ROIWV2 + ROIWV3 + \dots + ROIWV8) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

Overlap ROI :

$$VMAX = (\text{Maximum vertical output width}) \times 2 + GMRWT + GMTWT + GSDLY + 90$$

## 10.2. The register settings for HMAX

When Dual ADC is used, "Register setting" in the datasheet of IMX530/531/532 must be set the registers Initial Value.

| Address | bit   |                      | Register name | Initial Value | Setting value                                                            |                                         |                                          |                       |                          | Remarks |
|---------|-------|----------------------|---------------|---------------|--------------------------------------------------------------------------|-----------------------------------------|------------------------------------------|-----------------------|--------------------------|---------|
|         |       | to<br>21Ch<br>(540d) |               |               | 21Dh<br>(541d)<br>to<br>3B4h<br>(948d)                                   | 3B5h<br>(949d)<br>to<br>7F4h<br>(2036d) | 7F5h<br>(2037d)<br>to<br>D28h<br>(3368d) | D29h<br>(3369d)<br>to | Setting range<br>of HMAX |         |
|         |       | Chip ID = 08h        |               |               |                                                                          |                                         |                                          |                       |                          |         |
| 4Ch     | [7:0] |                      |               | 00h           | When Dual ADC is used, it must be set the registers 00h (Initial Value). |                                         |                                          |                       |                          |         |
| 4Eh     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 4Fh     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 74h     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 76h     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 77h     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 9Ch     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 9Eh     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |
| 9Fh     | [7:0] |                      |               | 00h           |                                                                          |                                         |                                          |                       |                          |         |

## 10.3. The register settings for SLVS-EC output 10 bit mode (IMX535, IMX536, IMX537 only)

When SLVS-EC output 10 bit mode is used, it is necessary to set the registers shown in the table below depending on the combination of data rate 4752 Mbps/lane and the number of lanes 8.

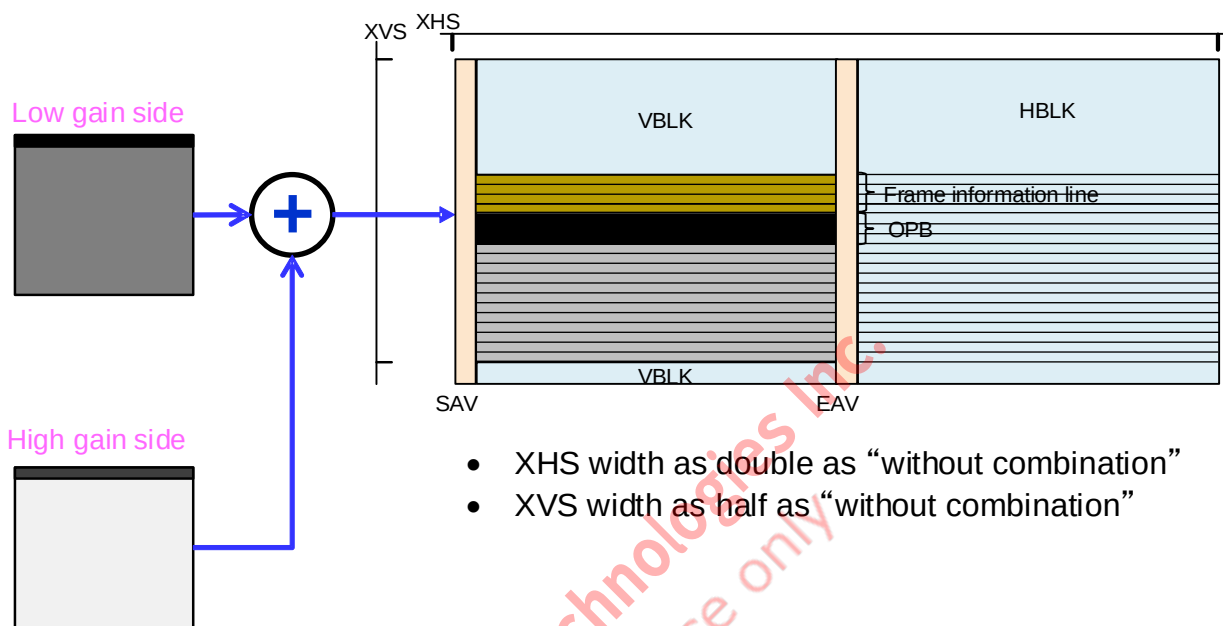
| Address       | bit   | Register name | 4752 Mbps/lane<br>8 lanes |
|---------------|-------|---------------|---------------------------|
| Chip ID = 08h |       |               |                           |
| 2Ch           | [7:0] |               | 64h                       |
| 2Eh           | [7:0] |               | 00h                       |
| 2Fh           | [7:0] |               | 00h                       |
| 54h           | [7:0] |               | 64h                       |
| 56h           | [7:0] |               | 20h                       |
| 57h           | [7:0] |               | 00h                       |
| 7Ch           | [7:0] |               | 64h                       |
| 7Eh           | [7:0] |               | 00h                       |
| 7Fh           | [7:0] |               | 00h                       |
| Chip ID = 12h |       |               |                           |
| 56h           | [7:0] |               | 23h                       |
| 96h           | [7:0] |               | 23h                       |
| Chip ID = 17h |       |               |                           |
| E6h           | [7:0] |               | 3Fh                       |
| F0h           | [7:0] |               | 95h                       |
| F2h           | [7:0] |               | 8Fh                       |
| F8h           | [7:0] |               | 93h                       |
| FAh           | [7:0] |               | 95h                       |
| Chip ID = 18h |       |               |                           |
| 04h           | [7:0] |               | 93h                       |
| 06h           | [7:0] |               | 95h                       |
| 0Ch           | [7:0] |               | 8Fh                       |

| Address       | bit   | Register name | 4752 Mbps/lane<br>8 lanes |
|---------------|-------|---------------|---------------------------|
| Chip ID = 18h |       |               |                           |
| 0Eh           | [7:0] |               | C0h                       |
| 14h           | [7:0] |               | 95h                       |
| 16h           | [7:0] |               | 8Fh                       |
| 34h           | [7:0] |               | 36h                       |
| 36h           | [7:0] |               | 95h                       |
| 3Ch           | [7:0] |               | 38h                       |
| 3Eh           | [7:0] |               | 97h                       |
| 44h           | [7:0] |               | 3Ah                       |
| 46h           | [7:0] |               | 99h                       |
| 4Ch           | [7:0] |               | 33h                       |
| 4Eh           | [7:0] |               | 39h                       |
| 54h           | [7:0] |               | 35h                       |
| 56h           | [7:0] |               | 3Ch                       |
| 5Ch           | [7:0] |               | 37h                       |
| 5Eh           | [7:0] |               | 3Eh                       |
| 6Ah           | [7:0] |               | A3h                       |
| 70h           | [7:0] |               | 9Dh                       |
| 76h           | [7:0] |               | 9Bh                       |
| 78h           | [7:0] |               | BFh                       |
| 98h           | [7:0] |               | 98h                       |
| 9Ah           | [7:0] |               | BEh                       |

## 11. Built-in combination Function (with built-in combination)

This function combines the line obtained at AD1 (low gain side) and the line obtained at AD2 (high gain side) and outputs one line at a time. The built-in combining is output as 12-bit data by gradation compression after D-Range is virtually extended by bit extension.

The frame rates of “with built-in combination” and “without combination” are almost same. In the “with built-in combination”, the number of output lines is halved compared to the “without combination”, but approximately twice the 1H time is required.

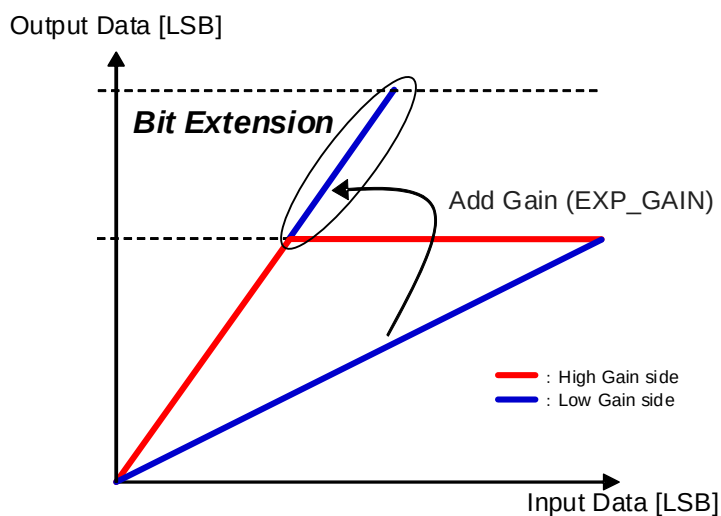


Built-in combining image

### 11.1. Bit extension (D-range extension)

The bits of data obtained with AD1 (low gain side) are extended.

At high gain, interpolation is performed by using data of the low gain side for the area where saturation occurs. D-Range is extended virtually.

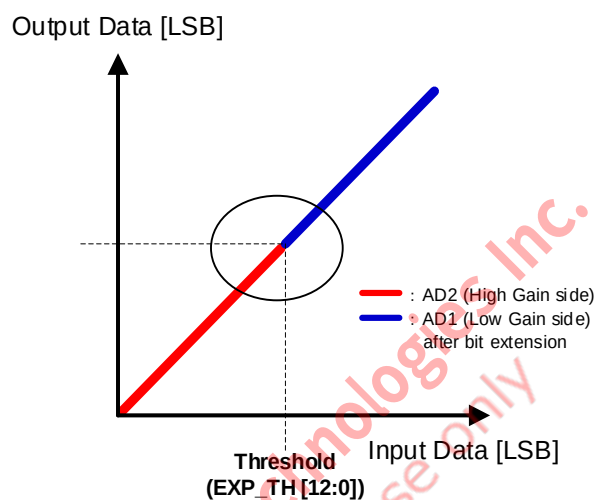


Bit extension

## 11.2. Threshold and Data selection

If the data obtained by AD2 (high gain side) has a value greater than or equal to the threshold register EXP\_TH [12:0] (Chip-ID = 04h, Address = 12h and 13h), bit-expanded AD1 (low gain side) data is selected.  
If the data is less than the threshold register, AD2 (high gain side) data is selected.

| Register      | Register details |                            |       | Initial value | Setting value                                                 |
|---------------|------------------|----------------------------|-------|---------------|---------------------------------------------------------------|
|               | Chip ID          | Address (I <sup>2</sup> C) | bit   |               |                                                               |
| EXP_TH [12:0] | 04h              | 12h (3212h)                | [7:0] | 0FFFh         | Threshold for High gain side and low gain side data switching |
|               |                  | 13h (3213h)                | [4:0] |               |                                                               |

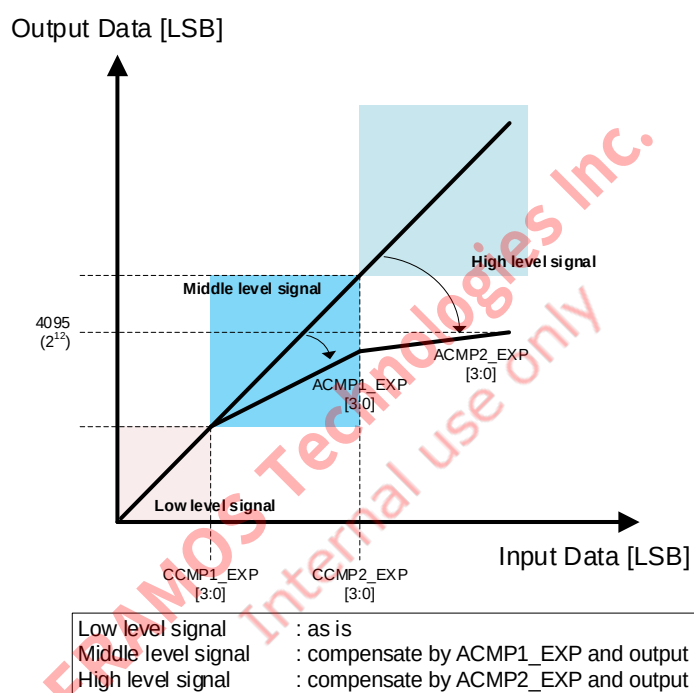


Threshold (EXP\_TH [12:0]) and Data selection

### 11.3. Gradation compression

After data selection, gradation compression processing is performed to adjust the output bit length to 12 bits. The compression position and the compression ratio starting from that position can be set up to 2 places.

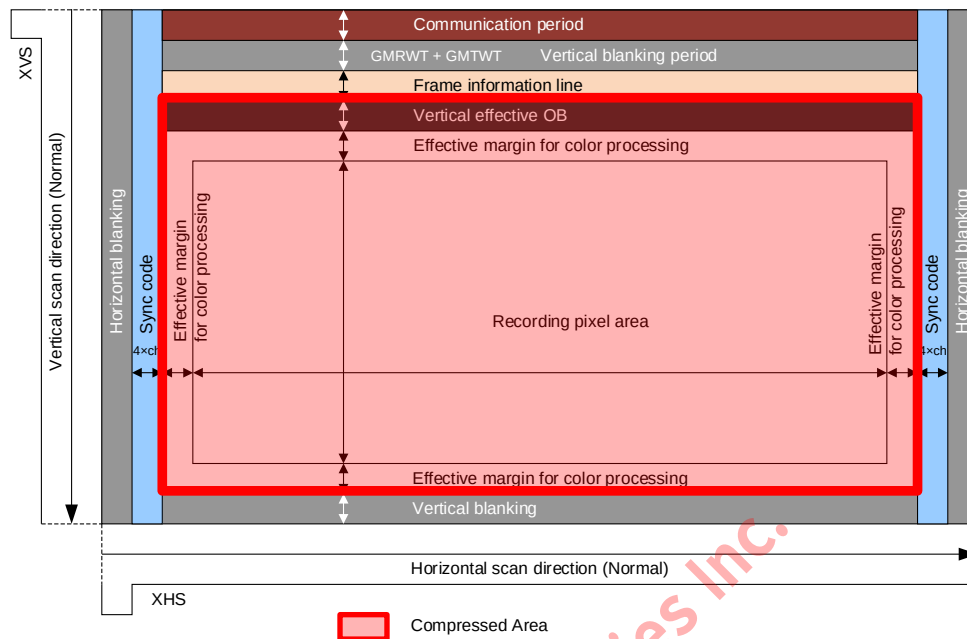
| Register        | Register details |                                     |       | Initial value | Setting value                 |
|-----------------|------------------|-------------------------------------|-------|---------------|-------------------------------|
|                 | Chip ID          | Address<br>( $\text{I}^2\text{C}$ ) | bit   |               |                               |
| CCMP1_EXP [3:0] | 04h              | 14h<br>(3214h)                      | [3:0] | 0h            | Gradation compression point 1 |
| CCMP2_EXP [3:0] |                  |                                     | [7:4] | 0h            | Gradation compression point 2 |
| ACMP1_EXP [3:0] |                  | 15h<br>(3215h)                      | [3:0] | 0h            | Gradation compression gain 1  |
| ACMP2_EXP [3:0] |                  |                                     | [7:4] | 0h            | Gradation compression gain 2  |



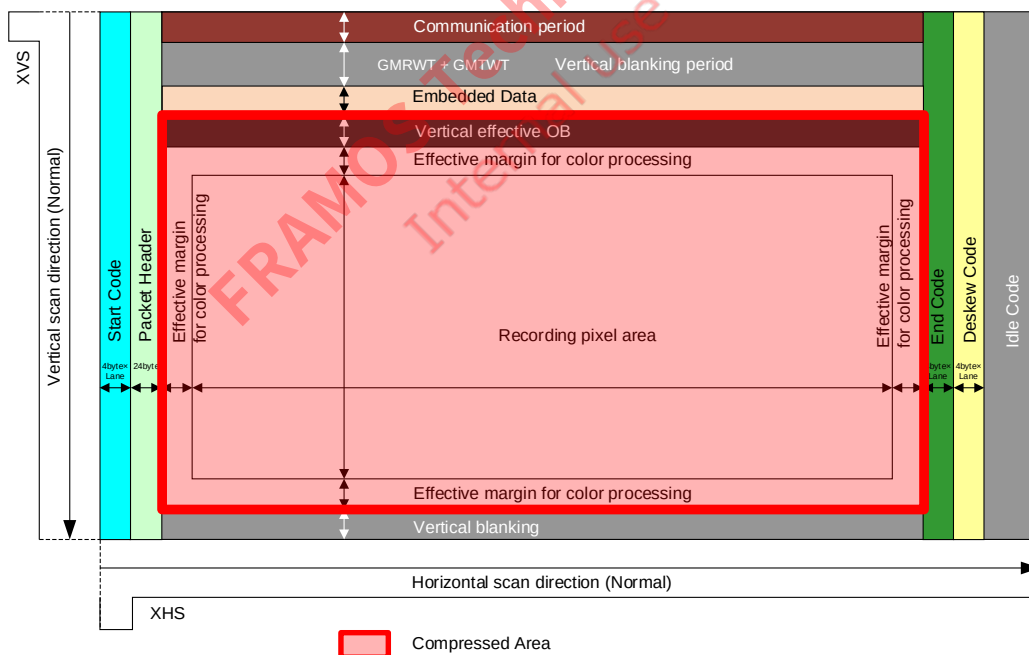
Gradation compression

#### 11.4. Area of Gradation compression

The area of Gradation compression is as below.



Area of Gradation compression (SLVS)



Area of Gradation compression (SLVS – EC)

#### 11.5. Supports A/D 12 bits / output 8 bits

In Dual ADC (with built-in combination), data of A/D 12 bits can be output as 8 bits.

In this case, after the 12-bit data is output using the built-in combining function described in this document, the data compressed to 8 bits is output by the gradation compression function different from that for built-in combining. Please refer to the separate application note for this gradation compression function.

## 12. Gain Adjustment Function (without combination / with built-in combination)

The normal register GAIN [8:0] (CID = 07h Address = 14h, 15h) is valid for both AD1 (low gain side) and AD2 (high gain side) outputs.

In Dual ADC mode, set the maximum analog gain to AGAIN [8:0] (CID = 07h Address = 04h, 05h). Up to 24 dB can be set to AGAIN [8:0]. Since the digital gain can be set up to 24 dB, the total gain setting range is 48 dB (AGAIN [8:0] + 24 dB).

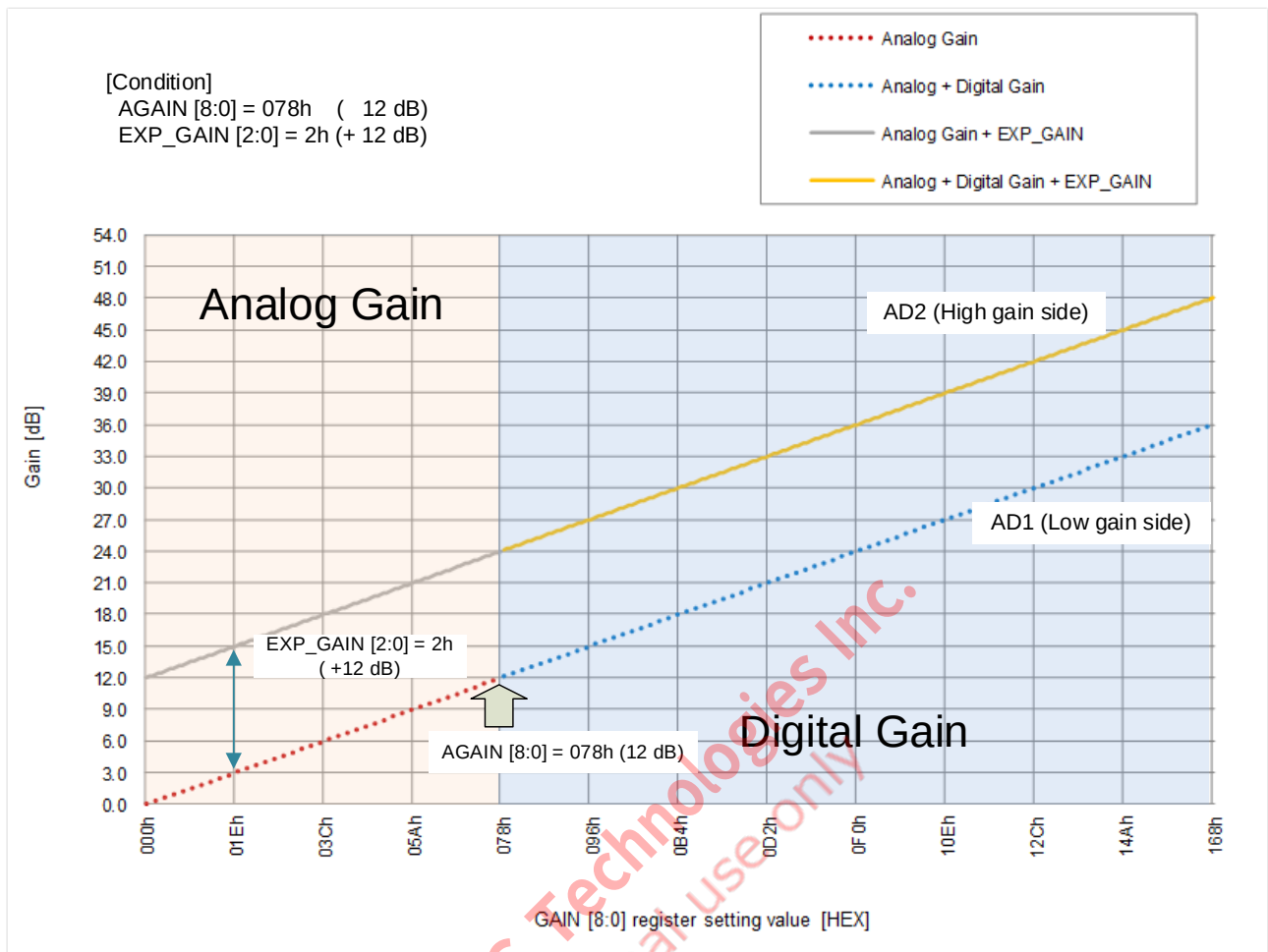
The high gain side adds the setting of EXP\_GAIN [2:0] (CID = 04h, Address = 10h) to the analog gain. The analog gain will be up to 24 dB, so set within the range where the value obtained by adding EXP\_GAIN [2: 0] does not exceed 24dB.

### Gain registers

| Register       | Register details |                                  |       | Initial value | Setting value                                                                                                                     |
|----------------|------------------|----------------------------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                | Chip ID          | Address<br>( ): I <sup>2</sup> C | bit   |               |                                                                                                                                   |
| EXP_GAIN [2:0] | 04h              | 10h<br>(3210h)                   | [2:0] | 00h           | Gain setting added to high gain side<br>0h: 0 dB   1h: +6 dB   2h: +12 dB   3h: +18 dB<br>4h: +24 dB   Others: Setting prohibited |
| AGAIN [8:0]    | 07h              | 04h<br>(3504h)                   | [7:0] | 0F0h          | Maximum analog gain setting<br>0 dB to 24 dB (240d)   0.1 dB Step                                                                 |
|                |                  | 05h<br>(3505h)                   | [0]   |               |                                                                                                                                   |
| GAIN [8:0]     |                  | 14h<br>(3514h)                   | [7:0] | 00h           | Gain setting<br>0 dB to “AGAIN + 24 dB”   0.1 dB Step                                                                             |
|                |                  | 15h<br>(3515h)                   | [0]   | 0h            | Maximum gain: 48 dB (480d)                                                                                                        |

The following shows the relationship between the total gain, analog gain, and digital gain in Dual ADC.

| Condition    | Analog gain   |                                                     | Digital gain  |                |
|--------------|---------------|-----------------------------------------------------|---------------|----------------|
|              | Low gain side | High gain side                                      | Low gain side | High gain side |
| GAIN < AGAIN | GAIN          | GAIN + EXP_GAIN<br>*Set "GAIN + EXP_GAIN ≤ 24dB".   | 0 dB          |                |
| GAIN ≥ AGAIN | AGAIN         | AGAIN + EXP_GAIN<br>*Set "AGAIN + EXP_GAIN ≤ 24dB". | GAIN – AGAIN  |                |



## Gain control

### 13. Mode Transition

The transition related to Twice Exposure mode is shown below. If perform the transition which is not described in table below, set via sensor standby.

#### 13.1. Mode Transitions

| Mode Transitions                           |   | Note                                       |
|--------------------------------------------|---|--------------------------------------------|
| Change of operation mode                   |   |                                            |
| Normal mode (XVS/XHS sync)                 | → | Dual ADC mode (VS/XHS sync)                |
| Dual ADC mode (VS/XHS sync)                | → | Normal mode (XVS/XHS sync)                 |
| Sequential Trigger mode                    | → | Dual ADC mode (Sequential Trigger mode)    |
| Dual ADC mode (Sequential Trigger mode)    | → | Sequential Trigger mode                    |
| Fast Trigger mode                          | → | Dual ADC mode (Fast Trigger mode)          |
| Dual ADC mode (Fast Trigger mode)          | → | Fast Trigger mode                          |
| Change of Readout mode                     |   |                                            |
| All-pixel                                  | → | ROI                                        |
| ROI                                        | → | All-Pixel                                  |
| ROI                                        | → | ROI                                        |
| Others                                     |   |                                            |
| With built-in combination in Dual ADC mode | → | Without Combination in Dual ADC mode       |
| Without Combination in Dual ADC mode       | → | With built-in combination in Dual ADC mode |

Via the standby state is necessary.

Via the standby state is unnecessary.  
Invalid frame: 0

Via the standby state is necessary.

### 14. Note

- It does not support driving with Dual Trigger mode.
- It does not support driving with Self Trigger mode.
- It does not support driving with Short Interval Shutter mode.
- It supports driving with Multi Frame Set output mode. (Only XVS/XHS sync)
- It supports driving with Multi Exposure Trigger mode. (Only Sequential Trigger mode and Fast Trigger mode)

## Revision History

| Version | Date           | Page                                                                 | Remarks                                                                                                        |
|---------|----------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Rev.0.1 | 2019 / 4 / 26  | –                                                                    | First Edition                                                                                                  |
| Rev.0.2 | 2019 / 6 / 28  | 1                                                                    | Modified Dual ADC driving method                                                                               |
|         |                | 3                                                                    | Modified the explanation of AD_SEL register<br>Modified the explanation of CMP_EN_EXP register                 |
|         |                | 6                                                                    | Modified the explanation of Overview                                                                           |
|         |                | 7                                                                    | Modified the title of section 5                                                                                |
|         |                | 12                                                                   | Modified the title of section 6                                                                                |
|         |                | 18                                                                   | Modified the title of section 7                                                                                |
|         |                | 20                                                                   | Modified the title of section 8                                                                                |
|         |                | 24                                                                   | Modified the title of section 9                                                                                |
|         |                | 30                                                                   | Modified the title of section 10<br>Modified the explanation of Built-in combination Function                  |
|         |                | 33                                                                   | Modified the explanation of “Supports A/D 12 bits / output 8 bits”                                             |
|         |                | 34                                                                   | Modified the title of section 11                                                                               |
|         |                | 36                                                                   | Modified the Mode Transitions                                                                                  |
| Rev.1.0 | 2020 / 11 / 25 | All pages                                                            | Add the types IMX536-AAQJ and IMX532                                                                           |
|         |                | 1                                                                    | Added Chapter “Product type”                                                                                   |
|         |                | 5                                                                    | Delete the MONOSEL register                                                                                    |
|         |                | 6 to 8                                                               | Added the registers                                                                                            |
|         |                | 10 to 14                                                             | Update IMX530 list                                                                                             |
|         |                | 15 to 24                                                             | Add the types IMX531 and IX532 to Readout Drive mode                                                           |
|         |                | 25, 27, 29, 30                                                       | Add GAINLY, AD_SEL, ADBIT, ODBIT, LLBLANK and Chip ID=08h/Address=E8h registers to Drive mode setting          |
|         |                | 31 to 42                                                             | Add the types IMX531 and IX532 to Drive mode setting                                                           |
|         |                | 45 to 48                                                             | Update IMX530 list                                                                                             |
|         |                | 49 to 56                                                             | Add the types IMX531 and IX532 to Readout Drive mode                                                           |
|         |                | 57, 59, 61, 62                                                       | Update IMX530 list                                                                                             |
|         |                | 58, 60                                                               | Update the Image Drawing and Drive Timing Chart                                                                |
|         |                | 63 to 74                                                             | Add the types IMX531 and IX532 to Drive mode setting                                                           |
|         |                | 76, 77                                                               | Add the description of registers                                                                               |
|         |                | 78                                                                   | Delete the section “Register setting value”                                                                    |
| Rev.2.0 | 2021 / 3 / 18  | 1, 2, 9, 10,<br>27 to 56,<br>75 to 110,<br>126 to 149,<br>168 to 203 | Added: IMX535 series                                                                                           |
|         |                | 7                                                                    | Added: Added the explanation of the register setting values of CCMP1_EXP, CCMP2_EXP, ACMP1_EXP, and ACMP2_EXP. |

| Version | Date          | Page                                                             | Remarks                                                                                                                                                                                                                                                                             |
|---------|---------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rev.2.0 | 2021 / 3 / 18 | 18, 49, 89                                                       | Correction: Frame rate value                                                                                                                                                                                                                                                        |
|         |               | 207                                                              | Correction: Maximum analog gain<br>30dB → 24dB<br>Delete: From the relationship table of total gain, analog gain, and digital gain<br>The condition of "AGAIN + EXP_GAIN ≥ 30dB" has been deleted.<br>Added: Constraint that the value obtained by adding EXP_GAIN is 24 dB or less |
|         |               | 9, 209                                                           | Correction: Gain size<br>GAIN [9:0] → GAIN [8:0]                                                                                                                                                                                                                                    |
|         |               | 111, 204                                                         | Added: SLVS-EC Output register setting in 10-bit mode                                                                                                                                                                                                                               |
|         |               | 58, 60, 64,<br>66, 70, 71,<br>151, 153,<br>157, 159,<br>163, 165 | Correction: blanking size                                                                                                                                                                                                                                                           |
| Rev.3.0 | 2021 / 7 / 2  | 6 to 28                                                          | Added the registers                                                                                                                                                                                                                                                                 |
|         |               | 77, 80, 83,<br>86, 92                                            | Update "Register setting value"                                                                                                                                                                                                                                                     |
| Rev.4.0 | 2022 / 4 / 28 | 12 to 22                                                         | Correction: Description of the value to specify for the ROI area<br>"Set the value of multiple of 16" to "Set the value of multiple of 8"                                                                                                                                           |
|         |               | 24                                                               | Correction: Register setting values and explanations for CCMP1_EXP and CCMP2_EXP                                                                                                                                                                                                    |
| Rev.5.0 | 2022 / 7 / 8  | 200, 203,<br>206, 209,<br>215, 221                               | Correction: Register setting value of "Chip ID = 07h"                                                                                                                                                                                                                               |